

Rust Core

Draft edition – language fundamentals for programmers coming
from another language

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Preface

0.1 Draft edition

Rust Core is a **draft edition** of the Vibe Learning language track. Chapters are usable end-to-end, but the text is not final: expect edits to examples, cross-links, and the production checklist (Chapter 20). When in doubt, confirm against the [official Rust Book](#) (2024 edition) and compiler output.

0.2 Who these notes are for

You already program in at least one language. You do not need a tutorial that starts with `for` loops. You need a **paradigm map**: how Rust thinks about memory, types, concurrency, and errors — and how to write **idiomatic** Rust instead of transliterating syntax from whatever you know today.

Backend developers, CLI authors, and curious systems programmers are all welcome. **Rust Core** is the language-fundamentals track inside **Vibe Learning**. Part V applies ideas to `std I/O`; Parts I-IV stand on their own.

Where tables compare **Java** and **Python**, treat them as **optional anchors** — handy if you know those languages, not required to follow the notes.

0.3 Why Rust is a different bet

Most languages make you pick a poison you already know: stay close to the metal and spend years chasing memory bugs, or buy safety from a **GC** or a **VM** and pay latency, pauses, and opaque runtime behavior. Rust

breaks that menu. **Ownership** and **borrow checking** reject huge classes of use-after-free, data race, and aliasing mistakes at **compile time** — with no garbage collector, no mandatory runtime, and no surrender of layout, syscalls, or `no_std` control. That is not a faster C or a stricter Java. It is a different contract between you, the compiler, and the machine.

The contract scales across surfaces that used to demand different stacks. The same language can run on an **ARM64** cloud instance, serve HTTP with **Axum**, embed as a **WebAssembly** module in the browser or beside **Node** via **N-API** / **FFI**, ship a desktop shell with **Tauri**, and still compile firmware for **embedded** targets or a daemon that talks to hardware. You are not maintaining four mental models for four deployment shapes — you are reusing types, tests, and habits.

At runtime Rust stays **lean**: release builds are native **machine code**, not bytecode behind another interpreter. At compile time the language is **expressive**: **algebraic enums**, **traits** instead of inheritance tax, **generics** and **iterators** that optimize like hand-written loops, **macros** when the type system needs a lever. Small binary and rich syntax are not opposites here — zero-cost abstractions and monomorphization are the point.

The industry vote is already in the tree, not in slide decks. **Firefox** shipped memory-safe components from the Servo line; **Windows** and **Android** move security-critical paths to Rust; Chromium hardens parsers at the Rust boundary. In the **Linux kernel**, Rust entered in **6.1** (2022); at the **2025 Kernel Maintainers Summit** it graduated from *experimental* to a **core kernel language** alongside C and assembly — with production drivers and subsystems on the path maintainers expect to keep. You are not learning a hype cycle. You are learning the language those merges already assume.

That is why these notes exist: not to sell Rust in the abstract, but to hand you the paradigm map before the compiler enforces it.

0.4 Production standards — what compile time does not teach

Most Rust material stops when the program builds. Production does not. Reviewers ask whether your error types belong in a public API, whether that `.clone()` hides a borrow smell, whether two `u8` parameters can be

swapped without the compiler noticing, and whether a workspace crate still shares dependencies correctly after the last refactor. Those questions rarely appear in language tutorials. They surface in merge requests, on-call threads, and code you maintain for years under enterprise SLAs.

Chapter 20: Production Rust standards closes that gap. It is a review checklist — typed boundaries, panic-free paths, deliberate allocation, workspace and test conventions — distilled from my own work shipping Rust in **production and enterprise** teams. The patterns there are not generic best-practice quotes recycled from docs. They are habits I apply after every diff: anti-patterns that compiled cleanly but failed in staging, conventions that kept multi-crate repos reviewable, and the bar I use before calling code merge-ready.

Parts I-IV give you the paradigm map. Chapter 20 gives you the production bar. Read it when you are ready to write Rust that survives code review, not just cargo check.

0.5 What you will not find here

- A line-by-line clone of *The Rust Book* (excellent; use it as reference — current stable edition is **2024**, Rust 1.90+)
- Framework churn (Axum vs Actix debates)
- Long proofs about memory models
- Long prosaic-style text (density over narrative padding)

0.6 What these notes are (and are not)

These are **lecture notes**, not a textbook. Treat each chapter like dense slides from a course: a map of ideas, runnable examples, and pointers for self-study — not exhaustive reference pages you read cover to cover once and shelve.

These notes are:

- paradigm maps for programmers coming from any mainstream language
- optional **Java** / **Python** comparison tables where they clarify a habit
- **Playground** snippets you can run, break, and fix
- edge-case drills through compiler errors — Rust is about restrictions; fluency is understanding the compiler's logic, not just syntax

- hooks for drill and review (Afterparty prompts, crosslinks, optional deep dives)

These notes are not:

- a hand-holding tutorial that explains every keyword in order
- a substitute for the official book or compiler documentation

AI drafts code; you still read, review, and steer. Success depends on three skills:

- you **understand** what the machine is doing
- you **feel** when a design is wrong
- your **technological thinking** is sharp enough to steer the result

These notes are written for that workflow.

0.7 Try it in the Rust Playground

Most examples are tagged **Playground** — paste them into the online editor and click **Run**:

<https://play.rust-lang.org/>

Tweak values, remove a semicolon, borrow after a move. The compiler errors are part of the lesson. You do not need a local install to start Part I.

Use the playground	Use Cargo locally
Syntax, ownership, types	Files, Command, serial ports
Quick experiments (< 50 lines)	External crates (tokio, serialport)
Sharing a link in chat or notes	Integration tests, multi-file crates

Rules for playground snippets: one file with `fn main()`, **std only**, no filesystem or subprocess APIs. Full details: Playground Guide.

0.8 How to work through a chapter

1. **Read once for the map** — skim headings and tables; do not memorize every line.
2. **Run the Playground example** — confirm output, then break it on purpose.

3. **Prompt what is still fuzzy** — open a **new chat for this chapter**, paste the [starter context](#), then a micro-prompt or Afterparty drill.
4. **Move on** — revisit later via AI Prompt Index IDs (**P001** onward).

0.9 Afterparty: aim, importance, and how to use

At the end of every chapter you will find **Afterparty** — numbered, copy-paste prompts for an AI tutor.

These notes give you a **map**, not an encyclopedia. No book — not even a full reference — can hold everything Rust can do or everything you might need in production. The space of possible knowledge is larger than any single track. Memorizing it all is neither required nor realistic.

Afterparty is how you choose your depth. Each prompt is a Lego block for **practice**: one angle on the chapter — a quiz, an error to decode, a comparison, a mini-design. Paste it, push back, verify in the playground, stop when it clicks — or keep going until it sticks. **You** set the pace; the book does not pretend to exhaust the topic.

Compare that to **Skills Lego blocks** — capability you assemble over time outside any one chapter: reading a crate, wrapping FFI safely, designing an error enum for a gateway, tuning Tokio under load. Those skills are **optional extensions**. They stack from prompts, side projects, and real code — not from reading cover to cover. Afterparty starts the stack; Skills blocks are what you add when a job or curiosity demands more.

This track ships **prompts**, not a fixed skill tree. That is intentional: the capacity available to you (AI tutor, docs, crates, production code) is greater than what fits in these pages. Prompts are the interface. Depth is your decision.

0.9.1 Aim

Afterparty drills the same concepts from different angles: quizzes, error decoding, comparisons, mini-design exercises. Each prompt is a complete sentence you paste into a chat — no setup required.

0.9.2 Importance

Afterparty is not a shortcut around learning Rust. It builds the skill that matters in an AI-assisted workflow:

- **interrogate** generated answers instead of accepting them
- **catch** ownership and type mistakes before they ship
- **stress-test** mental models that prompts alone cannot replace

Use the model as a sparring partner. You stay responsible for correctness and idiomatic style.

0.9.3 How to use

One chapter, one chat. Start a fresh conversation for each chapter. Keep Afterparty prompts in that thread; open a new chat when you move on — old ownership context will confuse async topics.

Session flow (after [reading and running examples](#)):

1. Paste the **starter context** below once.
2. Paste **one** Afterparty prompt; verify any code before the next.
3. Push back on vague answers; ask for exact compiler errors.
4. Repeat step 2 in the same chat until done.

Starter context — paste once at the top of each chapter chat:

```
You are my Rust tutor. I am working through
    Rust Core.
I already program in: [your language(s) ---
    e.g. Python, C++, Go].
```

Answer rules:

- Rust edition 2024, stable toolchain unless I ask otherwise
- Prefer single-file Playground snippets (fn main, std only) unless I say Cargo
- Quote compiler errors literally; explain ownership/borrow moves step by step
- Do not invent crates or APIs; if unsure, say so
- I will verify your code --- keep snippets short and runnable

Current chapter: [number and title --- e.g.
Chapter 1 Paradigm shift]
Topics I just read: [one line from the chapter
hook, or paste a section heading]

When to start a new chat instead of continuing:

Situation	Action
Finished the chapter’s Afterparty	New chat for the next chapter
Model repeats wrong advice after you corrected it twice	New chat + shorter starter context
You jumped to a unrelated topic (e.g. Tokio while still on lifetimes)	New chat scoped to one chapter
Context is huge and answers feel generic	New chat; paste only the section you are stuck on

What good answers look like: concrete Rust code, named compiler errors (E0382, E0502), and “this fails because ...” — not hand-wavy “Rust is strict about memory.” **Your job** is to run the snippet and say “that did not compile — here is the error” until the explanation matches reality.

0.9.4 Where to find prompts

Each chapter ends with its own block. All prompts are numbered globally in appendix/AI_PROMPT_INDEX.md (**P001** onward) so you can reuse them later without rereading the chapter.

0.10 When something is unclear, prompt it

You do **not** need to wait for Afterparty. Any paragraph, table row, compiler error, or half-understood example is fair game — especially the parts that feel unclear after a first read.

Micro-prompt recipe:

1. **Quote** the passage, snippet, or error text.
2. **State** what you expected to happen.
3. **Ask** for a concrete answer: the exact compiler message, one sentence with an analogy from a language you know (e.g. Java or Python), or a fixed 5-line snippet.

Example you can paste today:

“In Chapter 2, the notes say to prefer `&str` in parameters and `String` when storing. I am building a config parser. Explain why that split matters — give one function signature for `parse_line` and show what breaks if I use `String` everywhere.”

You remain the **reader and reviewer**. The model drafts explanations; you verify against the playground or `cargo check`. That habit turns lecture notes into understanding faster than rereading the same paragraph five times.

0.11 Toolchain: rustup and Cargo

Install Rust with **rustup** (toolchain manager) and build with **Cargo** (package manager, build system, test runner):

```
curl --proto '=https' --tlsv1.2 -sSf
  https://sh.rustup.rs | sh
cargo new my_app --bin
cd my_app && cargo run
```

Concept	In Rust
Package manager	Cargo — deps, build, test, run
Manifest	<code>Cargo.toml</code> — name, version, edition, dependencies
Lockfile	<code>Cargo.lock</code> — pinned versions for reproducible builds
Entry point	<code>fn main()</code> in <code>src/main.rs</code>
Crate registry	crates.io

Every binary crate follows the same shape: one `Cargo.toml`, one `src/main.rs`, commands through `cargo run` and `cargo test`. Prefer that workflow over invoking `rustc` directly.

`env!("CARGO_PKG_NAME")` embeds a **compile-time** string from `Cargo.toml` into your binary — not the same as reading OS environment variables at runtime (`std::env::var`).

Cargo only — first project locally:

```
# Cargo.toml
[package]
```

```
name = "hello_rust"
version = "0.1.0"
edition = "2024"

// src/main.rs
fn main() {
    println!("Hello from {}",
        env!("CARGO_PKG_NAME"));
}
```

One binary, one manifest. Commit `Cargo.lock` for applications (CLIs, services); library crates often omit it from version control.

0.12 Conventions

How the notes are formatted — not Rust syntax itself.

Convention	Meaning
Playground / Cargo only	Every code block is tagged. Playground = one <code>fn main()</code> , <code>std</code> only, play.rust-lang.org . Cargo only = needs <code>Cargo.toml</code> , <code>filesystem</code> , <code>subprocess</code> , or external crates. Details: STYLE_GUIDE.md , Playground Guide .
Chapter shape	Hook -> sections with examples -> Idiom spotlight -> Go deeper -> See also -> Afterparty .
Levels	Many chapters label examples Level 1 . . . N . Run the snippet, then read What happened before the next level.
Crosslinks	Relative paths (<code>08_errors_and_testing.md</code>). Chapter numbers match <code>CONTENTS.md</code> .
Java / Python columns	Optional in tables — skip if you do not need the analogy.
<code>unwrap()</code> in examples	Avoided in teaching snippets unless the section is about <code>panic</code> or <code>Result</code> (Chapter 8).
Prompt IDs	Afterparty lines are catalogued as P001+ in AI Prompt Index for reuse across chats.

Convention	Meaning
Edition	Rust 2021 , stable toolchain, unless a note says otherwise.

0.13 Reading ahead

Some syntax appears in examples before its dedicated chapter:

- **?** — “If this fails, return the error to the caller.” Explained fully in Chapter 8: Errors and testing. Until then, read it as a placeholder, not something you must master on day one.

0.14 See also

- CONTENTS.md — full map
- Playground Guide — rules and stand-ins
- AI Prompt Index — all Afterparty prompts (**P001+**)
- Java/Python/Rust map — optional one-page cheat sheet
- Chapter 1: Paradigm shift — start here after this page

Chapter 1

Paradigm Shift

1.1 Hook

Many mainstream languages let you **share** data freely: references, object graphs, mutable globals — **Java** and **Python** are familiar examples. Rust says: **one owner at a time**, checked at compile time.

That rule replaces a GC and prevents most data races. Learn to read compiler errors as hints, not obstacles.

1.2 Compiled vs interpreted

Aspect	Java	Python	Rust
Execution	Bytecode on JVM	Interpreter / bytecode	Native machine code
Type checks	Mostly compile-time	Runtime	Compile-time
Memory	GC	GC (refcount + cycle GC)	Ownership + deterministic drop

Rust has **no GC**. When a value's owner goes out of scope, `drop` runs immediately. No stop-the-world pauses — valuable for long-running loops and low-latency tools.

1.3 Ownership vs garbage collection

Python: `b = a` makes two names point at one list; mutating through `b` affects `a`.

Java: references to the same object; GC reclaims when unreachable.

Rust: `let s2 = s1` for a `String` **moves** ownership. `s1` is invalid afterward unless you borrow or clone.

```
// Playground
fn main() {
    let s1 = String::from("hello");
    let s2 = s1; // move
    println!("{}", s2);
    // println!("{}", s1); // would not compile
}
```

Think of a move as **renaming** the value, not copying.

1.3.1 Move edge cases (what the compiler catches)

Situation	Java / Python	Rust
Reassign name to new value	old object GC'd when unreachable	old owner dropped when <code>let s1 = ...</code> overwrites <code>s1</code>
Pass to function by value	reference still in caller	move — caller binding dead unless returned
Use after move	often still works	compile error

```
// Playground --- uncomment one failing line at
    a time
fn main() {
    let s1 = String::from("plc_a");
    let s2 = s1; // move: s1 -> s2
    // println!("{}", s1);
    // ERROR: use of moved value: `s1`

    let mut log = String::from("start");
    log = String::from("replaced");
    // drop old "start" buffer, bind new owner
```

```
println!("{}", log);  
}
```

Wrong — use heap value after moving into a function:

```
// Playground --- does not compile  
fn consume(s: String) {  
    println!("{}", s);  
}  
  
fn main() {  
    let label = String::from("plc1");  
    consume(label);  
    println!("{}", label);  
    // ERROR: borrow of moved value: `label`  
}
```

Idiomatic fixes: borrow for read (`consume_borrow(&label)`), take and return (`fn transform(s: String) -> String`), or `.clone()` when you truly need two owners (sparingly).

1.4 Zero-cost abstractions

Rust’s iterators, traits, and generics are designed to compile down to code as tight as hand-written C **when you use release builds** (`cargo build --release`). You do not pay for “elegant” APIs at runtime the way heavy OOP patterns can cost in Java.

1.5 Fearless concurrency (preview)

The same borrow rules that prevent use-after-free also prevent **data races** in safe Rust: you cannot mutate shared state from two threads without synchronization (`Mutex`, channels, atomics — Part III). The compiler enforces this; you do not rely on discipline alone.

1.6 Stack and heap

Ownership makes more sense once you see **where** values live. Rust uses two main memory regions:

- **Stack** — per-function frames, very fast allocate/deallocate (pointer bump). Size known at compile time.
- **Heap** — shared pool for data whose size is unknown or may grow. Slower; requires an explicit allocator and later **free** (in Rust: **drop** when the owner leaves scope).

In many GC-managed languages (Java, Python, C#, ...) you rarely think about stack vs heap: the runtime puts much of the data on the heap and cleans up later. Rust puts **small, fixed-size data on the stack by default** and uses the heap only when the type needs it (String, Vec, and similar).

Aspect	Java	Python	Rust
int / small number	often stack (local) or heap (object)	heap (PyObject)	stack (i32, Copy)
growable text	heap (String object)	heap (str object)	stack struct pointing at heap buffer
who frees heap?	GC	GC	owner's drop at end of scope
local variable cost	reference push	name binding	often zero-cost stack slot

1.6.1 Stack: fast and scoped

Each function call gets a **stack frame**. Locals like i32, bool, and tuples of Copy types sit there. When the function returns, the frame is popped — no separate “free” step.

```
// Playground
fn stack_demo() -> i32 {
    let a: i32 = 10;
    let b: i32 = 32;
    a + b
    // a and b die when this function returns
}

fn main() {
    println!("{}", stack_demo());
}
```

`a` and `b` die when `stack_demo` returns — their stack slots vanish with the frame. You still see **42** because `a + b` is not asking the caller to keep `a` or `b`. It **computes a new value**, and that value becomes the **return value** before the frame is destroyed.

Order matters: Rust evaluates `a + b`, places **42** in the return slot (for `i32`, a bitwise **copy**), then pops the frame and drops `a` and `b`. `main` receives the copied **42** and passes it to `println!`. The locals are gone; the **result** escaped via `return`.

“Die” means the **bindings** and their stack slots are destroyed — not that every value computed inside the function disappears. Only what you **return** (or otherwise move out) survives past the closing brace. With heap-backed types like `String`, the same rule applies: the owner binding dies, but the data can live on if ownership **moves** into the caller’s binding.

Nested blocks shrink live ranges — useful for [borrowing](#) below:

```
// Playground
fn main() {
    let outer = 1;
    {
        let inner = 2;
        println!("{}", outer, inner);
    } // inner dropped here
    println!("outer still live: {}", outer);
}
```

1.6.2 Heap: growable data with an owner

Types like `String` and `Vec` store **metadata on the stack** (pointer, length, capacity) and **payload on the heap**. The stack part is small and fixed; the heap part can grow.

Mental picture for `let s = String::from("hi")`:

stack		heap
+-----+		+-----+
s: String	--ptr----->	h i
len: 2		+-----+
cap: 2		
+-----+		

When `s` goes out of scope, Rust runs `drop` on `String`, which frees the heap buffer. No GC scan — cost is **predictable** and tied to scope.

```
// Playground
fn main() {
    let x: i32 = 42;
    // entirely on stack
    let s = String::from("hi");
    // stack handle -> heap bytes
    let r = &s;
    // borrow: no heap copy --- see
    // [References](#references-borrowing-and-de
    // referencing)
    println!("{}", x, r);
} // s dropped here -> heap "hi" freed
```

1.6.3 Moves and the heap

Assigning one `String` to another **moves** the stack handle (pointer/len/cap) to the new owner; the heap buffer is **not** copied. That is why `s1` becomes invalid after `let s2 = s1` — there is only one owner responsible for freeing that heap memory.

```
// Playground
fn main() {
    let s1 = String::from("hello");
    let s2 = s1;
    // move handle; heap buffer stays in place
    println!("{}", s2);
    // println!("{}", s1);
    // error: s1 no longer owns the heap data
}
```

Java would copy a **reference** (two refs, one object). Python would bind another name to the same object. Rust **transfers responsibility** — fewer copies, stricter rules.

1.6.4 Copy on stack vs heap-backed types

1.6.4.1 Why Rust splits the world this way

Rust’s default is **move** (transfer ownership). **Copy** is a narrow, opt-in exception — not an arbitrary rule:

Every value has exactly one owner who is responsible for cleaning up any resources it holds.

If assignment silently duplicated a `String` or `Vec` handle, you would get **two owners for one heap allocation**. When both go out of scope, Rust would free the same memory twice — a classic double-free. `C` and `C++` leave that to discipline; Rust makes the dangerous case **illegal at compile time**.

Copy is allowed only when a bitwise duplicate is **semantically identical** to the original: the type is self-contained, has no `Drop` that must run once, and duplicating it never shares external memory. Integers and floats are just stack bits — cheap and safe to copy. Heap-backed types **own** external memory, so the compiler **moves** unless you explicitly `.clone()`.

Rust treats ownership transfer as the normal case (handing someone a key) and silent duplication as a special case for values that cannot alias. Assignment feels “strict” compared to languages with implicit sharing because Rust prefers **provable correctness** over silent aliasing.

1.6.4.2 Which types are Copy vs move?

Copy types (assignment duplicates bits; both variables stay valid):

Category	Examples
Integer scalars	<code>i8</code> , <code>i16</code> , <code>i32</code> , <code>i64</code> , <code>i128</code> , <code>isize</code> , <code>u8</code> , <code>u16</code> , <code>u32</code> , <code>u64</code> , <code>u128</code> , <code>usize</code>
Floating point	<code>f32</code> , <code>f64</code>
Other scalars	<code>bool</code> , <code>char</code>
Unit type	<code>()</code>
Fixed-size arrays	<code>[T; N]</code> when <code>T</code> : Copy (e.g. <code>[u8; 4]</code>)
Tuples	<code>(T1, T2, . . .)</code> when every element is Copy
References	<code>&T</code> , <code>&mut T</code> (the reference itself is copied; it still <i>points to</i> the same data)
Raw pointers	<code>*const T</code> , <code>*mut T</code>

Category	Examples
Function pointers	<code>fn(...) -> ...</code>
<code>NonNull</code> , <code>NonZero*</code>	Many niche std types that wrap a plain integer

Move types (assignment transfers ownership; the source is invalid afterward unless you borrow):

Category	Examples
Growable / owned heap data	<code>String</code> , <code>Vec<T></code> , <code>Box<T></code> , <code>HashMap<K, V></code> , <code>HashSet<T></code> , <code>BTreeMap</code> , ...
Shared / interior mutability	<code>Rc<T></code> , <code>Arc<T></code> , <code>RefCell<T></code> , <code>Mutex<T></code> , <code>RwLock<T></code>
Most user-defined structs/enums	Unless you add <code>#[derive(Copy, Clone)]</code> and every field is <code>Copy</code>
Types with custom <code>Drop</code>	File handles, network sockets, anything that must run cleanup once

Rules of thumb:

1. **Primitives on the stack -> usually `Copy`.**
2. **Anything that owns heap memory or runs `Drop` -> move by default.**
3. **`Copy` and `Drop` are mutually exclusive** — a type cannot implement both.
4. **`.clone()` is the explicit, potentially expensive deep copy** when you truly need two independent heap values.

You can check any type in the playground or docs: `Copy` is a trait; if a type implements it, assignment copies; otherwise it moves.

```
// Playground
fn main() {
    let a = 5;
    let b = a;
    // Copy --- two independent i32 on stack
    println!("{}", a, b);

    let v1 = vec![1, 2, 3];
    let v2 = v1;
```

```

// move --- v1 invalid; one owner for heap
// array
println!("{:?}", v2);

let s1 = String::from("hi");
let s2 = s1.clone();
// explicit deep copy --- both valid, two
// heap buffers
println!("{}", s1, s2);
}

```

1.6.5 References, borrowing, and dereferencing

Moves and `.clone()` are not the only way to use data. A **reference** is a borrow: you get access without becoming the owner. Rust has two safe reference forms:

Type	Access	Alias rule (preview)
<code>&T</code>	read-only	many <code>&T</code> borrows at once
<code>&mut T</code>	read + write	one <code>&mut</code> at a time, no overlapping <code>&T</code>

Use the `&` operator to **create** a reference from an owner:

```

// Playground
fn main() {
    let s = String::from("sensor_a");
    let r = &s;
    // r: &String --- immutable borrow; s still
    // owns the heap buffer

    let mut count = 0;
    let m = &mut count;
    // m: &mut i32 --- exclusive borrow for
    // mutation
    *m += 1; // writes through m into count (1)

    // Works: after *m += 1, the mutable borrow
    // of count is over (last use of m)
}

```

```
println!("{}", r, count); // sensor_a 1
}
```

Walk through the two borrows separately:

- **r and s:** `r` is a read-only handle to `s`. The heap buffer still has one owner (`s`); `r` only lets you read it. Many immutable borrows of the same value can coexist.
- **m and count:** `count` must be `mut` before you can take `&mut count`. The `*m += 1` line **dereferences** `m` and updates the owner's slot in place — `count` becomes 1 without moving it.

The closing `println!` uses `count` directly, not `m`. That compiles because the **mutable borrow ended** when `m` was last used (`*m += 1`). The binding `m` still exists, but until you use `m` again, `count` is free for a read.

Try adding `m` to the same `println!` — it fails:

```
// Playground --- does not compile
fn main() {
    let s = String::from("sensor_a");
    let r = &s;

    let mut count = 0;
    let m = &mut count;
    *m += 1;

    // ERROR E0502: cannot borrow `count` as
    // immutable because it is also borrowed
    // as mutable
    println!("{}", r, m, count);
}
```

`println!` expands to code that must **use every argument** for the duration of the call. Passing `m` (`&mut i32`) **re-opens** the exclusive borrow of `count`. Passing `count` in the same macro asks for an **immutable** read of the owner while that mutable borrow is active. Rust rejects the overlap — you cannot read and exclusively borrow `count` at once.

Fixes (smallest first):

1. **Read through one path** — `println!("{}", r, *m, count)`

- still fails for the same reason (count plus active `m`). Use `println!("{}", {}"`, `r`, `*m`) or `println!("{}", {} {}"`, `r`, `count`) — not both `m` and `count`.
2. **End the borrow with a block** — { `let m = &mut count; *m += 1;` } then `println!("{}", {} {}"`, `r`, `count`).
 3. **Drop `m` explicitly** — `drop(m)`; before printing `count` (less idiomatic than a block; Chapter 10 covers `drop` in depth).

Java / Python: two variables can reach the same object; neither owns it in the Rust sense. **Rust:** the **owner** (`s`, `count`) stays put; `r` and `m` are temporary handles the compiler checks (Chapter 5).

References are **Copy**: `let r2 = r1` copies the **pointer**, not the heap data. There is still **one owner** of the buffer.

1.6.5.1 Dereferencing with `*`

If `r`: `&i32`, then `*r` is the `i32` **at** that address. The unary `*` operator **dereferences** — it follows the reference to the underlying value.

Operator	Reads as	Example
<code>&</code>	“borrow this”	<code>&s</code> , <code>&mut n</code>
<code>*</code>	“value behind this reference”	<code>*m</code> , <code>*m += 1</code>

Rust **auto-derefs** in many everyday spots: `println!("{}", r)` works with `&i32`, and method calls like `r.abs()` reach through the reference for you. You still write explicit `*` when you need the **inner value** in an expression — especially **assigning or mutating through `&mut T`**:

```
// Playground
fn bump(n: &mut i32) {
    *n += 1;
    // without *, you would try to reassign the
    // reference itself
}

fn main() {
    let x = 10;
    let r = &x;
    let sum = x + *r;
    // explicit: add the i32 behind r
```

```
println!("{}", sum);

let mut ticks = 0;
bump(&mut ticks);
println!("ticks = {}", ticks);
}
```

Read-only borrow: `*r` gives you a copy when `T: Copy` (like `i32`). You cannot write `*r = 5` if `r: &i32` — mutating requires `&mut`.

Wrong — mutate through &:

```
// Playground --- does not compile
fn main() {
    let x = 10;
    let r = &x;
    // *r = 20;
    // ERROR: cannot assign to `*r`, which is
    // behind a `&` reference
    println!("{}", r);
}
```

Mutable borrow: `*mut` = value updates the **owner's** data in place. That is how a function increments a caller's counter or fills a buffer without taking ownership.

```
// Playground
fn main() {
    let mut frame = [0u8; 4];
    let header = &mut frame[0..2];
    header[0] = 0xDE;
    // slice methods auto-deref; index writes
    // through &mut
    header[1] = 0xAD;
    println!("{:02X?}", frame);
}
```

1.6.5.2 References vs raw pointers (name only)

The type table above lists **raw pointers** `*const T` and `*mut T`. They also use `*` to dereference, but only inside **unsafe** code — common in embedded

and FFI, not day-one application Rust. Safe code uses `&T` / `&mut T`; the compiler enforces borrow rules instead of trusting you.

Rules of thumb:

- 1. Need read-only access without moving? -> `&T`
- 2. Need in-place mutation of the owner? -> `&mut T` (one at a time)
- 3. Callee should take ownership (store, send, transform-and-return)?
-> pass by value — **move**
- 4. Caller must reuse the same allocation (loop buffer, counter, in-place edit)? -> `&mut T`

1.6.5.3 Move vs `&mut`: who keeps the value? (rules 3-4)

These look similar in other languages — “pass something to a function and let it change” — but Rust splits them on **ownership**.

Aspect	Move	Mutable borrow
	<code>fn f(s: String)</code>	<code>fn f(s: &mut String)</code>
Ownership	Transfers to the callee	Stays with the caller
Caller after the call	Original binding is invalid	Original binding is still valid , often changed
Heap buffer	Same buffer, new owner	Same buffer, same owner
Callee’s job	Own, transform, drop, or return the value	Temporarily mutate, then give control back
Typical use	“Take this and finish with it”	“Tweak my value in place”

Move: the function becomes the owner. If it takes `String` by value, the caller’s `label` is gone after the call — like handing over the key. The callee may mutate and drop it, or return it to transfer ownership.

`&mut`: the caller keeps ownership. The function gets a **loan** — exclusive access for the call — to edit the existing data. When the call returns, the borrow ends and the caller reads the updated value.

```
// Playground
fn consume(mut s: String) {
    s.push('!');
    // callee owns s; mutates its own local
    // owner
} // s dropped here unless returned
```

```

fn append_bang(s: &mut String) {
    s.push('!');
    // mutates caller's buffer through the
    // borrow
}

fn main() {
    let label = String::from("plc1");

    // Move path: ownership leaves main
    let moved = label;
    consume(moved);
    // println!("{}", moved);
    // error: moved value

    // Mut borrow path: caller keeps label
    let mut label = String::from("plc1");
    append_bang(&mut label);
    // lend exclusive access for one call
    println!("{}", label);
    // plc1! --- same owner, updated heap data
}

```

Use a move when the callee **should** take ownership — sending a message on a channel, storing in a struct field, or returning a transformed value. After `consume(moved)`, `main` no longer frees that buffer.

Use `&mut` when the caller must **reuse** the same allocation — a `String` label in a loop, a reusable `Vec<u8>` frame buffer, or a tick counter. `&mut` mutates in place with no handoff and no `.clone()`.

For stack Copy types like `i32`, “move” is a bitwise copy — both sides can still use their copy. The distinction matters most for **heap-backed** owners (`String`, `Vec`, ...) where move means the source name dies.

5. **Need to assign or arithmetic on the pointee?** -> often explicit `*`
6. **Need an independent duplicate?** -> `.clone()` or pass by value — not a longer-lived borrow

1.6.6 Borrow checker edge cases

Rust's alias rules are strict. These patterns look reasonable if you come from a language with implicit sharing (Java, Python, JavaScript, ...). The compiler rejects them **before** run time.

1. Immutable borrow blocks mutation of the owner

```
// Playground --- does not compile
fn main() {
    let mut s = String::from("plc");
    let r = &s;
    // shared read loan starts
    s.push('!');
    // ERROR: cannot borrow `s` as mutable
    // while `r` is live
    println!("{}", r);
}
```

Fix: end the read borrow before mutating — nested block, or `println!` first so `r` is not used after `push`:

```
// Playground
fn main() {
    let mut s = String::from("plc");
    {
        let r = &s;
        println!("{}", r);
    } // r ends here
    s.push('!');
    println!("{}", s);
}
```

2. Only one `&mut` at a time

```
// Playground --- does not compile
fn main() {
    let mut v = vec![1, 2, 3];
    let a = &mut v;
    let b = &mut v;
    // ERROR: cannot borrow `v` as mutable more
    // than once
}
```



```

    a.push(4);
    println!("{:?}", b);
}

```

3. & and &mut cannot overlap

```

// Playground --- does not compile
fn main() {
    let mut n = 0;
    let r = &n;
    let m = &mut n;
    // ERROR: cannot borrow as mutable while
    // immutable borrow exists
    println!("{:?}", r, m);
}

```

4. Cannot move while borrowed

```

// Playground --- does not compile
fn main() {
    let s = String::from("plc");
    let r = &s;
    let moved = s;
    // ERROR: cannot move `s` because it is
    // borrowed
    println!("{:?}", r);
}

```

References are Copy, but they **point at** an owner. Moving the owner while a borrow is active would leave a dangling & — same class of bug as use-after-free, ruled out at compile time.

5. Returning a reference to a local (preview)

```

// Playground --- does not compile
fn broken() -> &str {
    let s = String::from("tmp");
    &s
    // ERROR: `s` does not live long enough ---
    // returned ref would dangle
}

```

The owner dies at the end of broken; the caller cannot hold `&str` afterward. Return owned `String`, or borrow from the caller's data (Chapter 5).

6. Two `&mut` element borrows on one `Vec`

```
// Playground --- does not compile
fn main() {
    let mut frame = vec![0u8; 4];
    let a = &mut frame[0];
    let b = &mut frame[1];
    // ERROR: second mutable borrow of `frame`
    *a = 1;
    *b = 2;
    println!("{:?}", frame);
}
```

Even when indices differ, `&mut frame[i]` borrows the **whole** `Vec` mutably — the compiler does not track “this element only.” Two element refs at once are two mutable borrows of the same owner.

Fix — mutate by index (no simultaneous refs):

```
// Playground
fn main() {
    let mut frame = vec![0u8; 4];
    frame[0] = 1;
    frame[1] = 2;
    println!("{:?}", frame);
}
```

Fix — `split_at_mut` for two non-overlapping subslices:

```
// Playground
fn main() {
    let mut frame = vec![0u8; 4];
    let (left, right) = frame.split_at_mut(2);
    left[0] = 1;
    right[0] = 2;
    println!("{:?}", frame);
}
```

Each subslices borrow is disjoint; the compiler can prove they do not alias.

Related trap — `&mut` to whole `Vec` while an element borrow is still live:

```
// Playground --- does not compile
fn main() {
    let mut frame = vec![0u8; 4];
    let a = &mut frame[0];
    let whole = &mut frame;
    // ERROR: `frame` already borrowed through
    // `a`
    whole.push(5);
    *a = 1;
}
```

End the element borrow (drop `a`'s scope) before `push` or other calls that need `&mut` to the entire collection.

1.6.7 When the compiler says no

Common errors in this chapter:

Error (typical wording)	You probably did	Smallest fix
use of moved value	let <code>s2 = s1</code> , or passed <code>String</code> by value	borrow <code>&s1</code> , or clone, or use return value
borrow of moved value	used binding after <code>consume(s)</code>	<code>consume(&s)</code> or receive owned return
cannot borrow as mutable more than once	two <code>&mut</code> to same value	one <code>mut</code> borrow; restructure loop
cannot borrow as mutable while immutable borrow is active	let <code>r = &s</code> then <code>s.push</code>	shrink <code>r</code> 's scope with <code>{ }</code>
cannot move out of ... because it is borrowed	move owner while <code>&owner</code> exists	drop borrows first, then move
cannot assign to <code>*r</code> behind <code>&</code> reference	<code>*r = 5</code> on <code>&i32</code>	use <code>&mut</code>
does not live long enough	return <code>&</code> to local	return <code>String</code> or borrow from caller

Read errors **top to bottom** — the first note is usually the root cause; later notes are often follow-on noise.

1.6.8 Why this matters for long-running programs

Control loops and serial parsers often run for hours. **Stack allocation is essentially free.** Heap allocation is fine when you need it, but freeing at a known scope avoids GC pauses and makes worst-case latency easier to reason about — the same theme as [ownership vs GC](#) above.

1.7 Idiom spotlight

Let the type system carry intent. Prefer `Result` and `Option` over sentinel values (`null`, `-1`, magic strings). Formalized in Chapter 6 and Chapter 8.

1.8 Go deeper

- [The Rust Book — Understanding Ownership](#)
- [Functional Rust — Option basics](#)

1.9 See also

- Chapter 4: Iterators — `iter` vs `into_iter` and borrow errors in chains
- Chapter 5: Lifetimes — when references must be named in types
- Chapter 14: Multithreading

1.9.1 Afterparty

1.9.1.1 Ownership, moves, and `drop`

1. **Move drill** — “Give 6 tiny Rust snippets mixing `String`, `Vec`, and `i32`. I predict ok or compile error; you reveal answers, quote the error message, and give the smallest fix.”
2. **Use-after-move decoder** — “Show one `println!` after a move that fails to compile. I explain the error in plain English; you refine my explanation and show the three fixes: borrow, `.clone()`, or restructure scope.”
3. **Return transfers ownership** — “Write a function `fn take(s: String) -> String` and a `main` that calls it. I trace who owns the heap buffer at each line, including after the return.”
4. **Move into a call** — “Snippet: `process(build_label())` where `build_label() -> String`. Explain when the heap allocation happens,

when ownership moves into `process`, and when `drop` runs if `process` takes `String` by value.”

5. **Scope and drop** — “Give a nested-block snippet with two `Strings` and one `Vec`. I mark the exact line where each heap buffer is freed vs where stack slots disappear; you correct and explain drop order.”
6. **Single-owner principle** — “State Rust’s rule ‘every value has exactly one owner’ in one sentence, then give one `String` example where breaking that rule would double-free. Use the key-handoff metaphor.”

1.9.1.2 Stack, heap, and memory layout

7. **Stack vs heap quiz** — “For 10 declarations (`i32`, `bool`, `[u8; 4]`, `String`, `Vec<i32>`, `&str`, `&String`, (`i32`, `f64`), `Box<i32>`, `()`), I say stack-only, heap involved, or both; you draw the pointer picture for any I miss.”
8. **String layout sketch** — “For `let s = String::from("hi")`, I describe stack fields (`ptr`, `len`, `cap`) and heap bytes; you correct and extend to `let v = vec![1, 2, 3]`.”
9. **Stack frame drill** — “I give a 3-function call chain with `i32` locals and one `String` passed by value. Trace stack frame push/pop, when each slot dies, and when the `String` heap buffer is dropped.”
10. **Nested block live ranges** — “Snippet with `{ let inner = ... }` inside `main`. I list which bindings are alive on each line; you explain why shorter live ranges matter for borrowing later.”
11. **Borrow without heap copy** — “Show `let s = String::from("x")`; `let r = &s`; — I explain what is on stack vs heap and why `r` does not duplicate the buffer; you add one line that would fail because of move/borrow conflict.”

1.9.1.3 References and dereferencing (& and *)

12. **& vs &mut pick** — “Five tasks (log a label, increment a tick counter, parse into a temp struct, share read-only config, swap two buffers in place). I choose pass-by-value, `&T`, or `&mut T`; you explain owner count and alias rules.”
13. **Create the borrow** — “Given `let mut n = 10`; and `let s = String::from("plc")`; I write the types and expressions for one `&` and one `&mut` borrow; you verify the owner is still usable afterward.”
14. **When is * required?** — “Four expressions mixing `&i32`, `&mut i32`, and `println!`. I mark where explicit `*r` is needed vs where auto-deref

handles it; you correct with compiled examples.”

15. **Mutate through *** — “Fill in `fn reset(n: &mut u32) { ... }` and a `main` that calls it. I must use `*` to zero the caller’s value; you show a broken version that tries to reassign `n` instead.”
16. **Reference copy vs move** — “After `let r1 = &s; let r2 = r1; vs let s2 = s1;` I compare heap owner count, which bindings stay valid, and whether heap data was copied.”
17. **Type of *r** — “Bindings `r: &i32, m: &mut i32, b: Box<i32>`. I name the type of `*r`, `*m`, and whether `*m = 5` mutates the owner; you extend with one `&T` where `*r = 5` fails.”
18. **Borrow blocks mutation** — “Snippet: `immutable let r = &s; then s.push('!')`. I explain the compile error; you fix by shrinking `r`’s scope with a nested block.”
19. **Automation counter** — “Sketch a 1 kHz loop with `ticks: u64` and a helper `fn maybe_rollover(t: &mut u64)`. I write the call site and one `*t` mutation inside the helper; you review without full thread code.”

1.9.1.4 Move vs mutable borrow

20. **Move or &mut quiz** — “Six tasks (append to a reusable log line, send a message on a channel, fill a frame `Vec<u8>` in a loop, store a device name in a struct, transform-and-return a label, increment a counter). I pick `fn take(T) move vs fn tweak(&mut T)`; you explain who owns the heap data after the call.”
21. **After the call** — “Two functions: `consume(String)` and `append_bang(&mut String)`. I trace which bindings in `main` are valid **after** each call and whether the heap buffer was dropped, reused, or returned.”
22. **Fix the signature** — “Show code that moves a `String` into a helper but then tries to `println!` it in `main`. I explain the error and choose the smallest fix: change to `&mut String`, restructure with a return value, or `.clone()` — you rank the idiomatic options.”
23. **Loop reuse** — “A serial parser reuses `let mut buf = Vec::with_capacity(256)` every tick. I explain why `parse_frame(buf)` (move) is wrong and write `parse_frame(&mut buf)` instead; you add one line showing the caller reading updated length after parse.”
24. **Transform-and-return** — “Task: uppercase a `String` and give it back. I write `fn upper(s: String) -> String` and call site; you contrast with an anti-pattern that takes `&mut String` when ownership should

transfer.”

25. **i32 vs String move** — “Same pattern with `fn add_one(n: i32)` and `fn add_bang(s: String)`: I explain why the caller can still use `n` after the call but not `s`; you map each to Copy vs heap move.”
26. **Call it twice** — “Snippet that needs to pass the same `String` to two helpers in one scope. I show why two by-value calls fail, then fix with `&str/&mut` borrows or one move plus `.clone()`; you flag the smell.”

1.9.1.5 Copy, move, and `.clone()`

27. **Copy eligibility quiz** — “Quiz me on 12 types (`i32`, `String`, `&str`, `Vec<i32>`, `[u8; 8]`, `(i32, String)`, `Box<i32>`, `fn()`, `bool`, `char`, `Rc<i32>`, struct with only `i32` fields). Copy, move, or ‘Copy only if derived’? Cite the rule each time.”
28. **Copy vs Drop paradox** — “Why can’t a type be both Copy and Drop? Use a `File` or socket handle: show what goes wrong if assignment bitwise-copies the handle.”
29. **Semantic copy test** — “For `i32`, `&T`, and `String`: if I duplicate the stack bits, is the result always semantically identical? When does it fail, and what does Rust do instead?”
30. **Double-free thought experiment** — “Hypothetical: `String` were Copy. Walk line-by-line through `let a = ...; let b = a; }` end of block — show both drops freeing the same address. Contrast with real move semantics.”
31. **When to derive Copy** — “I put `#[derive(Copy, Clone)]` on a struct with a `String` field — explain the error. Then give three struct shapes: safe to derive Copy, must stay move-only, and where `.clone()` belongs in the public API.”
32. **Reference is Copy** — “Four snippets: `let r2 = r1` for `&String` vs `let s2 = s1` for `String`, plus one with `&mut`. I predict ok/error; you count owners of the heap buffer after each line.”
33. **.clone() judgment** — “Five scenarios (store config `String`, pass into `fn` twice, cache key in a map, loop append, return from `fn`). For each, say move, borrow, or `.clone()` — and flag when `.clone()` is a design smell.”

Chapter 2

Types and Expressions

2.1 Hook

Rust's type system is not ceremony. The compiler uses it to check ownership, catch bugs, and keep code predictable **before** you run anything. This chapter covers built-in types, inference, and expression-style control flow.

2.2 Integer types

Rust has **fixed-width** signed and unsigned integers. Pick the width that matches your domain (protocol field, array index, counter).

Type	Width	Typical use
i8 ... i128	8-128 bit signed	Binary protocols, embedded registers
u8 ... u128	8-128 bit unsigned	Bytes, sizes, hashes
isize / usize	pointer-sized	indexing, len(), collection sizes

```
// Playground
fn main() {
    let port: u16 = 502;
    // Modbus-style port --- explicit width
    let count = 42;
```



```
// infers i32 by default
let byte = 0xFFu8;
let big: i64 = 1_000_000;
println!("{}", port, count, byte,
           big);
}
```

Defaults: integer literals infer i32 unless you suffix (42u64) or annotate.

For wire formats and hardware, **always** choose an explicit width. Do not rely on “whatever the platform uses.”

Overflow: in debug builds, arithmetic that overflows **panics**. In release, integers wrap (two’s complement).

Use `checked_add`, `saturating_add`, or `wrapping_add` when the policy matters.

2.3 Floating point, bool, and char

Type	Notes
f32, f64	IEEE floats; f64 is the default inference
bool	exactly true or false — not an integer in safe Rust
char	one Unicode scalar value (4 bytes), e.g. ' (crab) '
()	unit type — “no meaningful value”; used as empty return

```
// Playground
fn main() {
    let pi: f64 = 3.14159;
    let ok = true;
    let crab = '(crab)';
    let unit: () = (); // only value of type ()
    println!("{}", pi, ok, crab,
              unit);
}
```

`char` is not a UTF-8 byte — it is a single code point.

Text strings use `str` / `String` (below); ownership of those buffers is Chapter 1.

2.4 Inference and annotations

The compiler infers types when unambiguous. Add annotations or suffixes when it is not:

```
// Playground
fn main() {
    let xs = vec![1, 2, 3];    // Vec<i32>
    let ys: Vec<u8> = Vec::new();
    // empty vec needs a hint
    let n = 10u32;
    let ratio = n as f64 / 3.0;
    // cast integer to float
    println!("{:?} {:?}", ys, xs, ratio);
}
```

Casts use `as` between numeric types (`u16 as u32`). **Conversions** that can fail (string parsing, narrowing) use methods like `.parse()` or `TryFrom` — covered in Chapter 8.

2.5 Tuples and arrays

Tuples group fixed types; access by index (`.0`, `.1`):

```
// Playground
fn main() {
    let pair: (u16, &str) = (502, "modbus");
    println!("port {} proto {}", pair.0,
            pair.1);
}
```

Arrays `[T; N]` have compile-time fixed length and stack storage when `T` is `Copy`:

```
// Playground
fn main() {
    let frame: [u8; 4] = [0xDE, 0xAD, 0xBE,
                        0xEF];
    println!("len {} first {:02X}",
            frame.len(), frame[0]);
}
```

Growable `Vec<T>` and slice ownership return in Chapter 1 and Chapter 11.

2.6 Slices

A **slice** is a **view** into contiguous memory: pointer plus length, no ownership.

The type is written `&[T]` — a borrow of `[T]`. `[T]` is **unsized**: the compiler does not know its length at compile time.

Form	Meaning
<code>&[T]</code>	borrowed view of <code>T</code> elements
<code>&[u8]</code>	byte buffer view — wire payloads, file chunks, <code>Vec<u8></code> data
<code>&str</code>	UTF-8 text slice — same fat-pointer idea, different element type (<code>u8</code> with UTF-8 invariant)

Create slices from arrays, vectors, or other slices with **range syntax**:

```
// Playground
fn main() {
    let frame: [u8; 6] = [0x01, 0x02, 0x03,
                          0x04, 0x05, 0x06];
    let payload: &[u8] = &frame[2..5];
    // bytes at indices 2, 3, 4
    let all: &[u8] = &frame[..];
    // whole array as slice

    let nums = vec![10, 20, 30, 40];
    let mid = &nums[1..3];
    // &[i32] borrowing the Vec's buffer

    println!("payload {:?} all {:?} mid {:?}",
            payload, all, mid);
}
```

Indexing: `[i]` on a slice is a direct access. Out-of-range indices **panic** at runtime.

Prefer `.get(i) -> Option<T>` when the index comes from user input, parsed fields, or unproven loop math.

```
// Playground
fn main() {
    let frame: [u8; 4] = [0xDE, 0xAD, 0xBE,
                          0xEF];
    let i = 2;
    match frame.get(i) {
        Some(byte) => println!("byte {:02X}",
                               byte),
        None => println!("index {} out of
                        range", i),
    }
}
```

Half-open ranges match for loops: `&xs[0..3]` includes indices 0, 1, 2. An empty slice is valid: `&xs[0..0]`.

2.7 Strings and UTF-8

Rust strings are **UTF-8** byte sequences, not fixed-width characters. That keeps them compact and compatible with the web and most modern protocols.

It also means **byte index != character index**.

Type	Role
<code>str</code>	unsized UTF-8 text — always used as <code>&str</code> (borrowed slice)
<code>&str</code>	borrowed view into valid UTF-8 (literal, substring, or <code>String</code> borrow)
<code>String</code>	owned, growable UTF-8 buffer on the heap

```
// Playground
fn main() {
    let literal: &str = "fixed";
    // stored in read-only segment
    let owned = String::from("growable");
    // heap-backed owner
```

```

let also: &str = &owned;
// borrow the String as &str

let mut label = String::new();
label.push_str("port=");
label.push_str("502");

println!("{}", literal, owned, also);
println!("{}", label);
}

```

Literals have type `&str`. `String::from`, `.to_string()`, and `format!` allocate when you need an owned buffer.

`push_str` / `push` extend a `String` in place. Use them when building output in a loop — log lines, CSV rows, protocol text fields.

2.7.1 `&str` vs `String` in APIs

Callee needs	Parameter type
Read or compare text only	<code>&str</code>
Store text beyond the call	<code>String</code> (or return owned from caller)
Accept literal or owned caller data	<code>&str</code> — callers pass <code>"literal"</code> or <code>&my_string</code>

```

// Playground
fn log_label(label: &str) {
    println!("label={}", label);
}

fn store_tag(tag: String) -> String {
    tag // owner returns owned text
}

fn main() {
    let name = String::from("sensor_a");
    log_label("startup");      // &str literal
    log_label(&name);
    // &String coerces to &str
}

```

```

    let saved = store_tag(name);
    println!("{}", saved);
}

```

Rust **coerces** `&String` to `&str` at call sites. So `fn f(s: &str)` is the default for read-only string parameters.

Take `String` when the function must **own** the data — to store it, spawn a thread, or key a map.

2.7.2 Length, iteration, and slicing text

- `.len()` on `&str` / `String` is **byte** length, not grapheme or char count.
- `.chars()` iterates Unicode scalar values (`char`); use when you care about characters.
- `.bytes()` iterates raw `u8` values — useful next to binary parsers.
- `.lines()` splits on line endings — common for config and serial-line protocols.

```

// Playground
fn main() {
    let s = "naïve"; // 'ï' is two UTF-8 bytes
    println!("bytes={} chars={}", s.len(),
            s.chars().count());

    for ch in s.chars() {
        print!("{}", ch);
    }
    println();

    for line in "a\nb\n".lines() {
        println!("line: {}", line);
    }
}

```

String slicing uses the same range syntax as byte slices. Ranges must fall on **UTF-8 character boundaries**.

Slicing at a bad byte index **panics**:

```

// Playground
fn main() {

```

```

let s = "hello";
let hello = &s[0..5];
// ok --- ASCII, one byte per char

let emoji = "hi (crab)";
let hi = &emoji[0..2]; // ok
// let bad = &emoji[0..3];
// panic --- splits the crab emoji
println!("{}", hello, hi);
}

```

At protocol and config boundaries, treat **&[u8]** as opaque bytes on the wire. Treat **&str** as validated human-readable text — config keys, error messages, JSON string fields after parsing.

Do not cast arbitrary bytes to **&str** without validation. Use **std::str::from_utf8** or **String::from_utf8** when converting (Chapter 8).

2.7.3 Common **&str** helpers (no allocation)

Method	Use
<code>.starts_with / .ends_with</code>	protocol prefixes, file extensions
<code>.strip_prefix / .strip_suffix</code>	peel "CMD:" off a serial line
<code>.split / .split_once</code>	CSV-ish fields, key=value pairs
<code>.trim</code>	whitespace around user input
<code>.parse::<T>()</code>	"502".parse::<u16>() -> Result

```

// Playground
fn main() {
    let line = " PORT=502 ";
    let trimmed = line.trim();
    if let Some(rest) =
        trimmed.strip_prefix("PORT=") {
        match rest.parse::<u16>() {
            Ok(port) => println!("port {}",
                                port),
            Err(e) => println!("bad port: {}",

```

```

        e),
    }
}
}

```

2.8 Variables and mutability

```

// Playground
fn main() {
    let x = 10;      // immutable binding
    let mut y = 20;  // mutable binding
    y += x;
    println!("{}", y);
}

```

Default **let** is **immutable**. Use **mut** only where the value changes — it documents intent and helps the borrow checker later.

2.9 Control flow as expressions

```

// Playground
fn main() {
    let n = 7;
    let label = if n % 2 == 0 { "even" } else {
        "odd" };
    println!("{}", n, label);

    for i in 0..3 {
        println!("i = {}", i);
    }

    let mut sum = 0;
    let mut k = 0;
    while k < 3 {
        sum += k;
        k += 1;
    }
}

```



```
println!("sum = {}", sum);
}
```

- **if** is an expression — both branches must return the same type.
- **0..3** is half-open (0, 1, 2); **0..=3** is inclusive (0 through 3).
- **loop**, **while**, and **for** cover the usual iteration shapes; **for** over a range is idiomatic for indexed passes. Iterator pipelines (**map**, **filter**, **collect**) are Chapter 4.

2.10 match preview

Full power in Chapter 6; here, matching integers:

```
// Playground
fn main() {
    let code = 404;
    let msg = match code {
        200 => "ok",
        404 => "not found",
        _ => "other",
    };
    println!("{}", msg);
}
```

_ is the wildcard arm. **match** must be **exhaustive** — every possible value covered (easy for integers with **_**; enums need every variant).

2.11 Idiom spotlight

Name your widths at boundaries. Inside a function, inference is fine. At protocol edges, file headers, and public APIs, use explicit types (**u16**, **[u8; 8]**). That way refactors cannot silently change layout.

Borrow text, own when storing. Prefer **&str** and **&[u8]** in function parameters. Return or store **String** / **Vec<u8>** only when the data must outlive the caller. At binary/text boundaries, keep **&[u8]** for **bytes** and **&str** for **validated UTF-8**. Do not mix them silently.

2.12 Go deeper

- [The Rust Book — Data Types](#)
- [The Rust Book — String slices](#)

2.13 See also

- Preface — rustup and Cargo setup
- Chapter 1: Ownership and borrowing
- Chapter 4: Iterators
- Chapter 6: Enums and pattern matching

2.13.1 Afterparty

2.13.1.1 Scalars and inference

1. **Integer pick** — “Quiz me: for 8 scenarios (Modbus port, byte buffer, collection index, money cents, timestamp millis, hash output, loop counter, enum discriminant), I pick `u8/u16/u32/u64/i32/usize`; you correct and explain overflow risk.”
2. **char vs byte** — “Explain why `'A'` is not the same type as `65u8`. Show one valid `char` literal and one invalid escape; mention UTF-8 only when discussing `str/String`.”

2.13.1.2 Compound types and text

3. **Tuple vs array** — “When do I use `(u16, u16)` vs `[u16; 2]` for a coordinate pair? Give one idiomatic example of each.”
4. **Array bounds** — “Snippet with `[u8; 4]` and an index from user input. I explain compile-time vs runtime safety; you show bounds checking with `.get()` vs `[i]`.”
5. **&str vs String API** — “Five function signatures for logging labels. I choose `&str` or `String` for each; you explain ownership and allocation cost.”
6. **Parse drill** — “Give `let s = \"502\"`; — I write two ways to get `u16` (unwrap version and proper `Result` version); you grade idiomatic error handling.”

2.13.1.3 Slices and string idioms

7. **Slice from array** — “Given `[u8; 8]` with a 2-byte header and 6-byte payload, I write range expressions for header, payload, and full as `&[u8]`; you check half-open bounds.”
8. **.get vs [i]** — “Three indexing scenarios (fixed compile-time index, user-supplied index, loop variable). I pick `[i]` or `.get(i)` for each; you explain panic risk.”
9. **UTF-8 length trap** — “For `\"naïve\"` and `\"hi (crab)\"`, I predict `.len()` vs `.chars().count()`; you show one bad string slice that panics and the safe fix (`.chars()` or careful byte ranges).”
10. **Coercion quiz** — “Five call sites passing `&str`, `String`, `&String`, and a literal into `fn log(msg: &str)`. I say ok or what coercion happens; you quote the effective type.”
11. **Empty slice** — “Explain whether `&frame[0..0]` is valid, what `.len()` returns, and how `if slice.is_empty()` reads in a parser guard.”
12. **Lines and config** — “Multiline config string with comments and blank lines. I sketch a loop using `.lines()` and `.trim().starts_with('#')`; you extend with one `strip_prefix` for `KEY=` rows.”

2.13.1.4 Control flow and match

13. **Range quiz** — “For half-open vs inclusive ranges, I predict output of four for loops using `..` and `..=`; you correct with printed values.”
14. **if expression types** — “Show an `if` where branches return different types and fail to compile. I explain the error; you fix with a unified type (same type in both arms or wrap in `enum` preview).”
15. **match warm-up** — “Extend a `match` on HTTP status codes with `3xx`, `4xx`, `5xx` groupings using range patterns; I write arms; you check exhaustiveness.”
16. **Loop choice** — “Four iteration tasks (infinite retry with `break`, consume iterator, index `0..len`, early exit on condition). I pick `loop/while/for`; you confirm idiomatic choice.”

Chapter 3

Functions and Methods

3.1 Hook

You already know how to call functions. In Rust, every signature is a **contract** about ownership and types. A parameter is not “just a reference”; it tells you whether the callee **borrow**s, **mutably borrow**s, or **takes** the value.

This chapter covers standalone functions, methods, and return-type patterns you will reuse everywhere.

3.2 Scope — a brief tour

Function signatures, receivers, and return patterns — not full generics or `async`.

This chapter covers	Deferred
Signatures, ownership in params, <code>impl</code> methods	Full trait system -> Ch 7
<code>impl</code> Trait return preview, <code>mem::take</code>	Async fns -> Ch 16
Result signature preview	Error design -> Ch 8

3.3 Function signatures

A Rust function names its inputs and output explicitly:

```
// Playground
fn add(a: i32, b: i32) -> i32 {
    a + b
}

fn main() {
    println!("{}", add(2, 3));
}
```

Piece	Meaning
fn name	declare a function
(a: i32, b: i32)	parameters with types
-> i32	return type
{ ... }	body; the last expression (no semicolon) is the return value

Java / Python contrast

Aspect	Java	Python	Rust
Return	return x; or last statement	return x optional	last expression returns; use return for early exit
Missing return	void	implicit None	-> () unit type
Overloading	same name, different params	duck typing	no overloading — use different names or generics (Chapter 7)

3.4 Parameters and ownership

Parameters follow the same ownership rules as let bindings (Chapter 1):

```
// Playground
fn consume(s: String) {
    println!("{}", s);
} // s dropped here

fn borrow(s: &str) {
```

```
println!("{}", s);
}

fn main() {
    let label = String::from("sensor");
    borrow(&label);
    consume(label);
    // borrow(&label);
    // ERROR: label was moved into consume
}
```

Parameter type	Caller after call
T	value moved unless T: Copy
&T	still owns; shared borrow
&mut T	still owns; exclusive borrow

Rule of thumb: take &str / &[u8] when you only read; take owned String / Vec when you store or send the data elsewhere.

3.5 Expression bodies and early return

Omit braces for a single expression:

```
// Playground
fn double(x: i32) -> i32 {
    x * 2
}

fn abs_diff(a: i32, b: i32) -> i32 {
    if a > b { a - b } else { b - a }
}

fn main() {
    println!("{}", double(5), abs_diff(10,
        3));
}
```

Use return when you exit before the end:

```
// Playground
fn first_positive(v: &i32) -> Option<i32> {
    for &n in v {
        if n > 0 {
            return Some(n);
        }
    }
    None
}

fn main() {
    println!("{:?}", first_positive(&[-1, 0, 4,
    2]));
}
```

Adding a semicolon after the last expression **suppresses** the return value and gives ():

```
// Playground --- does not compile if return
    type is i32
fn broken() -> i32 {
    42;
    // semicolon turns this into a statement,
    // not the return value
    // ERROR: expected i32, found ()
}
```

3.6 The unit type () and diverging functions

Functions that do side effects only often return **unit**:

```
// Playground
fn log_line(msg: &str) {
    println!("[log] {}", msg);
} // implicit -> ()

fn main() {
    log_line("ready");
}
```

Some functions **never return** — they diverge with `panic!`, infinite loops, or `std::process::exit`. Their return type is `!` (never type):

```
// Playground
fn fatal(msg: &str) -> ! {
    panic!("{}", msg);
}

fn main() {
    // fatal("stop");
    // would never reach code below
    println!("ok");
}
```

You will see `!` rarely in application code. Knowing it exists explains why `match` arms can have different “no return” branches.

3.7 Methods and associated functions

Rust splits **data** (struct / enum) from **behaviour** (impl blocks). Methods take `&self`, `&mut self`, or `self`; **associated functions** have no `self` (like static methods):

```
// Playground
struct Gauge {
    value: f64,
}

impl Gauge {
    // associated function --- call as
    // Gauge::new(0.0)
    fn new(value: f64) -> Self {
        Self { value }
    }

    // method --- call as gauge.read()
    fn read(&self) -> f64 {
        self.value
    }
}
```



```

fn set(&mut self, v: f64) {
    self.value = v;
}

fn reset(self) -> Self {
    Self { value: 0.0 }
}

fn main() {
    let mut g = Gauge::new(12.5);
    println!("{}", g.read());
    g.set(99.0);
    let zero = g.reset();
    println!("{}", zero.read());
}

```

Receiver	Meaning
<code>&self</code>	borrow for read
<code>&mut self</code>	exclusive borrow for mutation
<code>self</code>	consume self (move out)

Java: instance methods inside classes. **Python:** functions in a class with `self`. **Rust:** explicit `impl` blocks — no inheritance chain.

3.8 Multiple `impl` blocks and privacy

You can split `impl` for clarity; only items marked `pub` are visible outside the module (Chapter 9):

```

// Playground
pub struct Counter {
    n: u32,
}

impl Counter {
    pub fn new() -> Self {
        Self { n: 0 }
    }
}

```

```

    }

    pub fn bump(&mut self) {
        self.n += 1;
    }

    pub fn get(&self) -> u32 {
        self.n
    }
}

fn main() {
    let mut c = Counter::new();
    c.bump();
    println!("{}", c.get());
}

```

Field `n` stays private; callers use methods — same encapsulation idea as Java private fields with public getters.

3.9 Generic functions (preview)

One function body can work for many types when they share a capability (**trait bound**). Full treatment is in Chapter 7; the shape is:

```

// Playground
use std::fmt::Display;

fn show_twice<T: Display>(x: T) {
    println!("{}", x, x);
}

fn main() {
    show_twice(42);
    show_twice("hello");
}

```

The compiler **monomorphizes** — generates `show_twice_i32`, `show_twice_str`, etc. — so you pay no runtime dispatch cost un-

less you ask for `dyn Trait`.

3.10 Functions that return `Result` (preview)

Fallible operations return `Result` instead of throwing. Chapter 8 covers this in depth; the signature pattern is:

```
// Playground
fn parse_port(s: &str) -> Result<u16,
    std::num::ParseIntError> {
    s.parse::<u16>()
}

fn main() {
    match parse_port("8080") {
        Ok(p) => println!("port {}", p),
        Err(e) => println!("bad port: {}", e),
    }
}
```

Inside a `fn -> Result<...> body`, `?` propagates errors upward. You will use it after Chapter 6 and Chapter 8.

3.11 `impl Trait` in return position

Iterator pipelines chain adapters (`map`, `filter`, `take`, ...). The **concrete** type of that chain is often long and brittle — e.g. `Take<IntoIter<f64>>` — and it changes every time you add or remove a step. Writing that type in the signature is noisy and couples callers to your implementation.

`impl Trait` in the return position names **what callers need** (here: “something iterable that yields `f64`”) while the compiler keeps track of the real adapter type. You get static dispatch with no heap allocation — unlike `Box<dyn Iterator<...>>`.

```
// Playground
fn top_readings(readings: &[f64], n: usize) ->
    impl Iterator<Item = f64> + '_ {
    let mut sorted: Vec<f64> =
        readings.to_vec();
```

```

        sorted.sort_by(|a, b|
            b.partial_cmp(a).unwrap());
        sorted.into_iter().take(n)
    }

fn main() {
    let vals = [1.2, 3.4, 2.1, 5.0];
    let top: Vec<_> = top_readings(&vals,
        2).collect();
    println!("{:?}", top);
}

```

Reading the signature:

Piece	Meaning
<code>impl Iterator<Item = f64></code>	returns <i>some</i> type that implements Iterator and yields f64; callers can <code>.collect()</code> , <code>.sum()</code> , etc. without knowing the concrete adapter
<code>+ '_</code>	the returned iterator may borrow from readings; <code>'_</code> asks the compiler to infer that lifetime from the <code>&[f64]</code> parameter

Inside the body, `sorted.into_iter().take(n)` is one specific iterator type. You never spell it out — Rust monomorphizes a version of `top_readings` for that exact chain at compile time.

One concrete type per function: `impl Trait` here does **not** mean “any iterator.” The compiler picks **one** concrete return type for the entire function, and every return path must produce exactly that type. Branches that build different adapters fail:

```

// Playground --- does not compile
fn pick(flag: bool) -> impl Iterator<Item =
    i32> {
    if flag {
        vec![1, 2].into_iter()
        // Vec<i32>::IntoIter
    } else {
        [3, 4].into_iter()
    }
}

```

```

        // std::array::IntoIter<i32, 2> ---
        // different type
    }
}

```

When you truly need different iterator shapes per path, erase the type: `Box<dyn Iterator<Item = f64>>` (heap + dynamic dispatch) or an enum wrapping each variant. See [Function edge cases](#) below and Chapter 4: Iterators.

3.12 Draining with `mem::take`

The problem: A struct accumulates data in a field — here, bytes in a `Vec`. Eventually you want to **hand that data to the caller** and start fresh, but you only have `&mut self`. You cannot write `out.extend(self.inner)` — that would **move** `inner` out of `buf` while `buf` still exists, leaving a hole the compiler rejects.

The aim: Move the accumulated `Vec` to the caller **and** leave `buf` in a valid, reusable state (empty, ready for the next batch). `std::mem::take` is the standard idiom for that (Chapter 10).

```

// Playground
struct Buffer {
    inner: Vec<u8>,
}

impl Buffer {
    fn drain_into(&mut self, out: &mut Vec<u8>)
    {
        out.extend(std::mem::take(&mut
            self.inner));
    }
}

fn main() {
    let mut buf = Buffer { inner: vec![1, 2, 3]
    };
    let mut batch = Vec::new();
}

```

```
println!("before: buf.inner={:?}
    batch={:?}", buf.inner, batch);
buf.drain_into(&mut batch);
println!("after:  buf.inner={:?}
    batch={:?}", buf.inner, batch);
// before: buf.inner=[1, 2, 3]  batch=[]
// after:  buf.inner=[]        batch=[1,
    2, 3]
}
```

What `take` does: it replaces the field with `T::default()` and **returns** the old value.

Step	buf.inner	value returned by take
before call	[1, 2, 3]	—
take(&mut self.inner)	[] (empty Vec, via Default)	vec![1, 2, 3] (moved out)
out.extend(...)	[]	appended into caller’s batch

`buf` is still a valid `Buffer` — just with an empty `inner`. You can keep using it; no need to drop and recreate the struct.

Alternatives and why they differ:

Approach	Effect
fn drain(self) -> Vec<u8>	moves the whole <code>Buffer</code> — caller owns the vec, but consumes the struct
self.inner.clear()	empties in place — caller never gets ownership of the old allocation
mem::take(&mut self.inner)	caller gets the <code>Vec</code> (and its capacity); struct stays alive, field reset to empty

When you see this: `log/event` batch flushes, connection write buffers handed to I/O, any “accumulate, then hand off the batch” API.

3.13 where clauses — readable bounds

When generic bounds clutter the signature, move them to a `where` block:

```
// Playground
use std::fmt::Display;

fn log_pair<T, U>(a: T, b: U)
where
    T: Display,
    U: Display,
{
    println!("{}", a, b);
}

fn main() {
    log_pair(502, "ready");
}
```

Same contract as `fn log_pair<T: Display, U: Display>(...)` — pick whichever reads cleaner.

3.14 Error constructor factories

Parse failures are usually **enum variants** with named fields — e.g. `BadValue { field, raw }`. You could build them inline at every call site:

```
ParseError::BadValue { field: "port".into(),
    raw: raw.into() }
```

That gets repetitive, and it is easy to swap `field` and `raw`. Production enums add **named constructors** — small functions on `impl ParseError` — so call sites stay short and consistent:

```
ParseError::bad_value("port", raw)
// same variant, clearer intent
```

Those constructors take `impl Into<String>` instead of `String`. Callers pass `&str` literals (`"port"`), borrowed slices (`raw`), or owned `String`; the constructor calls `.into()` inside and stores owned strings in the enum. No `.to_string()` clutter at every error site.

The example wires both ideas together: `missing_field` and `bad_value` are the factories; `parse_port_field` calls them instead of spelling variants:

```

// Playground
#[derive(Debug)]
// compile-time trait impl --- not a Java
// annotation or Python decorator; see Ch17
enum ParseError {
    MissingField { field: String, record:
        String },
    BadValue { field: String, raw: String },
}

impl ParseError {
    #[inline]
    fn missing_field(field: impl Into<String>,
        record: impl Into<String>) -> Self {
        Self::MissingField {
            field: field.into(),
            record: record.into(),
        }
    }

    #[inline]
    fn bad_value(field: impl Into<String>, raw:
        impl Into<String>) -> Self {
        Self::BadValue {
            field: field.into(),
            raw: raw.into(),
        }
    }
}

fn parse_port_field(raw: Option<&str>) ->
    Result<u16, ParseError> {
    let raw = raw.ok_or_else(||
        ParseError::missing_field("port",
            "config"))?;
    raw.parse()
        .map_err(|_|
            ParseError::bad_value("port", raw))

```



```

}

fn main() {
    for raw in [Some("502"), Some("oops"),
                None] {
        match parse_port_field(raw) {
            Ok(p) => println!("ok: {p}"),
            Err(ParseError::MissingField {
                field, record }) => {
                println!("missing field {field}
                        in {record}")
            }
            Err(ParseError::BadValue { field,
                raw }) => {
                println!("bad value for
                        {field}: {raw}")
            }
        }
    }
}

```

Piece in the example	What it demonstrates
fn missing_field(...) / fn bad_value(...) on impl ParseError	named constructors — hide variant struct syntax
impl Into<String> on parameters	callers pass "port" or raw (&str); .into() runs inside the constructor
ParseError::missing_field("port", "config")	factory call when the field is absent (None)
ParseError::bad_value("port", raw)	factory call when the value fails to parse
match in main	consumers still see the enum variants; factories are only for building errors

Use this in library crates with **thiserror** enums (Chapter 8) — the constructors stay even when `Display` is derived.

#[derive(...)] preview: the attribute auto-generates trait implementations at **compile time** (impl Debug for ParseError { ... }). It does not run at call time and is not runtime metadata. Std derives (Debug, Clone, ...), ecosystem derives (Serialize, Error, ...), and when custom derives

are worth it — Chapter 17: Derive attributes.

3.15 Config-holder structs (lightweight builders)

You do not always need a builder crate. A small struct that **holds configuration** and exposes one method per use case is enough:

```
// Playground
struct RequestBuilder {
    timeout_ms: u64,
    retries: u32,
    endpoint: String,
}

impl RequestBuilder {
    fn new(endpoint: impl Into<String>) -> Self {
        Self {
            timeout_ms: 5000,
            retries: 3,
            endpoint: endpoint.into(),
        }
    }

    fn with_timeout(mut self, ms: u64) -> Self {
        self.timeout_ms = ms;
        self
    }

    fn build_poll_body(&self, device_id: u32)
        -> String {
        format!(
            "GET
            /devices/{device_id}?timeout={}&
            retries={}",
            self.timeout_ms, self.retries
        )
    }
}
```

```

    }
}

fn main() {
    let req =
        RequestBuilder::new("gateway.local")
            .with_timeout(1000)
            .build_poll_body(7);
    println!("{}", req);
}

```

Reach for **new** + **chainable setters** + **one build_* method** when parameters are fixed at construction time but the output shape varies. Full builder crates pay off when you have many optional fields and validation at build time.

3.16 Extension traits — add behaviour without newtypes

When you need a helper on a type you do not own, define a **trait in your crate** and implement it for the foreign type you control the orphan rule for — or implement for your wrapper. Common pattern: extend **Option** with domain-specific ? helpers (Chapter 8) and extend **&str** with string normalizers (Chapter 13).

```

// Playground
trait TrimOrEmpty {
    fn trim_or_empty(&self) -> &str;
}

impl TrimOrEmpty for str {
    fn trim_or_empty(&self) -> &str {
        self.trim()
    }
}

fn label(raw: &str) -> &str {
    raw.trim_or_empty()
}

```

```

}

fn main() {
    println!("{}", label(" sensor-1 "));
}

```

Implement for **str** with **&self**, not for **&str** with **self** by value. When the method returns **&str**, the compiler must tie that borrow to the receiver — **&self** makes that relationship explicit. Call sites still write `raw.trim_or_empty()` on a **&str**; method resolution handles the rest.

Keep extension traits **small and focused** — one concern per trait, not a grab-bag of unrelated methods.

3.16.1 Function edge cases

Wrong — mismatched `impl Trait` return arms: see [impl Trait in return position](#) for the full explanation. Minimal repro:

```

// Playground --- does not compile
fn pick(_use_a: bool) -> impl Iterator<Item =
    i32> {
    if _use_a {
        vec![1, 2].into_iter()
    } else {
        [3, 4].into_iter()
        // different concrete iterator type
    }
}

```

const fn preview: Rust can evaluate some functions at compile time (`const fn add(a: i32, b: i32) -> i32 { a + b }`). Full const evaluation rules are in Chapter 17.

3.17 When the compiler says no

Common errors in this chapter:

Error (typical)	Cause	Fix
expected T, found ()	semicolon on last expression	remove ; or add return

Error (typical)	Cause	Fix
use of moved value	took <code>String</code> by value, caller uses again	borrow with <code>&str</code> or clone
cannot borrow as mutable type annotations needed	<code>&mut</code> while <code>&</code> still alive generic <code>T</code> unknown	shrink borrow scope turbofish or annotate call site

3.18 Go deeper

- [The Rust Book — Functions](#)
- [Methods](#)

3.19 See also

- Chapter 2: Types and expressions — types in signatures
- Chapter 1: Ownership — move vs borrow in parameters
- Chapter 4: Iterators — functions as pipeline steps
- Chapter 7: Structs, traits, and generics — `impl`, traits, generics
- Chapter 8: Errors and testing — `Result` and `?`
- Chapter 17: Metaprogramming — `#[derive]` syntax, not annotations/decorators

3.19.1 Afterparty

3.19.1.1 Signatures and ownership

1. **Parameter audit** — “Five function signatures for logging: `&str`, `String`, `&String`, `Cow<str>`, `impl AsRef<str>`. I pick one per use case; you explain ownership cost.”
2. **Move vs borrow** — “Snippet calls `process(s)` then uses `s` again. I explain the error and show two fixes (`&s`, `clone`).”

3.19.1.2 Methods and `impl`

3. **`impl` block** — “Struct `Timer` with `start`, `elapsed`, `reset`. I write `impl`; you check `&self` vs `&mut self`.”
4. **Associated `fn`** — “When is `Type::new()` idiomatic vs `Default::default()`? One example each.”

5. **Consume self** — “Method `fn into_inner(self) -> Vec<u8>` — why must it take `self` by value?”

3.19.1.3 Returns and control flow

6. **Semicolon trap** — “Three tiny functions: one returns `i32` correctly, two fail due to `;`. I fix them.”
7. **Early return** — “Rewrite nested `if` in a parser as early return `None / ?` style.”
8. **Unit vs value** — “Which functions should return `()` vs `bool` vs `Option<T>`? Three CLI helper names, I choose.”

3.19.1.4 Generics and errors preview

9. **Generic bounds** — “Fix `fn max(a: T, b: T)` without bounds; add `T: Ord` or `PartialOrd`.”
10. **Result signature** — “Design `fn read_config(path: &str) -> Result<Config, ...>`; list error variants, no body.”

3.19.1.5 impl Trait and drain

11. **impl Iterator return** — “Write `top_n(vals: &[f64], n: usize) -> impl Iterator<Item = f64>` — explain why two different iterator types in `if` arms fail.”
12. **mem take drain** — “Buffer struct drains `Vec<u8>` to caller via `mem::take` — show before/after inner field.”
13. **where clause** — “Rewrite cluttered `fn f<T: A + B + C>(x: T)` with `where` block — same behaviour.”
14. **Self return** — “Method `fn into_inner(self) -> Vec<u8>` on wrapper — why `Self` not concrete type name?”
15. **const fn** — “One `const fn` port validator `fn is_valid(p: u16) -> bool` — what can and cannot run at compile time?”
16. **Error constructors** — “Add `missing_field(name, record)` on a parse error enum with `impl Into<String>`; compare to spelling the variant at each site.”
17. **Config holder** — “HTTP poll client: struct holds timeout/retries/endpoint; one `build_request(device_id)` — no builder crate.”
18. **Extension trait** — “Add `TrimOrEmpty` for `&str`; use in three parser call sites vs free functions.”

Chapter 4

Iterators

4.1 Hook

Iterators are a shared pattern, not a Rust quirk. Walk a sequence once, transform or filter each step, then produce a result — without index loops.

Stack	How the pattern shows up
Java	<code>Stream<T></code> — <code>map</code> , <code>filter</code> , <code>collect</code>
Python	<code>for x in items</code> , list comprehensions, generators
C# / LINQ	<code>IEnumerable</code> , <code>.Select()</code> , <code>.Where()</code>
JavaScript	<code>Array.prototype.map</code> / <code>filter</code> / <code>reduce</code>
SQL	<code>SELECT</code> over rows (set-oriented, not index loops)
Unix	pipes — one program's output is the next program's input stream
Rust	Iterator trait — lazy adapters + consumers

Rust's version: the **Iterator** trait — lazy `.map()` / `.filter()` chains that run only when you `.collect()` or call another consumer. See Chapter 11 for the collection types these pipelines walk.

4.2 for is syntax sugar

Chapter 2 introduced `for i in 0..3` and `loop / while`. Behind `for`, Rust calls **IntoIterator** — “turn this value into something we can walk”:

```
// Playground
fn main() {
    for i in 0..3 {
        println!("{}", i); // 0, 1, 2
    }

    let words = vec!["modbus", "tcp"];
    for w in &words {
        println!("{}", w); // borrows each &str
    }
}
```

What you write	What happens
for x in 0..10	range implements IntoIterator
for x in &vec	vec.iter() — borrow each element
for x in vec	vec.into_iter() — consume the Vec

Ranges (`..`, `..=`) are iterators. So are `.lines()` and `.chars()` on `str` from Chapter 2.

4.3 Three ways to walk a Vec

Method	You get	Vec after loop
.iter()	Iterator<Item = &T>	still usable
.iter_mut()	Iterator<Item = &mut T>	still usable, elements may change
.into_iter()	Iterator<Item = T>	moved — do not use v afterward

```
// Playground
fn main() {
    let mut v = vec![1, 2, 3];
}
```



```

let sum_borrow: i32 = v.iter().sum();
println!("sum = {} vec = {:?}", sum_borrow,
    v);

for n in v.iter_mut() {
    *n *= 10;
}
println!("after mut: {:?}", v);

let v2 = vec![4, 5];
for n in v2.into_iter() {
    println!("consumed item {}", n);
}
// println!("{:?}", v2);
// error: v2 was moved by into_iter
}

```

Rule of thumb: default to `.iter()` when you only need to read. Use `.iter_mut()` to update in place. Use `into_iter()` / `for x in vec` when the collection itself should be consumed.

4.3.1 `iter` vs `into_iter` — what changes

The iterator method picks the **element type** the whole pipeline sees:

Call	Item type (for <code>Vec<T></code>)	Who owns heap data after the chain?
<code>v.iter()</code>	<code>&T</code>	still <code>v</code>
<code>v.iter_mut()</code>	<code>&mut T</code>	still <code>v</code>
<code>v.into_iter()</code>	<code>T</code>	moved out of <code>v</code> ; <code>v</code> is empty/unusable

Common habit in Java/Python-style loops: iterating a list does not destroy the list. In Rust, `for x in v` is `into_iter` — it moves the **whole collection** into the loop, not just the elements you touch in the body.

Borrow the collection — use it again after the loop:

```

// Playground
fn main() {

```

```

let v = vec![String::from("a"),
             String::from("b")];

for s in &v {
    println!("{}", s);
    // each step: &String (borrow)
}
println!("{:?}", v);
// ok --- v was never moved
}

```

Consume by value — the `vec` is gone after the loop, even when the body does nothing with each item:

```

// Playground --- does not compile
fn main() {
    let v2 = vec![1, 2, 3];

    for n in v2 {
        // empty body --- still consumes v2 via
        // into_iter()
        let _ = n;
        // i32 is Copy, but the Vec itself is
        // not
    }
    println!("{:?}", v2);
    // ERROR: v2 moved into the for loop
}

```

`for n in v2` desugars to `IntoIterator::into_iter(v2)` — **self takes ownership of the vec**. The loop body never runs `println!`, but the iterator still walks every slot and pulls each `i32` out (a cheap copy for `Copy` types). When the loop ends, **v2 is moved**, not the individual numbers you skipped printing.

Same rule with a conditional body — using only one element does not leave the rest behind:

```

// Playground --- does not compile
fn main() {
    let v2 = vec![String::from("a"),

```

```
String::from("b"), String::from("c")];

for s in v2 {
    if s == "b" {
        println!("{}", s);
        // only this one is printed
    }
    // "a" and "c" were still moved out of
    // the vec on their turns
    // and dropped here when `s` goes out
    // of scope
}
println!("{:?}", v2);
// ERROR: v2 moved into the for loop
}
```

Each loop step **moves the next element out of the vec** into `s`. Non-Copy values like `String` cannot be put back; unprinted items are dropped at the end of that iteration. The `vec` is empty and unusable when the loop finishes — same as if you had printed every item.

Loop form	What moves	Use <code>v</code> after the loop?
<code>for x in &v</code>	nothing — borrows elements	yes
<code>for x in &mut v</code>	nothing — mutably borrows elements	yes
<code>for x in v</code>	the whole <code>v</code> into the iterator; each element on its turn	no

Fix when you need the collection back: iterate a borrow — `for x in &v` — or clone what you need before the loop.

Wrong pipeline — owned vs borrowed mismatch:

```
// Playground --- does not compile; read the
// explanation below
fn main() {
    let nums = vec![1, 2, 3];
    // Goal: Vec<i32> of doubled values --- but
```

```

    iter yields &i32
let doubled: Vec<i32> =
    nums.iter().collect();
// ERROR: expected i32, found &i32 (or
    similar mismatch on collect)
}

```

Fix: either borrow through the chain and copy at the end, or consume:

```

// Playground
fn main() {
    let nums = vec![1, 2, 3];
    let doubled: Vec<i32> =
        nums.iter().map(|&x| x * 2).collect();
    // peel & with |&x|
    let also: Vec<i32> =
        nums.into_iter().map(|x| x *
            2).collect(); // nums moved
    println!("{:?} {:?}", doubled, also);
}

```

Second pass on the same data:

```

// Playground
fn main() {
    let nums = vec![1, 2, 3];
    let sum: i32 = nums.iter().sum();
    println!("{}", sum);
    let max = nums.iter().fold(i32::MIN, |acc,
        &x| acc.max(x)); // ok --- borrow again

    let v = vec![10, 20];
    let total: i32 = v.into_iter().sum();
    // let again: i32 = v.into_iter().sum();
    // ERROR: v already moved
    println!("{}", total);
}

```

`into_iter()` **consumes** the collection once. `.iter()` lets you run many consumers (sum, then max, then print) as long as you do not also move `v`.

4.3.2 for forms — quick trap sheet

You write	Typical mistake
for x in v	Using v after the loop — v was moved into the loop
for x in &v	Expecting x to be T — x is &T (often fine for println!)
for x in &mut v	Forgetting *x to mutate the element
for x in 0..v.len() with v[i]	Out-of-bounds panic if v shrinks while looping — prefer .iter()

4.4 Lazy iterators

Adapters return **another iterator** — they do not allocate a full intermediate Vec unless you .collect().

```
// Playground
fn main() {
    let nums = vec![1, 2, 3, 4, 5];
    let doubled_evens: Vec<i32> = nums
        .iter()
        .filter(|&&x| x % 2 == 0)
        .map(|&x| x * 2)
        .collect();
    println!("{:?}", doubled_evens); // [4, 8]
    println!("nums still {:?}", nums);
}
```

.filter and .map take **closures** (|x| ...). How closures capture variables (Fn, FnMut, FnOnce) is covered in Chapter 12 — Closures.

4.4.1 Why closures see &&i32 (double reference)

Two & layers stack — one from the iterator, one from .filter():

1. .iter() yields &i32 — a borrow of each element in the vec.
2. .filter() passes & + that item to your closure — it only checks the element, it does not take it.

& + &i32 = &&i32

That is the whole story for `v.iter().filter(...)`. The table below shows how the same rule plays out for `.map()` (which takes the item directly, without an extra `&`):

Chain	Item	<code>.filter</code> closure gets	<code>.map</code> closure gets
<code>v.iter()</code>	<code>&i32</code>	<code>&&i32</code>	<code>&i32</code>
<code>v.into_iter()</code>	<code>i32</code>	<code>&i32</code>	<code>i32</code>

Three ways to write the same filter — pick one style and stay consistent:

Closure param	Type of binding	Works for % 2?
<code>\ x\ </code>	<code>x: &&i32</code>	yes — <code>*/%</code> auto-deref through references
<code>\ \&x\ </code>	<code>x: &i32</code>	yes — one layer peeled
<code>\ \&\&x\ </code>	<code>x: i32</code>	yes — two layers peeled; most explicit

Mixing `.iter()` with a closure that expects owned `i32` without peeling any reference layer is a frequent compile error.

The `Iterator` trait lives in `std` and defines `.next() -> Option<Item>`. Calling `.iter()` on standard types covers most code — when you need a custom walk, see [Implementing Iterator](#) below and Chapter 7 — associated types.

4.5 Common adapters

Adapter	Effect
<code>.map(f)</code>	transform each item
<code>.filter(p)</code>	keep items where <code>p</code> is true
<code>.enumerate()</code>	(index, item) pairs
<code>.zip(other)</code>	pair items until one iterator ends
<code>.take(n) / .skip(n)</code>	first <code>n</code> / skip first <code>n</code>
<code>.chain(other)</code>	append another iterator

```
// Playground
fn main() {
```

```

let a = ['x', 'y', 'z'];
let b = [1, 2, 3];

for (i, ch) in a.iter().enumerate() {
    println!("{}", i, ch);
}

for (ch, n) in a.iter().zip(b.iter()) {
    println!("{}", ch, n);
}

let first_two: Vec<_> =
    a.iter().take(2).collect();
println!("{}", first_two);
}

```

On text, remember Chapter 2 helpers: `"a\nb\n".lines()`, `"hi".chars()` — each returns an iterator over parts of the string.

4.5.1 Adapter edge cases (empty, short, uneven)

Situation	What happens
<code>.zip</code> on unequal lengths	stops when either iterator ends — silent truncation
<code>.take(100)</code> on 3 items	yields 3, no error
<code>.skip(10)</code> on 3 items	yields nothing
<code>.chain</code>	fused walk; still lazy until a consumer runs

```

// Playground
fn main() {
    let ids = vec![1u16, 2];
    let vals = vec![10u16, 20, 30];
    // longer than ids
    let pairs: Vec<_> =
        ids.iter().zip(vals.iter()).collect();
    println!("{}", pairs);
    // [(1, 10), (2, 20)] --- third value
    // dropped
}

```

```

let empty: Vec<i32> = vec![];
let n: i32 = empty.iter().sum();
// 0 --- not an error
let hit = empty.iter().find(|&&x| x > 0);
// None
println!("sum={} find={:?}", n, hit);
}

```

Wrong — two mutable walks at once:

```

// Playground --- does not compile
fn main() {
    let mut v = vec![1, 2, 3];
    for a in v.iter_mut() {
        for b in v.iter_mut() {
            // ERROR: cannot borrow `v` as
            // mutable more than once
            *a += *b;
        }
    }
}

```

Rust forbids overlapping `&mut` borrows, including through nested `iter_mut()` loops. Fix: index by position, use `.enumerate()`, or collect indices first — patterns from Chapter 1.

4.6 Consumers — finishing the pipeline

A **consumer** drains the iterator and produces a value (or side effect).

Consumer	Result
<code>.collect()</code>	build <code>Vec</code> , <code>String</code> , <code>HashMap</code> , ...
<code>.sum()</code> / <code>.product()</code>	numeric fold
<code>.count()</code>	number of items
<code>.find(p)</code>	<code>Option<Item></code> — first match
<code>.any(p)</code> / <code>.all(p)</code>	<code>bool</code>
<code>.fold(init, f)</code>	single accumulated value


```
// Playground
fn main() {
    let nums = vec![3, 1, 4, 1, 5];
    let sum: i32 = nums.iter().sum();
    let max = nums.iter().fold(i32::MIN, |acc,
        &x| acc.max(x));
    let has_even = nums.iter().any(&x| x % 2
        == 0);
    println!("sum={} max={} has_even={}", sum,
        max, has_even);
}
```

4.6.1 collect and type hints

`collect()` can build many collection types. Sometimes Rust needs help:

```
// Playground
fn main() {
    let nums = vec![1, 2, 3];
    let doubled: Vec<i32> =
        nums.iter().map(&x| x * 2).collect();
    let also: Vec<i32> = nums.iter().map(&x| x
        * 2).collect::<Vec<_>>();
    println!("{:?} {:?}", doubled, also);
}
```

`Vec<_>` asks the compiler to infer the element type. Prefer an explicit binding type (`let v: Vec<i32> = ...`) when `collect` is the only clue.

Wrong — collect without enough type context:

```
// Playground --- does not compile
fn main() {
    let nums = vec![1, 2, 3];
    let doubled = nums.iter().map(&x| x *
        2).collect();
    // ERROR: type annotations needed --- many
    // types implement FromIterator
}
```

Wrong — collecting references, then dropping the owner:

`drop(value)` (from `std::mem::drop`) **ends the value's lifetime early** — it runs cleanup (same as when a variable goes out of scope at `}`) and frees the owned data. Values normally drop automatically at scope end; you call `drop` only when you need to release something *before* the closing brace. Here the example tries to destroy `lines` while `tags` still holds `&str` slices pointing into it:

```
// Playground --- does not compile
fn main() {
    let lines: Vec<String> =
        vec![String::from("PORT=502")];
    let tags: Vec<&str> = lines.iter().map(|s|
        s.as_str()).collect();
    drop(lines);
    // explicit early destroy --- would free
    // the strings `tags` points at
    // println!("{:?}", tags);
    // would dangle --- often fails earlier:
    // tags borrows `lines`, so drop(lines) is
    ERROR while tags is alive
}
```

Owned collection after parsing is the safe default: `.map(|s| s.clone())` or collect into `Vec<String>` before dropping the source buffer. Lifetimes formalize this in Chapter 5.

4.6.2 Consumer edge cases

Call	Empty input	Notes
<code>.sum()</code>	0 for numbers	type must implement <code>Sum</code>
<code>.product()</code>	1 for multiplication	
<code>.count()</code>	0	
<code>.find(...)</code>	None	not a panic — use <code>if let</code> / <code>match</code>
<code>.max()</code> / <code>.min()</code>	None	
<code>.any</code> / <code>.all</code>	false / true	empty all is vacuously true

```
// Playground
fn main() {
    let ports: Vec<u16> = vec![];
```

```

let ok = ports.iter().all(|&p| p >= 1 && p
    <= 65535);
println!("all in range? {}", ok);
// true on empty --- know your domain rule

let logs = vec!["ok", "ERROR: timeout"];
if let Some(line) = logs.iter().find(|&s|
    s.contains("ERROR")) {
    println!("first error line: {}", line);
}
}

```

4.7 Implementing Iterator

Most code uses `.iter()` on collections. When you own the walk logic — numeric ranges, non-empty lines, frame parsing — implement `Iterator` yourself.

4.7.1 Inclusive range counter

Walk integers `1..=5` without building a `Vec` first — useful for sample indices, tick counts, or any stepped sequence:

```

// Playground
struct RangeCounter {
    next: u16,
    end: u16,
}

impl Iterator for RangeCounter {
    type Item = u16;

    fn next(&mut self) -> Option<Self::Item> {
        if self.next > self.end {
            return None;
        }
        let n = self.next;
        self.next += 1;
        Some(n)
    }
}

```

```

    }
}

fn main() {
    let values: Vec<u16> = RangeCounter { next:
        1, end: 5 }.collect();
    let sum: u16 = RangeCounter { next: 1, end:
        4 }.sum(); // 1 + 2 + 3 + 4
    println!("values={:?} sum={}", values, sum);
}

```

type `Item = u16` is an **associated type** — see Chapter 7. Adapters like `.map` and `.sum` use it to know what flows through the pipeline.

4.7.2 Non-empty line iterator

Skip blank lines while walking config text:

```

// Playground
struct NonEmptyLines<'a> {
    inner: std::str::Lines<'a>,
}

impl<'a> Iterator for NonEmptyLines<'a> {
    type Item = &'a str;

    fn next(&mut self) -> Option<Self::Item> {
        loop {
            let line = self.inner.next()?;
            let trimmed = line.trim();
            if !trimmed.is_empty() {
                return Some(trimmed);
            }
        }
    }
}

fn main() {
    let cfg = "PORT=502\n\nHOST=127.0.0.1\n";
    let keys: Vec<_> = NonEmptyLines { inner:

```

```

        cfg.lines() }
        .filter_map(|line|
            line.split('=').next())
        .collect();
    println!("{:?}", keys);
}

```

State lives in the struct fields. `next` returns `None` when the inner iterator is exhausted — that is how every iterator signals “done.”

4.7.3 Implementing Iterator edge cases

Empty range — immediate `None`:

```

// Playground
struct RangeCounter {
    next: u16,
    end: u16,
}

impl Iterator for RangeCounter {
    type Item = u16;

    fn next(&mut self) -> Option<Self::Item> {
        if self.next > self.end {
            return None;
        }
        let n = self.next;
        self.next += 1;
        Some(n)
    }
}

fn main() {
    let scan = RangeCounter { next: 5, end: 4 };
    // empty range: start already past end
    let n = scan.count();
    println!("{}", n); // 0
}

```

Infinite range — always pair with `.take(n)`:

```
// Playground
struct Counter {
    n: u64,
}

impl Iterator for Counter {
    type Item = u64;
    fn next(&mut self) -> Option<u64> {
        let v = self.n;
        self.n += 1;
        Some(v)
    }
}

fn main() {
    let first_five: Vec<_> = Counter { n: 0 }
        .take(5).collect();
    println!("{:?}", first_five);
    // [0, 1, 2, 3, 4]
}
```

IntoIterator vs Iterator: collections implement `IntoIterator` so for `x` in `vec` works. Your type can implement both — `Iterator` for `.next()`, and `IntoIterator` with `type Item = Self`; `type IntoIter = Self` when consuming `self` in a `for` loop is natural.

4.8 When the compiler says no

Common errors in this chapter:

Error message (typical)	Likely cause	Smallest fix
value used after move	for <code>x</code> in <code>v</code> or <code>into_iter()</code> then <code>v</code> again	for <code>x</code> in <code>&v</code> or <code>.iter()</code>
cannot borrow as mutable more than once	nested <code>iter_mut</code>	one <code>mut</code> borrow at a time; restructure
type annotations needed for <code>collect</code>	ambiguous target collection	let <code>x: Vec<T> = ...</code> or <code>turbofish</code>

Error message (typical)	Likely cause	Smallest fix
expected T, found &T in collect	.iter() but collecting owned values	\&x in map or use into_iter()
borrow of moved value	.into_iter().sum() then reuse v	clone input or use .iter()
lifetimes / may not live long enough	Vec<&str> pointing into dropped String	own the strings

Read the **first** error top-down; iterator chains confuse the borrow checker, but the fix is usually “wrong walk mode (**iter** vs **into_iter**)” or “references outlive owner.”

4.9 Java / Python contrast

Aspect	Java	Python	Rust
Lazy pipeline	Stream (often from collection)	generator / itertools	Iterator adapters
Eager list comp	—	[f(x) for x in xs]	.iter().map(f).collect()
Consume collection	stream from list	for x in list	into_iter() moves
Index loop	for (int i=0; ...)	for i in range(len)	prefer .iter().enumerate()

4.10 Idiom spotlight

Prefer iterator chains over index loops. Use `.iter()`, `.enumerate()`, or `.windows(2)` instead of `for i in 0..v.len()` with `v[i]`. Sliding windows: Chapter 11.

4.11 Go deeper

- [The Rust Book — Iterators](#)
- [Functional Rust — iterator topics](#)

4.12 See also

- Chapter 1: Ownership and borrowing — `iter` vs `into_iter`

- Chapter 2: Types and expressions — ranges, `for`, string iterators
- Chapter 5: Lifetimes — borrows inside iterator chains
- Chapter 7: Structs, traits, and generics — `type Item`, associated types
- Chapter 11: Collections — `Vec`, `HashMap`, `.windows`
- Chapter 12: Closures — `Fn` traits for adapters

4.12.1 Afterparty

4.12.1.1 `for`, ranges, and three walk modes

1. **for desugar quiz** — “Four `for` loops (`0..n`, `&vec`, `vec`, `&mut vec`). I say which calls `iter`, `into_iter`, or `range`; you correct and show owner state after the loop.”
2. **iter vs into_iter** — “Give 4 snippets using `Vec`; I predict whether `v` is usable after the loop; you explain move vs borrow.”
3. **iter_mut drill** — “Task: double every element in `Vec<f64>` in place without allocating a new `Vec`. I write the loop; you review `*n` and borrow rules.”
4. **Range vs collect** — “When is `for i in 0..n` better than `(0..n).collect::? Give two automation examples (retry count vs materializing indices).”`

4.12.1.2 Adapters and lazy pipelines

5. **Adapter chain** — “Task: parse lines, trim, keep non-empty, parse as `u16`. I sketch `.lines().map(...).filter(...).collect()`; you refine.”
6. **zip pairs** — “Two `Vec<u16>`: register IDs and values. Build `Vec<(u16, u16)>` with `.zip()`; I write it; you handle length mismatch policy.”
7. **take and skip** — “Paginate a log: skip first 100 lines, take next 20. I use `.skip().take()` on `.lines()`; you show one pitfall if the source is not recomputed.”
8. **chain iterators** — “Concatenate header rows and body rows (two `&[u8]` slices) without copying bytes into one array first — sketch with `.iter().chain()`.”
9. **enumerate vs index** — “Same sum-over-evens task twice: index `for` vs `.enumerate()`. Compare readability and bounds-check risk.”
10. **Lazy vs eager** — “Explain when `filter().map()` allocates vs when `.collect()` forces work. One example with `println!` in `map` showing evaluation order.”

4.12.1.3 Consumers and types

11. **collect turbofish** — “Three `collect()` calls that fail without hints — I add type annotation or turbofish; you verify.”
12. **find and Option** — “Wire scan: `Vec<&str>` of lines, find first containing `ERROR`. I return `Option<&str>` with `.find()`; you contrast with `.filter().next()`.”
13. **fold vs sum** — “Compute max and count in one pass with `.fold()` vs calling `.max()` and `.len()` separately — when is fold worth it?”
14. **any and all** — “Validate a batch: all ports in `1..=65535`, any line starts with `#`. I write `.all()` / `.any()` predicates; you fix one double-reference mistake.”

4.12.1.4 Errors, ownership, and automation

15. **Moved Vec mistake** — “Show code that does `for x in v` then uses `v` again. I explain the error and fix with `.iter()` or `clone`; you rank fixes.”
16. **Borrow in chain** — “Snippet: build `Vec<&str>` from `String` lines then drop the `String`. I explain why it fails; you fix with owned `String` or different lifetime design.”
17. **Modbus-style scan** — “List of raw register values `Vec<u16>`: filter evens, map to `f64` scale 0.1, sum. I write the iterator chain; you check overflow and types.”
18. **Zero-cost check** — “Does `nums.iter().filter(...).map(...).sum()` allocate intermediate `Vecs`? Answer for `--release` and what to measure conceptually.”

4.12.1.5 Compile errors and edge cases

19. **Trap sheet drill** — “Give 6 snippets mixing `for x in v`, `for x in &v`, `into_iter`, and `collect` type errors. I predict compile ok or fail and why; you show the fixed line.”
20. **&&i32 decoder** — “One `.iter().filter().map()` chain: I label the closure parameter type at each step; you correct and show `|x|`, `|\&x|`, and `|\&\&x|` versions that compile.”
21. **Empty iterator policy** — “Three tasks (`sum`, `find`, `all`) on possibly empty `Vec`. I state the result and whether it is a domain bug; you correct (e.g. empty `all` is true).”
22. **zip truncation** — “IDs `len 5`, values `len 100` — I write `zip collect`; you explain silent loss and sketch `zip + length check` or `enumerate` on the

longer vec.”

4.12.1.6 Custom iterators

23. **RangeCounter impl** — “Implement `Iterator` for integers 1..`n`; collect to `Vec` and sum with `.sum()` — show `type Item` and `fn next` only.”
24. **Skip blanks** — “Config string with empty lines — write `NonEmptyLines` iterator that trims and skips `' '`; collect keys before `=`.”
25. **Infinite take** — “Counter from 0 without end — why must you `.take(n)` before `.collect()`? Show hang vs bounded collect.”
26. **IntoIterator pair** — “Same struct: implement `Iterator` and `IntoIterator` so both `scan.next()` and `for p in scan` work — minimal impl blocks.”
27. **Stateful parser** — “Byte buffer iterator yielding complete 4-byte frames; partial frame stays in struct — sketch `next()` state machine.”
28. **Capstone iterator** — “CSV line iterator: split fields, parse col 2 as `u16`, filter `> 0` — custom struct + one consumer chain.”

Chapter 5

Lifetimes

5.1 Hook

In GC-managed languages (**Java**, **Python**, **C#**, ...), a collector keeps heap objects alive while any reference can reach them. You rarely ask whether a pointer is still valid.

Rust has no GC. Every **&T** must not outlive the value it borrows. **Lifetimes** are the compiler's way of proving that — usually without you writing any syntax.

5.2 Scope — a brief tour

Lifetimes are compile-time labels on references — not runtime values. This chapter covers elision, struct borrows, and **'static** traps. HRTB, GATs, and Pin are deferred.

This chapter covers	Deferred
'a on functions and structs, elision	Higher-ranked lifetimes (HRTB)
'static, owned vs borrowed fields	Pin and self-referential structs
T: 'a bounds preview	Full struct chapter -> Ch 7

5.3 References always have a lifetime

Every `&T` is valid for some span of code. If you return a reference to a local variable, that reference dies when the stack frame ends.

The compiler rejects that pattern.

```
// Playground --- this pattern is OK: reference
    does not outlive owner
fn first_word(s: &str) -> &str {
    let bytes = s.as_bytes();
    for (i, &item) in bytes.iter().enumerate() {
        if item == b' ' {
            return &s[0..i];
        }
    }
    s
}

fn main() {
    let sentence = String::from("hello world");
    let word = first_word(&sentence);
    println!("{}", word);
}
```

5.4 What lifetimes are for

Aim: guarantee you never use a borrow after its owner is gone.

Chapter 1 said each `&T` must not outlive what it points to. That rule is easy inside one function.

It gets hard when you **return** a reference or **store** one in a struct. The caller cannot see your local variables, so the compiler needs a contract on the signature.

Bad (won't compile): return a pointer into memory that dies when the function returns.

```
// Playground --- uncomment to see the error
fn broken() -> &str {
    let s = String::from("tmp");
}
```

```

    &s
    // error: `s` dropped here; return would
    // dangle
}

```

Good: return a slice that still lives in the caller’s `String`:

```

fn first_word(s: &str) -> &str { /* ... */ }
// return borrows from `s`, not from locals

```

A **lifetime** is the span of code where a borrow is valid. You do not set it at runtime — the compiler checks it at compile time.

Lifetime syntax on functions and structs answers one question:

“Which input (or owner) must still be alive while this reference exists?”

5.4.1 How 'a helps (it’s just a label)

When several references in one signature must live **together**, give that span a name — 'a, 'b, 'input, whatever reads clearly.

Same idea as naming a generic type `T`: the name is for humans and the compiler. **'a is not magic.**

```

fn longest<'a>(x: &'a str, y: &'a str) -> &'a
    str {
    if x.len() >= y.len() { x } else { y }
}

```

Read this as: “The returned `&str` is valid while **both** `x` and `y` are valid.” Those borrows share one lifetime 'a.

If `s1` is dropped while something still holds the return from `longest(&s1, &s2)`, the program must not compile.

5.4.2 What if you omit <'a>?

The body does not change — you still return either `x` or `y`. The lifetime is **not** something you execute at runtime; it is a **signature** contract for the compiler. Try writing `longest` without any lifetime parameters:

```
fn longest(x: &str, y: &str) -> &str {
    // does not compile
    if x.len() >= y.len() { x } else { y }
}
```

Rust rejects this with **missing lifetime specifier** (or similar).

Reason: the return might point into **x** or **y**. Those borrows can come from **different owners** with **different lifetimes**. The compiler will not guess.

With lifetimes	Without lifetimes
-> &'a str tied to x and y	Compiler does not know which input the return borrows from
Caller must keep both s1 and s2 alive while using the result	No check that you drop the wrong String too early
Elision cannot help: two reference inputs + one reference output	Signature is incomplete

Compare `first_word(s: &str) -> &str`: one borrowed input, one returned reference. Elision can assume “output lives as long as s.”

`longest` has **two** borrowed inputs, so elision stops. You must write `'a` (or `'long`, etc.) yourself.

```
// Playground
fn longest<'a>(x: &'a str, y: &'a str) -> &'a
    str {
    if x.len() >= y.len() { x } else { y }
}

fn main() {
    let s1 = String::from("long");
    let s2 = String::from("x");
    let r = longest(&s1, &s2);
    // drop s1;
    // error: r might still point into s1
    println!("{}", r);
}
```

Without `<'a>` on the function, this `main` could compile even when `r` still points at freed memory.

Lifetimes exist so that mistake is caught on the **function definition**, before any caller runs.

Situation	What you are telling the compiler
<code>fn first<'a>(s: &'a str) -> &'a str</code>	Return borrows from s only
<code>fn longest<'a>(x: &'a str, y: &'a str) -> &'a str</code>	Return may point into x or y; both must stay alive
<code>fn pick<'a, 'b>(x: &'a str, y: &'b str) -> &'a str</code>	Return borrows from x only; y can die earlier

The only lifetime name with a **fixed** meaning is **'static**: valid for the whole program (e.g. string literals).

Do not slap **'static** on normal borrows to silence errors.

Mental model: owner = suitcase, &T = claim ticket. Lifetimes prove you never read the ticket after the suitcase was thrown away.

5.5 Elision (why you rarely write lifetimes)

Often the compiler infers the contract above — **lifetime elision**. You write the short form; Rust fills in **'a** for you:

```
// Playground --- elision expansion (signatures
//   only; not runnable alone)
// fn len(s: &str) -> usize
//
// means: fn len<'a>(s: &'a str) -> usize ---
// no return reference, so no tie needed
//
// fn first_word(s: &str) -> &str
//
// means: return borrows from `s` (elision ties
// input and output)
```

Three **elision rules** cover most everyday signatures (the compiler applies them automatically):

1. Each reference parameter gets its own lifetime parameter (`&str -> &'a str, ...`).

2. If there is exactly **one** input lifetime, it is assigned to **all** output lifetimes (fn foo<'a>(x: &'a str) -> &'a str).
3. If there are multiple input lifetimes but one is **&self** or **&mut self**, the lifetime of **self** is assigned to all output lifetimes (methods only).

When elision fails — several references in and out, or an ambiguous return — write 'a / 'b explicitly. Any name works.

```
fn longest<'long>(x: &'long str, y: &'long str)
    -> &'long str {
    if x.len() >= y.len() { x } else { y }
}
```

5.6 Named lifetimes on structs

You have not met **structs** in depth yet (Chapter 7 covers **struct**, **impl**, and **traits**). For lifetimes, you only need this much:

A **struct** groups named fields into one type — like a labeled tuple or a small Java/Python data class:

```
struct Point {
    x: f64,
    y: f64,
}
```

When a field is a **borrow** (&str, &[u8], ...) instead of an owned value (String, Vec), the struct does not own that memory. It only holds a pointer into someone else's buffer.

The same question applies as for **longest**:

“Which owner must still be alive while this struct exists?”

Put 'a on the **struct** (not just on a function) when a field stores a reference:

```
// Playground
struct Excerpt<'a> {
    text: &'a str,
    // borrows from outside; struct does not
    // own the bytes
}
```



```
fn main() {
    let novel = String::from("Call me Ishmael.
        Once upon a time...");
    let first =
        novel.split('.').next().unwrap_or("");
    let e = Excerpt { text: first };
    println!("{}", e.text);
}
// `e` and `first` dropped here; then `novel`
// can be dropped
```

Read `Excerpt<'a>` like `longest<'a>`: **'a is the lifetime of the borrow in text**. The struct may not outlive the data `text` points into — here, `novel`.

Piece	Role
<code>novel: String</code>	Owner of the heap text
<code>first: &str</code>	Borrow into <code>novel</code>
<code>e: Excerpt<'a></code>	Bundles that borrow; 'a ties <code>e.text</code> to <code>novel</code> 's lifetime

5.6.1 What if the struct omits `<'a>`?

```
struct Excerpt {    // does not compile
    text: &str,
}
```

Same idea as `fn longest(x: &str, y: &str) -> &str` without lifetimes. The compiler sees a reference inside the type but **no contract** for how long it is valid.

Error: **missing lifetime specifier** on `text`.

Practical rule: struct fields that are references almost always need a lifetime parameter on the struct. Fields that are owned (`String`, `u32`, `Vec<..>`) do not.

```
// Playground --- uncomment `drop` to see the
    error
struct Excerpt<'a> {
```

```

    text: &'a str,
}

fn main() {
    let novel = String::from("Call me
        Ishmael.");
    let first =
        novel.split('.').next().unwrap_or("");
    let e = Excerpt { text: first };
    // drop(novel);
    // error: `e.text` still borrows `novel`
    println!("{}", e.text);
}

```

Prefer **owned** fields in public APIs (String instead of &str) until you need zero-copy views.

Methods on structs (impl Excerpt { ... }) come in Chapter 7.

5.7 'static trap — formatted strings

Wrong — reference to temporary:

```

// Playground --- does not compile
fn label() -> &'static str {
    let s = format!("sensor-{}", 1);
    &s // ERROR: `s` dropped at end of function
}

```

Right — return owned:

```

// Playground
fn label() -> String {
    format!("sensor-{}", 1)
}

fn main() {
    println!("{}", label());
}

```

Only string literals and leaked allocations are 'static. format! always

produces an owned `String`.

5.8 Two lifetimes — when `'a` and `'b` diverge

Return borrows from **one** input only:

```
// Playground
fn first<'a, 'b>(x: &'a str, _y: &'b str) ->
    &'a str {
    x
}

fn main() {
    let a = String::from("left");
    let b = String::from("right");
    let r = first(&a, &b);
    drop(b);
    println!("{}", r);
    // OK --- r only borrows `a`
}
```

If the return could point into either argument, both inputs share one `'a` like longest.

5.9 `Config<'a>` — borrowed slices vs owned `fix`

```
// Playground
struct Config<'a> {
    host: &'a str,
    port: u16,
}

fn parse_line<'a>(line: &'a str) ->
    Option<Config<'a>> {
    let mut parts = line.split(':');
    let host = parts.next()?;
    let port: u16 = parts.next()?.parse().ok()?;
}
```

```

    Some(Config { host, port })
}

fn main() {
    let line = String::from("127.0.0.1:502");
    if let Some(cfg) = parse_line(&line) {
        println!("{}", cfg.host, cfg.port);
    }
}

```

Public APIs often use owned `String` fields instead — callers need not keep the original buffer alive.

5.9.1 Lifetime edge cases

T: 'a — generic container holding a borrow:

```

// Playground
struct Holder<'a, T: 'a> {
    value: &'a T,
}

fn main() {
    let n = 502u16;
    let h = Holder { value: &n };
    println!("{}", h.value);
}

```

T: 'a means “T must live at least as long as 'a.”

5.10 When the compiler says no

Common errors in this chapter:

Typical fixes:

- Return an **owned** `String` instead of `&str`.
- Take owned data as parameter and return owned data.
- Use `Rc/Arc` when shared ownership is real (Chapter 10).

5.11 Java / Python contrast

Aspect	Java	Python	Rust
dangling pointer	rare (GC + JIT)	possible C extensions	compile error
return inner ref	object on heap OK	OK if object lives	must tie lifetimes
self-referential struct	awkward	awkward	needs pins/advanced

5.12 Idiom spotlight

Return owned types from public APIs unless you are building a zero-copy internal API and can document lifetime ties.

String and Vec are easy. &str in returns often fights the borrow checker for newcomers.

5.13 Playground: two inputs, one shared lifetime

When the return must be valid for **either** argument, both references share one lifetime parameter. Any name works — here 'a.

```
// Playground
fn longest<'a>(x: &'a str, y: &'a str) -> &'a
    str {
    if x.len() >= y.len() { x } else { y }
}

fn main() {
    let s1 = String::from("longer");
    let s2 = String::from("x");
    println!("{}", longest(&s1, &s2));
}
```

5.14 Go deeper

- [The Rust Book — Validating References with Lifetimes](#)

- [Functional Rust — pattern matching & types](#)

5.15 See also

- Chapter 1: Ownership and borrowing
- Chapter 4: Iterators — `.iter()` borrows reinforce lifetime thinking
- Chapter 7: Structs, traits, and generics — full `struct / impl` coverage (this chapter only needs borrowed fields + `'a`)

5.15.1 Afterparty

1. **Error archaeology** — “I paste a ‘lifetime may not live long enough’ error; walk me through owner vs reference diagram.”
2. **Return type choice** — “For API `fn title(book: &Book) -> ???` compare `&str` vs `String` trade-offs for a library.”
3. **Struct lifetime** — “Design `ConfigParser` holding `&str` slices into input buffer — when is it sound vs use owned `String`?”
4. **Elision quiz** — “Add explicit lifetimes to 4 function signatures where elision fails.”
5. **Fix mine** — “I return `&String` built inside function; show three idiomatic fixes ranked by simplicity.”

5.15.1.1 Lifetimes in practice

6. **static trap** — “Function returns `&str` from `format!` — show error and owned fix.”
7. **two lifetimes** — “Write `fn first<'a, 'b>(x: &'a str, y: &'b str) -> &'a str` — drop `y` while result lives.”
8. **Config struct** — “Parse `host:port` into `Config<'a>` — when must caller keep line alive?”
9. **Owned refactor** — “Same parser returning owned `Config { host: String, port: u16 }` — tradeoffs in 3 bullets.”
10. **T: 'a bound** — “Explain `struct Holder<'a, T: 'a> { value: &'a T }` — what fails if `T` is shorter-lived?”
11. **Elision fail** — “Four signatures where elision works vs fails — I label each.”
12. **Iterator borrow** — “Collect `Vec<&str>` from `String` lines — why drop order matters (link Ch 4).”

Chapter 6

Enums and Pattern Matching

6.1 Hook

Java uses `null` for “missing” and often throws exceptions for failures. **Python** uses `None` and exceptions, or informal unions (`int | str`) without the type checker enforcing branches. If you know other languages, map their “missing value” and error habits to the table below.

Rust encodes “one of several possibilities” in the **type system**:

Idea	Java	Python	Rust
Missing value	<code>null</code>	<code>None</code>	<code>Option<T></code>
Failure	exception	exception	<code>Result<T, E></code>
Several shapes	class hierarchy	duck typing / Union	<code>enum</code> + <code>match</code>
Must handle all cases	runtime surprise	runtime surprise	compile error if <code>match</code> is incomplete

`match` is not a fancy `switch`. It **destructures** data, and the compiler insists you cover every variant. That is how Rust replaces null checks and many runtime type errors with reviewable compile-time rules (Chapter 1).

Chapter 2 previewed `match` on integers. This chapter applies the same machinery to `Option`, `Result`, and your own `enums`.

6.2 Option<T> — no null

Option is an enum with two variants: Some(T) or None.

```
// Playground
fn divide(a: f64, b: f64) -> Option<f64> {
    if b == 0.0 { None } else { Some(a / b) }
}

fn main() {
    match divide(10.0, 2.0) {
        Some(v) => println!("{}", v),
        None => println!("undefined"),
    }
}
```

Callers must decide what None means — log, default, propagate, or abort. Rust will not let you treat Option like a plain T without handling None (unless you explicitly force it — see edge cases below).

6.2.1 if let, while let, and let ... else

When only **one** variant matters:

```
// Playground
fn main() {
    let maybe_port = Some(502u16);
    if let Some(p) = maybe_port {
        println!("port {}", p);
    }

    let raw = "8080";
    let Ok(n) = raw.parse:::<u16>() else {
        println!("bad port");
        return;
    };
    println!("parsed {}", n);
}
```

let ... else (Rust 1.65+) is idiomatic for “parse or bail early” without nested match.

6.2.2 Option combinators (before heavy match)

Method	Use when
<code>.map(f)</code>	transform inner value if Some
<code>.and_then(f)</code>	chain fallible steps (f returns Option)
<code>.unwrap_or(default)</code>	supply fallback
<code>.ok_or(err)</code>	turn None into Err for Result chains

```
// Playground
fn main() {
    let s = "502";
    let port = s.trim().parse:::<u16>().ok();
    // Option<u16>
    let doubled = port.map(|p| p * 2);
    println!("{:?}", doubled);
}
```

Full error typing for Result chains: Chapter 8.

6.2.3 Option edge cases and compiler traps

Wrong — treat **Option** like nullable Java without branching:

```
// Playground --- does not compile
fn main() {
    let maybe: Option<i32> = Some(5);
    // let x: i32 = maybe;
    // ERROR: expected i32, found Option<i32>
    let x = maybe.unwrap();
    // compiles --- panics on None at runtime
}
```

Approach	None behavior	Idiom
match / if let	you handle it	preferred
unwrap() / expect("...")	panic	prototypes, tests, “impossible” invariants
unwrap_or / unwrap_or_else	default	config with safe fallback
? in fn -> Option	propagate None	chaining parsers

Wrong — **unwrap** in production paths:

```
// Playground --- runs, then may panic in
production
fn read_port(s: &str) -> u16 {
    s.parse().ok().unwrap()
    // panic if config line is wrong
}
```

Prefer `match`, if `let`, or `Result` with `?` so a bad config file returns an error instead of crashing a long-running service.

Exhaustive `match` on `Option`:

```
// Playground --- does not compile if you omit
an arm
fn main() {
    let x = Some(1);
    match x {
        Some(n) => println!("{}", n),
        // None => {}
        // ERROR: non-exhaustive patterns:
        // `None` not covered
    }
}
```

6.3 `Result<T, E>` — errors as values

`Result` is also a two-variant enum: `Ok(T)` or `Err(E)`.

```
// Playground
fn parse_positive(s: &str) -> Result<i32,
    &'static str> {
    let n: i32 = s.parse().map_err(|_| "not a
        number")?;
    if n > 0 { Ok(n) } else { Err("not
        positive") }
}

fn main() {
    println!("{:?}", parse_positive("42"));
    println!("{:?}", parse_positive("-1"));
```

```
println!("{:?}", parse_positive("oops"));
}
```

The `?` operator propagates `Err` early — like `throw`, but **typed** and visible in the signature. Custom error types, **thiserror**, and errors as enums: Chapter 8.

6.3.1 Result edge cases

Wrong — ignore the error arm:

```
// Playground --- does not compile
fn main() {
    let r: Result<i32, &str> = Ok(1);
    match r {
        Ok(n) => println!("{}", n),
        // Err(e) => ...
        // ERROR: non-exhaustive patterns
    }
}
```

Wrong — unwrap on Result in library code:

```
// Playground
fn load_timeout(s: &str) -> u32 {
    s.parse().unwrap()
    // panic on "abc" --- use ? or match instead
}
```

Idiomatic boundary pattern:

```
// Playground
fn parse_port(s: &str) -> Result<u16, String> {
    s.trim().parse::<u16>().map_err(|e|
        format!("port: {}", e))
}

fn main() {
    match parse_port("502") {
        Ok(p) => println!("ok {}", p),
        Err(msg) => eprintln!("config error:
            {}", msg),
    }
}
```

```
}  
}
```

Where to match vs where to use ?: errors have to stop somewhere. Split your code into two layers:

Layer	Where	What to do	Why
Boundary	main, CLI entry, HTTP handler, test harness	match (or if let) on Result — print, log, set exit code, return HTTP 400	No caller above you to propagate to; you decide the user-facing outcome
Interior	parsers, validators, load_config, anything called by other Rust code	? to return Err early; keep the happy path flat	Callers choose how to handle failure; helpers stay reusable

In the example above, `parse_port` is **interior** — it returns `Result<u16, String>` and lets the caller decide. `main` is the **boundary** — it matches once and turns `Err(msg)` into `eprintln!(...)`. If `main` used `?` instead, it would need its own `-> Result<..., ...>` signature and something *above* `main` to handle failure (Rust binaries often use a `main -> Result` wrapper for that — Chapter 8).

Rule of thumb: library and helper functions **return** `Result`; the outermost layer **consumes** `Result` and converts it to human output or process exit status. Do not `unwrap` in the middle; do not `match` on every line inside a ten-function call stack.

6.4 Custom enums — sum types

Your own `enum` lists allowed states or message kinds. Each variant can be a `unit`, `tuple`, or `struct` shape.

```
// Playground  
enum Status {  
    Idle,  
    Running { rpm: u32 },  
    Fault { code: u8 },  
}
```

```

fn describe(s: &Status) -> String {
    match s {
        Status::Idle => "idle".into(),
        Status::Running { rpm } =>
            format!("running at {}", rpm),
        Status::Fault { code } =>
            format!("fault {}", code),
    }
}

fn main() {
    let s = Status::Running { rpm: 1500 };
    println!("{}", describe(&s));
}

```

6.4.1 Enum shapes (pattern matching preview)

Variant syntax	Example	Pattern in match
Unit	Idle	Status::Idle
Tuple	Ping(u32)	Message::Ping(ms)
Struct	Fault { code }	Status::Fault { code }

```

// Playground
enum Frame {
    Heartbeat,
    Data { len: u16, payload: [u8; 4] },
}

fn opcode(f: &Frame) -> u8 {
    match f {
        Frame::Heartbeat => 0x00,
        Frame::Data { len, .. } => {
            if *len > 0 { 0x01 } else { 0xFF }
        }
    }
}

```

```
fn main() {  
    let f = Frame::Data { len: 2, payload: [1,  
        2, 0, 0] };  
    println!("op={:02X}", opcode(&f));  
}
```

.. in a pattern ignores fields you do not need (Chapter 2 ranges use the same token for a different feature — context disambiguates).

6.5 Pattern matching mechanics

6.5.1 Exhaustive match

For an enum, **every variant** needs an arm (or a catch-all that is intentionally broad).

```
// Playground  
enum Mode { Auto, Manual }  
  
fn label(m: Mode) -> &'static str {  
    match m {  
        Mode::Auto => "auto",  
        Mode::Manual => "manual",  
    }  
}  
  
fn main() {  
    println!("{}", label(Mode::Auto));  
}
```

6.5.1.1 Why -> &'static str?

The return type is a **borrowed string slice** (&str), not an owned String. Each match arm returns a **string literal** — "auto", "manual" — which is stored in the program binary and lives for the **entire run**. Rust types that as &'static str: a reference valid for the **'static** lifetime (see Chapter 5 — Lifetimes).

Return expression	Lifetime meaning	Who must stay alive?
"auto" (literal)	'static	nobody — bytes are in the binary
s.as_str() from caller's String	usually 'a tied to s	the owner String
format!(...)	cannot return &str at all	need owned String

So `label` does **not** borrow from `m` or from any local variable. It only returns pointers to immortal literals. That is why the signature can be `&'static str` without a generic `'a` parameter.

What if you call `label(Mode::Auto)` many times?

```
label(Mode::Auto);
label(Mode::Auto);
label(Mode::Auto);
```

Nothing new is allocated. The bytes for `"auto"` are written into the **program binary once** at compile time. Every call returns another `&str` — a pointer and length pair — pointing at **those same bytes**:

Rust Reference — String literal: “A string literal is a string stored directly in the final binary, and so will be valid for the `'static` duration.” Its type is `&'static str`.

Rust Reference — String literal expressions: the expression’s type is `&'static str`; its value is “a reference to a **statically allocated str** containing the UTF-8 encoding of the represented string.”

The Rust Book — What is Ownership?: “In the case of a string literal, we know the contents at compile time, so the text is **hardcoded directly into the final executable**.”

Rust By Example — 'static lifetime: a string literal “is stored in the **read-only memory of the binary**”; when the binding goes out of scope, “the data **remains in the binary**.”

str primitive — std docs: string literals are `&'static str` and “guaranteed to be valid for the duration of the **entire program**.”

binary: ... | a u t o | ...

^
|_____ all calls return &str here

Per call	What happens
Memory for "auto" text	unchanged — one copy for the whole process
Return value	a fresh &str value (reference), not a fresh string on the heap
Cost	tiny — match + return a pointer; no String allocation, no copying of "auto"

You can call `label` in a loop, print and discard the result, or store many `&'static str` bindings — they all refer to the same immortal "auto" / "manual" slices. That is why `'static` is safe here: the data outlives every call site, every loop iteration, and every variable that holds the returned reference.

Wrong — return &str pointing at a local String:

```
// Playground --- does not compile
fn bad_label(m: Mode) -> &str {
    let s = format!("{:?}", m);
    // owned String on stack
    s.as_str()
    // ERROR: `s` does not live long enough ---
    // returned ref would dangle
}
```

Idiomatic alternatives when text is built at run time:

```
// Playground
enum Mode { Auto, Manual }

fn label_owned(m: Mode) -> String {
    match m {
        Mode::Auto => "auto".to_string(),
        Mode::Manual => "manual".to_string(),
    }
}

fn main() {
```



```
println!("{}", label_owned(Mode::Auto));
}
```

What if you call `label_owned(Mode::Auto)` many times?

Yes — **each call allocates fresh heap memory**, even though the source text is the same `"auto"` literal.

Per call to <code>label_owned</code>	What happens
<code>.to_string()</code>	copies <code>"auto"</code> bytes from the binary into a new heap <code>String</code>
Return value	an owned <code>String</code> — separate allocation every time
After use	when that <code>String</code> is dropped, the heap block is freed

```
for _ in 0..1000 {
    let s = label_owned(Mode::Auto);
    // 1000 separate heap allocations
    println!("{}", s);
    // each `s` dropped at end of iteration
}
```

Contrast with `label -> &'static str`: a thousand calls still point at the **one** `"auto"` slice in the binary — no heap copy per call.

For this enum, `label` is the lighter choice. Prefer `&'static str` for fixed catalog strings; reach for `String` when you need ownership (append, cross a thread, store in a struct) or build text at run time — Chapter 5 covers lifetimes, Chapter 2 covers `String` vs `&str`.

Add a variant — compiler forces updates:

```
// Playground --- does not compile until you
    add `Mode::Safe => ...`
enum Mode { Auto, Manual, Safe }

fn label(m: Mode) -> &'static str {
    match m {
        Mode::Auto => "auto",
        Mode::Manual => "manual",
```

```

    }
    // ERROR: non-exhaustive patterns: `Safe`
    // not covered
}

```

After you add `Mode::Safe => "safe"`, the match compiles again — that is the safety net when protocols or state machines grow.

That is a **feature**: refactors cannot silently forget a new PLC mode or protocol opcode.

6.5.2 Wildcard `_` — power and footgun

On integers, `_` means “everything else.” On enums, `_` means “any variant I did not list”:

```

// Playground
fn severity(code: u16) -> &'static str {
    match code {
        0..=99 => "info",
        _ => "unknown",
        // catches 100+ --- fine for integers
    }
}

```

Footgun on enums: if you write `_` instead of listing variants, **adding a new variant will not force you to update this match**. The compiler assumes you meant to lump new cases into `_`. Prefer explicit arms for domain enums you control. Reserve `_` for “truly don’t care” cases or integer ranges.

6.5.3 `match` on references — borrow vs move

```

// Playground
enum Status {
    Idle,
    Running { rpm: u32 },
    Fault { code: u8 },
}

fn describe(s: &Status) -> String {

```

```

match s {
  Status::Idle => "idle".into(),
  Status::Running { rpm } =>
    format!("running at {}", rpm),
  Status::Fault { code } =>
    format!("fault {}", code),
}

fn main() {
  let s = Status::Running { rpm: 100 };
  match &s {
    Status::Running { rpm } =>
      println!("rpm {}", rpm),
    _ => println!("other"),
  }
  println!("still own s: {}", describe(&s));
}

```

`match &s` borrows; `match s` moves `s` into the match (often wrong unless you intentionally consume).

Wrong — move field out of non-Copy enum:

```

// Playground --- does not compile
enum Packet { Text(String) }

fn main() {
  let p = Packet::Text(String::from("hi"));
  match p {
    Packet::Text(s) => println!("{}", s),
    // moves s out of p
  }
  // use p again
  // ERROR: use of partially moved value: `p`
}

```

Use `ref` in the pattern to borrow inside fields: `Packet::Text(ref s)` or `match &p`.

6.5.4 Match guards

Add a boolean condition on an arm:

```
// Playground
fn classify(v: i32) -> &'static str {
    match v {
        n if n < 0 => "negative",
        0 => "zero",
        n if n > 100 => "high",
        _ => "normal",
    }
}

fn main() {
    println!("{}", classify(150));
}
```

Guards do not replace exhaustiveness — you still need arms that cover all cases (often with `_`).

6.5.5 if let vs match

Use if let / while let	Use match
one variant matters	several variants or mixed handling
quick peel Some / Ok	equal weight per branch
while let Some(line) = iter.next()	full Result + Err mapping

Long chains of `if let` on the same value are a smell — switch to `match` for clarity.

6.6 Java / Python contrast (deeper)

Pitfall	Java / Python habit	Rust idiom
Forgot null check	NPE / AttributeError at runtime	Option + match / ?
Swallowed exception	empty catch	match on Err or explicit ?
Stringly-typed states	"IDLE", "RUN"	enum Mode { Idle, Run }

Pitfall	Java / Python habit	Rust idiom
Unchecked union	isinstance ladders	one match on enum

Python match (3.10+) looks like Rust but matches object structure; Rust match ties to **algebraic types** and ownership.

6.7 Service-shaped example

```
// Playground
enum ReadOutcome {
    Value(u16),
    Timeout,
    CrcFault,
}

fn handle(r: ReadOutcome) {
    match r {
        ReadOutcome::Value(v) =>
            println!("register = {}", v),
        ReadOutcome::Timeout =>
            eprintln!("retry"),
        ReadOutcome::CrcFault => eprintln!("bus
            fault"),
    }
}

fn main() {
    handle(ReadOutcome::Value(42));
    handle(ReadOutcome::CrcFault);
}
```

Adding `ReadOutcome::Disconnected` later without updating `handle` -> **compile error**, not a silent log gap.

6.8 Advanced patterns

Patterns for parsers and config loaders — same domain as the service example above.

6.8.1 Slice patterns — split a command line

Parse "GET /path HTTP/1.1" into verb and path without index math:

```
// Playground
fn parse_request(line: &str) -> Option<(&str,
    &str, &str)> {
    match
        line.split_whitespace().collect::<Vec<_>
        >().as_slice() {
        [method, path, version] =>
            Some((*method, *path, *version)),
        _ => None,
    }
}

fn main() {
    let line = "GET /api/status HTTP/1.1";
    if let Some((m, p, v)) =
        parse_request(line) {
        println!("{}", m, p, v);
    }
}
```

For paths with spaces, collect into an owned String or match [method, rest @ .., version] and join rest into a local String before returning.

6.8.2 matches! — enum check without full match

Before — verbose match:

```
// Playground
#[derive(Debug)]
enum Mode { Auto, Manual, Off }

fn is_active(mode: &Mode) -> bool {
```

```

match mode {
    Mode::Auto | Mode::Manual => true,
    Mode::Off => false,
}
}

```

After — **matches!** macro:

```

// Playground
#[derive(Debug)]
enum Mode { Auto, Manual, Off }

fn is_active(mode: &Mode) -> bool {
    matches!(mode, Mode::Auto | Mode::Manual)
}

fn main() {
    println!("{}", is_active(&Mode::Auto));
}

```

Use `matches!` for boolean guards. Keep full match when arms return different values.

6.8.3 if let chains — bind host and port together

Parse `host:port` from a config line:

```

// Playground
fn parse_endpoint(line: &str) -> Option<(&str,
    u16)> {
    let mut parts = line.split(':');
    let host = parts.next()?.trim();
    if host.is_empty() {
        return None;
    }
    let port_str = parts.next()?;
    if parts.next().is_some() {
        return None; // reject host:port:extra
    }
    let port: u16 = port_str.parse().ok()?;
    if port == 0 {

```

```

        return None;
    }
    Some((host, port))
}

fn main() {
    println!("{:?}",
        parse_endpoint("127.0.0.1:502"));
}

```

Each condition must pass for the binding to succeed — good for multi-step parsing without nested match.

6.8.4 @ bindings — match and bind in one arm

Accept only valid port numbers while keeping the value:

```

// Playground
fn classify_port(p: u16) -> &'static str {
    match p {
        n @ 1..=1023 => "system",
        n @ 1024..=49151 => "registered",
        _ => "other",
    }
}

fn main() {
    println!("{}", classify_port(502));
}

```

6.8.5 Advanced pattern edge cases

Wrong — fixed-length slice on variable input:

The pattern `[method, path]` means **exactly two** elements — no more, no less. The code compiles, but real input often has a different length:

Input	<code>[method, path]</code> matches?	What happens
<code>["GET", "/only"]</code>	yes	prints <code>GET /only</code>

Input	[method, path] matches?	What happens
["GET", "/", "only", "HTTP/1.1"]	no — three elements	falls through to <code>_</code> -> "bad request"
["GET"]	no — one element	"bad request"
[]	no	"bad request"

So a valid three-token request line is rejected even though the first two tokens are fine. Rust does not warn — the `_` arm makes the match exhaustive.

```
// Playground --- logic bug, not compile error
fn main() {
    let words: Vec<&str> = vec!["GET", "/", "only"];
    match words.as_slice() {
        [method, path] => println!("{}", method, path),
        _ => println!("bad request"),
    }
}
```

Fix when you need at least a method and one path token — allow extra trailing tokens with `..`:

```
// Playground
fn main() {
    for words in [
        &["GET", "/", "only"][..],
        &["GET", "/", "only", "HTTP/1.1"][..],
        &["GET"][..],
    ] {
        match words {
            [method, path, ..] => println!("ok: {} {}", method, path),
            _ => println!("bad request"),
        }
    }
    // ok: GET /only
    // ok: GET /only      <- third token
    // ignored, not rejected
}
```

```
// bad request      <- only one token
}
```

Fix when the path itself can span multiple tokens — bind the remainder as a sub-slice with @ ...:

```
match words {
  [method, path @ ..] if !path.is_empty() => {
    println!("{}", method, path);
    // path: ["/only"] or ["/only",
    // "HTTP/1.1"]
  }
  _ => println!("bad request"),
}
```

Or skip pattern length entirely and check explicitly: if words.len() >= 2 { ... }.

Empty slice:

```
// Playground
fn main() {
  let empty: &[&str] = &[];
  match empty {
    [] => println!("empty"),
    [first, ..] => println!("first={}",
      first),
  }
}
```

6.9 When the compiler says no

Common errors in this chapter:

Error (typical)	Cause	Fix
non-exhaustive patterns	missing enum variant or None/Err	add arm or intentional _
expected T, found Option<T>	treated Option as value	match / if let / ?
use of partially moved value	field moved out in match	match &e, ref in pattern, or clone

Error (typical)	Cause	Fix
cannot move out of ... behind a reference	match on & but pattern moves	ref / ref mut in pattern
unreachable pattern	arm duplicated or shadowed by _ above	reorder arms; remove redundant _
? in fn without Result return	? only propagates in Result/Option fns	change return type or use match

6.10 Idiom spotlight

match on Result at boundaries, ? inside. At main or API edge, convert to user-facing messages; inside libraries, propagate with ?.

Prefer explicit enum variants over strings for device and protocol states. Let the compiler remind you when the protocol adds a code.

Avoid unwrap in long-running services — use Option/Result so bad input is data, not a panic.

6.11 Go deeper

- [The Rust Book — Enums](#)
- [The Rust Book — Patterns and Matching](#)
- [Option basics](#) — examples 041-044
- [Result basics](#) — examples 045-048

6.12 See also

- Chapter 2: Types — integer match, if expressions, String vs &str
- Chapter 5: Lifetimes — 'static, returning &str, why bad_label fails
- Chapter 4: Iterators — find returns Option
- Chapter 8: Errors — ?, custom Error, testing Result
- Chapter 7: Traits — traits on enums, impl blocks
- Chapter 17: Metaprogramming — matches! macro

6.12.1 Afterparty

6.12.1.1 Option and null habits

1. **Null replacement** — “Translate 5 Java methods returning null into Option Rust; explain callsite changes.”
2. **unwrap audit** — “Paste 20 lines with 4 unwrap() calls; I mark panic risk; you rewrite with match or ?.”
3. **Combinator chain** — “Parse port Option<u16>, double if Some, default 502 — I write map/unwrap_or; you add and_then version.”
4. **let-else port** — “Rewrite nested match on parse() into let Ok(x) = ... else { ... }; preserve behavior.”

6.12.1.2 Result and propagation

5. **Result railway** — “Chain parse -> validate -> compute with ?; I fill blanks, you verify.”
6. **Err arm missing** — “Show match on Result without Err arm; I quote error and fix; add boundary eprintln! pattern.”
7. **unwrap vs ?** — “Same parser twice: unwrap vs fn -> Result with ?; compare panic risk and signature honesty.”

6.12.1.3 Enums and exhaustive match

8. **Exhaustive match** — “Enum with 4 variants; I write match; you add variant Safe and show compile error until I fix.”
9. **Wildcard footgun** — “Explain why _ on your own enum hides new variants; show explicit arms vs _ refactor story.”
10. **Partial move** — “enum with String field: match moves field; I fix with ref or match &e; you show error text.”
11. **State machine** — “TCP/serial states as enum; connect/send/close return Result<(), IllegalTransition>.”
12. **Python Union** — “Python int | str parameter -> Rust enum + match; no dyn.”

6.12.1.4 Patterns and style

13. **if let vs match** — “Three tasks: one variant, two variants, five variants — I pick if let or match each time.”
14. **Match on ref** — “Given owned Status, I choose match s vs match &s; predict move errors; you diagram ownership.”

15. **Guard drill** — “Classify i32 with guards (<0 , $==0$, >100); I write arms; you check exhaustiveness.”
16. **Opcode table** — “Design enum for 3 frame types + match returning u8 opcode; add fourth type as compile-break exercise.”

6.12.1.5 Errors and automation

17. **ReadOutcome extend** — “Add Disconnected to automation ReadOutcome; show all match sites compiler lists.”
18. **Config sentinel** — “Rewrite fn port() -> i32 returning -1 on failure to Optionu16 + match in main.”
19. **Checklist drill** — “Match 6 Chapter 6 compiler errors to snippets; I name fix (match arm, ref, ?, return type).”
20. **Java enum map** — “Java enum State { IDLE, RUN } with method — port to Rust enum + match + impl; contrast nullability.”

6.12.1.6 Advanced patterns

21. **Slice split** — “Parse POST /api/v1/run HTTP/1.1 with [method, path @ .., _ver] — I write; you handle single-token input.”
22. **matches refactor** — “Replace 6-arm match that returns bool with matches! — show before/after on Mode enum.”
23. **if let chain** — “Parse host:port:extra — chain should reject extra segments; fix my broken parser.”
24. **@ binding quiz** — “Three arms with @ ranges for port classes — I label which values hit which arm.”
25. **Exhaustive slice** — “[a, b] on Vec of len 1 or 3 — predict _ arm vs bug; suggest .. rest pattern.”
26. **matches vs match** — “When must you keep full match instead of matches!? Give one example returning String.”
27. ****Guard + @**** — “Match port n @ 1024.. 65535 with guard $n \% 2 == 0$ — sketch arm.”
28. **Capstone parse** — “Frame header [sync, len @ 1.. 255 , payload @ ..] from byte slice — sketch match arms only.”

Chapter 7

Structs, Traits, and Generics

7.1 Hook

Rust uses **structs** for data, **impl** for methods, **traits** for shared behaviour, and **enums** for closed alternatives. **Java** bundles data with inheritance; **Python** relies on duck typing (“if it quacks...” — optional comparison points. Rust favors composition over inheritance, all checked at compile time.

7.2 Structs and methods

```
// Playground
struct Sensor {
    id: u32,
    value: f64,
}

impl Sensor {
    fn new(id: u32, value: f64) -> Self {
        Self { id, value }
    }
}
```

```
fn scaled(&self, factor: f64) -> f64 {
    self.value * factor
}

fn main() {
    let s = Sensor::new(1, 25.0);
    println!("{}", s.scaled(0.5));
}
```

7.3 Traits — interfaces done right

If you know **Java** or **Python**, a **trait** names a **capability** any type can opt into — shared behaviour **without inheritance**, checked at compile time. The table below maps the habit; then the examples.

Why traits are a win

Benefit	Java / Python pain	Rust trait answer
Checked contracts	interface optional; duck typing fails at runtime	missing method = compile error
No hierarchy tax	deep trees, fragile super chains	flat types + impl blocks
Cross-crate reuse	can't add methods to String / third-party classes	implement your trait for your wrapper (orphan rule applies)
Performance	interface dispatch / dynamic checks	static dispatch by default (impl Trait -> monomorphization, no vtable)
Optional polymorphism	always reference types / ABCs	dyn Trait only when you need mixed-type collections

You will use traits everywhere: Display, Clone, Iterator, custom Measurable / Summary in automation code. Start with impl Trait for MyType and fn f(x: &impl Trait); reach for dyn Trait when you need mixed types in one collection (full detail in [Trait objects](#) below).

```
// Playground
trait Summary {
```

```

fn summarize(&self) -> String;
}

struct Reading { v: f64 }

impl Summary for Reading {
    fn summarize(&self) -> String {
        format!("reading: {}", self.v)
    }
}

fn print_summary(item: &impl Summary) {
    println!("{}", item.summarize());
}

fn main() {
    let r = Reading { v: 3.14 };
    print_summary(&r);
}

```

Java	Python	Rust
interface	informal protocol	trait
implements	“has method”	impl Trait for Type
default interface methods	mixin / ABC	trait default bodies

7.4 Enums, structs, and traits together

Chapter 6 gave you `enum` + exhaustive match. This section covers **how structs and traits attach to enums** — Rust’s substitute for a class hierarchy or a Python Union of unrelated types.

The idiomatic shape for automation and protocol code:

1. `enum` — closed set of states or message kinds (compiler tracks every variant).
2. `struct` — per-variant payload when a variant carries real data.
3. `trait` — shared behaviour across *different* types (`Reading`, `Alarm`, ...), or a uniform API over one enum.

4. `impl` — inherent methods on the enum *and/or* trait implementations that match inside.

```
// Playground
trait Measurable {
    fn value(&self) -> f64;
    fn unit(&self) -> &'static str;
}

struct Temperature {
    celsius: f64,
}

struct Pressure {
    bar: f64,
}

enum SensorReading {
    Temp(Temperature),
    Press(Pressure),
    Skipped { reason: String },
}

impl Measurable for SensorReading {
    fn value(&self) -> f64 {
        match self {
            SensorReading::Temp(t) => t.celsius,
            SensorReading::Press(p) => p.bar,
            SensorReading::Skipped { .. } =>
                f64::NAN,
        }
    }

    fn unit(&self) -> &'static str {
        match self {
            SensorReading::Temp(_) => " degC",
            SensorReading::Press(_) => "bar",
            SensorReading::Skipped { .. } =>
                "n/a",
        }
    }
}
```

```

    }
}

impl SensorReading {
    fn is_valid(&self) -> bool {
        !matches!(self, SensorReading::Skipped
            { .. })
    }
}

fn log(item: &impl Measurable) {
    println!("{}", item.value(),
        item.unit());
}

fn main() {
    let r = SensorReading::Temp(Temperature {
        celsius: 21.5 });
    log(&r);
    println!("valid={}", r.is_valid());
}

```

What `!matches!(self, SensorReading::Skipped { .. })` does

- `matches!(value, pattern)` — standard-library **macro** (Chapter 17: Metaprogramming) that expands to a `match` returning `true` or `false`. It is syntax sugar for “does this value fit this pattern?” without binding variables you do not need.
- `SensorReading::Skipped { .. }` — matches the `Skipped` struct variant and **ignores** the `reason` field (`..` = “other fields don’t matter for this test”). Same pattern token as in Chapter 6.
- `!` — negates the result: `Temp` and `Press` -> `true`; `Skipped { .. }` -> `false`.

Equivalent without the macro:

```

// Playground
enum SensorReading {
    Temp,

```

```
    Press,
    Skipped { reason: String },
}

fn is_valid_longhand(r: &SensorReading) -> bool
{
    match r {
        SensorReading::Skipped { .. } => false,
        _ => true,
    }
}

fn main() {
    println!("{}",
        is_valid_longhand(&SensorReading::Temp))
    ;
}
```

Use matches! for readable guards and filters; use full match when each arm returns different data. More on declarative macros: Chapter 17 — macro_rules!.

7.4.1 Two impl blocks on one type

Block	Role	Java / Python analogy
impl SensorReading { ... }	Inherent methods — always available on SensorReading	methods on the class
impl Measurable for SensorReading { ... }	Trait impl — call only where Measurable is required	interface implementation

Both on the same enum is normal. Put variant-specific logic in inherent methods. Put cross-type contracts on traits.

7.4.2 Struct-in-enum vs enum-in-struct

Layout	Sketch	Reach for it when
Struct inside enum variant	enum Msg { Data(Frame), Ping }	variants own <i>different</i> shapes; match is the dispatcher
Enum field inside struct	struct Device { kind: Kind, addr: u8 }	all rows share the same fields; tag only selects behaviour
Unit variants only	enum Mode { Auto, Manual }	no payload — traits/methods ignore inner data

```
// Playground --- shared metadata + tagged kind
enum DeviceKind {
    Modbus,
    OpcUa,
}

struct Device {
    name: String,
    kind: DeviceKind,
}

impl Device {
    fn default_port(&self) -> u16 {
        match self.kind {
            DeviceKind::Modbus => 502,
            DeviceKind::OpcUa  => 4840,
        }
    }
}

fn main() {
    let d = Device {
        name: "plc-1".into(),
        kind: DeviceKind::Modbus,
    };
    println!("{}", d.name, d.default_port());
}
```

Prefer **enum + struct variants** when each variant would have been a subclass with different fields. Prefer **struct + enum field** when every instance shares the same columns and only behaviour differs.

7.4.3 Trait on enum: match is mandatory

Unlike a single struct, an enum trait body almost always uses `match self` (or `match &self`) — one arm per variant. That mirrors Chapter 6 exhaustiveness. Add a variant, and the compiler lists every `impl` and `match` you must update.

Default trait methods still work — override only where a variant differs:

```
// Playground
trait HasCode {
    fn code(&self) -> u8;
    /// Shared default formatting --- call this
    // from overrides to avoid recursion.
    fn default_label(&self) -> String {
        format!("code {}", self.code())
    }
    fn label(&self) -> String {
        self.default_label()
    }
}

enum Fault {
    OverTemp(u8),
    CommLost,
}

impl HasCode for Fault {
    fn code(&self) -> u8 {
        match self {
            Fault::OverTemp(c) => *c,
            Fault::CommLost => 0xFF,
        }
    }

    fn label(&self) -> String {
```

```
match self {
    Fault::CommLost => "communication
        lost".into(),
    other => other.default_label(),
    // trait default body, not this
    // override
}

}

}

fn main() {
    println!("{}", Fault::CommLost.label());
    println!("{}",
        Fault::OverTemp(0x0A).label()); //
    default: "code 10"
}
```

7.4.3.1 Calling the trait’s default from your override

You want two behaviours in one label override:

Variant	Desired output
CommLost	custom string — "communication lost"
OverTemp	shared default — "code {n}" from the trait

The trait exposes **default_label** for the shared path. In the override, `other.default_label()` calls **that trait method’s body** — not your label override:

Call in impl HasCode for Fault	What runs
<code>other.default_label()</code>	trait default: <code>format!("{}", other.code())</code>
<code>other.label()</code>	this override again -> infinite recursion on OverTemp

That is why the trait names the fallback `default_label` instead of reusing `label`. Same idea as Java’s `HasCode.super.label()` — call the default implementation, not your override.

Footgun: never write `other.label()` inside `fn label` unless you mean to recurse. Use `other.default_label()`, or inline `format!("code {}", other.code())`.

Wrong — “call the override again” (infinite recursion on `OverTemp`):

```
// Playground --- stack overflow at runtime for
// OverTemp
impl HasCode for Fault {
    fn label(&self) -> String {
        match self {
            Fault::CommLost => "communication
                                lost".into(),
            other => other.label(),
            // ERROR path: calls THIS override
            // again, forever
        }
    }
}
```

The `OverTemp` arm above is the bug — `other.label()` never reaches the trait default.

7.4.3.2 One `impl` block per trait (not `impl A, B for T`)

No — Rust does not allow multiple traits in one `impl` header:

```
// Playground --- does not compile
// impl HasCode, Display for Fault { ... }
// ERROR: expected `for`, found ``,``
```

A type may implement **many** traits, but each trait gets its **own** `impl` block:

```
// Playground
use std::fmt;

trait HasCode {
    fn code(&self) -> u8;
}

enum Fault {
    OverTemp(u8),
```

```

CommLost,
}

impl HasCode for Fault {
    fn code(&self) -> u8 {
        match self {
            Fault::OverTemp(c) => *c,
            Fault::CommLost => 0xFF,
        }
    }
}

impl fmt::Display for Fault {
    fn fmt(&self, f: &mut fmt::Formatter<'_,_>)
        -> fmt::Result {
        write!(f, "fault 0x{:02X}", <Self as
            HasCode>::code(self))
    }
}

fn main() {
    println!("{}", Fault::CommLost);
}

```

Where **multiple traits appear together** is on **bounds**, not on **impl**:

Location	Example	Meaning
Generic parameter	<code>fn show<T: HasCode + Display>(x: T)</code>	T must implement both traits
where clause	<code>where T: HasCode + Display</code>	same, spelled out
Trait inheritance	<code>trait Loud: Display { ... }</code>	every Loud type must also implement Display

So: `impl HasCode for Fault` and `impl Display for Fault` are two separate blocks; `T: HasCode + Display` means “T implements both.”

7.4.4 `#[derive(...)]` on enums and structs

Enums and structs in the same model often share derives:


```
// Playground
#[derive(Debug, Clone, PartialEq, Eq)]
enum Command {
    Stop,
    SetSpeed(u32),
}

#[derive(Debug, Clone, PartialEq, Eq)]
struct SetSpeedLog {
    at: u64,
    rpm: u32,
}

impl Command {
    /// Turn a command into a timestamped log
    /// row when it carries speed data.
    fn to_log(&self, at: u64) ->
        Option<SetSpeedLog> {
        match self {
            Command::SetSpeed(rpm) =>
                Some(SetSpeedLog { at, rpm:
                    *rpm }),
            Command::Stop => None,
        }
    }
}

fn main() {
    let c = Command::SetSpeed(1500);
    assert_eq!(c, Command::SetSpeed(1500));

    let log =
        c.to_log(1_700_000_000).expect("SetSpeed
        maps to a log row");
    assert_eq!(log, SetSpeedLog { at:
        1_700_000_000, rpm: 1500 });
    println!("{:?}", log);
}
```

Two shapes for one domain:

Type	Role	Example
Command (enum)	wire / control message	Stop, SetSpeed(1500)
SetSpeedLog (struct)	persisted audit row	{ at: timestamp, rpm: 1500 }

Command::to_log converts between them: SetSpeed(rpm) -> Some(SetSpeedLog { ... }); Stop -> None (nothing to log).

Both types carry the same `#[derive(Debug, Clone, PartialEq, Eq)]` — commands and the rows they produce usually need the same test/debug tooling. `#[derive]` generates trait impls at compile time (not runtime annotations); details in Chapter 17: Derive attributes.

Derive constraint: `PartialEq/Eq` on an enum requires every payload type to support it. A variant with `f64` breaks `Eq` — use integers, drop `Eq`, or model floats differently (see edge cases below).

7.4.5 dyn Trait — mixed types in one collection

So far, traits use `static dispatch`: `fn print_summary(item: &impl Summary)` and `impl Measurable` for `SensorReading` are resolved at **compile time** — the compiler generates specialized code per concrete type (no vtable).

Use **dyn Trait** when several **different struct types** must sit in the **same collection** or pass through one function parameter type. **Yes — this is Rust’s runtime polymorphism**, closest to a **Java interface reference** or Python duck typing through a shared protocol — but Rust makes you opt in explicitly; it is not the default for every trait call.

Approach	Type in signature	Dispatch	Typical use
<code>impl Trait</code> / generics	<code>&impl Measurable</code>	static — monomorphized	helper called with one concrete type at a time
<code>&dyn Trait</code>	borrowed trait object	dynamic — vtable lookup	<code>&[&dyn Measurable]</code> — stack values, mixed types

Approach	Type in signature	Dispatch	Typical use
Box<dyn Trait>	owned trait object on heap	dynamic	Vec<Box<dyn Measurable>> — store plug-ins for process lifetime

Java habit vs Rust:

Java	Rust	Same idea?
	dyn	
interface Measurable { double value(); }	trait Measurable { fn value(&self) -> f64; }	shared contract
class TempSensor implements Measurable	impl Measurable for TempSensor	type opts in
Measurable s = new TempSensor();	let s: Box<dyn Measurable> = Box::new(TempSensor { ... });	variable typed as interface/trait, holds concrete type
List<Measurable> sensors = List.of(new TempSensor(), new PressSensor());	Vec<Box<dyn Measurable>> = vec![Box::new(...), Box::new(...)];	mixed concrete types in one collection
s.value() — virtual dispatch via vtable	s.value() on &dyn Measurable — vtable lookup at runtime	yes — dynamic dispatch

Key differences from Java:

Java	Rust
Interface references are the usual polymorphism style	impl Trait / generics first — dyn only when you need type erasure in a collection or factory
Every object reference is already a pointer	dyn Measurable is unsized (!Sized) — must live behind &, Box, Arc, ...
List<Measurable> homogenizes references automatically	Vec<Box<dyn Measurable>> — Box required because TempSensor and PressSensor have different sizes
Inheritance + implements	no inheritance — flat structs + separate impl blocks
Most interface methods work in collections	only object-safe traits become dyn (see below)

A **dyn Measurable** value is a **fat pointer** (like a Java object reference carrying type info for virtual calls): address of the concrete sensor **plus** a vtable pointer for **impl Measurable**. Because each struct has a different size, **dyn Measurable** cannot sit on the stack by itself — hence **&dyn** or **Box<dyn>**.

```
// Playground
trait Measurable {
    fn value(&self) -> f64;
}

struct TempSensor { celsius: f64 }
struct PressSensor { bar: f64 }

impl Measurable for TempSensor {
    fn value(&self) -> f64 { self.celsius }
}
impl Measurable for PressSensor {
    fn value(&self) -> f64 { self.bar }
}

fn average(sensors: &[&dyn Measurable]) -> f64 {
    let sum: f64 = sensors.iter().map(|s|
        s.value()).sum();
    sum / sensors.len() as f64
}

fn main() {
    let t = TempSensor { celsius: 20.0 };
    let p = PressSensor { bar: 1.2 };

    // borrow different struct types through
    // one trait interface
    println!("avg = {}", average(&[&t, &p]));

    // owning heterogeneous Vec --- each
    // element is Box<dyn Measurable>
    let bank: Vec<Box<dyn Measurable>> = vec![
        Box::new(TempSensor { celsius: 22.0 }),
```

```

        Box::new(PressSensor { bar: 0.9 })),
    ];
    for s in &bank {
        println!("{}", s.value());
    }
}

```

Object-safe traits only: not every trait can become dyn Trait. Methods must use `&self` (or `&mut self`) — the vtable lists instance methods with a fixed shape. Traits with associated functions without `self`, generic methods, or `-> Self` returns often fail — see the edge case below and Object safety.

For a **closed set** of variants (`SensorReading::Temp | Press | ...`), an **enum + match** is usually better than `Vec<Box<dyn _>>` — no vtable, exhaustiveness checking. Use `dyn` when the set of concrete types is **open** (plug-ins, drivers loaded at runtime).

7.4.6 Enums + traits edge cases and compiler traps

Wrong — non-exhaustive match in trait impl (add a variant, forget an arm):

```

// Playground --- does not compile
enum Status { Idle, Running, Fault }

trait Label {
    fn label(&self) -> &'static str;
}

impl Label for Status {
    fn label(&self) -> &'static str {
        match self {
            Status::Idle => "idle",
            Status::Running => "running",
            // Status::Fault => ...
            // ERROR: non-exhaustive patterns
        }
    }
}

```

Wrong — partial move: extract payload, then reuse self:

```
// Playground --- does not compile
enum Packet {
    Raw(String),
    Empty,
}

impl Packet {
    fn dump(self) {
        if let Packet::Raw(s) = self {
            println!("{}", s.len());
        }
        // ERROR: use of partially moved value:
        `self`
        // Packet::Empty arm never ran, but
        `Raw` variant moved `self`
        let empty = matches!(self,
            Packet::Empty);
        println!("empty={}", empty);
    }
}
```

Fix: take `&self`, clone the `String` when you need ownership, or handle all variants in one match.

Wrong — `match self` by value in one method, then call another method on `self`:

```
// Playground --- does not compile
impl Packet {
    fn describe(self) -> String {
        match self {
            Packet::Raw(s) => format!("raw {}
                bytes", s.len()),
            Packet::Empty => "empty".into(),
        }
    }

    fn tag(self) -> &'static str {
        match self {
            Packet::Raw(_) => "RAW",

```

```

        Packet::Empty => "EMPTY",
    }
}

fn full(self) -> String {
    format!("{: {}}", self.tag(),
        self.describe()) // ERROR: `self`
        moved
    }
}

```

Idiomatic fix: `fn full(&self) -> String` and match on references, or merge into a single match.

Wrong — orphan rule (external trait + external type):

```

// Playground --- does not compile
// impl std::fmt::Display for Vec<u8> { ... }
// ERROR: impl doesn't apply to type defined
// outside of crate

```

You can `impl Display` for `YourEnum`, or wrap `Vec<u8>` in a newtype struct `Frame(pub Vec<u8>)`; and implement there.

Wrong — trait not object-safe (cannot build `dyn Trait`):

The working `Measurable` trait above uses `fn value(&self)`. This trait adds `fn read() -> f64` **without** `self` — the vtable cannot list it, so `dyn` fails:

```

// Playground --- does not compile
trait RawReading {
    fn value(&self) -> f64;
    fn read() -> f64;
    // no `self` --- not object-safe
}

// let items: Vec<Box<dyn RawReading>> =
//     vec![];
// ERROR: trait `RawReading` is not dyn
//     compatible

```

Fix: make every callable method take `&self`, split static helpers into free functions, or skip `dyn` and use an `enum` / generics instead. More traps (`-> Self`, generic methods, `dyn` by value) — Object safety below.

Wrong — Eq on enum with f64 payload:

```
// Playground --- does not compile
#[derive(Eq, PartialEq)]
enum Sample {
    Analog(f64),
    Digital(bool),
}
// ERROR: the trait bound `f64: Eq` is not
    satisfied
```

Use integer fixed-point, ordered floats (`ordered-float`), or derive only `PartialEq`.

Trap	Symptom	Idiom
New enum variant	errors in every match / trait impl	treat as feature — update all arms
Partial move in match	“use of partially moved value”	match <code>&self</code> , clone, or one combined match
Trait impl without match	wrong for multi-variant enums	one arm per variant
dyn Trait collection	“not dyn compatible”	object-safe trait, or enum of variants instead of heterogenous <code>Vec<Box<dyn _>></code>
Orphan rule	“impl doesn’t apply to type outside crate”	newtype wrapper or own trait

7.4.7 Idiom spotlight (enums + traits)

Model closed sets as `enum`, not strings. Put data in struct variants; implement traits with exhaustive `match`.

New variant `->` compiler updates your checklist — no forgotten `else` when the PLC adds a fault code.

Prefer `&self` on enums unless you intentionally consume the value.

7.5 Generics

```
// Playground
fn largest<T: PartialOrd>(list: &[T]) -> &T {
    let mut max = &list[0];
    for item in &list[1..] {
        if item > max { max = item; }
    }
    max
}

fn main() {
    let nums = vec![3, 1, 4, 1, 5];
    println!("{}", largest(&nums));
}
```

7.6 Trait bounds and where

```
// Playground
use std::fmt::Display;

fn show<T: Display>(x: T) {
    println!("{}", x);
}

fn main() {
    show(42);
    show("text");
}
```

7.7 Trait objects (dyn Trait)

Above introduced `&dyn Trait` and `Box<dyn Trait>`. This section expands on fat pointers, factories, `impl` vs `dyn` vs `enum`, and object-safety traps.

Generics and `impl Trait` pick the concrete type at **compile time** — monomorphized, no vtable.

dyn Trait is runtime polymorphism through a vtable — like Java interface references or Python duck typing at runtime. Prefer **impl Trait** when types are known at compile time; use **dyn Trait** for mixed-type collections.

7.7.1 What you actually store

A trait object is a **wide pointer** (fat pointer):

Part	Points to
Data pointer	the concrete value (En, Fr, ...)
Vtable pointer	that type's impl Greeter method table

Because size varies by concrete type, **dyn Greeter** is **dynamically sized** (!Sized). It almost always lives **behind a pointer**:

Form	Owns value?	Typical use
&dyn Trait	no — borrow	pass one of several types into fn notify(x: &dyn Trait)
&mut dyn Trait	no — mutable borrow	plug-in you mutate in place
Box<dyn Trait>	yes — heap	Vec<Box<dyn Trait>> of mixed types
Arc<dyn Trait>	yes — shared	callbacks/handlers shared across threads (Send + Sync)

```
// Playground
trait Greeter {
    fn greet(&self) -> String;
}

struct En;
struct Fr;

impl Greeter for En {
    fn greet(&self) -> String { "Hello".into() }
}
impl Greeter for Fr {
    fn greet(&self) -> String {
```

```
        "Bonjour".into() }
    }

    fn announce(g: &dyn Greeter) {
        println!("{}", g.greet());
    }

    fn main() {
        announce(&En);
        // no Box --- stack value, borrowed as
        // trait object
        announce(&Fr);

        let voices: Vec<Box<dyn Greeter>> =
            vec![Box::new(En), Box::new(Fr)];
        for v in &voices {
            println!("{}", v.greet());
        }
    }
}
```

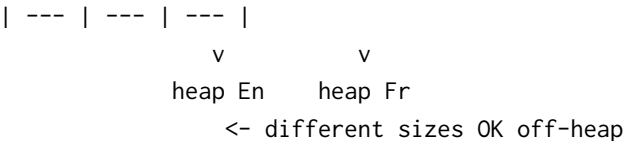
Why Vec needs Box<dyn Greeter>:

A Vec stores elements **back-to-back in memory**. Every slot must be the **same byte width** so the CPU can jump to index *i* with **base + i * stride**.

What you try	Problem
Vec<En> mixed with Fr	En and Fr may have different sizes — one array cannot hold both inline
Vec<dyn Greeter>	dyn Greeter is unsized — Rust does not know how many bytes one slot needs
Vec<Box<dyn Greeter>>	works — each slot is exactly one Box (same size every time)

Box::new(En) and Box::new(Fr) move the concrete values to the **heap**. The Vec only stores identical Box<dyn Greeter> handles — each a **fat pointer** (data address + vtable address), same width for every element:

```
Vec slot: [ Box<dyn> | Box<dyn> | Box<dyn> ]
          <- fixed-size slots
```



That is what “homogenize” means here: the collection stores **uniform handles**, not the varying struct bodies. Java does this implicitly — every `Measurable` reference in an `ArrayList` is one pointer width. Rust makes the heap indirection explicit with `Box`.

7.7.2 When `dyn Trait` is idiomatic

Situation	Prefer	Why
Function called with one of a few types , known when writing code	<code>fn f(x: &impl Greeter)</code>	static dispatch, zero vtable
Vec / registry of plug-ins loaded at runtime	<code>Vec<Box<dyn Handler>></code>	one collection, mixed concrete types
Return type depends on runtime input (factory)	<code>Box<dyn Greeter></code>	caller only knows the trait
Closed protocol (fixed set of variants)	<code>enum + match</code>	no vtable; exhaustiveness — see enums + traits
Shared handler across threads	<code>Arc<dyn AlarmSink + Send + Sync></code>	cheap clone, thread-safe ref count

Automation sketch — alarm sinks (open set of outputs):

```
// Playground
trait AlarmSink {
    fn emit(&self, code: u8, msg: &str);
}

struct LogSink;
struct MetricsSink;

impl AlarmSink for LogSink {
    fn emit(&self, code: u8, msg: &str) {
        println!("[{}] {}", code, msg);
    }
}
```

```

impl AlarmSink for MetricsSink {
    fn emit(&self, code: u8, msg: &str) {
        println!("metric alarm_code={} len={}",
            code, msg.len());
    }
}

fn raise(sinks: &[&dyn AlarmSink], code: u8,
    msg: &str) {
    for s in sinks {
        s.emit(code, msg);
    }
}

fn main() {
    let log = LogSink;
    let metrics = MetricsSink;
    raise(&[&log, &metrics], 0x07, "over-temp");
}

```

Borrowed `&dyn AlarmSink` avoids heap allocation when callers already own `LogSink` / `MetricsSink` on the stack. Use a `Vec<Box<dyn AlarmSink>>` when sinks are constructed dynamically and stored for the process lifetime.

Factory returning erased type:

```

// Playground
trait Greeter {
    fn greet(&self) -> String;
}

struct En;
struct Fr;

impl Greeter for En {
    fn greet(&self) -> String {
        "Hello".into()
    }
}

impl Greeter for Fr {
    fn greet(&self) -> String {
        "Bonjour".into()
    }
}

```

```
    }
}

fn greeter_for(lang: &str) -> Box<dyn Greeter> {
    match lang {
        "fr" => Box::new(Fr),
        _ => Box::new(En),
    }
}

fn main() {
    let g = greeter_for("fr");
    println!("{}", g.greet());
}
```

Why -> Box<dyn Greeter> and not -> impl Greeter?

impl Greeter in return position means: “this function returns **one** concrete type, picked at compile time — the same type on **every** return path.”

match arm	Concrete type returned
"fr" => ...	Fr
_ => ...	En

Two different types -> no single hidden concrete type -> -> impl Greeter **does not compile**.

Return type	Works here?	Why
-> impl Greeter	no	each arm must return the same struct (En <i>or</i> Fr, not both)
-> Box<dyn Greeter>	yes	erases to one pointer type; runtime picks En or Fr
-> GreeterKind (enum)	yes	closed set — enum { En(En), Fr(Fr) } + match at call site

Use **Box<dyn Trait>** when the caller only needs the trait interface. Use an **enum** when the variant set is fixed in your crate and you want exhaustiveness without heap allocation.

7.7.3 impl Trait vs dyn Trait vs enum

```
// Playground --- same behaviour, three Rust
// styles

trait Driver {
    fn poll(&self) -> u32;
}

struct Modbus;
struct OpcUa;
impl Driver for Modbus { fn poll(&self) -> u32
    { 502 } }
impl Driver for OpcUa { fn poll(&self) -> u32 {
    4840 } }

// 1) Static dispatch --- best default
fn port_static(d: &impl Driver) -> u32 {
    d.poll() }

// 2) Runtime erasure --- plug-in list
fn ports_dynamic(drivers: &[&dyn Driver]) ->
    Vec<u32> {
    drivers.iter().map(|d| d.poll()).collect()
}

// 3) Closed sum --- best when variants are
// fixed in your crate
enum Device {
    Modbus,
    OpcUa,
}
impl Device {
    fn poll(&self) -> u32 {
        match self {
            Device::Modbus => Modbus.poll(),
            Device::OpcUa => OpcUa.poll(),
        }
    }
}
```

```
fn main() {
    assert_eq!(port_static(&Modbus), 502);
    assert_eq!(ports_dynamic(&[&Modbus,
        &OpcUa]), vec![502, 4840]);
    assert_eq!(Device::OpcUa.poll(), 4840);
}
```

7.7.4 Object safety — not every trait can be dyn

A trait is **dyn compatible** (object-safe) only if the vtable can list a fixed set of methods. Common **object-unsafe** patterns:

Trait shape	Why it breaks dyn
fn read() -> f64 — no self	no receiver to dispatch on
fn convert<T>(&self, x: T) — generic method	vtable cannot hold infinite monomorphizations
fn clone(&self) -> Self	return type is concrete Self, unknown at call site
trait Foo: Sized	trait object itself is !Sized

Wrong — not object-safe:

```
// Playground --- does not compile
trait Reader {
    fn read() -> f64; // no `self`
}

// fn load(r: &dyn Reader) { ... }
// ERROR: the trait `Reader` is not dyn
// compatible ... consider marking as
`#[allow(...)]`
```

Wrong — generic method on trait:

```
// Playground --- does not compile
trait Parse {
    fn parse<T: std::str::FromStr>(&self, s:
        &str) -> Option<T>;
}
```



```
// let p: &dyn Parse = &MyParser;
// ERROR: trait `Parse` is not dyn compatible
```

Fix for `-> Self`: return a boxed trait object or associated type with a bound, or use static dispatch (impl Clone on generic T instead of dyn Clone-style patterns). Standard library Clone is not dyn-compatible for this reason. You clone concrete types or use Box<dyn Display> etc. for other traits.

Cross-reference: the same object-safety trap appears in enums + traits edge cases above.

7.7.5 More edge cases and compiler traps

Wrong — dyn Trait by value on the stack:

```
// Playground --- does not compile
trait Greeter { fn greet(&self) -> String; }
struct En;
impl Greeter for En { fn greet(&self) -> String
    { "Hi".into() } }

// let g: dyn Greeter = En;
// ERROR: the size for values of type `dyn
// Greeter` cannot be known
```

Use Box<dyn Greeter>, &dyn Greeter, etc.

Wrong — heterogeneous Vec without homogenizing pointer:

```
// Playground --- does not compile
// let v: Vec<dyn Greeter> = vec![En, Fr];
// ERROR: the size for values of type `dyn
// Greeter` cannot be known
```

Lifetime on borrowed trait objects — the reference must not outlive the concrete value:

```
// Playground --- OK inside one function:
    `en`/`fr` live long enough
trait Greeter {
    fn greet(&self) -> String;
}
```

```

struct En;
struct Fr;
impl Greeter for En {
    fn greet(&self) -> String {
        "Hello".into()
    }
}
impl Greeter for Fr {
    fn greet(&self) -> String {
        "Bonjour".into()
    }
}

fn pick_in_place(use_fr: bool) {
    let en = En;
    let fr = Fr;
    let g: &dyn Greeter = if use_fr { &fr }
        else { &en };
    println!("{}", g.greet());
}

fn main() {
    pick_in_place(true);
}

```

Returning &dyn Greeter from locals — two compile errors:

Inside one function, borrowing locals works — `pick_in_place` above keeps `en/fr` alive for the whole call. Returning that borrow to the **caller** fails: `en/fr` are dropped when broken returns, so the trait object would dangle.

```

// Playground --- does not compile (include
    Greeter / En / Fr from above, or full copy
    below)
trait Greeter {
    fn greet(&self) -> String;
}
struct En;
struct Fr;
impl Greeter for En { fn greet(&self) -> String

```

```
    { "Hello".into() } }
impl Greeter for Fr { fn greet(&self) -> String
    { "Bonjour".into() } }

fn broken(use_fr: bool) -> &dyn Greeter {
    let en = En;
    let fr = Fr;
    if use_fr { &fr } else { &en }
    // ERROR E0106: missing lifetime specifier
    // on `&dyn Greeter`
    // Rust asks: how long does this borrow
    // live? You must name a lifetime.
}

fn main() {}
```

If you add a lifetime (fn broken<'a>(...) -> &'a dyn Greeter), the compiler reaches the real problem:

ERROR E0515: cannot return value referencing
local variable `en` / `fr`

The return type promises a borrow, but en and fr are **owned by broken and destroyed at }** — nothing for the caller to borrow from.

Situation	Works?	Pattern
Use &dyn Greeter inside the same function	yes	pick_in_place — locals outlive the borrow
Return &dyn Greeter to caller from locals	no	E0106, then E0515
Return owned trait object to caller	yes	-> Box<dyn Greeter> — greeter_for above

Fix: when the factory **creates** the value, return **Box<dyn Greeter>** (or pass a caller-owned buffer in). Use **&dyn Greeter** only when borrowing something the **caller** already owns.

Auto traits on trait objects: Box<dyn AlarmSink + Send + Sync> when the handler crosses thread boundaries; omit when single-threaded.

Trap	Symptom	Idiom
Unsize dyn	“size cannot be known at compile time”	&, Box, Arc, never bare dyn value
Non-object-safe trait	“trait X is not dyn compatible”	fix trait API, or use enum / generics
-> impl Trait factory	arms return different types	Box<dyn Trait> or enum
Closed protocol	stringly plug-in names	enum beats dyn for speed + exhaustiveness
Dangling &dyn	borrow checker on return	Box<dyn Trait> or borrow from caller’s value

7.8 Trait decomposition — small interfaces

Large “god traits” are hard to mock and test. Split I/O boundaries into **focused traits** that each do one job. The same orchestration code runs in production, CLI, and tests by swapping implementations:

```
// Playground
use std::io::{self, Read};

trait ConfigLoader {
    fn load_text(&self, path: &str) ->
        io::Result<String>;
}

trait PortValidator {
    fn validate(&self, port: u16) -> Result<(),
        &'static str>;
}

struct FileConfigLoader;

impl ConfigLoader for FileConfigLoader {
    fn load_text(&self, path: &str) ->
        io::Result<String> {
        std::fs::read_to_string(path)
    }
}
```

```

struct PrivilegedPortGuard;

impl PortValidator for PrivilegedPortGuard {
    fn validate(&self, port: u16) -> Result<(),
        &'static str> {
        if port < 1024 {
            Err("port below 1024")
        } else {
            Ok(())
        }
    }
}

fn startup_port<L: ConfigLoader, V:
    PortValidator>(
    loader: &L,
    validator: &V,
    path: &str,
) -> io::Result<u16> {
    let text = loader.load_text(path)?;
    let port: u16 =
        text.trim().parse().map_err(|e|
            io::Error::new(io::ErrorKind::InvalidDat
                a, e))?;
    validator.validate(port).map_err(|msg|
        io::Error::new(io::ErrorKind::InvalidInp
            ut, msg))?;
    Ok(port)
}

fn main() {
    let loader = FileConfigLoader;
    let guard = PrivilegedPortGuard;
    // startup_port(&loader, &guard,
        "port.txt") in a real project
    let _ = (&loader as &dyn ConfigLoader,
        &guard as &dyn PortValidator);
    println!("traits split for test doubles");
}

```

}

Split	Role
ConfigLoader	where bytes come from (file, env, mock)
PortValidator	business rule isolated from I/O
generic orchestrator	one function, many backend pairs

In async services, the same split applies with `async fn` traits and the `async_trait` crate — see Chapter 8 — trait mocks.

7.9 Transform traits — parse once, map to domain

Keep **wire format** separate from **domain records**. A parser yields raw rows; a transformer maps them to what you persist:

```
// Playground
struct RawReading {
    tag: String,
    value: f64,
}

struct StoredReading {
    tag: String,
    value_milli: i64,
}

trait ToStored {
    fn to_stored(&self) -> StoredReading;
}

impl ToStored for RawReading {
    fn to_stored(&self) -> StoredReading {
        StoredReading {
            tag: self.tag.clone(),
            value_milli: (self.value *
                1000.0).round() as i64,
```

```

    }
}

fn ingest(rows: &[RawReading]) ->
    Vec<StoredReading> {
    rows.iter().map(|r| r.to_stored()).collect()
}

fn main() {
    let raw = RawReading {
        tag: "temp".into(),
        value: 22.5,
    };
    println!("{:?}", ingest(&[raw]).len());
}

```

Each source (Modbus, OPC-UA, CSV) can implement the same transform trait on its row type. The runner stays generic over “anything that becomes `StoredReading`”.

7.10 Associated types and supertraits

You already implement traits with methods. Many std traits also declare an **associated type** — one output type fixed per `impl`, not a separate generic parameter on the trait itself.

7.10.1 `Iterator::Item` — one output type per iterator

Chapter 4 — Implementing Iterator defines a port scanner. The associated type pins what `.next()` yields:

```

// Playground
struct PortScan {
    next_port: u16,
    end: u16,
}

impl Iterator for PortScan {

```

```

type Item = u16;
// associated type --- fixed for this impl

fn next(&mut self) -> Option<Self::Item> {
    if self.next_port > self.end {
        return None;
    }
    let p = self.next_port;
    self.next_port += 1;
    Some(p)
}

fn main() {
    let ports: Vec<u16> = PortScan { next_port:
        502, end: 505 }.collect();
    let sum: u16 = PortScan { next_port: 502,
        end: 504 }.sum();
    println!("ports={:?} sum={}", ports, sum);
}

```

type `Item = u16` tells every adapter (`.map`, `.sum`, `.collect`) what flows through the pipeline. Change it to `String` and the same `impl` body stops compiling — callers and adapters depend on that one associated type.

7.10.2 Custom trait with type `Output`

When a trait's return type varies **per implementor** but stays **one type per implementor**, use an associated type instead of a generic parameter on the trait:

```

// Playground
trait Summarizable {
    type Output;
    fn summarize(&self) -> Self::Output;
}

struct TempReading(f64);
struct PressReading(f64);

```



```

impl Summarizable for TempReading {
    type Output = String;
    fn summarize(&self) -> String {
        format!("temp={:.1}C", self.0)
    }
}

impl Summarizable for PressReading {
    type Output = String;
    fn summarize(&self) -> String {
        format!("press={:.0}kPa", self.0)
    }
}

fn log_reading<T: Summarizable>(r: &T)
where
    T::Output: std::fmt::Display,
{
    println!("{}", r.summarize());
}

fn main() {
    log_reading(&TempReading(22.5));
    log_reading(&PressReading(101.3));
}

```

7.10.3 Associated type vs generic trait parameter

Style	Example	When
Associated type	trait Iterator { type Item; ... }	One output type per implementor (u16 ports, &str lines)
Generic param	trait Storage<T> { fn get(&self) -> T; }	Caller picks T at each call site — many operations per implementor
Both in std	FromStr with type Err	Error type is fixed per parsed type, like Item for iterators

Iterator could have been `trait Iterator<T> { fn next(&mut self) -> Option<T>; }` — but then every function taking “some iterator” would need an extra type parameter. Associated types keep call sites clean.

7.10.4 Supertraits — stacking requirements

A **supertrait** requires another trait as a prerequisite. Your type must implement both:

```
// Playground
use std::fmt::Display;

trait Loggable: Display {
    fn log(&self) {
        println!("[LOG] {}", self);
    }
}

struct Port(u16);
impl Display for Port {
    fn fmt(&self, f: &mut
        std::fmt::Formatter<'_,_>) ->
        std::fmt::Result {
        write!(f, "Port({})", self.0)
    }
}
impl Loggable for Port {}

fn emit(p: &impl Loggable) {
    p.log();
}

fn main() {
    emit(&Port(502));
}
```

`Loggable: Display` means “anything loggable must also be displayable.” The default method can call `Display` formatting inside the trait.

7.10.5 UFCS when two traits share a method name

Wrong — ambiguous method call:

```
// Playground --- does not compile
trait A { fn id(&self) -> u32; }
trait B { fn id(&self) -> u32; }
struct Tag;
impl A for Tag { fn id(&self) -> u32 { 1 } }
impl B for Tag { fn id(&self) -> u32 { 2 } }

fn main() {
    let t = Tag;
    // println!("{}", t.id());
    // ERROR: multiple applicable items in scope
    println!("A={} B={}", A::id(&t), B::id(&t));
}
```

Use **UFCS** — `TraitName::method(&self)` — when two traits define the same method name.

7.11 Idiom spotlight

Default: `impl Trait` / generics (static dispatch). Use `dyn Trait` for runtime plug-in lists or return-type erasure — and `enum` when the variant set is closed.

`&dyn Trait` when callers own values; `Box<dyn Trait>` for owning heterogeneous collections.

Split large traits into small getters/savers; **enum-dispatch** fixed backends; **transform traits** bridge wire format -> domain.

7.12 Go deeper

- [The Rust Book — Structs and Methods](#)
- [The Rust Book — Generics, Traits, and Lifetimes](#)
- [The Rust Book — Trait Objects](#)
- [Records / structs — example 062](#)

7.13 See also

- Chapter 6: Enums and pattern matching — `match`, exhaustiveness, `Option/Result`
- Chapter 4: Iterators — custom `Iterator` and type `Item`
- Chapter 11: Collections
- Chapter 17: Derive attributes — `#[derive]` syntax, `std/ecosystem/custom`
- Chapter 16: Async traits (advanced)

7.13.1 Afterparty

7.13.1.1 Structs and inherent `impl`

1. **new vs default** — “When is `Sensor::new` idiomatic vs `Default` + field update? One automation example each.”
2. **Method receiver** — “Same logic three ways: `fn f(self)`, `fn f(&self)`, `fn f(&mut self)` on a struct; I predict what calls compile.”

7.13.1.2 Enums, structs, and traits together

3. **Enum trait `impl`** — “Add variant `Fault` to `Status`; show every compile error until trait `impl` and inherent methods match.”
4. **Struct vs enum layout** — “Modbus/OpcUa device registry: justify `struct Device { kind: Enum }` vs `enum Device { Modbus(...), OpcUa(...) }`; sketch types.”
5. **SensorReading port** — “Python `Union[TempReading, PressReading, Skipped]` -> Rust enum + struct payloads + `Measurable` trait; show match in `impl`.”
6. **Two `impl` blocks** — “Same type: add inherent `is_valid()` and trait `Summary`; explain which call sites see which methods.”
7. **Partial move fix** — “Given enum `Packet { Raw(String), Empty }`, reproduce ‘partially moved value’ and rewrite with `&self` or one `match`.”
8. **`matches!` drill** — “Rewrite `!matches!(self, Skipped { .. })` as longhand `match`; then back to `matches!`; link to `macro_rules` conceptually.”

7.13.1.3 Default trait methods and UFCS

9. **Default override** — “Trait `HasCode` with default `label()`; override for one enum variant only; use `HasCode::label(self)` for the rest.”

10. **Recursion trap** — “Show infinite recursion when override calls `self.label()` instead of `HasCode::label(self)`; fix it.”
11. **One trait per impl** — “Show `impl HasCode, Display for T` compile error; split into two blocks; add `fn show<T: HasCode + Display>(x: T).`”

7.13.1.4 `#[derive]` and shared models

12. **Command + log row** — “enum `Command` + struct `SetSpeedLog` + `to_log()`; list which derives each type needs and why.”
13. **Eq on floats** — “Add `Analog(f64)` variant; show `#[derive(Eq)]` failure; fix with integer fixed-point or `PartialEq` only.”

7.13.1.5 Generics and trait bounds

14. **Generic bounds** — “Fix compiler error: `T` needs `Display + Clone`; minimal bound set on `fn duplicate_and_print<T>(x: T).`”
15. **largest pitfalls** — “Why does `largest` need non-empty slice? Add `Option` return or document panic; compare to Java generics erasure story.”
16. **where clause** — “Rewrite `fn f<T: A + B + C>(x: T)` with a `where` block; same signature, longer trait list.”

7.13.1.6 `impl Trait` vs `dyn Trait` vs `enum`

17. **dyn vs impl quiz** — “Four scenarios (plug-in `Vec`, single helper `fn`, closed protocol, factory return) — pick `impl`, `dyn`, or `enum` each time.”
18. **AlarmSink registry** — “Implement `&[&dyn AlarmSink]` for log + metrics; then refactor to `enum Sink` if set is closed — compare trade-offs.”
19. **Factory return** — “`greeter_for(lang) -> Box<dyn Greeter>` vs `-> impl Greeter` — show why `impl` fails when arms return `En` and `Fr`.”
20. **Driver three ways** — “Same `poll()` behaviour with `&impl Driver`, `&[&dyn Driver]`, and `enum Device`; benchmark story without running code.”
21. **Box vs borrow** — “When is `&dyn Trait` enough vs `Box<dyn Trait>` required? `Vec` of mixed types + dangling return examples.”

7.13.1.7 Object safety and dyn traps

- 22. **Object safety audit** — “Mark each trait dyn-safe or not: no-self method, generic method, `-> Self, trait Foo: Sized.`”
- 23. **Reader fix** — “Trait with `fn read() -> f64` fails as dyn; redesign for `&dyn Reader` or use enum.”
- 24. **Clone not dyn** — “Why no `Box<dyn Clone>` pattern for heterogenous clone list; suggest enum or generic `T: Clone` instead.”
- 25. **Send + Sync** — “Alarm handler shared across threads: write type as `Arc<dyn AlarmSink + Send + Sync>`; what breaks if handler holds `Rc`?”
- 26. **Unsize trap** — “Show three snippets that fail: `let g: dyn Greeter = En, Vec<dyn Greeter>`, returning `&dyn` from locals; fix each.”

7.13.1.8 Orphan rule and cross-crate patterns

- 27. **Orphan rule** — “Why `impl Display for Vec<u8>` fails; fix with `newtype struct Frame(pub Vec<u8>) + impl Display for Frame.`”
- 28. **External trait** — “Wrap third-party struct; implement your trait on the wrapper; call from automation main.”

7.13.1.9 Capstone drills

- 29. **Checklist drill** — “Match 8 Chapter 7 compiler errors to snippets (non-exhaustive match, partial move, orphan, not dyn compatible, unsize, moved self, `Eq+f64`, multi-trait impl).”
- 30. **PLC message model** — “Design full model: enum frames, struct payloads, two traits, one dyn registry for sinks, derive list — no code over 80 lines.”
- 31. **Refactor story** — “Start with `Vec<Box<dyn Driver>>`; protocol closes to two devices; refactor to enum; list what the compiler now catches.”

7.13.1.10 Associated types and supertraits

- 32. **Item type quiz** — “Change `PortScan`’s type `Item` from `u16` to `(u16, bool)` — list every call site in a `.map().collect()` chain that breaks and why.”
- 33. **Summarizable design** — “Add `Summarizable` with type `Output = String` for three sensor structs; one `fn report(r: &impl Summarizable)` — no duplicate return types in the trait.”
- 34. **Associated vs generic** — “Same cache API twice: `trait Get<T>` vs `trait Get { type Value; }` — when is each painful at call sites?”

35. **Supertrait bounds** — “Trait `Exportable`: `Display` + `Debug` with default `fn export` — show impl for `Port(u16)`; what fails if `Display` is missing?”
36. **UFCS fix** — “Type implements `A` and `B`, both define `name()` — show ambiguous call error and UFCS fix with `A::name(&x)`.”

Chapter 8

Errors and Testing

8.1 Hook

Many languages use exceptions for control flow (**Java** checked/unchecked throws, **Python** raise, ...). Rust splits **recoverable** (**Result**) from **unrecoverable** (**panic!**). Long-running code should treat errors as data, not surprises.

Why **panic!** / **unwrap()** in production is a bad idea

A **panic** stops the thread — often the whole process. Use it for bugs, not for expected failures like bad config or offline devices.

Expected failure (data)	Panic (crash)
Log, alert operator, retry next poll	Whole gateway or PLC bridge down until someone restarts it
Return Err to caller — choose fallback port or safe state	No chance to enter safe mode or drain the queue
Error is typed and testable in the signature	Failure is a surprise at a random unwrap() line

At service boundaries, use **? on Result** — not **unwrap** on ports, config, or field I/O. Full panic mechanics below; see also Chapter 6 — avoid **unwrap** in automation.

8.2 Panic, unwind, and why it is not Result

When Rust **panics** (`panic!`, `failed assert!`, `unwrap` on `Err/None`), the runtime treats the situation as **non-recoverable for this thread**. What happens next depends on the panic strategy:

Strategy	Behaviour	Typical target
Unwind (default on many hosts)	Walk the stack backwards ; run Drop on each frame ; then thread dies or error propagates	desktop, server
Abort (<code>panic = "abort"</code>)	Immediate process exit — no Drop, no cleanup	embedded, some release builds

Unwind is Rust’s structured stack retreat — closer to C++ exceptions or Java unwinding than to returning `Err`. It is **not** “return an error value”.

```
normal Result path:  open()? ->
                     Err(io::Error) -> caller matches, logs,
                     retries
panic path:          unwrap() -> panic!
                     -> unwind stack -> thread/process gone
```

8.2.1 Problems unwind creates in production

1. **No typed error in the signature** — callers cannot match on what went wrong; the thread just stops.
2. **Drop runs under panic** — destructors flush buffers, close handles, release locks. If **Drop also panics**, Rust **aborts the whole process** (double panic).
3. **Partial cleanup story** — during unwind, some resources are dropped and some code never runs. Invariants may be left mid-update unless you designed for it.
4. **Unattended automation** — a Modbus poll loop that panics on one bad frame **stops all future polls**. A `Result` lets you log, skip, and continue.
5. **catch_unwind is a niche tool** — it can catch unwinding panics in isolated blocks (e.g. foreign callbacks). It **must not** replace `Result` for business logic. It misses **abort panics**, has **UnwindSafe** bounds, and adds complexity.

```
// Playground --- illustration only; not
    idiomatic error handling
use std::panic;

fn main() {
    let outcome = panic::catch_unwind(|| {
        panic!("simulated bug");
    });
    println!("caught panic: {:?}",
        outcome.is_err());
}
```

For PLC gateways and serial bridges: **Result at boundaries, panic for programmer mistakes.**

8.3 Error propagation

Chapter 6 introduced `Result<T, E>`. This section is the **railway**: each step returns `Result`; `?` exits early on `Err` and passes the success value forward. No unwind, no panic.

[Diagram omitted in print edition — see the web repository.]

Layered gateway load (playground — uses manual `AppError` defined below):

```
// Playground
#[derive(Debug)]
enum AppError {
    Parse(String),
    OutOfRange(u16),
}

fn read_file(path: &str) -> Result<String,
    AppError> {
    // stand-in for std::fs::read_to_string in
    demos
    if path.is_empty() {
        Err(AppError::Parse("empty
            path".into()))
    }
}
```

```
    } else {
        Ok("8080".into())
    }
}

fn parse_port(text: &str) -> Result<u16,
    AppError> {
    text.trim()
        .parse()
        .map_err(|_|
            AppError::Parse(text.into()))
}

fn validate_port(port: u16) -> Result<u16,
    AppError> {
    if port < 1024 {
        Err(AppError::OutOfRange(port))
    } else {
        Ok(port)
    }
}

fn load_gateway_config(path: &str) ->
    Result<u16, AppError> {
    let text = read_file(path)?;
    let port = parse_port(&text)?;
    validate_port(port)?;
    Ok(port)
}

fn run() -> Result<u16, AppError> {
    load_gateway_config("config.toml")
}

fn main() {
    match run() {
        Ok(port) => println!("gateway on {}",
            port),
    }
}
```

```

        Err(e) => eprintln!("startup failed:
            {:?}" , e),
    }
    println!("{:?}",
        parse_port("8080").and_then(validate_por
            t));
}

```

Each `?` is an **early return** of `Err` — same idea as Java `throw`, but the error type is visible in every signature. The `match` in `main` is the **boundary**; `run` and helpers bubble errors with `?`.

8.3.1 Three propagation tools

Tool	Use when	Effect on <code>Err</code>
<code>?</code>	Inside <code>fn -> Result<_, E></code>	return <code>Err</code> immediately (via <code>From</code> if types differ)
<code>.map</code> / <code>.map_err</code>	Transform value or error in place	stay in <code>Result</code> ; no early exit from current function
<code>.and_then</code>	Next fallible step on success	chain; equivalent to <code>?</code> inside a closure

```

// Playground
fn parse_doubled(s: &str) -> Result<u32,
    String> {
    s.trim()
        .parse::<u32>()
        .map_err(|e| e.to_string())
        .map(|p| p * 2)
}

fn main() {
    println!("{:?}", parse_doubled("21"));
}

```

`.and_then(validate_port)` in the gateway example chains two fallible steps without a separate function body.

8.3.2 Internal vs boundary

Layer	Who	Pattern
Internal	library helpers, poll loop body	? bubble Err up — no printing here
Boundary	main, HTTP handler, PLC supervisor	one match or if let Err(e) = run() — log, exit code, retry

In the gateway example above: `read_file` / `parse_port` / `validate_port` / `load_gateway_config` are **internal**; main's `match run()` is the **boundary**.

8.3.3 Example: read first line from a file (Cargo only)

```
// Cargo project --- needs std::fs
use std::fs::File;
use std::io::{self, Read};

fn read_first_line(path: &str) ->
    Result<String, io::Error> {
    let mut f = File::open(path)?;
    let mut buf = String::new();
    f.read_to_string(&mut buf)?;
    Ok(buf.lines().next().unwrap_or("")).to_string()
}

fn main() {
    match read_first_line("Cargo.toml") {
        Ok(s) => println!("{}", s),
        Err(e) => println!("error: {}", e),
    }
}
```

Line	What it does
-> Result<String, io::Error>	Success = first line; failure = std I/O error (no panic).
File::open(path)?	Propagate open failure — return path , not unwind.
f.read_to_string(&mut buf)?	Same for read errors.

Line	What it does
<code>unwrap_or("")</code>	Empty file -> ""; not a panic path.
<code>match in main</code>	Boundary — process keeps running after logging.

Playground equivalent:

```
// Playground
fn parse_config(s: &str) -> Result<u32, String>
{
    s.trim().parse::<u32>().map_err(|e|
        e.to_string())
}

fn main() {
    println!("{:?}", parse_config("8080"));
    println!("{:?}", parse_config("oops"));
}
```

What ? desugars to:

```
let mut f = match File::open(path) {
    Ok(v) => v,
    Err(e) => return Err(e),
};
```

8.3.4 Propagation edge cases

Wrong — ? in a function that does not return **Result** (or **Option**):

```
// Playground --- does not compile
fn broken(s: &str) -> u32 {
    s.parse::<u32>()?
    // ERROR: the `?` operator can only be used
    // in a function that returns `Result` or
    // `Option`
}
```

? on **Option** inside **fn -> Option** (Chapter 6):

```
// Playground
fn first_word(s: &str) -> Option<&str> {
    let word = s.split_whitespace().next()?;
    // None -> return None
    Some(word)
}
```

Wrong — error type mismatch with ?:

```
// Playground --- does not compile
enum AppError { Io(std::io::Error) }

fn read(path: &str) -> Result<String, AppError>
{
    let mut buf = String::new();
    std::fs::File::open(path)?.read_to_string(&mut buf)?; // ERROR: `` can't convert
    `io::Error` to `AppError`
    Ok(buf)
}
```

Fix: `map_err`, `From impl`, or `thiserror #[from]` — see below.

Wrong — `unwrap` where ? belongs:

```
// Playground --- runs, panics on bad input
(unwind path)
fn load_port(s: &str) -> u16 {
    s.parse().unwrap()
}
```

Idiomatic poll loop — log and continue:

```
// Playground
fn poll_port(s: &str) {
    match s.parse::<u16>() {
        Ok(p) => println!("poll ok {}", p),
        Err(e) => eprintln!("bad config, skip
            tick: {}", e),
    }
}
```

```
fn main() {
    poll_port("502");
    poll_port("oops");
}
```

Helper	On failure	Production use
?	return Err	library / internal propagation
match / if let	branch	boundaries, loops, recovery
unwrap() / expect("...")	panic , unwind	tests, prototypes, truly impossible
unwrap_or / unwrap_or_else	default value	safe fallback (not for I/O errors)

8.4 Std library errors — fine for learning, weak for production

Std error types are **correct** and **zero-dependency** — perfect for tutorials and tiny scripts. Real automation pain comes from **ergonomics and maintainability**, not from `io::Error` being wrong.

Std type	Problem in production automation
<code>io::Error</code>	Opaque; <code>ErrorKind</code> matching is brittle; hard to attach <i>which config file</i> or <i>which device</i>
<code>String</code> / <code>&str</code>	Not matchable by variant; context lost when re-wrapped; easy to typo the same message twice
<code>ParseIntError</code> etc.	Tiny; no path/key context; every callsite needs manual <code>map_err</code>
Manual impl <code>Error</code>	Boilerplate: <code>Display</code> , <code>source()</code> , <code>From</code> for each wrapped type — easy to forget

Stage	Error type	Tooling
Playground / first week	<code>String</code> , <code>&str</code> , manual <code>enum</code>	none
Library crate / gateway core	<code>enum</code> + <code>thiserror</code>	<code>derive Error</code> , <code>#[from]</code> , <code>#[error("...")]</code>

Stage	Error type	Tooling
Binary / CLI / service main	<code>anyhow::Result</code>	<code>.context("while loading config")</code> chains

Honest take: add `thiserror` to any library or gateway core you will maintain. Use std errors while learning; switch before you ship.

8.5 Custom errors

8.5.1 Manual `enum` — learn the model

Real automation crates combine variants (parse, I/O, device timeout) instead of a bare `String`:

```
// Playground
#[derive(Debug)]
enum AppError {
    Parse(String),
    OutOfRange(u32),
}

fn set_port(s: &str) -> Result<u16, AppError> {
    let p: u16 = s.parse().map_err(|_|
        AppError::Parse(s.into()))?;
    if p < 1024 {
        Err(AppError::OutOfRange(p as u32))
    } else {
        Ok(p)
    }
}

fn main() {
    println!("{:?}", set_port("8080"));
    println!("{:?}", set_port("80"));
    println!("{:?}", set_port("oops"));
}
```

Line	What it does
enum AppError	Closed set of failure modes — match at boundaries.
map_err(...)	Turn parse failure into your variant with context.
?	Propagate Err(AppError::Parse(...)) early.
if p < 1024 { Err(...) }	Business rule as data — no panic.

What manual production code still needs (what `thiserror` generates for you):

```
// Playground --- conceptual boilerplate; use
    thiserror instead
use std::fmt;

impl fmt::Display for AppError {
    fn fmt(&self, f: &mut fmt::Formatter<'_>)
        -> fmt::Result {
        match self {
            AppError::Parse(s) => write!(f,
                "invalid port '{s}'"),
            AppError::OutOfRange(p) =>
                write!(f, "port {p} below
                    privileged range"),
        }
    }
}

impl std::error::Error for AppError {}
```

Add `source()` when wrapping `io::Error` — another ~10 lines per variant. **thiserror** generates this for you.

Chaining with `?` across error types — implement `From`:

```
// Playground
use std::num::ParseIntError;

#[derive(Debug)]
```

```

enum AppError {
    Parse(String),
    OutOfRange(u32),
}

impl From<ParseIntError> for AppError {
    fn from(_: ParseIntError) -> Self {
        AppError::Parse("invalid
            integer".into())
    }
}

fn set_port_from(s: &str) -> Result<u16,
    AppError> {
    let p: u16 = s.parse()?;
    // ``?`` converts via From
    if p < 1024 {
        Err(AppError::OutOfRange(p as u32))
    } else {
        Ok(p)
    }
}

fn main() {
    println!("{:?}", set_port_from("8080"));
}

```

8.5.2 thiserror in production — count it in

For libraries and gateway cores, **thiserror** replaces manual `Display + Error + From` with one derive. `#[derive(Error)]` is an **ecosystem derive attribute** — same `#[derive]` syntax as `Debug`, implemented by a proc-macro crate, not a runtime annotation (Chapter 17).

Cargo.toml:

```

[dependencies]
thiserror = "2"

```

Cargo project example:

```
// Cargo project
use thiserror::Error;

#[derive(Debug, Error)]
pub enum AppError {
    #[error("failed to read config at {path}")]
    ConfigRead {
        path: String,
        #[source]
        source: std::io::Error,
    },

    #[error("invalid port '{raw}'")]
    ParsePort {
        raw: String,
        #[source]
        source: std::num::ParseIntError,
    },

    #[error("port {0} below privileged range")]
    OutOfRange(u16),

    #[error("device timeout after {ms} ms")]
    Timeout { ms: u64 },

    #[error("serial error")]
    Serial(#[from] SerialError),
}

#[derive(Debug, Error)]
pub enum SerialError {
    #[error("framing error")]
    Framing,
    #[error("crc mismatch")]
    Crc,
}
```

Attribute / line	What it does
<code>#[derive(Error)]</code>	Generates <code>Display</code> + <code>std::error::Error</code> + <code>source()</code> chain
<code>#[error("...")]</code>	Operator-facing message template — no manual <code>fmt</code>
<code>#[source]</code>	Underlying error preserved for logs / <code>error.chain()</code>
<code>Serial(#[from] SerialError)</code>	Auto <code>From<SerialError></code> — use <code>?</code> on serial helpers

Same railway, less noise:

```
// Cargo project --- with thiserror AppError
    above
use std::fs;

pub fn load_gateway_config(path: &str) ->
    Result<u16, AppError> {
    let text =
        fs::read_to_string(path).map_err(|source|
            AppError::ConfigRead {
                path: path.into(),
                source,
            })?;
    let port = text
        .trim()
        .parse()
        .map_err(|source| AppError::ParsePort {
            raw: text.clone(),
            source,
        })?;
    if port < 1024 {
        return Err(AppError::OutOfRange(port));
    }
    Ok(port)
}
```

With `#[from]` on `io::Error` variant you could shorten further. Trade explicit context vs brevity.

8.5.3 anyhow for binaries (sidebar)

Use **anyhow** in **main** and binary crates — not in library public APIs:

```
// Cargo project --- binary crate
// anyhow = "1"

use anyhow::Context;

fn load_gateway_config(_path: &str) ->
    Result<u16, std::io::Error> {
    // stand-in for lib helper from the
    // thiserror example above
    Ok(8080)
}

fn run() -> anyhow::Result<> {
    let port =
        load_gateway_config("config.toml")
        .context("loading gateway config")?;
    println!("port {}", port);
    Ok(())
}

fn main() -> anyhow::Result<> {
    run()
}
```

Pairing: use a **thiserror** enum in **lib** for typed errors for callers. Use **anyhow** in **bin** for context chains at the top. See [errors and enums](#) for matching variants at the boundary.

8.5.4 Crate-local **Result** alias

Large modules repeat the same error type on every signature. A **type alias** fixes the error side once:

```
// Playground
#[derive(Debug)]
pub enum GatewayError {
    Timeout,
```

```

    Parse(String),
}

pub type Result<T> = std::result::Result<T,
    GatewayError>;

pub fn poll() -> Result<f64> {
    Ok(42.0)
}

fn main() {
    println!("{:?}", poll());
}

```

Use `pub type Result<T> = std::result::Result<T, MyError>` in library roots — not in public APIs that re-export `std::result::Result` ambiguously. Document the alias in crate docs.

8.5.5 Transparent `#[from]` variants

Wrap subsystem errors without a custom message — `#[error(transparent)]` forwards `Display` and `source()`:

```

// Cargo project --- conceptual with thiserror
use thiserror::Error;

#[derive(Debug, Error)]
pub enum AppError {
    #[error("config read failed")]
    Config { source: std::io::Error },

    #[error(transparent)]
    Serial(#[from] SerialError),
}

#[derive(Debug, Error)]
pub enum SerialError {
    #[error("framing error")]
    Framing,
}

```

```
fn read_frame() -> Result<(), AppError> {
    Err(SerialError::Framing.into())
    // ``?`` works on SerialError helpers
}

fn main() {
    let _ = read_frame();
}
```

8.5.6 Error aggregation for batch work

Stream and batch pipelines often collect **per-item failures** and report them together instead of failing silently on the first error. A small trait keeps the merge logic reusable:

```
// Playground
#[derive(Debug)]
struct ItemError {
    line: usize,
    msg: String,
}

#[derive(Debug)]
enum BatchError {
    Multiple { errors: Vec<ItemError> },
}

trait FromMultipleErrors<E> {
    fn from_multiple(errors: Vec<E>) -> Self;
}

impl FromMultipleErrors<ItemError> for
    BatchError {
    fn from_multiple(errors: Vec<ItemError>) ->
        Self {
        Self::Multiple { errors }
    }
}
```



```

fn flush_errors(errors: Vec<ItemError>) ->
    Result<(), BatchError> {
    if errors.is_empty() {
        Ok(())
    } else {
        Err(BatchError::from_multiple(errors))
    }
}

fn main() {
    let errs = vec![
        ItemError { line: 3, msg: "bad
            port".into() },
        ItemError { line: 9, msg: "missing
            tag".into() },
    ];
    println!("{:?}", flush_errors(errs));
}

```

Format the **Display** message for Multiple variants to list each sub-error — operators need the full list, not “something failed”.

8.6 Extension traits on Option

Hand-written parsers repeat `ok_or_else` for required fields. An **extension trait** on `Option<T>` centralizes the error shape (pattern from Chapter 3 — Extension traits):

```

// Playground
#[derive(Debug)]
enum ParseError {
    MissingField { name: String },
}

trait RequiredField<T> {
    fn required(self, name: &str) -> Result<T,
        ParseError>;
}

```

```

impl<T> RequiredField<T> for Option<T> {
    fn required(self, name: &str) -> Result<T,
        ParseError> {
        self.ok_or_else(||
            ParseError::MissingField {
                name: name.to_string(),
            })
    }
}

fn parse_unit_id(fields:
    &std::collections::HashMap<&str, &str>) ->
    Result<u8, ParseError> {
    let raw =
        fields.get("unit_id").copied().required(
            "unit_id"?;
    raw.parse::<u8>()
        .map_err(|_| ParseError::MissingField {
            name: "unit_id".into() })
    }

fn main() {
    let mut m =
        std::collections::HashMap::new();
    m.insert("unit_id", "7");
    println!("{:?}", parse_unit_id(&m));
}

```

Name constructors on the error enum (`ParseError::missing_field(...)`) keep call sites even shorter (Chapter 3).

8.6.1 Custom error edge cases

Wrong — panic inside error path:

```

// Playground --- anti-pattern
#[derive(Debug)]
enum AppError {
    BadPort(String),

```

```

}

fn bad(s: &str) -> Result<u16, AppError> {
    Ok(s.parse:::<u16>()).unwrap_or_else(|_|
        panic!("bad port {}", s))) // should
        use map_err, not panic
}

fn main() {
    let _ = bad("502"); // ok path
    // bad("oops");
    // runtime panic --- defeats Result
}

```

Match at boundary for operator-facing messages:

This is the **typed** version of the boundary pattern from Chapter 6: helpers return structured errors; `main` turns each variant into a message an operator can act on.

```

// Playground
#[derive(Debug)]
enum AppError {
    Parse(String),
    OutOfRange(u32),
}

fn set_port(s: &str) -> Result<u16, AppError> {
    let p: u16 = s
        .parse()
        .map_err(|_|
            AppError::Parse(s.into()))?;
    if p < 1024 {
        Err(AppError::OutOfRange(p as u32))
    } else {
        Ok(p)
    }
}

fn main() {

```

```
match set_port("80") {
  Ok(p) => println!("listening on {}", p),
  Err(AppError::OutOfRange(p)) =>
    eprintln!("port {} needs root", p),
  Err(AppError::Parse(s)) =>
    eprintln!("not a port: {:?}", s),
}
```

Two layers in one example:

Layer	Function	Role
Interior	set_port	validate input, return Result<u16, AppError> — uses ? and map_err, no printing
Boundary	main	match on Result — pick the operator message per variant

What set_port("80") does step by step:

1. "80".parse() -> Ok(80) — valid number, so no Parse error.
2. 80 < 1024 -> Err(AppError::OutOfRange(80)) — business rule: privileged ports need root.
3. main matches OutOfRange(80) -> prints port 80 needs root.

Why typed variants beat Result<T, String>:

Aspect	AppError enum	plain String
Caller distinguishes failure kinds	match on Parse vs OutOfRange	string compare / contains
Compiler checks exhaustiveness	must handle every variant	easy to miss a case
Refactor	rename variant -> compile errors at call sites	silent string drift

Interior code **names the failure** (Parse, OutOfRange); boundary code **explains it to a human**. Keep Display / thiserror messages on the enum for logging; save custom wording for main / CLI / HTTP when the audience differs.

8.7 Errors and enums

Your error type **is an enum** — same machinery as `Option`, `Result`, and domain enums in Chapter 6. Use `match` in helpers and `#[derive(Error)]` instead of hand-written trait impls (Chapter 7).

8.7.1 Variant shapes for errors

Shape	Example	Use
Unit	<code>Timeout</code>	simple tag
Tuple	<code>OutOfRange(u16)</code>	single payload
Struct	<code>ConfigRead { path, source }</code>	context + <code>#[source]</code>
<code>#[from]</code> wrapper	<code>Serial(#[from] SerialError)</code>	transparent propagation

```
// Playground --- error enum mirrors Ch6 enum
    shapes
#[derive(Debug)]
enum GatewayError {
    Timeout,
    // unit
    OutOfRange(u16),
    // tuple
    ConfigRead { path: String },
    // struct
}
```

8.7.2 Error enum vs domain enum — keep them separate

Type	Models	match purpose
Command (domain)	what the PLC sends	dispatch behaviour
AppError (error)	what went wrong	recovery / logging / exit codes

Do not merge domain commands and error variants into one enum — they have different lifecycles.

8.7.3 Recovery by variant

Automation often **branches on error kind** without panicking:

```
// Playground --- conceptual poll with typed
recovery
#[derive(Debug)]
enum AppError {
    Timeout { ms: u64 },
    ParsePort { raw: String },
    DeviceOffline,
}

fn poll_device() -> Result<f64, AppError> {
    Err(AppError::Timeout { ms: 500 })
}

fn handle_poll() {
    match poll_device() {
        Ok(v) => println!("stored {}", v),
        Err(AppError::Timeout { .. }) =>
            eprintln!("retry next tick"),
        Err(AppError::ParsePort { raw }) =>
            eprintln!("fix config: {}", raw),
        Err(AppError::DeviceOffline) =>
            eprintln!("alert operator"),
    }
}

fn main() {
    handle_poll();
}
```

Variant	Typical recovery
Timeout	retry, backoff
ParsePort	alert config team, skip startup
DeviceOffline	alarm, safe state

8.7.4 Nested error enums

Subsystem errors as their own enum, embedded in the top-level error:

```
// Playground
#[derive(Debug)]
enum SerialError {
    Framing,
    Crc,
}

#[derive(Debug)]
enum AppError {
    Serial(SerialError),
    Timeout { ms: u64 },
}

fn read_frame() -> Result<(), AppError> {
    Err(AppError::Serial(SerialError::Crc))
}

fn main() {
    if let
        Err(AppError::Serial(SerialError::Crc))
        = read_frame() {
        eprintln!("crc fail --- request
            retransmit");
    }
}
```

Same pattern as **struct-in-enum** in Chapter 7.

8.7.5 Errors-and-enums edge cases

Exhaustiveness — add a variant, update every **match**:

```
// Playground --- does not compile until you
    handle DeviceOffline everywhere
#[derive(Debug)]
enum AppError {
    Timeout { ms: u64 },
```

```

    DeviceOffline, // new
}

fn describe(e: &AppError) -> &'static str {
    match e {
        AppError::Timeout { .. } => "timeout",
        // AppError::DeviceOffline =>
            "offline", // ERROR: non-exhaustive
    }
}

```

source() vs **matching the enum** — match **AppError** variants for recovery policy. Inspect **source()** only when you need the underlying `io::Error` kind for logging.

#[non_exhaustive] on public library error enums allows adding variants without breaking downstream match exhaustiveness. Downstream must keep a wildcard arm.

Wrong — **Result** inside an **enum variant** (usually an anti-pattern):

```

// Playground --- awkward; prefer Result at the
    fn level
enum Messy {
    Pending(Result<u16, AppError>),
}

```

Model outcomes with **fn -> Result<T, E>** or a dedicated domain enum. Do not nest **Result** inside unrelated enums.

Trap	Idiom
String errors in lib API	enum + thiserror
Mixed domain + error variants	two enums
New error variant	compiler lists every match — treat as feature
Need underlying I/O kind	#[source] + logging, not string formatting

8.8 panic! vs recover — decision table

Mechanism	Control flow	Cleanup	Caller can react?
Result / Option	normal return	you decide in match	yes
panic! / unwrap	unwind or abort	Drop on unwind; abort skips it	no (except catch_unwind niche)

Use	When
Result	Missing file, bad input, timeout, CRC fail, device NAK
Option	Optional value, “not found” without error detail
panic! / assert!	Logic bug, violated invariant, test failure
expect("...")	Same as unwrap but message names the invariant (still panics)

Libraries should not panic on bad user input. Binaries may use `main() -> Result<(), E>` and print once at the top:

```
// Playground --- CLI pattern (conceptual)
fn run() -> Result<(), AppError> {
    let port = set_port("8080");
    println!("port {}", port);
    Ok(())
}

fn main() {
    if let Err(e) = run() {
        eprintln!("fatal: {:?}", e);
        std::process::exit(1);
    }
}
```

Exit code + message at **one** boundary — not scattered unwraps.

8.8.1 Panic and unwind edge cases

Panic in Drop -> abort:

```
// Playground --- do not do this; conceptual
struct Bad;
```

```
impl Drop for Bad {
    fn drop(&mut self) {
        panic!("drop panicked");
    }
}
// let _b = Bad;
// drop triggers panic; if already panicking ->
    abort
```

Thread panic without handler — spawned thread panics can **join as Err**. If ignored, the runtime may **abort the whole program** on drop of an unjoined panicking thread.

assert! in production paths — failed assert is a panic (unwind/abort), not a graceful Result. Prefer explicit checks returning Err.

Trap	Symptom	Idiom
unwrap on config/I/O	process exit on bad deploy	? + log at boundary
? wrong return type	compile error	change signature or map_err / From
panic in Drop	double panic -> abort	never panic in Drop
catch_unwind as policy	fragile, not for business errors	use Result
empty file vs missing file	unwrap_or("") vs open()?	separate “not found” from “empty” if it matters

8.9 Testing

Tests use **assert!** / **assert_eq!** — a failed test **panics** (that is intentional; CI catches it). Production error paths should not rely on panics for expected cases.

Cargo only:

```
// src/lib.rs or same file with #[cfg(test)]
pub fn add(a: i32, b: i32) -> i32 {
    a + b
}

pub fn parse_port(s: &str) -> Result<u16,
```

```
String> {
  let p: u16 = s.parse().map_err(|e|
    e.to_string());
  Ok(p)
}

#[cfg(test)]
mod tests {
  use super::*;

  #[test]
  fn adds() {
    assert_eq!(add(2, 2), 4);
  }

  #[test]
  fn parse_valid_port() {
    assert_eq!(parse_port("502").unwrap(),
      502);
  }

  #[test]
  fn parse_rejects_non_numeric() {
    assert!(parse_port("oops").is_err());
    // prefer is_err over unwrap in
    // negative tests
  }
}
```

What each piece does:

Piece	Role
pub fn	tests in mod tests import via use super::*
#[cfg(test)]	module compiled only when running cargo test — zero cost in release binary
assert_eq!	equality check; fails test with diff on mismatch
parse_port("502").unwrap()	OK in tests — failure means test broken, not field conditions

Piece	Role
<code>is_err()</code>	checks error path without panicking the test runner

```
cargo test
```

Also: **doc tests** in `///` examples and **integration tests** in `tests/*.rs` (external crate view of your API).

8.9.1 Testing edge cases

Table-driven tests — one function, many inputs:

```
// Playground
fn parse_port(s: &str) -> Result<u16, String> {
    let p: u16 = s.parse().map_err(|e:
        std::num::ParseIntError|
        e.to_string());
    Ok(p)
}

#[cfg(test)]
mod table {
    use super::parse_port;

    #[test]
    fn ports() {
        for (input, ok) in [("502", true),
            ("0", true), ("oops", false)] {
            assert_eq!(parse_port(input).is_ok(),
                ok, "input={input}");
        }
    }
}

fn main() {}
```

Testing Result errors — use `match`, `.unwrap_err()`, or crates like `assert_matches` instead of panicking on unexpected `Ok`.

8.9.2 Trait-based mocks — test orchestration, not I/O

Production code depends on **traits** (Chapter 7). Tests supply a **mock struct** that implements the same trait and records calls:

```
// Playground
use std::cell::RefCell;

trait DeviceWriter {
    fn write_reading(&self, tag: &str, value:
        f64);
}

struct LiveWriter;

impl DeviceWriter for LiveWriter {
    fn write_reading(&self, tag: &str, value:
        f64) {
        println!("live {}={}", tag, value);
    }
}

struct MockWriter {
    pub calls: RefCell<Vec<(String, f64)>>,
}

impl DeviceWriter for MockWriter {
    fn write_reading(&self, tag: &str, value:
        f64) {
        self.calls.borrow_mut().push((tag.to_str
            ing(), value));
    }
}

fn ingest<W: DeviceWriter>(w: &W, tag: &str,
    value: f64) {
    w.write_reading(tag, value);
}

#[cfg(test)]
```

```

mod mock_tests {
  use super::*;

  #[test]
  fn records_one_write() {
    let mock = MockWriter {
      calls: RefCell::new(Vec::new()),
    };
    ingest(&mock, "temp", 22.5);
    assert_eq!(mock.calls.borrow().len(),
              1);
  }
}

fn main() {}

```

Use **RefCell** or **Mutex** in mocks when the trait takes **&self** but tests need interior mutability. Async traits use the same pattern with **#[async_trait]** and **#[tokio::test]**.

8.9.3 Golden-file and shared parser tests

Parsers fit a **shared test helper**: read input fixture, run the parser, compare to expected output. One helper covers every source format:

```

// Playground --- pattern sketch
fn parse_line(line: &str) -> Result<(String,
  f64), String> {
  let (tag, val) = line
    .split_once('=')
    .ok_or_else(|| "missing
      '='".to_string())?;
  Ok((tag.into(), val.parse().map_err(|e|
    e.to_string())?))
}

#[cfg(test)]
mod golden {
  use super::parse_line;

```

```
#[test]
fn sample_lines() {
    let inputs = ["temp=22.5",
                  "press=101.3"];
    let expected: Vec<_> = inputs
        .iter()
        .map(|s| parse_line(s).unwrap())
        .collect();
    assert_eq!(expected.len(), 2);
}
```

For large nested structs, **pretty_assertions::assert_eq** (dev-dependency) prints colored diffs on failure — worth it in data-heavy tests.

8.9.4 Comparable test DTOs — normalize before equality

Golden comparisons often fail on **benign differences** (whitespace, sort order). Build a **test-only comparable type** that normalizes:

```
// Playground
#[derive(Debug, PartialEq)]
struct Reading {
    tag: String,
    value: f64,
}

#[derive(Debug, PartialEq)]
struct ComparableReading {
    tag: String,
    value_milli: i64,
}

impl ComparableReading {
    fn from_reading(r: Reading) -> Self {
        Self {
            tag: r.tag.trim().to_lowercase(),
            value_milli: (r.value *

```

```

        1000.0).round() as i64,
    }
}

#[cfg(test)]
mod compare {
    use super::*;

    #[test]
    fn normalizes_whitespace_and_float() {
        let a =
            ComparableReading::from_reading(Reading {
                tag: " Temp ".into(),
                value: 22.499,
            });
        let b =
            ComparableReading::from_reading(Reading {
                tag: "temp".into(),
                value: 22.5,
            });
        assert_eq!(a, b);
    }
}

fn main() {}

```

8.9.5 Test-only derives with `cfg_attr`

Production types may skip **Deserialize** or **PartialEq** to keep dependencies lean. Tests opt in with **cfg_attr**:

```

// Playground
#[derive(Debug, Clone, PartialEq)]
#[cfg_attr(test, derive(serde::Deserialize))]
struct RegionOffsets {
    start: u64,
}

```



```

    end: u64,
}

fn main() {
    let _ = RegionOffsets { start: 0, end: 10 };
}

```

Requires `serde` as a **dev-dependency** when you deserialize fixtures in tests only.

8.9.6 Async tests

Stream and parser tests that use `.await` need a runtime:

```

// Cargo project --- needs tokio with "macros"
+ "rt"
#[tokio::test]
async fn delayed_ok() {
    tokio::time::sleep(std::time::Duration::from
        _millis(1)).await;
    assert_eq!(2 + 2, 4);
}

```

See Chapter 16 for `#[tokio::test]` and spawn patterns.

8.10 Idiom spotlight

Libraries: `enum` + **thiserror** — typed variants, `#[from]` for `?`, `#[error("...")]` for messages. **Never `Result<T, String>` in a public API.**

Binaries: **anyhow** at main for context chains; map **thiserror** errors at the boundary.

Propagation: `?` inside helpers; **one** match or `main() -> Result` at the edge. Expected failures are **data**; panics **unwind** (or **abort**) — wrong tool for retry loops on a factory floor.

Batch pipelines: aggregate per-item errors; **extension traits** on **Option** for required fields. **Mock traits** in tests — same interface as production I/O.

8.11 Go deeper

- [The Rust Book — Error Handling](#)
- [The Rust Book — Writing Automated Tests](#)
- [Result railway](#)
- [Unit test patterns — 744+](#)
- [thiserror docs](#)

8.12 See also

- Chapter 6: Result enum
- Chapter 7: Enums + traits — same enum machinery for errors
- Chapter 19: I/O errors
- Chapter 17: Derive attributes — `#[derive(Error)]`, annotations vs decorators

8.12.1 Afterparty

8.12.1.1 Error propagation

1. **? chain** — “Refactor nested match on Results to ? railway; explain each desugared return Err.”
2. **map / and_then** — “Same parser with `.map_err + .map` vs `.and_then(validate)`; when to use each.”
3. **Wrong return type** — “Show ? compile error in `fn -> u32`; fix signature and boundary match.”
4. **Poll loop** — “Rewrite `unwrap Modbus config parse` to log-and-continue; process must survive bad line.”
5. **Internal vs boundary** — “Mark 8 functions in a gateway crate: bubble with ? or handle in main?”

8.12.1.2 Panic, unwind, and production

6. **Unwind vs Result** — “Diagram stack for `open()`? vs `unwrap()` on missing file; who runs Drop?”
7. **panic audit** — “Mark 10 `unwrap/expect` sites: keep (test/invariant) vs Result (I/O/config).”
8. **Drop double panic** — “Explain why panicking in Drop aborts; sketch safe cleanup pattern.”

9. **catch_unwind scope** — “When is `catch_unwind` appropriate vs abuse? One automation anti-example.”
10. **abort strategy** — “Embedded firmware: `panic = abort` — what cleanup is skipped vs unwind?”

8.12.1.3 Std errors, `thiserror`, anyhow

11. **Std limits** — “List 4 pain points of `io::Error` + `String` errors in a Modbus gateway; no FUD.”
12. **Manual vs `thiserror`** — “Same `AppError` twice: hand-written `Display/Error/From` vs `#[derive(Error)]`; count lines saved.”
13. **`thiserror` design** — “Design `GatewayError` with `ConfigRead`, `ParsePort`, `Timeout`, `Serial` sub-enum; full derive attrs.”
14. **anyhow vs `thiserror`** — “Pick for automation binary vs library crate; show `main -> anyhow::Result + .context()`.”
15. **Why not `String`** — “Explain why `pub fn connect() -> Result<(), String>` is weak; propose enum API.”

8.12.1.4 Errors and enums

16. **Error enum design** — “Design `AppError` for config + serial + timeout; variant shapes + recovery match.”
17. **Recovery match** — “Poll loop: `Timeout -> retry`, `ParsePort -> alert`, `DeviceOffline -> safe state` — sketch match.”
18. **Nested `SerialError`** — “Add `AppError::Serial(SerialError)`; show propagation with `#[from]`.”
19. **Exhaustive trap** — “Add `DeviceOffline` variant; quote non-exhaustive match errors until fixed.”

8.12.1.5 Custom errors and boundaries

20. **Boundary pattern** — “`run() -> Result` + main maps to exit code; no unwrap in between.”
21. **`map_err` drill** — “Convert `ParseIntError -> AppError::Parse` with and without `From / #[from]`.”

8.12.1.6 Testing

22. **Table-driven ports** — “Tests for `parse_port`: valid, zero, non-numeric, too large for `u16`.”

23. **Integration test** — “Sketch `tests/config_load.rs` that expects `Err` on missing file without panicking.”

Part I

Crates, Memory, and Data

Chapter 9

Modules, Paths, and Crates

9.1 Hook

Packages and modules group code by namespace in **Java**, **Python**, and many other stacks. Rust uses **modules** inside a **crate** — one compiled unit. The module tree splits projects across files, controls visibility, and exposes a public API.

Most of this chapter is **Cargo only** — you need a filesystem and **cargo build**. Playground snippets show the syntax in a single file; multi-file layout is the production pattern.

9.2 Scope — a brief tour

Module tree, **pub**, and Cargo layout — not publishing or semver policy.

This chapter covers	Deferred
Module tree, pub , use, workspaces	cargo publish , semver policy
<code>#[cfg]</code> and Cargo <code>[features]</code>	Full cross-compilation matrix
Unit and integration tests	Fuzzing, property tests
<code>///</code> docs and cargo doc	Custom rustdoc themes

9.3 Crate, package, and workspace

Term	Meaning
Crate	One library (rlib) or binary (bin) the compiler builds from a root file
Package	One Cargo.toml + source that produces one or more crates
Workspace	Several packages in one repo sharing one Cargo.lock

A default binary package looks like:

```
my_app/
  Cargo.toml
  src/
    main.rs      # crate root for the binary
```

A library + binary in one package:

```
my_lib/
  Cargo.toml
  src/
    lib.rs      # library crate root --- `pub`
                API lives here
    main.rs     # binary uses the library via
                `use my_lib:....`
    config.rs   # submodule file
```

Run `cargo build` from the package directory. The **crate name** in `Cargo.toml` is the root for `use crate_name:....` in dependent packages.

9.4 Declaring modules

Two ways to add a module:

Inline — code in the same file:

```
// Playground
mod math {
  pub fn double(x: i32) -> i32 {
    x * 2
  }
}
```

```

    }
}

fn main() {
    println!("{}", math::double(21));
}

```

File-backed — in `lib.rs` or `main.rs`:

```

// src/lib.rs or main.rs
mod config;
// loads src/config.rs or src/config/mod.rs

```

```

// src/config.rs
pub fn app_name() -> &'static str {
    "gateway"
}

```

Form	Use when
<code>mod foo { ... }</code>	tiny helpers, tests, one-off nesting
<code>mod foo; + foo.rs</code>	normal production split

9.5 Paths: `self`, `super`, `crate`

Paths mirror the module tree:

Prefix	Points to
<code>crate::</code>	root of this crate
<code>super::</code>	parent module
<code>self::</code>	current module (often omitted)
<code>foo::bar</code>	child module <code>foo</code> , item <code>bar</code>

```

// Playground
mod outer {
    pub fn tag() -> &'static str {
        "outer"
    }
}

```



```

pub mod inner {
    pub fn full() -> String {
        format!("{}", inner, super::tag())
    }
}

fn main() {
    println!("{}", outer::inner::full());
}

```

Java: `com.example.app.Config`. **Python:** `package.submodule`. **Rust:** explicit tree from crate root; no classpath scanning.

9.6 Visibility: the `pub` ladder

By default, items are **private** to their module. Mark what callers may use:

Visibility	Visible from
(none)	same module + child modules
<code>pub</code>	anywhere that can reach the path
<code>pub(crate)</code>	anywhere in this crate
<code>pub(super)</code>	parent module
<code>pub(in path::to::module)</code>	specific ancestor branch

```

// Playground
mod api {
    pub fn public_entry() -> i32 {
        helper()
    }

    fn helper() -> i32 {
        42
    }
}

fn main() {
    println!("{}", api::public_entry());
}

```

```
// api::helper(); // ERROR: private
}
```

Library design: keep fields private; expose `pub fn` constructors and accessors (Chapter 3). Hide internals so you can refactor without breaking callers.

9.7 use and re-exports

`use` brings names into scope so you do not repeat long paths:

```
// Playground
mod shapes {
    pub mod circle {
        pub fn area(r: f64) -> f64 {
            std::f64::consts::PI * r * r
        }
    }
}

use shapes::circle;

fn main() {
    println!("{}", circle::area(2.0));
}
```

Re-export a dependency's type under your crate's API:

```
// lib.rs pattern (Cargo only --- conceptual)
// pub use serde::Deserialize;
```

Callers write `use my_crate::Deserialize` instead of depending on path details you might change later.

9.8 Binary vs library crate

Crate	Root file	Typical role
Binary	<code>src/main.rs</code>	CLI entry, <code>fn main</code> , thin wiring
Library	<code>src/lib.rs</code>	reusable logic, most <code>pub API</code> , unit tests

Cargo only — binary calling library in the same package:

```
// src/lib.rs
pub fn greet(name: &str) -> String {
    format!("Hello, {name}")
}

// src/main.rs
fn main() {
    println!("{}", my_lib::greet("Rust"));
}
```

Replace `my_lib` with the `[package]` name from `Cargo.toml`. Keep `main` small: parse args, call library functions, map errors to exit codes (Chapter 8).

9.9 Tests as modules

Unit tests live in a nested module guarded by `#[cfg(test)]`:

```
// Playground
pub fn add(a: i32, b: i32) -> i32 {
    a + b
}

#[cfg(test)]
mod tests {
    use super::*;

    #[test]
    fn adds() {
        assert_eq!(add(2, 2), 4);
    }
}

fn main() {
    println!("{}", add(1, 1));
}
```

`cargo test` compiles and runs tests; normal `cargo build` skips them. Integration tests go in `tests/*.rs` at the package root (**Cargo only**).

9.10 Conditional compilation and Cargo features

Gate optional serial support behind a feature flag.

Cargo.toml:

```
# Cargo only
[package]
name = "gateway"
version = "0.1.0"
edition = "2024"

[features]
default = []
serial = []

[dependencies]
serialport = { version = "4", optional = true }
```

src/lib.rs:

```
// Cargo only --- conceptual
#[cfg(feature = "serial")]
pub mod serial_io {
    pub fn list_ports() -> Vec<String> {
        serialport::available_ports()
            .unwrap_or_default()
            .into_iter()
            .map(|p| p.port_name)
            .collect()
    }
}

#[cfg(not(feature = "serial"))]
pub mod serial_io {
    pub fn list_ports() -> Vec<String> {
        vec![]
    }
}
```

Build with `cargo build --features serial` to compile the real backend. Without the flag, the stub stays in the binary. The API path stays identical.

9.10.1 Common `cfg` attributes

```
// Playground
#[cfg(test)]
fn only_in_tests() -> i32 {
    42
}

fn main() {
    #[cfg(debug_assertions)]
    println!("debug build");

    println!("running");
}
```

Attribute	Effect
<code>#[cfg(test)]</code>	compiled only for cargo test
<code>#[cfg(debug_assertions)]</code>	debug builds only
<code>#[cfg(feature = "serial")]</code>	enabled when feature is on
<code>cfg!(feature = "serial")</code>	runtime bool — code still compiled

`#[cfg(...)]` removes code at compile time. `if cfg!(...) { ... }` keeps both branches in the binary.

9.10.2 Feature edge case — optional dependency

```
# Cargo only
tokio = { version = "1", optional = true,
    features = ["rt", "macros"] }

[features]
async = ["dep:tokio"]
```

Enabling `async` pulls in `tokio` with the listed features — one switch for callers.

9.11 Integration tests

Integration tests live in `tests/` at the package root. Each file is a separate crate that links your library:

```
gateway/
  Cargo.toml
  src/lib.rs
  tests/
    parse_config.rs
```

```
// Cargo only --- tests/parse_config.rs
// use gateway::parse_port;

// #[test]
// fn reads_port_from_line() {
//     assert_eq!(parse_port("502").unwrap(),
//         502);
// }
```

Run with `cargo test`. Unlike `#[cfg(test)] mod tests`, integration tests only see the **public** API — they simulate external callers.

9.11.1 Re-export edge case — prelude module

```
// Playground
mod api {
    pub mod inner {
        pub fn connect() -> i32 {
            502
        }
    }
    pub use inner::connect;
}

fn main() {
    println!("{}", api::connect());
}
```

`pub use` re-exports `connect` at the `api` level so callers skip `inner`.

9.12 Workspaces (brief)

When one repo holds multiple packages:

```
workspace/
Cargo.toml      # [workspace] members =
    ["core", "cli"]
core/
  Cargo.toml
  src/lib.rs
cli/
  Cargo.toml
  src/main.rs   # depends on core via path
                dependency
```

Each member has its own crate graph. Shared versions live in the workspace `Cargo.lock`. Start with one package. Split when a library clearly deserves reuse.

9.13 Orphan rule (crate boundary)

You may implement **your trait** for **your type**, or either side from **your crate**. You cannot add `impl Display for Vec<u8>` in your app — both trait and type are defined elsewhere. Fix with a **newtype** wrapper (Chapter 7).

9.14 Documentation comments

Public API items use `///` doc comments. Run `cargo doc --open` to browse generated HTML locally:

```
// Playground
/// Parses a port number from decimal text.
///
/// # Errors
/// Returns ParseIntError when the string is
    not a valid u16.
pub fn parse_port(s: &str) -> Result<u16,
    std::num::ParseIntError> {
    s.parse()
```

```
}

fn main() {
    println!("{}", parse_port("502").unwrap());
}
```

Use `# Examples`, `# Errors`, and `# Panics` sections for items callers depend on. Hide internal helpers with `#[doc(hidden)]` when re-exporting.

9.15 When the compiler says no

Common errors in this chapter:

Error (typical)	Cause	Fix
file not found for module <code>foo</code>	missing <code>foo.rs</code> or <code>foo/mod.rs</code>	create file or use inline <code>mod</code>
private item	forgot <code>pub</code> on <code>fn/struct</code>	add <code>pub</code> or use <code>pub(crate)</code>
unresolved import	wrong path or typo	use <code>crate::...</code> from crate root
found duplicate module	<code>mod foo</code> ; twice	one declaration per module

9.16 Idiom spotlight

Thin `main`, fat `library`. Put logic in `lib.rs` modules; test there with `mod tests`. The binary is just the process entry point.

9.17 Go deeper

- [The Rust Book — Modules](#)
- [Cargo workspaces](#)

9.18 See also

- Preface — `rustup` and Cargo setup
- Chapter 3: Functions and methods — `pub` on methods
- Chapter 8: Errors and testing — `mod tests`, `cargo test`
- Chapter 10: Smart pointers — `Arc` across modules

- Chapter 7: Traits — orphan rule

9.18.1 Afterparty

9.18.1.1 Layout and paths

1. **File tree** — “Design module tree for a CLI that reads config and runs commands. Directories + mod lines only, no bodies.”
2. **Nested mod.rs** — “Draw ui/theme/themes/ and ui/theme/palettes/ as a directory tree. When is foo/mod.rs better than foo.rs?”
3. **Path quiz** — “From `crate::service::worker::run`, how do I reach `crate::config::load`? Show use and fully qualified call.”
4. **use crate::** — “In `app/events.rs`, import `AppState` from `app/state.rs` and `PanelLocation` from `core/location.rs` — write both use lines.”
5. **super:: siblings** — “`dsp/flowgraph.rs` needs `silence::silenced` from a sibling file under `dsp/`. Show the use line (not a path from crate root).”
6. **lib vs bin** — “What belongs in `main.rs` vs `lib.rs` for a tool with 500 lines of logic?”
7. **Binary-only** — “No `lib.rs`: list top-level mod lines in `main.rs` for a TUI with `core`, `ui`, `app`, and `util` modules. Where does `pub(crate) fit`?”

9.18.1.2 Visibility

1. **pub audit** — “List items that should be `pub` vs `private` in a library crate exposing `Client::connect`.”
2. **pub(super)** — “`dsp/mod.rs` must call `flowgraph::run`, but `sdr.rs` must not reach it. Show mod `flowgraph`; and `pub(super) fn run` — who can call `run`?”
3. **Facade module** — “`ui/mod.rs` has mod `renderer`; and `pub use renderer::Renderer`; . Why not `pub mod renderer`?”
4. **Barrel re-export** — “`config/mod.rs` exposes `Station` and `load_stations` without callers writing `config::stations::`. Sketch `pub mod` + `pub use` lines.”
5. **pub(crate)** — “Name three helpers that should be `pub(crate)` in a binary crate (shared across `app/`, `ui/`, `main`) but must not be `pub`.”
6. **pub(crate) mod** — “`theme/mod.rs` keeps `palettes` visible inside the crate only. Show `pub(crate) mod palettes`; vs `pub mod palettes` — what breaks for external callers?”
7. **pub mod vs mod** — “In `browser/mod.rs`, `viewer_image` is `private`

but viewer is `pub mod`. What can code outside browser/ import?”

8. **Re-export dep** — “Sketch `pub use` so users see `my_crate::Error` but you wrap `thiserror` internally.”

9.18.1.3 Crates and workspaces

1. **Workspace split** — “Two crates: `core` library + `cli` binary. Write `Cargo.toml` dependency path only.”
2. **Integration test** — “Where does `tests/smoke.rs` live and how does it use the library?”
3. **Orphan fix** — “I want `Display` on `Vec` — show `newtype` wrapper module layout.”
4. **Lib name split** — “`Tauri` package `sdr_fm` uses `[lib] name = \"sdr_fm_lib\"` and `main` calls `sdr_fm_lib::run()`. Why not name the lib `sdr_fm`?”

9.18.1.4 Practice

1. **Domain layers** — “Split a file manager TUI into `core/` (no `ratatui`), `ui/` (drawing), `app/` (state + events). What must never live in `core/`?”
2. **Split monolith** — “Given one `main.rs` with `config` + `parser` + `runner`, name three modules and what each owns.”
3. **cfg test** — “Explain why `mod tests` uses `#[cfg(test)]` and use `super::*`.”
4. **Leaf tests** — “`dsp/mod.rs` and `core/settings.rs` each have `#[cfg(test)] mod tests`. Why co-locate tests in submodule files instead of only at `lib.rs`?”
5. **Capstone** — “Generate `src/` tree for `sensor_core` library + `sensor_cli` binary in one workspace; I implement.”

9.18.1.5 Features and cfg

1. **Feature flag** — “Add `serial` feature gating `mod serial_io` — write `Cargo.toml` `[features]` and one `#[cfg]` line.”
2. **cfg vs cfg!** — “Same debug log twice: `#[cfg(debug_assertions)]` block vs `if cfg!(debug_assertions)` — what stays in release binary?”
3. **Optional dep** — “Wire optional `tokio` behind feature `async` — show `[features]` and `dep:tokio` line.”
4. **Platform module** — “Sketch `#[cfg(target_os = \"linux\")] pub mod linux_home_trash`; in `core/mod.rs` — what happens on macOS

builds?”

5. **Platform stub fn** — “Same fn `process_is_root()` -> bool on Unix (real check) and Windows (always false). Show the `#[cfg(unix)] / #[cfg(not(unix))]` pair.”
6. **Empty stub fn** — “`disable_spellcheck()` runs real code on macOS and pub fn `disable_spellcheck() {}` elsewhere. Why compile the module on all targets instead of gating the whole file?”
7. **Target deps** — “Add `libc` only on Unix in `Cargo.toml`: show `[target.'cfg(unix)'.dependencies]` block.”

9.18.1.6 Integration and docs

1. **Integration layout** — “Draw tree for `tests/load_config.rs` calling public `load()` — what is invisible to the test?”
2. **pub use prelude** — “Users should call `my_crate::connect` not `my_crate::internal::connect` — show re-export.”
3. **Module docs** — “Top of `core/mod.rs` describes what the module owns. Show a `//!` inner doc comment (two lines) vs `///` on a single pub fn.”
4. **doc hidden** — “When to mark helper `#[doc(hidden)]` on a public re-export surface?”
5. **Capstone crate** — “Design gateway crate: `serial` feature, integration test, `///` on public parse fn — tree + TOML only.”

Chapter 10

Smart Pointers and Interior Mutability

10.1 Hook

Some languages treat almost everything as heap references (**Python** names, typical **Java** objects). Rust defaults to **stack + unique ownership**. Reach for **smart pointers** only when you need heap indirection, shared ownership, or mutation through a shared borrow.

Module layout for splitting code across files is Chapter 9. This chapter is about **memory patterns**.

10.2 Scope — a brief tour

Std smart-pointer patterns — not custom allocators or every `NonNull` API.

This chapter covers	Deferred
<code>Box</code> , <code>Rc</code> , <code>Arc</code>	Custom allocators, <code>NonNull</code>
<code>Cell</code> , <code>RefCell</code> , <code>Rc<RefCell<T>></code>	<code>Mutex/RwLock</code> depth -> Chapter 14
<code>Deref</code> coercion, <code>Drop</code>	<code>Pin/Unpin</code> -> Chapter 16, Chapter 18
<code>Weak</code> (break cycles)	Full graph GC designs

10.3 Box<T> — heap, single owner

Box allocates on the heap but keeps **one owner** — a unique pointer with known size for the type system:

```
// Playground
fn main() {
    let b = Box::new(42);
    println!("{}", b);
}
```

Use Box for:

- **Trait objects** — Box<dyn Trait> when types differ at runtime (Chapter 7)
- **Recursive types** — enum variants that contain themselves (e.g. AST nodes)
- **Large blobs** — move heavy data without copying the stack frame

10.3.1 Recursive enum (why Box is required)

A type cannot contain itself with infinite size. Indirection fixes that:

```
// Playground
#[derive(Debug)]
enum List {
    Cons(i32, Box<List>),
    Nil,
}

use List::{Cons, Nil};

fn main() {
    let list = Cons(1, Box::new(Cons(2,
        Box::new(Nil))));
    println!("{:?}", list);
}
```

What happened: Cons stores i32 plus a **Box<List>** (pointer-sized). The chain terminates at Nil. Without Box, the compiler rejects infinite size.

10.3.2 Box edge cases

Situation	What happens
Move Box	ownership moves; old binding unusable
*box_ref	dereference moves inner value out if T is not Copy
Drop last Box owner	heap freed immediately

Wrong — use after move:

```
// Playground --- does not compile
fn main() {
    let a = Box::new(1);
    let b = a;
    println!("{}", a); // ERROR: value moved
}
```

Move inner value out of Box:

```
// Playground
fn main() {
    let b = Box::new(String::from("hi"));
    let s: String = *b;
    // moves String out; b is empty/unusable
    println!("{}", s);
}
```

10.4 Rc<T> and Arc<T> — shared ownership

When many parts of the program need to **own** the same data (read-mostly), use reference counting:

Type	Thread-safe	Use
Rc<T>	no	single-thread graphs, shared config
Arc<T>	yes	Chapter 14 shared state

```
// Playground
use std::rc::Rc;

fn main() {
```

```
let a = Rc::new(String::from("shared"));
let b = Rc::clone(&a);
println!("{}", Rc::strong_count(&a));
println!("{}", a, b);
}
```

What happened: `Rc::clone(&a)` bumps the strong count — it does **not** deep-clone the `String`. Both `a` and `b` point at the same heap buffer.

10.4.1 clone on the smart pointer vs `.clone()` on the inner value

Call	Effect
<code>Rc::clone(&a)</code>	new handle, same allocation, count +1
<code>(*a).clone()</code>	deep copy of <code>T</code> , new allocation, count unchanged

Wrong habit from Java/Python: calling `.clone()` on the `String` inside when you only needed another handle:

```
// Playground
use std::rc::Rc;

fn main() {
    let a = Rc::new(String::from("config"));
    let b = Rc::clone(&a);
    // cheap --- shared handle
    let c = (*a).clone();
    // expensive --- duplicate string on heap
    println!("{}", a, b, c);
}
```

10.4.2 Cycles and Weak

`Rc` tracks a **strong count**: while it is > 0 , the heap allocation stays alive. Every `Rc::clone` increments it; every drop decrements it. When it hits zero, the value is freed.

The cycle problem: if `A` holds `Rc<B>` and `B` holds `Rc<A>`, each keeps the other's count ≥ 1 — neither drops, memory leaks forever.

Weak<T> is a **non-owning** handle. It does **not** bump the strong count, so it cannot keep the allocation alive by itself. Think of it as “I know where this value is, but I’m not an owner.”

Handle	Owns the data?	Keeps allocation alive?
Rc<T> (strong)	yes — shared ownership	yes — while any strong ref exists
Weak<T>	no — observer only	no — target may already be gone

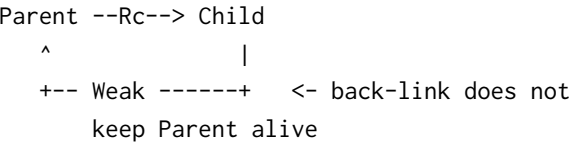
Two operations:

API	What it does
Rc::downgrade(&strong)	create a Weak from a live Rc — does not increase strong count
weak.upgrade()	try to get Option<Rc<T>> — Some if the value still exists, None if all strong refs were dropped

```
// Playground --- conceptual pattern (no cycle
    formed)
use std::rc::{Rc, Weak};

fn main() {
    let strong = Rc::new(42);
    let weak: Weak<i32> =
        Rc::downgrade(&strong);
    println!("upgrade = {:?}", weak.upgrade());
    // Some(42) while strong lives
    drop(strong);
    // strong count -> 0, value freed
    println!("after drop = {:?}",
        weak.upgrade()); // None --- target gone
}
```

Breaking a cycle: keep the “main” direction as Rc (parent -> child) and the back-link as Weak (child -> parent). When the parent is dropped, its strong refs go away; the child’s Weak upgrade returns None instead of pinning dead memory.



Typical tree pattern: parent: Rc<Node>, children: Vec<Rc<Node>>, parent_link: Weak<Node> on each child. The child can ask “is my parent still alive?” with parent_link.upgrade() without preventing the parent from being dropped.

10.4.3 Rc / Arc edge cases

Error / symptom	Cause	Fix
Rc in thread::spawn	not Send	Arc
Memory never freed	Rc cycle	Weak, redesign graph
strong_count stays > 0	leaked clone handle	audit who holds Rc

Arc uses atomic ref count — more CPU on clone/drop than Rc, but required for thread-safe sharing.

10.5 Cell and RefCell — interior mutability

Sometimes you need mutation **through a shared reference** — counters in single-thread code, or fields behind Rc.

Cell and **RefCell** enforce borrow rules **at runtime** instead of compile time:

Type	Access	Panics when
Cell<T>	.get() / .set() for Copy types	—
RefCell<T>	.borrow() / .borrow_mut()	double borrow_mut, borrow while mut borrow alive

10.5.1 Cell for small Copy values

```
// Playground
use std::cell::Cell;
```

```
fn main() {
    let n = Cell::new(0);
    n.set(n.get() + 1);
    println!("{}", n.get());
}
```

Cell has no `.borrow()` — values are copied in and out. Non-Copy types belong in `RefCell` (or redesign ownership).

10.5.2 RefCell basics

```
// Playground
use std::cell::RefCell;

fn main() {
    let total = RefCell::new(0);
    *total.borrow_mut() += 1;
    *total.borrow_mut() += 2;
    println!("{}", total.borrow());
}
```

10.5.3 Rc<RefCell<T>> — shared mutable graph on one thread

```
// Playground
use std::cell::RefCell;
use std::rc::Rc;

fn main() {
    let counter = Rc::new(RefCell::new(0));
    let c2 = Rc::clone(&counter);
    *counter.borrow_mut() += 1;
    *c2.borrow_mut() += 10;
    println!("{}", counter.borrow());
}
```

Do **not** share `RefCell` across threads — it is not Sync. For threads, use `Arc<Mutex<T>>` or channels (Chapter 14).

10.5.4 RefCell edge cases

Wrong — hold an immutable borrow, then borrow mutably:

```
// Playground --- panics at runtime (double
    borrow)
use std::cell::RefCell;

fn main() {
    let cell = RefCell::new(vec![1, 2, 3]);
    let r = cell.borrow();
    // immutable borrow active
    cell.borrow_mut().push(4);
    // panic: already borrowed
    drop(r);
}
```

Fix: shrink the immutable borrow scope with a nested block, or restructure so borrows do not overlap.

Wrong — re-enter `borrow_mut` while guard lives:

```
// Playground --- panics at runtime
use std::cell::RefCell;

fn bump_twice(cell: &RefCell<i32>) {
    let mut g1 = cell.borrow_mut();
    let mut g2 = cell.borrow_mut();
    // panic --- g1 still alive
    *g1 += 1;
    *g2 += 1;
}

fn main() {
    let cell = RefCell::new(0);
    bump_twice(&cell);
    // panics: `RefCell` already mutably
    // borrowed
}
```

Fix: finish one mutation, drop the guard, then borrow again — or use a single `borrow_mut` block.

Compile-time vs runtime: the borrow checker catches overlapping `&mut` at compile time. `RefCell` moves that check to **runtime** and **panics** on violation. Same logic, different moment you find the bug.

Pattern	Compile-time borrow	RefCell
Overlapping <code>&mut</code>	compile error	runtime panic
Cost	zero	small bookkeeping
Thread-safe	N/A for shared <code>RefCell</code>	no — use <code>Mutex</code>

10.6 Deref and deref coercion

Smart pointers implement **Deref** so you can call methods on the inner value:

```
// Playground
fn main() {
    let s = Box::new(String::from("hi"));
    println!("len = {}", s.len());
    // Deref: Box<String> -> &String -> &str
    // for .len()
}
```

Deref coercion converts references at call sites:

From	Often coerces to
<code>&String</code>	<code>&str</code>
<code>&Box<T></code>	<code>&T</code>
<code>&Rc<T></code>	<code>&T</code>

That is why `fn f(s: &str)` accepts `&String` and `Box<String>` after appropriate refs.

10.6.1 Deref edge cases

- Coercion applies to **references**, not owned values moving into `fn take(s: String)`.
- Chains stop at the target type the function expects — no infinite deref ladder in safe code.

- **Deref to str for custom newtypes** is a common pattern for string-like wrappers (Chapter 13).

10.7 Drop — cleanup on scope end

When an owner goes out of scope, Rust runs **drop** automatically (Chapter 1).

Type	When heap / cleanup runs
Box<T>	when Box dropped
Rc / Arc	when strong count hits 0
Custom Drop	before memory freed, on last owner

Custom cleanup:

```
// Playground
struct Logger;

impl Drop for Logger {
    fn drop(&mut self) {
        println!("Logger shut down");
    }
}

fn main() {
    let _log = Logger;
    println!("working...");
} // prints "Logger shut down" here
```

Never panic in Drop — a panicking destructor during unwind aborts the process (Chapter 8).

10.7.1 Drop order edge cases

Locals drop in **reverse declaration order** (last declared, first dropped):

```
// Playground
struct Tag(&'static str);

impl Drop for Tag {
```

```

fn drop(&mut self) {
    println!("drop {}", self.0);
}

fn main() {
    let _a = Tag("a");
    let _b = Tag("b");
    println!("running");
}
// prints: running, then drop b, then drop a

```

Rc drop: inner T drops only when the **last** Rc/Arc handle goes away — not when one clone drops.

```

// Playground
use std::rc::Rc;

struct Loud(i32);
impl Drop for Loud {
    fn drop(&mut self) {
        println!("drop Loud({})", self.0);
    }
}

fn main() {
    let a = Rc::new(Loud(1));
    let b = Rc::clone(&a);
    drop(a);
    println!("b still alive");
} // drop Loud(1) here when b goes out of scope

```

10.8 Choosing a pointer (quick table)

Need	Type
heap, one owner	Box<T>
shared read, one thread	Rc<T>
shared read, many threads	Arc<T>

Need	Type
mutate through shared ref, one thread	RefCell<T> / Rc<RefCell<T>>
mutate across threads	Arc<Mutex<T>> or channels
break Rc cycles	Weak<T>

10.9 When the compiler says no

Common errors in this chapter:

Error (typical)	Cause	Fix
Rc cannot be sent between threads	not Send	Arc
RefCell cannot be shared between threads	not Sync	Arc<Mutex<T>>
already borrowed (runtime)	overlapping RefCell borrows	shrink scope, restructure
recursive type has infinite size	direct self in enum	Box, Rc, or Arc
value moved out of Box	*b moved T	clone inner or keep in Box
T: Copy not satisfied for Cell	non-Copy in Cell	RefCell or owned mutation

10.10 Idiom spotlight

Reach for Box/Arc when ownership is genuinely shared or recursive — not by default. Most data should stay owned or borrowed.

Rc::clone is not .clone() on the data. Count handles; deep-copy only when semantics require a duplicate.

RefCell trades compile-time errors for runtime panics — use for single-thread patterns; prefer compile-time borrows when you can.

10.11 Go deeper

- [Smart pointers — Rust Book](#)
- [Interior mutability](#)

- [Reference cycles with Weak](#)

10.12 See also

- Chapter 9: Modules — crate layout
- Chapter 12: Closures — Fn traits with captured handles
- Chapter 14: Multithreading — Arc, Mutex, Send/Sync
- Chapter 7: Trait objects — Box<dyn Trait>
- Chapter 1: Ownership — drop and move

10.12.1 Afterparty

10.12.1.1 Box and heap ownership

1. **Box why** — “When is Box<T> better than Vec<T> on the stack? Two cases.”
2. **Recursive list** — “Draw memory for Cons(1, Box::new(Cons(2, Nil))) — stack vs heap boxes.”
3. **Move out of Box** — “What happens after let s = *box_string? When is that idiomatic vs keeping the Box?”
4. **Trait object box** — “Three plugin types implement Plugin — sketch Vec<Box<dyn Plugin>> factory; why not Vec<Plugin>?”

10.12.1.2 Rc, Arc, and Weak

5. **Rc cycle** — “Explain why Rc cycles leak; contrast with Python reference cycles and Weak fix.”
6. **Handle vs deep clone** — “Audit snippet with Rc<String> and both Rc::clone and (*rc).clone() — label cost of each.”
7. **Arc vs Rc** — “Thread spawn with Rc — show error; fix with Arc; explain atomic count overhead in one sentence.”
8. **Weak upgrade** — “Parent/child with Rc parent and Weak child back-ref — I sketch types; you explain cycle break.”
9. **strong_count debug** — “strong_count stays at 2 after I thought I dropped all refs — list 5 places handles hide.”

10.12.1.3 RefCell and interior mutability

10. **RefCell trap** — “Show double borrow_mut panic; fix with scoped borrows.”

11. **Immutable then mut** — “Hold `let r = cell.borrow()` and call `borrow_mut` — explain panic; fix with nested block.”
12. **Cell vs RefCell** — “Counter `u32` vs `Vec` cache — I pick `Cell` or `RefCell` each; you correct.”
13. **Rc RefCell graph** — “Two nodes share `Rc<RefCell<Node>>` — one updates field, one reads — sketch borrow rules on one thread.”
14. **Compile vs runtime** — “Same overlapping-mut pattern: show compile error with `&mut` and runtime panic with `RefCell` side by side.”

10.12.1.4 Deref, Drop, and threads

15. **Deref coercion** — “Why does `fn takes_str(s: &str)` accept `&String`, `&Box<String>`, and `&Rc<String>`? Trace steps.”
16. **Drop order** — “Three `Drop` structs in one function — I predict print order; you confirm reverse declaration rule.”
17. **Rc last handle** — “Two `Rc` clones dropped at different times — when does inner `Drop` run? Step through with `println` in `Drop`.”
18. **Drop panic** — “Explain double-panic abort if `Drop` panics during unwind — link to Ch8.”
19. **Arc Mutex sketch** — “Diagram thread-safe cache with `Arc<Mutex<HashMap>>` — no full code.”
20. **Pick pointer** — “Five scenarios (AST node, thread cache, GUI callback graph, config string, plugin list) — I pick `Box/Rc/Arc/RefCell/Weak`; you grade.”

10.12.1.5 Capstone

21. **Java heap map** — “Map Java ‘everything is reference’ to Rust ownership — when `Arc`, when plain `&`, when neither.”
22. **Refactor to smart ptr** — “I paste struct with `Box`, `Rc`, or raw `Vec` tree — you suggest minimal smart pointer fix and justify.”
23. **Leak hunt** — “Sketch `Rc` cycle in observer pattern; refactor one edge to `Weak` and explain count after each drop.”

Chapter 11

Collections

11.1 Hook

Lists and maps are workhorses in most languages (**Python** lists/dicts, **Java** `ArrayList`/`HashMap`, ...). Rust's standard collections are **generic**, **ownership-aware**, and pair with iterator pipelines from Chapter 4. Closures in `.sort_by` and `.retain` appear in Chapter 12.

11.2 Scope — a brief tour

Daily `std` collections — not third-party crates or custom allocators.

This chapter covers	Deferred
<code>Vec</code> , <code>HashMap</code> , <code>HashSet</code> , <code>VecDeque</code> , <code>BTreeMap</code> / <code>BTreeSet</code>	Persistent/immutable collections
<code>entry</code> , iterators, <code>.windows</code>	Full algorithm analysis
Ownership with <code>collect</code> / <code>insert</code>	Database/query engines

11.3 Choosing a collection

Type	Keys / order	Typical use
<code>Vec<T></code>	indexed, contiguous	default sequence, stack-like growth

Type	Keys / order	Typical use
VecDeque<T>	indexed, double-ended	queue, BFS, push/pop both ends
HashMap<K, V>	unordered keys	fast lookup, counts, caches
HashSet<T>	unique keys, unordered	membership, dedup
BTreeMap<K, V>	sorted keys	ordered map, range queries
BTreeSet<T>	sorted unique	ordered set

All live in `std::collections` except `Vec` (always available).

Java / Python contrast

Task	Java	Python	Rust
Growable list	ArrayList	list	Vec
Map	HashMap	dict	HashMap
Set	HashSet	set	HashSet
Queue	ArrayDeque	deque	VecDeque
Sorted map	TreeMap	sortedcontainers (stdlib: none)	BTreeMap

11.4 Vec<T>

```
// Playground
fn main() {
    let mut v = vec![10, 20, 30];
    v.push(40);
    println!("{:?}", v);
    println!("second = {}", v[1]);
}
```

Indexing with `[i]` **panics** out of bounds. Use `.get(i)` for `Option`.

Walk with `.iter()` / `.iter_mut()` / `.into_iter()` — Chapter 4.

Capacity: `Vec` over-allocates for amortized push. Use `Vec::with_capacity(n)` when you know size upfront.

11.4.1 Safe indexing and mutation

API	On missing index	On success
<code>v[i]</code>	panic	T or &T
<code>v.get(i)</code>	None	Some(&T)
<code>v.get_mut(i)</code>	None	Some(&mut T)

```
// Playground
fn main() {
    let v = vec![10, 20];
    println!("{:?}", v.get(5));
    // None --- no panic
    // println!("{}", v[5]);
    // panic if uncommented
}
```

11.4.2 Vec methods worth knowing

```
// Playground
fn main() {
    let mut v = vec![3, 1, 4, 1, 5];
    v.sort();
    // in-place sort
    v.dedup();
    // remove consecutive duplicates (sort
    // first!)
    v.retain(|&x| x % 2 == 0);
    // keep evens only
    v.extend([9, 8]);
    // append from iterator
    println!("{:?}", v);
}
```

What happened: `dedup` only removes **adjacent** duplicates — sort first if you need global dedup. `retain` takes a closure (Chapter 12).

11.4.3 Vec edge cases

Wrong — use `Vec` as a queue with `remove(0)`:

```
// Playground --- works but O(n) per pop front
fn main() {
    let mut v = vec![1, 2, 3];
    while !v.is_empty() {
        let front = v.remove(0);
        // shifts entire buffer left each time
        println!("{}", front);
    }
}
```

Fix: use `VecDeque` for FIFO (see below).

Wrong — hold reference into `Vec`, then mutate:

```
// Playground --- does not compile
fn main() {
    let mut v = vec![1, 2, 3];
    let r = &v[0];
    v.push(4);
    // ERROR: cannot borrow `v` as mutable
    // because it is also borrowed as immutable
    println!("{}", r);
}
```

Why: push may reallocate. The old reference would dangle. Finish using `r` before mutating.

Symptom	Cause	Fix
panic on <code>[i]</code>	out of bounds	<code>.get(i)</code> or check <code>len()</code>
slow queue on <code>Vec</code>	<code>remove(0)</code> is $O(n)$	<code>VecDeque</code>
borrow error on push	ref into element still alive	shrink ref scope
dedup leaves dupes	not sorted	sort then dedup

11.5 `HashMap<K, V>`

```
// Playground
use std::collections::HashMap;

fn main() {
```

```

let mut counts = HashMap::new();
for word in ["a", "b", "a"] {
    *counts.entry(word).or_insert(0) += 1;
}
println!("{:?}", counts);
}

```

Keys need **Eq + Hash**. Owned String keys are common. Lookup accepts &str via **Borrow** — `map.get("alice")` works when keys are String.

11.5.1 The entry API

`.entry(key)` returns an **Entry**. Use it to insert, update, or skip in one lookup:

```

// Playground
use std::collections::HashMap;

fn main() {
    let mut scores = HashMap::new();
    scores.insert("alice", 10);

    let e = scores.entry("bob").or_insert(0);
    *e += 5;

    scores.entry("alice").and_modify(|v| *v +=
        1).or_insert(0);

    println!("{:?}", scores);
}

```

Method	Effect
<code>or_insert(v)</code>	insert if missing, return &mut V
<code>and_modify(f)</code>	run f on existing value only
<code>or_default()</code>	insert <code>Default::default()</code> if missing
<code>or_insert_with(f)</code>	lazy insert — call f only if missing

Lazy insert avoids work when the key already exists:

```
// Playground
use std::collections::HashMap;

fn main() {
    let mut cache = HashMap::new();
    cache.entry("key").or_insert_with(|| {
        println!("building expensive value");
        vec![1, 2, 3]
    });
    cache.entry("key").or_insert_with(|| {
        println!("should not run");
        vec![99]
    });
}
```

What happened: the closure runs **once** — on first insert only.

11.5.2 HashMap edge cases

insert overwrites — returns the old value if any:

```
// Playground
use std::collections::HashMap;

fn main() {
    let mut m = HashMap::new();
    m.insert("port", 502);
    let old = m.insert("port", 503);
    println!("old = {:?}, now = {:?}", old,
        m.get("port"));
}
```

Wrong — double lookup (get then insert):

```
// Playground --- compiles but wasteful /
    awkward for mutation
use std::collections::HashMap;

fn bump(m: &mut HashMap<String, i32>, key:
    &str) {
    if m.contains_key(key) {
```

```

        *m.get_mut(key).unwrap() += 1;
    } else {
        m.insert(key.to_string(), 1);
    }
}

```

Idiomatic: `*m.entry(key.to_string()).or_insert(0) += 1;`

Borrow while iterating: you cannot mutably insert while iterating the same map. Collect keys first, or use `entry` outside the loop.

Error / symptom	Cause	Fix
key type needs Hash	custom struct key	<code>#[derive(Hash, Eq, PartialEq)]</code>
cannot borrow map mutably	iterator or get ref still alive	drop borrows first
silent overwrite	insert on existing key	check return or use <code>entry</code>
f32 as key	floats not Hash in std	integer keys or <code>newtype</code>

11.6 HashSet<T>

Unique values with set algebra:

```

// Playground
use std::collections::HashSet;

fn main() {
    let a: HashSet<_> = [1, 2,
        3].into_iter().collect();
    let b: HashSet<_> = [3, 4,
        5].into_iter().collect();
    let union: HashSet<_> =
        a.union(&b).copied().collect();
    let inter: HashSet<_> =
        a.intersection(&b).copied().collect();
    println!("union {:?} inter {:?}", union,
        inter);
}

```


Operation	Method	Notes
membership	set.contains(&x)	O(1) average
dedup	collect:: <hashset<_>>()< td=""><td>order lost</td></hashset<_>>()<>	order lost
difference	a.difference(&b)	in a not in b

Dedup preserving order — `HashSet` loses order. For stable unique order, use a `Vec + HashSet` seen-set, or `sort + dedup`.

11.7 VecDeque<T>

Efficient push/pop at **front and back**:

```
// Playground
use std::collections::VecDeque;

fn main() {
    let mut q = VecDeque::new();
    q.push_back(1);
    q.push_back(2);
    q.push_front(0);
    println!("pop front = {:?}", q.pop_front());
    println!("rest = {:?}", q);
}
```

Operation	Vec	VecDeque
push_back	O(1) amortized	O(1)
pop_front	O(n) with remove(0)	O(1)
random index	O(1)	O(1)

Use as a **FIFO queue** or **BFS frontier**. Use `Vec` when you only grow at the end.

11.8 BTreeMap and BTreeSet

Sorted keys — useful for ordered iteration and range scans:

```
// Playground
use std::collections::BTreeMap;

fn main() {
    let mut map = BTreeMap::new();
    map.insert(30, "c");
    map.insert(10, "a");
    map.insert(20, "b");
    for (k, v) in &map {
        println!("{}", k, v);
    }
}
```

Keys need **Ord**. Iteration is always **sorted by key**.

11.8.1 Range queries

```
// Playground
use std::collections::BTreeMap;

fn main() {
    let mut map = BTreeMap::new();
    for k in [100, 150, 200, 250] {
        map.insert(k, format!("reg-{}", k));
    }
    for (k, v) in map.range(120..=220) {
        println!("{}", k, v);
    }
}
```

What happened: prints keys **150** and **200** — inclusive range `120..=220`.

Need	Pick
fastest single lookup	HashMap
sorted walk or range	BTreeMap
min/max key always handy	BTreeMap (first_key_value, last_key_value)

BTreeMap is $O(\log n)$ per op; HashMap is $O(1)$ average. Extra cost buys

order.

11.9 Iterators and collection methods

Collection methods delegate to iterators — see Chapter 4:

```
// Playground
fn main() {
    let nums = vec![1, 2, 3, 4];
    let sum: i32 = nums.iter().sum();
    let evens: Vec<_> =
        nums.iter().filter(|&&x| x % 2 ==
            0).collect();
    println!("{}", sum, evens);
}
```

11.9.1 .windows, .chunks, .enumerate

```
// Playground
fn main() {
    let samples = vec![1.0, 1.1, 2.5, 2.4];
    let rising: Vec<_> = samples
        .windows(2)
        .filter(|w| w[1] > w[0])
        .collect();
    println!("rising pairs {:?}", rising);

    let bytes = vec![0xDE, 0xAD, 0xBE, 0xEF];
    for chunk in bytes.chunks(2) {
        println!("{:02X?}", chunk);
    }
}
```

Adapter	Yields
.windows(n)	overlapping slices of length n
.chunks(n)	non-overlapping chunks
.enumerate()	(index, item)

11.9.2 `sort_by` and custom order

```
// Playground
fn main() {
    let mut items = vec![("b", 3), ("a", 1),
        ("c", 2)];
    items.sort_by(|a, b| a.1.cmp(&b.1));
    // sort by numeric field
    println!("{:?}", items);
}
```

Requires **Ord** on compared fields or a custom comparator closure (Chapter 12).

11.10 Collecting into collections

`collect()` builds many types when the element type is known:

```
// Playground
use std::collections::{HashMap, HashSet};

fn main() {
    let pairs = vec![("a", 1), ("b", 2)];
    let map: HashMap<_, _> =
        pairs.into_iter().collect();
    let uniq: HashSet<_> = vec![1, 1,
        2].into_iter().collect();
    println!("{:?} {:?}", map, uniq);
}
```

Duplicate keys in `collect` to `HashMap`: later pairs **overwrite** earlier ones — know your source data.

Wrong — ambiguous `collect`:

```
// Playground --- does not compile
fn main() {
    let nums = vec![1, 2, 3];
    let v = nums.iter().map(|&x| x *
        2).collect();
    // ERROR: type annotations needed
```

```
}

```

Fix: `let v: Vec<i32> = ... or .collect::<Vec<_>>().`

11.10.1 into_iter vs iter when building collections

Source walk	You get in pipeline	Source after
<code>.iter()</code>	<code>&T</code>	Vec still owned
<code>.into_iter()</code> / owned vec	<code>T</code>	Vec consumed

Building `HashMap<String, i32>` from owned strings requires **into_iter** or cloning keys.

11.11 When the compiler says no

Common errors in this chapter:

Error (typical)	Cause	Fix
T cannot be hashed	key missing Hash	derive or newtype
T: Ord not satisfied	BTreeMap key not ordered	derive Ord or use HashMap
cannot borrow as mutable	ref into Vec/HashMap alive	end borrow scope
type annotations needed	ambiguous collect()	turbofish or annotate
iterator invalidation	mutate map during for (k,v)	collect keys first
dedup ineffective	unsorted Vec	sort_unstable then dedup

11.12 Idiom spotlight

Iterator chains over index loops. `for i in 0..v.len()` plus `v[i]` is a smell when `.iter().enumerate()` or `.windows(2)` states intent.

.entry for HashMap updates — one lookup, not `contains_key` + `get_mut` + `insert`.

Pick the collection for the access pattern — queue -> `VecDeque`; sorted range -> `BTreeMap`; default list -> `Vec`.

11.13 Go deeper

- [Rust Book — Collections](#)
- [HashMap entry API](#)
- [Functional Rust — iterator topics](#)

11.14 See also

- Chapter 4: Iterators
- Chapter 12: Closures — `.sort_by`, `.retain`
- Chapter 13: Standard traits — `Borrow`, `Eq`, `Hash`
- Chapter 1: Borrowing
- Chapter 7: Generics

11.14.1 Afterparty

11.14.1.1 Choosing and comparing collections

1. **Pick collection** — “Five tasks (dedup, sorted range scan, FIFO queue, index by id, min-key lookup) — I pick `Vec`/`HashMap`/`BTree`/`VecDeque`/`HashSet` each.”
2. **Hash vs BTree** — “Same 10k insert + range scan workload — when `HashMap` wins vs `BTreeMap`; one sentence each.”
3. **Queue anti-pattern** — “Review `while !v.is_empty() { v.remove(0) }` — cost and fix with `VecDeque`.”

11.14.1.2 Vec drills

4. **Loop port** — “Rewrite C-style indexed loop as iterator chain; preserve behavior.”
5. **get vs index** — “Four access patterns — I pick `[i]` vs `.get(i)` vs `.get_mut` vs `if let Some`.”
6. **sort dedup** — “Dedup `[3,1,4,1,5]` wrong vs right — show sort + dedup pipeline.”
7. **retain vs filter** — “Remove evens in-place vs new `Vec` — compare `retain` and `filter().collect()`.”
8. **Borrow push trap** — “Explain `let r = &v[0]; v.push(1)` error; fix with `scope`.”

11.14.1.3 HashMap and HashSet

9. **entry drill** — “Word frequency from `Vec<&str>` using only `.entry` — no double lookup.”
10. **or_insert_with** — “Lazy cache: expensive `Vec` built once per key — sketch with `or_insert_with`.”
11. **HashMap merge** — “Two maps of scores — merge by max per key; iterator + entry style.”
12. **insert overwrite** — “Track old value on port remap 502 -> 503 using `insert return`.”
13. **Set ops** — “Tags on two records — union, intersection, difference with `HashSet`.”
14. **Stable dedup** — “Unique `String` lines preserving first-seen order — no `HashSet`-only `collect`.”

11.14.1.4 BTree, windows, collect

15. **BTree range** — “List keys in `BTreeMap<u32, _>` between 100 and 200 inclusive.”
16. **Windows** — “Detect rising edges in `Vec<f64>` with `.windows(2)`; extend to `.windows(3)` for slope.”
17. **chunks vs windows** — “Parse byte stream into 4-byte frames — `chunks(4)` vs `windows(4)` when?”
18. **collect types** — “Three `collect()` calls that need type hints — fix with `turbofish`.”
19. **Duplicate keys** — “`collect` to `HashMap` from duplicate-key pairs — predict final map; explain overwrite rule.”

11.14.1.5 Performance and capstone

20. **Performance myth** — “Do Rust iterators optimize to loops? When might they not?”
21. **Capacity hint** — “Read 1M lines into `Vec` — when `with_capacity` matters; rough sizing rule.”
22. **Capstone** — “Design in-memory store: register id `u16` -> last reading `f64`, need range scan by id — pick map type, list three API methods, no impl.”

Chapter 12

Closures and the Fn Traits

12.1 Hook

You already used closures in Chapter 4 — `|x| x * 2` inside `.map()`. A **closure** is an anonymous function that can **capture** variables from its scope. Rust classifies captures as **Fn**, **FnMut**, or **FnOnce**. That controls where you can store and call them.

12.2 Scope — a brief tour

Closures as callbacks — how they capture variables, which **Fn*** trait applies, and how to store them. Async closures and parallel thread pools wait for later chapters.

This chapter covers	Deferred
Syntax, capture modes, move, Fn / FnMut / FnOnce , common traps	Async closures -> Ch 16
<code>.map</code> , <code>.sort_by</code> , <code>.retain</code> ; impl Fn and <code>Box<dyn Fn()></code> storage	<code>rayon</code> -> Ch 14; <code>HRTB</code> , <code>Pin</code> -> Ch 16 / 18

12.3 Closure syntax

```
// Playground
fn main() {
```



```
let factor = 3;
let scale = |x| x * factor;
println!("{}", scale(10));
}
```

Form	Notes
<code>\ x\ x + 1</code>	parameter types often inferred
<code>\ x: i32\ -> i32 { x + 1 }</code>	explicit types when needed
<code>\ \ println!("tick")</code>	no parameters

Closures are **expressions** — you can pass them to functions, store them (with trait bounds), or return them (carefully).

12.4 Capture modes

The compiler decides how a closure uses outer variables:

Capture	Example	Closure trait (typical)
By shared borrow	<code>read factor</code>	<code>Fn</code>
By mutable borrow	<code>*count += 1</code>	<code>FnMut</code>
By move (ownership)	<code>move \ \ drop(s)</code>	<code>FnOnce</code>

```
// Playground
fn main() {
    let mut total = 0;
    let mut add = |n| total += n;
    // FnMut --- mutably borrows total
    add(5);
    add(2);
    println!("{}", total);
}
```

12.5 The move keyword

`move` forces the closure to **take ownership** of captured values. Use when the closure may **outlive** the current scope — threads, `spawn`, or storing in a struct:

```
// Playground
fn main() {
    let label = String::from("worker");
    let print = move || println!("{}", label);
    print();
    // println!("{}", label);
    // ERROR: label moved into closure
}
```

Chapter 14 uses `move` on closures passed to `thread::spawn`.

12.6 Fn, FnMut, FnOnce

These are **traits** implemented automatically for closures:

Trait	Call style	When
Fn()	call many times, shared &self	read-only capture
FnMut()	call many times, &mut self	mutates captured state
FnOnce()	consumes self on first call	moves captured values out

Subtrait chain: Fn implies FnMut implies FnOnce. A function expecting FnOnce accepts any of them; expecting Fn is the strictest.

```
// Playground
fn call_twice(f: impl Fn(i32) -> i32) {
    println!("{}", f(1));
    println!("{}", f(2));
}

fn main() {
    let double = |x| x * 2;
    call_twice(double);
}
```

12.7 Closures and iterator adapters

`.map`, `.filter`, and `.for_each` take closures. The iterator holds the closure and calls it per item. The closure must match the adapter's trait bound:

```
// Playground
fn main() {
    let nums = vec![1, 2, 3];
    let doubled: Vec<_> = nums.iter().map(|&x|
        x * 2).collect();
    println!("{:?}", doubled);
}
```

On `.iter()`, items are references — see [Chapter 4](#) — double reference.

Wrong — move closure with borrow after:

```
// Playground --- does not compile
fn main() {
    let name = String::from("log");
    let v = vec![1, 2];
    let _ = v.into_iter().map(move |_|
        name.len()); // name moved into closure
    // println!("{}", name); // ERROR
}
```

12.8 Closures vs function pointers

Named functions can coerce to `fn()` (function pointer) when they capture nothing:

```
// Playground
fn add_one(x: i32) -> i32 {
    x + 1
}

fn apply(f: fn(i32) -> i32, x: i32) -> i32 {
    f(x)
}

fn main() {
    println!("{}", apply(add_one, 5));
}
```

Capturing closures are a different type entirely:

What you write	Rust type	Fits in <code>fn(i32) -> i32</code> ?
<code>add_one</code> — plain function, no capture	<code>fn(i32) -> i32</code>	yes — function pointer
<code>\ x\ x + factor</code> — reads factor from scope	unique anonymous struct (implements <code>Fn</code>)	no — carries captured state

A `fn()` pointer is just an address — no room for captured variables. A closure that reads `factor` is a **small custom type** generated by the compiler, with a field holding `factor` (or a borrow of it). Each closure expression gets its **own** type, even when the syntax looks identical.

```
// Playground --- does not compile
fn apply_bad(f: fn(i32) -> i32, x: i32) -> i32 {
    f(x)
}

fn main() {
    let factor = 3;
    let scale = \|x\| x * factor;
    apply_bad(scale, 10);
    // ERROR: expected fn pointer, found closure
}
```

To accept a capturing closure, take `impl Fn` (or a generic `F: Fn`) — the compiler monomorphizes for that closure’s unique type:

```
// Playground
fn apply<F: Fn(i32) -> i32>(f: F, x: i32) ->
    i32 {
    f(x)
}

fn main() {
    let factor = 3;
    let scale = \|x\| x * factor;
    println!("{}", apply(scale, 10));
    // ok --- generic accepts any Fn
}
```

Use `fn()` for plain functions with no capture. Use `impl Fn / F: Fn` when

the callback might close over local data.

12.9 Returning closures

Returning `impl Fn()` works when the closure owns its data or only captures 'static references:

```
// Playground
fn make_adder(n: i32) -> impl Fn(i32) -> i32 {
    move |x| x + n
}

fn main() {
    let add5 = make_adder(5);
    println!("{}", add5(10));
}
```

Returning a closure that borrows local stack variables fails — lifetimes (Chapter 5) forbid dangling borrows.

12.10 Callback registry — storing closures

Filter log lines through pluggable rules. Each rule is a `Box<dyn Fn(&str) -> bool>`:

```
// Playground
fn main() {
    let mut rules: Vec<Box<dyn Fn(&str) ->
        bool>> = Vec::new();
    rules.push(Box::new(|line|
        line.contains("ERROR")));
    rules.push(Box::new(|line|
        line.starts_with("WARN")));

    let lines = ["INFO ok", "ERROR timeout",
        "WARN retry"];
    for line in lines {
        if rules.iter().all(|rule| rule(line)) {
            println!("blocked: {}", line);
        } else if rules.iter().any(|rule|
```

```

        rule(line)) {
            println!("matched: {}", line);
        }
    }
}

```

`Box<dyn Fn(...)>` homogenizes closures into one type for a `Vec`. Each closure may capture different data, but the **signature** must match. Prefer `impl Fn` in helpers that do not need storage. Boxing adds heap allocation and dynamic dispatch.

12.11 Sort and filter with closures

Chapter 11 methods take closures when the logic is local:

```

// Playground
#[derive(Debug)]
struct SensorReading {
    id: u32,
    value: f64,
    valid: bool,
}

fn main() {
    let mut readings = vec![
        SensorReading { id: 1, value: 22.1,
            valid: true },
        SensorReading { id: 2, value: -1.0,
            valid: false },
        SensorReading { id: 3, value: 21.8,
            valid: true },
    ];

    readings.retain(|r| r.valid);
    readings.sort_by(|a, b| {
        b.value
        .partial_cmp(&a.value)
        .unwrap_or(std::cmp::Ordering::Equal)
    })
}

```

```
});

println!("{:?}", readings);
}
```

`.retain` needs `FnMut` — it may mutate the vector while iterating. `.sort_by` compares pairs; use `sort_by_key` when you only need one field (`|r| r.id`).

12.12 Generic helpers — `impl Fn` vs boxing

Call a closure twice without boxing:

```
// Playground
fn apply_twice(f: impl Fn(i32) -> i32, x: i32)
    -> i32 {
    f(f(x))
}

fn main() {
    let scale = 3;
    let triple = |n| n * scale;
    println!("{}", apply_twice(triple, 2));
    // 18
}
```

`impl Fn` monomorphizes — the compiler generates a specialized copy per closure type. Use `Box<dyn Fn(> only when you need a homogeneous collection or trait object return type.`

12.13 Capture edge cases

for label in labels + move — this is OK:

Each loop iteration **owns** the next `String` from the `vec`. Moving it into one closure does not affect the next iteration — there is no reuse of the same binding.

```
// Playground
fn main() {
    let labels = vec![String::from("a"),
```

```

    String::from("b")]);
let mut fns: Vec<Box<dyn Fn()>> =
    Vec::new();
for label in labels {
    fns.push(Box::new(move ||
        println!("{}", label))); // ok ---
        label is fresh each iteration
    }
    for f in &fns {
        f();
    }
    // labels vec is consumed by the loop; fns
    // own the strings now
}

```

Wrong — iterate by reference, closure outlives the borrow:

```

// Playground --- does not compile
fn main() {
    let labels = vec![String::from("a"),
        String::from("b")];
    let mut fns: Vec<Box<dyn Fn()>> =
        Vec::new();
    for label in &labels {
        fns.push(Box::new(|| println!("{}",
            label)));
        // ERROR: closure may run after loop
        // --- `label` ref dies at end of
        // iteration
    }
}

```

Stored closures (`Box<dyn Fn()>`) can outlive the loop. A non-move closure **borrows** `label`; that reference ends when the iteration ends, but the closure still needs it later.

Fix — clone when iterating by reference:

```

// Playground
fn main() {
    let labels = vec![String::from("a"),

```



```

    String::from("b")]);
let mut fns: Vec<Box<dyn Fn()>> =
    Vec::new();
for label in &labels {
    let owned = label.clone();
    fns.push(Box::new(move ||
        println!("{}", owned)));
}
for f in &fns {
    f();
}
println!("labels still here: {:?}", labels);
// vec untouched
}

```

Wrong — use `label` after `move` in the same iteration:

```

// Playground --- does not compile
fn main() {
    let labels = vec![String::from("a")];
    let mut fns: Vec<Box<dyn Fn()>> =
        Vec::new();
    for label in labels {
        fns.push(Box::new(move ||
            println!("{}", label)));
        println!("{}", label);
        // ERROR: label moved into closure above
    }
}

```

RefCell + FnMut — shared counter:

```

// Playground
use std::cell::RefCell;

fn main() {
    let count = RefCell::new(0);
    let mut bump = |n| *count.borrow_mut() += n;
    bump(5);
    bump(2);
}

```

```
println!("{}", count.borrow());
}
```

Interior mutability (Chapter 10) lets a closure mutate state through a shared borrow.

12.14 Thread handoff — move on spawn

Closures passed to Chapter 14 `thread::spawn` must own their captures:

```
// Playground
use std::thread;

fn main() {
    let label = String::from("worker-1");
    let handle = thread::spawn(move || {
        println!("running {}", label);
    });
    handle.join().expect("thread panicked");
}
```

Without `move`, the closure would borrow `label`. The thread may outlive the stack frame. `move` transfers ownership into the closure.

12.15 Java / Python contrast

Aspect	Java	Python	Rust
Lambda	SAM interfaces	lambda, nested def	closure + Fn* traits
Capture	effectively final	full closure	borrow / move checked
Store in field	functional interface type	any callable	Box<dyn Fn()> or generic F: Fn()

12.16 When the compiler says no

Common errors in this chapter:

Error (typical)	Cause	Fix
expected Fn, found FnOnce	closure moves captured var	move + redesign, or clone before
cannot move out of ...	FnOnce consumed	call once or clone
borrowed after move	move closure took ownership	restructure scopes
FnMut in concurrent context	shared mutation	Mutex, channels (Chapter 14)
expected Fn, found FnOnce	closure consumes captured value on call	clone before capture, or use FnOnce bound
FnMut required	.retain / .sort_by mutates through closure	change bound to FnMut or use mut closure
type mismatch in Vec<Box<dyn Fn(>>>	closures with different signatures	same param/return types for every box

12.17 Idiom spotlight

Prefer `impl Fn` in helpers that only need to call a closure a few times. Use `move` when the closure crosses threads or outlives the stack frame.

12.18 Go deeper

- [The Rust Book — Closures](#)
- [Functional Rust — closure topics](#)

12.19 See also

- Chapter 4: Iterators — adapters that consume closures
- Chapter 11: Collections — `.retain`, `.sort_by`
- Chapter 7: Traits — trait bounds on generics
- Chapter 14: Multithreading — `move` + `spawn`

12.19.1 Afterparty

12.19.1.1 Capture and traits

1. **Fn quiz** — “Four closures: I label each `Fn` / `FnMut` / `FnOnce`; you correct and explain capture.”

2. **move drill** — “Thread spawn snippet missing `move` — show compile error and fix.”
3. **Iterator chain** — “`.filter` closure that uses `&config` — why `Fn` not `FnMut`?”
4. **RefCell bump** — “Closure mutates `RefCell<u32>` counter — which `Fn` trait and why?”

12.19.1.2 Types and storage

5. **fn vs closure** — “When can you pass `fn()` vs `impl Fn()` to the same helper?”
6. **Box dyn Fn** — “Store heterogeneous callbacks in a `Vec` — sketch trait object version.”
7. **Return closure** — “Write `make_multiplier(f: f64) -> impl Fn(f64) -> f64` and explain `move`.”
8. **Callback registry** — “Three log filters in `Vec<Box<dyn Fn(&str) -> bool>>` — all must match signature; I add a wrong one; you fix.”
9. **Loop move trap** — “Building `Vec<Box<dyn Fn()>>` in a `for` loop over `String` — show `move` error and clone fix.”

12.19.1.3 Collections and pipelines

10. **Double reference** — “Fix `.iter().filter(|x| ...)` type error on `Vec<String>` — show `|s|` vs `|&s|` patterns.”
11. **sort_by** — “Sort `Vec<(String, u32)>` by count descending with `sort_by` closure.”
12. **sort_by_key** — “Same sort with `sort_by_key` — when is key extraction cleaner?”
13. **retain valid** — “Drop invalid `SensorReading` rows in-place with `.retain` — `FnMut` bound.”
14. **for_each vs for** — “Same side-effect loop twice: `for` vs `.for_each(|...|)` — style tradeoffs.”

12.19.1.4 Errors and concurrency

15. **Fn bound too strict** — “Helper takes `impl Fn()` but caller passes closure that mutates — fix signature.”
16. **Thread move** — “spawn closure borrows `String` — show error without `move` and fix.”
17. **Send on Box Fn** — “When does `Box<dyn Fn() + Send>` matter for

thread pool callbacks?”

12.19.1.5 Capstone

18. **Capstone** — “Pipeline: read lines, filter non-empty, parse u16, sum — all with closures; I write; you review Fn bounds.”

Chapter 13

Standard Traits and Conversions

13.1 Hook

Rust's standard library is built on **traits** — shared behaviour you opt into with `impl` or `#[derive]`. This chapter covers formatting, comparison, conversion, and flexible-argument traits you see in almost every crate. Deep mechanics (`dyn`, object safety) stay in Chapter 7. Here the focus is **idioms**.

13.2 Scope — a brief tour

Formatting, comparison, and conversion traits — not custom `Formatter` depth or `Serde` derives.

This chapter covers	Deferred
Debug, Display, Default	Custom <code>Formatter</code> flags in depth
PartialEq, Eq, Hash, Ord	Float ordering edge cases (nominal)
From/Into, TryFrom/TryInto, FromStr	Full error-type design -> Chapter 8
AsRef, Borrow, Cow	<code>#[derive]</code> mechanics, <code>serde</code> /custom derives -> Chapter 17

13.3 Debug and Display

Debug — developer-oriented formatting via `{:?}`. Derive for most structs with `#[derive(Debug)]` — compile-time codegen, not a decorator (Chapter 17):

```
// Playground
#[derive(Debug)]
struct Point {
    x: i32,
    y: i32,
}

fn main() {
    let p = Point { x: 1, y: 2 };
    println!("{:?}", p);
    println!("{:#?}", p);
    // pretty-print with newlines
}
```

Display — user-facing formatting via `{}`. No derive — implement manually:

```
// Playground
use std::fmt;

struct Point {
    x: i32,
    y: i32,
}

impl fmt::Display for Point {
    fn fmt(&self, f: &mut fmt::Formatter<'_>)
        -> fmt::Result {
        write!(f, "({}, {})", self.x, self.y)
    }
}

fn main() {
    println!("{}", Point { x: 3, y: 4 });
}
```

Trait	Format	Typical use
Debug	{:?}", {:#?}	logs, tests, derive
Display	{}	CLI output, user messages

ToString: any type with `Display` gets `.to_string()` via a blanket impl — you rarely implement `ToString` yourself.

13.3.1 Debug vs Display — when which

Situation	Use
dbg!, tests, internal logs	Debug + derive
CLI status line, user error text	Display
Secret field in struct	manual Debug with redaction
Enum in production logs	derive Debug or compact custom

Redacted Debug — do not derive if logs would leak tokens:

```
// Playground
use std::fmt;

struct ApiConfig {
    pub name: String,
    api_key: String,
}

impl fmt::Debug for ApiConfig {
    fn fmt(&self, f: &mut fmt::Formatter<'_>)
        -> fmt::Result {
        f.debug_struct("ApiConfig")
            .field("name", &self.name)
            .field("api_key", &[REDACTED])
            .finish()
    }
}

fn main() {
    let c = ApiConfig {
```



```

        name: String::from("prod"),
        api_key: String::from("secret-123"),
    };
    println!("{:?}", c);
}

```

13.3.2 Formatting edge cases

Wrong — {} on type with only Debug:

```

// Playground --- does not compile
#[derive(Debug)]
struct OnlyDebug(i32);

fn main() {
    // println!("{}", OnlyDebug(1));
    // ERROR: `OnlyDebug` doesn't implement
    // Display
    println!("{:?}", OnlyDebug(1)); // ok
}

```

Newtype for orphan rule — implement Display on your wrapper, not on Vec<u8>:

```

// Playground
use std::fmt;

struct HexBytes(pub Vec<u8>);

impl fmt::Display for HexBytes {
    fn fmt(&self, f: &mut fmt::Formatter<'_>)
        -> fmt::Result {
        for b in &self.0 {
            write!(f, "{:02X}", b)?;
        }
        Ok(())
    }
}

fn main() {

```

```
println!("{}", HexBytes(vec![0xDE, 0xAD]));
}
```

13.3.3 Formatting flags — quick reference

Use these in `println!`, `format!`, and `write!` (Chapter 17 covers macro syntax):

```
// Playground
#[derive(Debug)]
struct Reading {
    id: u32,
    value: f64,
}

fn main() {
    let r = Reading { id: 1, value: 22.456 };
    println!("{:?}", r);
    // Reading { id: 1, value: 22.456 }
    println!("{:#?}", r);
    // pretty-printed multiline Debug
    println!("{:.2}", r.value);
    // 22.46 --- precision on floats
    println!("{:04}", r.id);
    // 0001 --- zero-padded width
}
```

Flag	Effect	Typical use
<code>{:?}</code>	Debug	logs, tests
<code>{:#?}</code>	pretty Debug	nested struct inspection
<code>{:.2}</code>	2 decimal places	sensor readings
<code>{:04}</code>	width 4, zero-pad	register ids
<code>{:X}</code> / <code>{:02X}</code>	hex	byte dumps (HexBytes above)

For CLI path helpers, accept `impl AsRef<Path>`. User-facing path formatting is in Chapter 19.

13.4 Default

Types with a sensible “empty” or “zero” value implement **Default**:

```
// Playground
#[derive(Default, Debug)]
struct Config {
    timeout_ms: u64,
    retries: u32,
}

fn main() {
    let c = Config {
        retries: 3,
        ..Default::default()
    };
    println!("{:?}", c);
}
```

Enums need `#[default]` on one variant (Rust 1.62+) to derive **Default**:

```
// Playground
#[derive(Default, Debug)]
enum Mode {
    #[default]
    Auto,
    Manual,
}

fn main() {
    println!("{:?}", Mode::default());
}
```

Use **Default** with struct update syntax (`..Default::default()`) and `HashMap::entry(...).or_default()`.

13.5 PartialEq, Eq, Hash, and Ord

Collection APIs from Chapter 11 require specific traits:

Trait	Enables
PartialEq	<code>==</code> , <code>!=</code>
Eq	full equivalence (no NaN-style holes)
Hash	HashMap / HashSet keys
Ord	BTreeMap / BTreeSet, sorting

```
// Playground
#[derive(Debug, Clone, PartialEq, Eq, Hash)]
struct UserId(u64);

fn main() {
    let a = UserId(1);
    let b = UserId(1);
    println!("{}", a == b, a);
}
```

Wrong — Eq with f64 field:

```
// Playground --- does not compile
// #[derive(Eq, PartialEq)]
// struct Sample { value: f64 }
// ERROR: binary operation `==` cannot be
//       applied to type `f64` in Eq context
```

Fix: use integer fixed-point, ordered-float crates, or derive only `PartialEq` (no `Eq`, no `Hash` on floats).

Goal	Minimum derives
test equality	PartialEq (+ Eq if valid)
hash map key	Eq + Hash
btree key / sort	Ord (implies Eq)

13.6 Clone and Copy (brief)

Trait	Meaning
Copy	bitwise duplicate on assign — no move (Chapter 1)
Clone	explicit <code>.clone()</code> — may allocate

Derive Clone on small config types; skip on unique handles (file descriptors, connection pools). `#[derive(Copy, Clone)]` together when all fields are Copy.

13.7 From and Into

From<T> defines conversion **into** your type. **Into<T>** is auto-implemented when you implement From:

```
// Playground
struct Port(u16);

impl From<u16> for Port {
    fn from(n: u16) -> Self {
        Port(n)
    }
}

fn main() {
    let p: Port = 8080.into();
    let q = Port::from(443);
    println!("{}", p.0, q.0);
}
```

Rule: implement **From** on your type — not **Into** on the source.

13.7.1 Chaining From without duplication

```
// Playground
struct Label(String);

impl From<String> for Label {
    fn from(s: String) -> Self {
        Label(s)
    }
}

impl From<&str> for Label {
    fn from(s: &str) -> Self {
```

```

        Label(s.to_string())
    }
}

fn main() {
    let a = Label::from("hello");
    let b = Label::from(String::from("world"));
    println!("{}", a.0, b.0);
}

```

Std library is full of From: `String::from("text")`, `Vec::from([1, 2])`, etc.

? **and From:** ? converts errors via From (Chapter 8). Implement `From<ParseIntError>` for `MyError` once. Then `parse()? works in fn -> Result<_, MyError>`.

13.8 TryFrom, TryInto, and FromStr

Fallible conversions use **TryFrom** / **TryInto** when validation can fail:

```

// Playground
use std::convert::TryFrom;

#[derive(Debug)]
struct Port(u16);

impl TryFrom<i32> for Port {
    type Error = &'static str;

    fn try_from(n: i32) -> Result<Self, Self::Error> {
        if (1..=65535).contains(&n) {
            Ok(Port(n as u16))
        } else {
            Err("port out of range")
        }
    }
}

```

```
fn main() {
    println!("{:?}", Port::try_from(8080));
    println!("{:?}", Port::try_from(0));
}
```

FromStr — parse from string slice (what `.parse()` calls):

```
// Playground
use std::str::FromStr;

#[derive(Debug)]
struct Port(u16);

impl FromStr for Port {
    type Err = std::num::ParseIntError;

    fn from_str(s: &str) -> Result<Self,
        Self::Err> {
        let n: u16 = s.parse()?;
        Ok(Port(n))
    }
}

fn main() {
    let p: Port = "8080".parse().unwrap();
    println!("{:?}", p);
}
```

Mechanism	Input	Error type
From / into()	known-valid value	none
TryFrom / try_into()	value needing check	associated Error
str::parse() / FromStr	text	FromStr::Err

When to use which: use `s.parse::<u16>()` for plain numbers. Use custom `FromStr` for **domain types** (`Port`, `HostPort`) with extra rules.

13.8.1 FromStr for structured values — paths and endpoints

Parse **one string into a sum type** at the boundary. Validation happens once; the rest of the code pattern-matches on variants:

```
// Playground
use std::str::FromStr;

#[derive(Debug, PartialEq)]
enum DevicePath {
    Tcp { host: String, port: u16 },
    Serial { port_name: String },
}

impl FromStr for DevicePath {
    type Err = String;

    fn from_str(s: &str) -> Result<Self,
        Self::Err> {
        if let Some(rest) =
            s.strip_prefix("tcp://") {
            let (host, port) = rest
                .split_once(':')
                .ok_or_else(|| format!("invalid
                    tcp path: {s}"))?;
            Ok(DevicePath::Tcp {
                host: host.into(),
                port: port.parse().map_err(|_|
                    format!("bad port in
                    {s}"))?,
            })
        } else if let Some(name) =
            s.strip_prefix("serial://") {
            Ok(DevicePath::Serial {
                port_name: name.into(),
            })
        } else {
            Err(format!("expected tcp:// or
```



```

        serial://, got {s}"))
    }
}

fn main() {
    let p: DevicePath =
        "tcp://plc.local:502".parse().unwrap();
    assert_eq!(
        p,
        DevicePath::Tcp {
            host: "plc.local".into(),
            port: 502,
        }
    );
    println!("{:?}", p);
}

```

Same idea for `s3://`, `file://`, or `host:port` — **FromStr** at the CLI/config edge, typed enum inside the core.

13.8.2 Display + FromStr round-trip for CLI enums

CLI tools often need **human-readable flags** that round-trip through strings. Implement both traits on the same enum:

```

// Playground
use std::fmt;
use std::str::FromStr;

#[derive(Debug, Clone, PartialEq)]
enum LogFormat {
    Json,
    Plain,
}

impl fmt::Display for LogFormat {
    fn fmt(&self, f: &mut fmt::Formatter<'_>)
        -> fmt::Result {
        match self {

```

```

        LogFormat::Json => write!(f,
            "json"),
        LogFormat::Plain => write!(f,
            "plain"),
    }
}

impl FromStr for LogFormat {
    type Err = String;

    fn from_str(s: &str) -> Result<Self,
        Self::Err> {
        match s {
            "json" => Ok(LogFormat::Json),
            "plain" => Ok(LogFormat::Plain),
            other => Err(format!("unknown log
                format: {other}")),
        }
    }
}

fn main() {
    let fmt = LogFormat::Json;
    let s = fmt.to_string();
    assert_eq!(LogFormat::from_str(&s).unwrap(),
        fmt);
    println!("{}", s);
}

```

Display strings should match what FromStr accepts — tests can assert `parse(display(x)) == x`.

13.8.3 Display on field-name enums — stable log labels

Small enums that name **schema fields** or **metric keys** implement Display so error and log messages stay consistent:

```
// Playground
use std::fmt;

enum MetricField {
    PollLatencyMs,
    ErrorCount,
}

impl fmt::Display for MetricField {
    fn fmt(&self, f: &mut fmt::Formatter<'_>)
        -> fmt::Result {
        match self {
            MetricField::PollLatencyMs =>
                write!(f, "poll_latency_ms"),
            MetricField::ErrorCount =>
                write!(f, "error_count"),
        }
    }
}

fn main() {
    let field = MetricField::PollLatencyMs;
    println!("metric {} exceeded threshold",
        field);
}
```

Prefer enums over string literals for field names — renames stay compile-time checked.

13.8.4 TryFrom edge cases

Wrong — **as** cast that silently truncates:

```
// Playground
fn main() {
    let big: i32 = 70000;
    let p = big as u16; // wraps --- no error
    println!("{}", p);
    // 14464 --- not what you want for ports
}
```

Fix: `u16::try_from(big)` or custom `TryFrom` with validation.

13.9 AsRef and Borrow

Flexible APIs accept many input forms **without allocating**:

Trait	Role
<code>AsRef<T></code>	cheap ref conversion (<code>String -> &str</code> , <code>Vec<u8> -> &[u8]</code>)
<code>Borrow<T></code>	like <code>AsRef</code> but tied to hashing/equality for map keys

```
// Playground
fn print_label<S: AsRef<str>>(s: S) {
    println!("{}", s.as_ref());
}

fn main() {
    print_label("literal");
    print_label(String::from("owned"));
}
```

Why `HashMap<String, V>` accepts `get(&str)`: keys implement `Borrow<str>`. You can look up with `&str` without allocating a `String`.

```
// Playground
use std::collections::HashMap;

fn main() {
    let mut m = HashMap::new();
    m.insert(String::from("alice"), 10);
    println!("{:?}", m.get("alice"));
    // &str borrows as key
}
```

13.9.1 AsRef vs &T parameter

Signature	Accepts	Notes
<code>fn f(s: &str)</code>	<code>&str</code> only	callers with <code>String</code> pass <code>&s</code>
<code>fn f(s: impl AsRef<str>)</code>	<code>&str</code> , <code>String</code> , <code>Cow<str></code> , ...	ergonomic public API
<code>fn f(s: &String)</code>	<code>&String</code> only	usually a smell

Same pattern: `impl AsRef<Path>`, `impl AsRef<[u8]>`.

13.10 Cow — clone on write

`Cow<'a, T>` (clone-on-write) is either **borrowed** or **owned**:

```
// Playground
use std::borrow::Cow;

fn normalize<'a>(s: Cow<'a, str>) -> Cow<'a,
    str> {
    if s.contains(' ') {
        Cow::Owned(s.replace(' ', "_"))
    } else {
        s
    }
}

fn main() {
    let borrowed =
        normalize(Cow::Borrowed("hello"));
    let owned = normalize(Cow::Borrowed("hello
        world"));
    println!("{}", borrowed, owned);
}
```

Cheap or expensive? It depends on the input — that is the point of `Cow`.

Call	Path	Cost
<code>normalize(Cow::Borrowed("hello"))</code>	<code>spaces -> return s unchanged</code>	cheap — one scan (contains), no heap allocation , still borrows original bytes
<code>normalize(Cow::Borrowed("hello").replace(" ", "_"))</code>	<code>spaces -> replace world"))</code>	pays — allocates a new String, copies/transforms every byte

Trace the cheap case: `Cow::Borrowed("hello")` goes in; `contains(' ')` is false; the function returns the same `Cow::Borrowed` — zero copies beyond the read-only scan.

Trace the expensive case: `replace(' ', "_")` always builds a **new** owned string — you cannot mutate a borrowed `&str` in place when the result might be longer or when the source is shared.

Compared to always returning `String`:

```
fn normalize_always(s: &str) -> String {
    s.replace(' ', "_")
    // allocates even when s has no spaces
}
```

Even "hello" gets a fresh heap allocation. With `Cow`, most inputs that need **no change** skip allocation entirely.

Rule of thumb: `Cow` is worth it when **many** call sites pass borrowed data and **only some** inputs need transformation. If every call already owns a `String` or always needs a new buffer, take/return `String` and skip `Cow`.

Variant	When
<code>Cow::Borrowed(&T)</code>	no allocation; return as-is
<code>Cow::Owned(T)</code>	had to mutate or build owned data

13.10.1 `into_owned()` — finish with a plain owned value

`Cow` is for **maybe** allocating. Once you know the caller needs an owned `String` or `Vec` long-term, call `.into_owned()` — it returns `T` without cloning when the data is already `Cow::Owned`:

```
// Playground
use std::borrow::Cow;

fn slugify<'a>(s: Cow<'a, str>) -> String {
    let cow = if s.contains(' ') {
        Cow::Owned(s.replace(' ', "-"))
    } else {
        s
    };
    cow.into_owned()
    // String --- cheap move if already Owned,
    // one clone if Borrowed
}

fn store_label(label: String) {
    println!("stored: {}", label);
}

fn main() {
    store_label(slugify(Cow::Borrowed("sensor
1"))); // allocates once inside slugify
    store_label(slugify(Cow::Borrowed("sensor-2"
))); // into_owned clones the &str
}
```

Call	If Cow was...	Cost
<code>.into_owned()</code>	<code>Borrowed(&T)</code>	clone T into owned
<code>.into_owned()</code>	<code>Owned(T)</code>	move T out — no extra clone

Use **into_owned()** at the boundary when the next API takes `String / Vec<T>`, not `Cow`.

13.10.2 ToOwned and .to_owned() on references

The **ToOwned** trait generalizes “borrowed view -> owned copy”. For **&str**, **.to_owned()** is the same idea as **.to_string()** — both produce a `String`:

```
// Playground
fn cache_key(prefix: &str, id: u32) -> String {
    format!("{}", prefix.to_owned(), id)
}

fn main() {
    let literal = "modbus";
    let owned = String::from("opcua");
    println!("{}", cache_key(literal, 1));
    println!("{}", cache_key(&owned, 2));
}
```

Typical implementations:

Borrowed	.to_owned() gives
&str	String
&[T] where T: Clone	Vec<T>
&Path	PathBuf
Cow<'a, T>	T (via .into_owned())

When to reach for .to_owned()

- You hold **&str** / **&[u8]** and the next step needs **String** / **Vec<u8>** for storage or Send across threads.
- You are **normalizing** borrowed input once at the boundary (trim, replace) before pushing into a collection.

When to skip it

- The value is **already** String — use it as-is; .to_owned() on &String allocates a **duplicate** (Chapter 20).
- You only **read** the data — take &str / impl AsRef<str> and borrow (AsRef above).

```
// Playground --- weak: double allocation
fn weak(label: &String) -> String {
    label.to_owned()
    // clones entire String when &str would
    // borrow
}
```



```
// Playground --- strong: borrow until you store
fn strong(label: &str) -> String {
    label.to_string()
    // or to_owned() --- one allocation when
    // storage is required
}

fn main() {
    let s = String::from("plc-1");
    println!("{}", weak(&s), strong(&s));
}
```

.clone() vs **.to_owned()**: on **&T**, prefer **.to_owned()** when you mean “give me an owned copy of this borrowed data”. On **String / Vec**, **.clone()** duplicates the owned value. The names signal intent to readers.

API pattern: accept **Cow<'a, str>** when you might normalize or pass through unchanged; return **String** (or call **.into_owned()** before storing). Common in parsers and HTTP header helpers.

Method	On what	Result	Cost
.to_owned()	borrowed &T	owned copy of T (&str -> String)	heap alloc + copy — same price as .clone() on owned data; skip if you can borrow
.clone()	already-owned T	duplicate of T (String -> String)	heap alloc + copy for String/Vec ; cheap for Copy types
.into_owned()	Cow<'a, T>	owned T	move if already Owned; alloc + copy if Borrowed

One line: borrow -> **.to_owned()**; own and keep two -> **.clone()**; Cow -> plain owned -> **.into_owned()**.

13.11 Deref — cheap access through smart pointers

Deref lets you treat **Arc**<T>, **Box**<T>, and newtypes like **T** for field access. In async code, tasks often hold **Arc**<Config> — destructure without cloning the whole struct:

```
// Playground
use std::ops::Deref;
use std::sync::Arc;

struct GatewayConfig {
    host: String,
    port: u16,
}

fn connect(cfg: Arc<GatewayConfig>) {
    let GatewayConfig { host, port, .. } =
        cfg.deref();
    println!("connecting to {}:{}", host, port);
}

fn main() {
    connect(Arc::new(GatewayConfig {
        host: "plc.local".into(),
        port: 502,
    }));
}
```

Rust also **deref-coerces** **&String** -> **&str** via **Deref**. Newtype wrappers often implement **Deref** to reach inner data (Chapter 10).

13.12 Extension traits on **&str**

Add string helpers as traits on **&str** (or your newtype) instead of a pile of free functions. Keeps call chains readable and each helper unit-testable:

```
// Playground
trait NormalizeWhitespace {
    fn normalize_whitespace(self) -> String;
}
```

```

}

impl NormalizeWhitespace for &str {
    fn normalize_whitespace(self) -> String {
        self.split_whitespace().collect::<Vec<_>
            >().join(" ")
    }
}

fn clean_label(raw: &str) -> String {
    raw.normalize_whitespace()
}

fn main() {
    println!("{}", clean_label(" over temp
    "));
}

```

Use a **state machine** or **split loop** for hot paths; reach for regex only when the pattern truly needs it. Pair with Cow<str> when most inputs need no allocation (Cow above).

13.13 Common derive sets (cheat sheet)

Type role	Typical derives
log / test DTO	Debug, Clone, PartialEq
config file	Debug, Clone, Deserialize, Default
map key	Eq, Hash, Clone, Debug
sortable record	Eq, Ord, PartialOrd, Debug
error enum	Debug, Error (thiserror) — Chapter 8

Do not derive Clone on unique handles. Do not derive Debug on secrets without redaction.

13.14 Java / Python contrast

Idea	Java	Python	Rust
toString	toString()	__str__	Display / Debug
equals / hashCode	equals, hashCode	__eq__, __hash__	PartialEq, Eq, Hash
parsing	constructors, valueOf	int(s)	FromStr, TryFrom, .parse()
flexible param	overloads	duck typing	AsRef, impl Trait

13.15 When the compiler says no

Common errors in this chapter:

Error (typical)	Cause	Fix
X doesn't implement Display	used {}	{:?} or impl Display
the trait bound Eq is not satisfied	f64 or partial type	fixed-point or PartialEq only
T: Hash not satisfied	key not hashable	derive Hash + Eq
conflicting From impls	two From to same type	one canonical path
orphan rule	impl Display for Vec<u8>	newtype wrapper
? can't convert error	missing From impl	From or map_err
Borrow / key mismatch	wrong lookup type	use Borrow target (&str for String keys)

13.16 Idiom spotlight

Implement From, accept **impl AsRef<str>** in public helpers. Callers pass literals or owned strings; you stay allocation-free when borrowing suffices.

Display for users, **Debug for developers** — derive Debug freely; invest in Display where humans read the output.

Validate at the boundary — TryFrom / FromStr at parse time; trust types inside the core.

Pair Display + FromStr on CLI enums; use **FromStr on path strings** to build sum types in one step.

13.17 Go deeper

- [The Rust Book — Useful Traits](#)
- [The Rust Book — Derivable Traits \(Appendix C\)](#)
- [From/Into](#)
- [Borrow trait](#)

13.18 See also

- Chapter 7: Structs, traits, and generics — orphan rule, `impl Trait`
- Chapter 8: Errors and testing — `From` in `?`, custom errors
- Chapter 11: Collections — trait bounds on keys
- Chapter 3: Functions — flexible parameters, first `#[derive(Debug)]` on errors
- Chapter 17: Derive attributes — `std` vs ecosystem vs custom derives
- Chapter 19: I/O — `AsRef<Path>`, CLI path/env patterns

13.18.1 Afterparty

13.18.1.1 Debug, Display, Default

1. **Debug vs Display** — “When derive `Debug` only vs implement `Display` for a CLI status line?”
2. **Redacted Debug** — “Struct with `api_key: String` — sketch manual `Debug` with `[REDACTED]`.”
3. **Pretty debug** — “Same struct with `{:#?}` vs `{:?}",` — when does pretty-print help in tests?”
4. **Display impl** — “Implement `Display` for `Port(u16)` showing `Port(8080)` — I write `fmt`, you review.”
5. **Default enum** — “Three-variant mode enum — derive `Default` with `#[default]` on `Auto`; show update syntax.”

13.18.1.2 Eq, Hash, Ord

6. **Derive set quiz** — “Map key, log line, sortable row, error enum — I list derives each needs.”
7. **Eq on floats** — “Struct with `f64` field — show `Eq` derive failure; three fixes.”
8. **HashMap key** — “Why does `UserId(String)` need `Eq + Hash` for `HashMap` keys?”

9. **Ord sort** — “Sort `Vec<MyRecord>` by timestamp field — bounds needed on `MyRecord`?”

13.18.1.3 From, TryFrom, FromStr

10. **From chain** — “String -> MyLabel via From; add `From<&str>` without duplicating logic.”
11. **TryFrom port** — “Port validation with custom enum error `OutOfRange` instead of `&str`.”
12. **parse vs TryFrom** — “When `s.parse::<u16>()` vs `u16::try_from(x)` vs custom `FromStr`?”
13. **FromStr type** — “Parse `host:port` into struct — sketch `FromStr` with split and two parse steps.”
14. **Silent cast trap** — “Show `70000i32` as `u16` vs `TryFrom` — predict values; argue for validation.”
15. **From in errors** — “Wire `ParseIntError` into `AppError` with `From so ? works` — list `impl` only.”

13.18.1.4 AsRef, Borrow, Cow

16. **AsRef drill** — “Rewrite three functions taking `&String` to `impl AsRef<str>`.”
17. **AsRef bytes** — “Function logging wire payload — signature with `impl AsRef<[u8]>`; accept `Vec`, `&[u8]`, array.”
18. **Borrow lookup** — “Explain `HashMap<String, V>.get(&str)` — role of `Borrow<str>`.”
19. **Cow API** — “Normalize slug: accept `Cow<str>`, return borrowed if already valid else owned.”
20. **into_owned** — “When caller needs `String` after your `Cow` helper — where call `into_owned()`?”
21. **to_owned vs clone** — “Three snippets: `&str->store`, `&String->store`, `already-String` — pick `to_owned`, `borrow`, or `move`; explain each.”

13.18.1.5 API design and capstone

22. **Newtype Display** — “Wrap `Vec<u8>` as `HexBytes` — implement `Display` without orphan violation.”
23. **Derive audit** — “Config with secrets + TOML — list safe vs unsafe derives.”
24. **Mini crate API** — “Public `HostPort { host, port }` with `Display`,

TryFrom<&str> for host:port — list impl blocks only.”

25. **Capstone** — “Design public API for `RateLimit` { max: u32, window_secs: u64 }: parsing, display, equality — traits only, no bodies.”

Part II

Concurrency

Chapter 14

Multithreading

14.1 Hook

Concurrency models differ by language. **Python** threads fight the GIL for CPU work. **Java** threads share the heap with locks. Rust threads are **OS threads**. **Safe Rust** blocks many data races at compile time via `Send` and `Sync`. Use channels and mutexes when sharing is real.

Threading is a **large topic** — scheduling, deadlocks, lock granularity, pools, atomics, `async`, and cross-process IPC each deserve their own deep dive. **This chapter is a starter only:** `spawn/join`, one channel pattern, and shared state with `Arc<Mutex<T>>`. Treat it as vocabulary and first working examples, not production concurrency design.

14.2 Scope — a brief tour

Intro to OS threads, channels, and `Arc<Mutex<T>>` — not pools, IPC, or lock-free depth.

This chapter covers	Deferred to See also / Afterparty
<code>thread::spawn</code> , <code>join</code> , <code>move</code> , <code>thread::scope</code> (brief)	detached threads, custom pools
<code>mpsc</code> channels	bounded channels, backpressure, worker pools
<code>Arc<Mutex<T>></code> , <code>RwLock</code> , <code>OnceLock</code> (brief)	<code>Condvar</code> , <code>Barrier</code> , <code>LazyLock</code> depth

This chapter covers	Deferred to See also / Afterparty
Send / Sync basics	RefCell across threads, pinning
One worker sketch (poll + channel)	rayon, custom thread pools, cross-process IPC
—	Lock-free atomics -> Chapter 15
—	Async concurrency -> Chapter 16

14.3 What multithreading is

A **thread** is an independent call stack scheduled by the OS. Your program can run multiple threads at once — or interleaved on one CPU.

Idea	Plain language
Thread	Its own stack; runs code in parallel with other threads
Why bother	Overlap I/O wait (serial port + network) or parallelize isolated work units
Rust twist	Ownership and borrowing apply across threads — the compiler rejects many sharing mistakes that become data races in C++/Java
Two safe patterns	Message passing (mpsc channels) or shared state (Arc<Mutex<T>>)

```
main thread: spawn worker --> worker runs -->
              join() waits --> use result
```

Arc exists largely because threads need **shared ownership** — Rc is single-threaded only.

14.4 Send and Sync — the thread boundary rules

When data crosses a thread boundary, the compiler checks two marker traits:

Trait	Meaning (simplified)
Send	Owning value may be moved to another thread
Sync	Shared &T may be used from multiple threads safely

Most types you write are both **Send** and **Sync**. Common exceptions:

Type	Send?	Sync?	Why
Rc<T>	no	no	ref count not thread-safe
RefCell<T>	yes*	no	interior mutability not thread-safe
Mutex<T>	yes if T: Send	yes if T: Send	lock enforces exclusive access

*Moving **RefCell** to another thread is allowed; sharing **&RefCell** across threads is not.

`thread::spawn` requires a **Send** closure with **'static** captures. Captured data must outlive the spawn and be safe to transfer. That is why **Rc** inside a spawned thread fails but **Arc** works (Chapter 10).

14.5 Examples: elementary -> hard

Work through the levels in order. After each snippet: **run it**, then read **what happened**.

14.5.1 Level 1 — Elementary: spawn and join

```
// Playground
use std::thread;
use std::time::Duration;

fn main() {
    let handle = thread::spawn(|| {
        thread::sleep(Duration::from_millis(10))
    });
}
```

```

    42
    });
    println!("joined = {:?}", handle.join());
}

```

What happened:

- Main spawns a child thread; child sleeps 10 ms, returns 42.
- **joined = Ok(42)** — `join()` waits for the child and wraps its return value in `Ok`.
- If the child **panics**, `join()` returns **Err(...)** (not a normal return) — see Chapter 8 — panic and unwind and Level 6 below.

14.5.2 Level 2 — Elementary: move into the thread

Data used inside the closure must be **owned** or **'static**. Usually you **move** values into the new thread:

```

// Playground
use std::thread;

fn main() {
    let msg = String::from("poll");
    let handle = thread::spawn(move || {
        println!("{msg}");
    });
    handle.join().unwrap();
}

```

What happened:

- Prints **poll** — the child owns `msg` inside its closure.
- **move** transfers ownership from main into the thread; `msg` is **not** usable in main after spawn.

Wrong — use `msg` after `move`:

```

// Playground --- does not compile
use std::thread;

fn main() {
    let msg = String::from("poll");

```

```

let handle = thread::spawn(move ||
    println!("{msg}"));
println!("{msg}");
// ERROR: borrow of moved value: `msg`
handle.join().unwrap();
}

```

Fix: **msg.clone()** before spawn if both sides need a copy. That **duplicates the String on the heap**. Large buffers can get costly. Cheaper patterns: return data via **join()** or a **channel**, or share with **Arc** (Level 4) when many threads need one allocation.

14.5.3 Level 3 — Medium: mpsc channel

Multi-producer, single-consumer — send messages instead of sharing mutable state:

```

// Playground
use std::sync::mpsc;
use std::thread;

fn main() {
    let (tx, rx) = mpsc::channel();
    thread::spawn(move || {
        tx.send(1).unwrap();
        tx.send(2).unwrap();
    });
    for val in rx {
        println!("got {}", val);
    }
}

```

What happened:

- Prints **got 1** then **got 2** — order preserved (FIFO).
- **move** on the closure: **tx** ownership moves to the worker; main keeps **rx**.
- When all **tx** senders are **dropped**, the **for val in rx** loop **ends** — channel closed.
- **Message passing** often beats shared mutable state for work queues and command streams.

14.5.4 Level 4 — Medium: `Arc<Mutex<T>>` shared counter

When threads must **mutate the same data**, wrap it in **Mutex** (exclusive access) and **Arc** (shared ownership):

```
// Playground
use std::sync::{Arc, Mutex};
use std::thread;

fn main() {
    let counter = Arc::new(Mutex::new(0));
    let mut handles = vec![];

    for _ in 0..4 {
        let c = Arc::clone(&counter);
        handles.push(thread::spawn(move || {
            let mut n = c.lock().unwrap();
            *n += 1;
        }));
    }
    for h in handles {
        h.join().unwrap();
    }
    println!("count = {}",
        *counter.lock().unwrap());
}
```

Piece	Role
<code>Arc</code>	Reference-counted handle — many threads hold the same allocation
<code>Mutex</code>	Only one thread mutates inner <code>T</code> at a time
<code>lock().unwrap()</code>	Block until lock acquired; panic if mutex is poisoned
<code>Arc::clone(&counter)</code>	Cheap handle copy — points at same <code>Mutex</code>

`Arc::clone` vs cloning the inner value — easy to confuse:

Call	What it copies	Cost
<code>Arc::clone(&counter)</code>	Pointer + atomic ref-count only — same heap Mutex	Cheap — $O(1)$, no data duplicate
<code>(*counter.lock().unwrap()).clone()</code> (<code>clone()</code> on <code>T</code> when <code>T: Clone</code> (e.g. whole <code>String</code> , <code>Vec</code>)		Expensive — full deep copy of protected data

In the loop, `Arc::clone(&counter)` gives each thread a **handle** to the **one** shared counter. Calling `.clone()` on the inner value copies the data under the lock. That is the wrong tool for “many threads, one counter.” See Chapter 10 — Arc vs deep clone.

What happened:

- Prints **count = 4** — each of four threads incremented once; no lost updates.
- Without **Mutex**, sharing `&mut` across threads would not compile — Rust blocks the data race at compile time.
- **Lock poisoning**: if a thread **panics while holding the lock**, others get **PoisonError** on `lock()` — see Lock poisoning below.

14.5.5 Level 5 — Hard: Send trap — Rc cannot enter a thread

Types must implement **Send** to be **moved** into another thread. **Rc** is not **Send**:

```
// Playground --- does not compile
use std::rc::Rc;
use std::thread;

fn main() {
    let data = Rc::new(0);
    thread::spawn(move || {
        println!("{data}");
    });
    // ERROR: `Rc<i32>` cannot be sent between
    threads safely
}
```

What happened:

- Compiler **rejects** the spawn — Rc ref-count is not thread-safe.
- **Fix:** use **Arc::new(0)** (atomic ref-count) — same pattern as Level 4.

```
// Playground
use std::sync::Arc;
use std::thread;

fn main() {
    let data = Arc::new(0);
    let d = Arc::clone(&data);
    thread::spawn(move ||
        println!("{d}")).join().unwrap();
    println!("main still has {data}");
}
```

What happened: compiles and prints **0** twice — Arc is **Send + Sync** when the inner type allows it. **Arc::clone(&data)** is cheap (ref-count bump); main and the worker share **one** i32 on the heap, not two copies.

14.5.6 Level 6 — Hard: join without unwrap + sensor poll worker

6a. Handle panics at the boundary (Chapter 8 — don't unwrap production join in unattended services):

```
// Playground
use std::thread;

fn main() {
    let handle = thread::spawn(|| {
        // simulate failure: panic!("device
        fault");
        100
    });
    match handle.join() {
        Ok(reading) => println!("ok {reading}"),
        Err(_) => eprintln!("worker panicked
        --- log and continue supervisor
        loop"),
    }
}
```



```
}
```

What happened:

- **Ok(100)** -> prints **ok 100**.
- If you uncomment the panic -> **Err(_)** arm runs; **main survives** — contrast with panic in the worker killing the whole process if unhandled.

6b. Poll worker sketch — poll thread, main logs readings:

```
// Playground
use std::sync::mpsc;
use std::thread;
use std::time::Duration;

fn main() {
    let (tx, rx) = mpsc::channel();

    thread::spawn(move || {
        for id in 1..=3 {
            thread::sleep(Duration::from_millis(
                5));
            tx.send(format!("sensor-{{id}}:
                {:.1}", 20.0 + id as
                f64)).unwrap();
        }
        // tx dropped here --- channel closes
    });

    for reading in rx {
        println!("log {{}}", reading);
    }
    println!("poll loop ended");
}
```

What happened:

- Prints **log sensor-1: 21.0, sensor-2: 22.0, sensor-3: 23.0**, then **poll loop ended**.
- Worker **produces**; main **consumes** — classic **message-passing work queue** without shared mutable state.

- Dropping `tx` when the worker finishes closes the channel; main's for reading in `rx` exits cleanly.

14.6 RwLock — many readers, rare writers

Sensor cache: many threads read, one thread reloads config:

```
// Playground
use std::collections::HashMap;
use std::sync::{Arc, RwLock};

fn main() {
    let cache: Arc<RwLock<HashMap<String,
    f64>>> =
        Arc::new(RwLock::new(HashMap::from([("temp".into(), 22.5)])));

    let reader = Arc::clone(&cache);
    let t = std::thread::spawn(move || {
        let map = reader.read().expect("lock
        poisoned");
        println!("read temp={}", map["temp"]);
    });
    t.join().unwrap();

    {
        let mut map =
            cache.write().expect("lock
            poisoned");
        map.insert("temp".into(), 23.1);
    }
    let map = cache.read().unwrap();
    println!("after reload temp={}",
        map["temp"]);
}
```

`.read()` allows concurrent readers. `.write()` exclusive for updates. If a thread panics while holding a lock, the lock is **poisoned** — `.expect` or `match` on the `Result`.

14.7 OnceLock — initialize once

Lazy global config without hand-rolled double-checked locking:

```
// Playground
use std::sync::OnceLock;

struct Settings {
    port: u16,
}

static CONFIG: OnceLock<Settings> =
    OnceLock::new();

fn settings() -> &'static Settings {
    CONFIG.get_or_init(|| Settings { port: 502
    })
}

fn main() {
    println!("port={}", settings().port);
    println!("same={}", settings().port);
}
```

First call runs the closure; later calls return the same reference. For stack-local one-time init, `std::sync::LazyLock` (Rust 1.80+) works similarly without `static`.

14.8 `thread::scope` — borrow stack data into threads

Parallel checksum over chunks — each thread borrows `&chunks[i]` safely:

```
// Playground
use std::thread;

fn checksum(chunk: &[u8]) -> u32 {
    chunk.iter().map(|&b| b as u32).sum()
}
```

```
fn main() {
    let chunks: Vec<u8; 4> = vec![[1, 2, 3,
        4], [5, 6, 7, 8], [9, 10, 11, 12]];

    thread::scope(|s| {
        let handles: Vec<_> = chunks
            .iter()
            .map(|chunk| s.spawn(||
                checksum(chunk)))
            .collect();

        let total: u32 =
            handles.into_iter().map(|h|
                h.join().unwrap()).sum();
        println!("total={}", total);
    });
}
```

scope guarantees threads join before returning — so borrows from `chunks` cannot dangle. Plain `spawn` cannot borrow local `Vec` elements without 'static data.

14.8.1 Sync primitive edge cases

RwLock vs Mutex: use `RwLock` when reads dominate; `Mutex` is simpler when writes are frequent or critical sections are tiny.

14.8.2 Lock poisoning — what it is

If a thread **panics while holding** a `Mutex` or `RwLock`, Rust marks that lock **poisoned**. The panic might have stopped mid-update — the protected value may be **inconsistent** (half-written counter, broken struct invariant).

Step	What happens
Thread panics inside <code>lock()</code> / <code>write()</code> guard	Lock flagged poisoned; guard drops during unwind
Another thread calls <code>lock()</code> / <code>read()</code> / <code>write()</code>	Returns <code>Err(PoisonError<...>)</code> , not a normal guard
You call <code>.unwrap()</code> on that Result	Panics again — “poisoned lock”

Poisoning is a **warning**, not automatic repair. The next thread must decide: fail fast, rebuild state, or recover data deliberately.

```
// Playground --- pattern only; poison happens
    when a holder panics
use std::sync::{Arc, Mutex};

fn main() {
    let data = Arc::new(Mutex::new(0));
    // if some thread panicked while holding
    // this mutex:
    match data.lock() {
        Ok(guard) => println!("value = {}",
                               *guard),
        Err(poisoned) => {
            eprintln!("lock poisoned ---
                       rebuilding or using recovered
                       value");
            let guard = poisoned.into_inner();
            // still gives you MutexGuard<T>
            println!("recovered value = {}",
                       *guard);
        }
    }
}
```

Practical rule: treat poison like a serious fault in automation services — log it, drop bad cache, or restart the worker. Use `into_inner()` only when you accept the data may be corrupt. See Chapter 8 — panic and unwind.

14.9 Techniques at a glance

Popular primitives — one line each. Details for starred rows live in After-party or linked chapters.

Technique	One-line use	Where
<code>thread::spawn / join</code>	Start OS thread; wait for result	Levels 1-2, 6
<code>move closures</code>	Transfer ownership into thread	Levels 2-4

Technique	One-line use	Where
<code>mpsc</code>	Message passing, work queues	Levels 3, 6b
<code>Arc<Mutex<T>></code>	Shared mutable state, short locks	Level 4
Send	Safe to move <code>T</code> to another thread	Level 5
Sync	Safe to share &T across threads	Level 4 (Arc shares &Mutex)
<code>RwLock</code>	Many readers, rare writers	Section above
<code>rayon</code> / thread pools	CPU-bound parallelism	Afterparty / Go deeper
<code>atomics</code>	Lock-free counters, flags	Chapter 15
<code>async</code> / <code>.await</code>	Many concurrent I/O waits	Chapter 16

Send / Sync in practice:

- Most plain types (`i32`, `String`, `Vec`) are **Send** and **Sync**.
- **Rc**, **RefCell** — not **Sync** (single-thread interior mutability).
- **Raw pointers** — neither unless you enforce safety yourself (`unsafe`).
- `Arc<Mutex<T>>` — share **Sync** handle; mutation only inside `lock()`.

14.10 Idiom spotlight

Prefer channels for work queues and command streams; use `Mutex` for short critical sections. Long-held locks hurt poll latency (a Modbus cycle can miss its slot).

I/O-bound services: one thread blocked on serial read, another on network publish — overlap waits without fighting the GIL.

Handle `join()` like `Result` at the supervisor boundary — worker panic is data, not necessarily process death.

14.11 Go deeper

- [The Rust Book — Fearless Concurrency](#)
- [Mutex basics](#) — 986
- [Thread pool](#) — 923
- [Arc threads](#) — 109

14.12 See also

- Chapter 8: Errors and panic — `join()` after worker panic, why `unwrap` in loops is risky
- Chapter 10: Arc and smart pointers — `Rc` vs `Arc`
- Chapter 12: Closures — `move` and `Fn` traits for `spawn`
- Chapter 15: Atomics — lock-free counters, `OnceLock` vs hand-rolled `init`
- Chapter 16: Async — when threads are not the right tool

14.12.1 Afterparty

Use these for worker pools and topics not covered above.

14.12.1.1 Spawn, move, join

1. **Race quiz** — “Which snippets are data races in C++ but rejected by Rust compiler?”
2. **Move fix** — “Show `spawn` without `move` that fails; fix with `move` or `clone`.”
3. **Join panic** — “Worker panics; rewrite `main` to match `join()` and keep supervisor alive.”
4. **Detached threads** — “Why is `mem::forget(handle)` after `spawn` dangerous? Better pattern?”

14.12.1.2 Channels

5. **Channel design** — “Worker pool with `mpsc`: I describe throughput; sketch thread count + channel shape.”
6. **Drop tx footgun** — “Main exits before worker sends — diagram who holds `tx/rx`.”
7. **Multiple producers** — “Clone `tx` to two workers; main receives merged stream — sketch code.”
8. **Bounded vs unbounded** — “When does unbounded `mpsc` blow memory in automation? Bounded alternative?”

14.12.1.3 Mutex and shared state

9. **Mutex vs RwLock** — “Read-heavy sensor cache — pick primitive and why.”
10. **Hold lock briefly** — “Refactor bad code that calls network I/O while holding `Mutex` lock.”

11. **Deadlock sketch** — “Two mutexes, lock order A then B vs B then A — show hang scenario.”
12. **Poison recovery** — “Thread panics holding lock; show `PoisonError` and `into_inner()` recovery.”
13. **Send fix** — “I try to move `Rc` into thread; show fix with `Arc`.”
14. **RefCell trap** — “Why `Arc<RefCell<T>>` is not `Sync`; fix pattern for shared mutation.”

14.12.1.4 Production and automation

15. **PLC gateway layout** — “Sketch poll thread + command channel + main supervisor; no code over 40 lines.”
16. **Lock latency** — “Modbus cycle 20 ms — max time holding mutex for register cache update?”
17. **Level ladder recap** — “Explain Levels 1-6 in one paragraph each for a Java teammate.”

14.12.1.5 `RwLock`, `OnceLock`, and scope

18. **`RwLock` cache** — “Read-heavy sensor map — sketch `Arc<RwLock<HashMap>>`; when does write starve readers?”
19. **Mutex vs `RwLock`** — “Same cache with 50% writes — pick `Mutex` or `RwLock` and justify in two sentences.”
20. **`OnceLock` init** — “`get_or_init` fails second init with different value — show `OnceLock` behaviour.”
21. **Scope borrow** — “Parallel sum over `Vec<u8;512>` with `thread::scope` — why plain `spawn` fails on `&chunk`.”
22. **Poison recovery** — “Writer thread panics holding `RwLock` — show poisoned `read()` error and recovery options.”
23. **Capstone sync** — “Design: lazy config (`OnceLock`), shared cache (`RwLock`), scoped batch workers — list types only.”

Chapter 15

Atomics and Lock-Free Basics

15.1 Hook

Lock-free primitives appear in many runtimes (**Java** `java.util.concurrent.atomic.*`, **Python** atomics behind the GIL in CPython, ...). Rust exposes `std::sync::atomic` for lock-free counters and flags when mutex overhead is too high. **Memory ordering** is part of the contract.

15.2 Scope — a brief tour

Practical atomics for counters and flags — not lock-free data structures or formal memory-model proofs.

This chapter covers	Deferred to See also / Afterparty
AtomicBool, AtomicUsize, load / store / fetch_add / compare_exchange	Lock-free queues, stacks, hazard pointers
Practical Ordering subset	Full formal memory-model proofs
Data races + happens-before intuition	Formal verification
Shared flags/counters in threads and async	crossbeam, hand-rolled lock-free DS
When not to use atomics	SIMD atomics, portable-atomic edge cases

15.3 Memory ordering — Ordering cheat sheet

Every atomic operation takes an **Ordering**. It answers two questions:

1. **Is this read/write on the atomic word indivisible?** — yes for **all** orderings.
2. **Do other threads see my non-atomic writes in a predictable order?** — only for **some** orderings.

Ordering	Valid on	What it synchronizes	Plain language
Relaxed	load, store, RMW	This atomic word only — no ordering for other memory	“Count reliably; I don’t publish other data through this op.”
Acquire	load, RMW (success)	Reads after this load see writes that happened before a Release on the same atomic	Reader: “When I see the new flag/version, I see the data the writer prepared.”
Release	store, RMW (success)	Writes before this store are visible to threads that later Acquire -load the same atomic	Writer: “I finish writing config, then I publish the flag/version.”
AcqRel	RMW only (fetch_add, compare_exchange, ...)	Both — successful RMW releases prior writes and acquires prior releases	“Update and publish (or consume and publish) in one atomic step.”
SeqCst	all ops	Global total order among all SeqCst ops — strongest, easiest mental model, often slightly slower	“I want one strict worldwide timeline when in doubt.”

Handoff pattern (used in Levels 1 and 4):

```
writer: write config buffer ->
        atomic.store(..., Release)
reader: atomic.load(Acquire) -> read config
        buffer    // sees writer's data
```

`compare_exchange` has two orderings (Level 3):

```
counter.compare_exchange_weak(current, current
    + 1, Ordering::AcqRel, Ordering::Relaxed)
//
    ^ success          ^ failure (retry path)
```

Use **Relaxed** on failure when you only loop and retry; use **AcqRel** on success when the RMW participates in a handoff.

Quick pick:

Scenario	Ordering
Packet / poll / frame counter — nothing else depends on it	Relaxed
Shutdown flag, version number, “data ready” bit + other shared fields	Release (writer) + Acquire (reader)
CAS / fetch_add that caps or publishes in one step	AcqRel on success
Low-frequency flag, want simplest model	SeqCst everywhere on that atomic

Official reference: [std::sync::atomic::Ordering](#).

15.4 Where atomics fit — threads, async, and Chapter 14

Atomics solve **concurrent read/write of one machine word** without a **Mutex lock**. They appear in **OS-thread** code (Chapter 14) and **async** services (Chapter 16). Same types, same rules.

Context	Shared pattern	Atomics role
OS threads (Ch14 Level 4)	Arc<AtomicUsize> across thread::spawn	Hot counter without mutex contention
Async tasks (Ch16)	Arc<AtomicBool> read by many tokio::spawn tasks	Cooperative shutdown without locking the runtime
Both	Arc wraps the atomic — Arc::clone is cheap	Metrics: Relaxed; flags with dependent data: Release / Acquire

Ch14: thread::spawn --> Arc<AtomicUsize>

```
    fetch_add
Ch16: tokio::spawn    --> Arc<AtomicBool>
    load/store
```

Key message: Async does **not** remove concurrency. Tasks on a thread pool still share memory. Atomics are **Send + Sync**. Wrap them in **Arc** when many tasks or threads need the same counter or flag.

Atomics do **not** replace channels (message streams) or **Mutex** (multi-field invariants). One atomic word is not a lock-free entire struct.

15.5 Data races — why atomics exist

A **data race** (informal): two threads access the **same memory**, at least one **writes**, with **no synchronization**. In C/C++ that is **undefined behavior**.

Kind	Example	Safe Rust?
Prevented at compile time	two &mut to same String in two threads	compile error
Non-atomic shared counter	static mut COUNTER += 1 in two threads	unsafe UB — do not do this
Atomic operations	fetch_add(1, Ordering::Relaxed)	defined — each op is indivisible for that word
Visibility bug (ordering)	flag set with Relaxed but reader never sees prior writes to other fields	defined atomic op, wrong logic — use Release/Acquire

Two different problems atomics address:

Problem	Non-atomic counter += 1	fetch_add on AtomicUsize
Lost updates	Thread A and B both read 5, both write 6 -> count is 6 , not 7	Hardware read-modify-write is one indivisible step — no lost increments
Where it shows up	C shared mut, static mut + unsafe in Rust	Atomic* with any Ordering

Ordering is a separate issue: even when increments are not lost, **Relaxed** only guarantees atomicity of **that one word**. It does **not** guarantee that other threads see writes you made **before** updating a flag or counter.

Use case	Ordering
Standalone metric (packet count, sample tally)	Relaxed — usually enough
Flag that means “other data is ready” (config buffer, shutdown)	Release on writer, Acquire on reader — see Levels 4-5

So: **fetch_add** fixes lost increments; **Release/Acquire** fix visibility of *other* memory. You need both ideas when a flag publishes state beyond the atomic itself.

Conceptual anti-pattern (UB — not safe Rust):

```
// Playground --- do not write this;
    illustrates why atomics exist
// static mut COUNTER: u32 = 0;
// thread::spawn(|| unsafe { COUNTER += 1 });
// thread::spawn(|| unsafe { COUNTER += 1 });
// two unsynchronized writes = data race
    (undefined behavior)
```

Same job as Chapter 14 Level 4 `Arc<Mutex<usize>>` — atomics trade **expressiveness** for **speed** on simple ops.

15.6 Examples: elementary -> hard

Work through in order. After each snippet: **run it**, then read **what happened**.

15.6.1 Level 1 — Elementary: `AtomicBool` shutdown flag

Worker loops until main sets **shutdown** with **Release**; worker reads with **Acquire**:

```
// Playground
use std::sync::atomic::{AtomicBool, Ordering};
use std::sync::Arc;
use std::thread;
use std::time::Duration;
```

```
fn main() {
    let shutdown =
        Arc::new(AtomicBool::new(false));
    let flag = Arc::clone(&shutdown);

    let worker = thread::spawn(move || {
        while !flag.load(Ordering::Acquire) {
            thread::sleep(Duration::from_millis(
                5));
            // simulate poll work
        }
        println!("worker stopped");
    });

    thread::sleep(Duration::from_millis(20));
    shutdown.store(true, Ordering::Release);
    worker.join().unwrap();
}
```

What happened — step by step:

Step	Who	Action
1	main	Creates shutdown = false, spawns worker with a clone of the same Arc
2	worker	Loops: load(Acquire) -> still false -> sleep 5 ms, repeat
3	main	Sleeps 20 ms, then store(true, Release)
4	worker	Next load(Acquire) -> true -> exits loop, prints worker stopped
5	main	join() returns — worker finished cleanly

Why atomics and these orderings?

Not a plain bool: two threads reading/writing the same bool without synchronization is a **data race** (undefined behavior in C; blocked or unsafe in Rust). AtomicBool makes each load/store safe.

Not Relaxed (for this pattern): Relaxed only promises “this one flag updates correctly.” It does **not** promise that the worker sees **other** data main wrote earlier (logs flushed, config updated, port number set). For a **stop signal that goes with other setup**, use a matched pair:

Who	Code	Simple meaning
main	<code>store(true, Release)</code>	“I finished my other work — now I turn on shutdown.”
worker	<code>load(Acquire)</code>	“If I see shutdown on, I also see that other work.”

Picture it like a doorbell:

1. Main prepares the package (writes config, flushes logs, ...).
2. Main rings the bell — `store(true, Release)`.
3. Worker hears the bell — `load(Acquire)`.
4. Worker opens the door and uses the package.

Steps 1-3 must stay in that order across threads. **Release + Acquire** is how Rust asks the CPU for that guarantee. (Formal name: **happens-before** — main’s earlier writes **happen before** the worker’s reads that follow the `Acquire`.)

In this tiny example the flag is the only shared value, so Relaxed would often still stop the loop. We show **Release / Acquire** because real services almost always mean “stop **and** read the state I prepared first” — Level 4 adds a **config + version** to make that explicit.

Same layout with `Arc<AtomicBool>` inside `tokio::spawn` (Chapter 16): tasks call `load(Acquire)` between `.await` points instead of `thread::sleep`.

15.6.2 Level 2 — Elementary: `fetch_add` counter

Lock-free increment — port of Ch14 Mutex counter:

```
// Playground
use std::sync::atomic::{AtomicUsize, Ordering};
use std::sync::Arc;
use std::thread;

fn main() {
    let n = Arc::new(AtomicUsize::new(0));
```

```
let mut handles = vec![];

for _ in 0..4 {
    let c = Arc::clone(&n);
    handles.push(thread::spawn(move || {
        c.fetch_add(1, Ordering::Relaxed);
    }));
}
for h in handles {
    h.join().unwrap();
}
println!("{}", n.load(Ordering::Relaxed));
}
```

Piece	Role
AtomicUsize	One shared counter word — no Mutex
fetch_add(1, Relaxed)	Atomic read-modify-write; Relaxed OK for metrics-only counter
Arc::clone(&n)	Cheap handle — same atomic on heap (Chapter 14)

What happened:

- Prints **4** — four threads each added 1; **no lost updates** (contrast non-atomic static mut).
- **Relaxed** does not synchronize **other** memory — fine here because **only** the atomic is touched.

15.6.3 Level 3 — Medium: `compare_exchange` — cap a counter

Compare-and-swap (CAS): update only if current value matches expected — retry on contention:

```
// Playground
use std::sync::atomic::{AtomicUsize, Ordering};
use std::sync::Arc;
use std::thread;
```



```

const MAX: usize = 100;

fn try_inc(counter: &AtomicUsize) {
    loop {
        let current =
            counter.load(Ordering::Relaxed);
        if current >= MAX {
            return;
        }
        if counter
            .compare_exchange_weak(current,
                                   current + 1, Ordering::AcqRel,
                                   Ordering::Relaxed)
            .is_ok()
        {
            return;
        }
        // spurious failure --- retry
    }
}

fn main() {
    let n = Arc::new(AtomicUsize::new(98));
    let mut handles = vec![];

    for _ in 0..4 {
        let c = Arc::clone(&n);
        handles.push(thread::spawn(move ||
            try_inc(&c)));
    }
    for h in handles {
        h.join().unwrap();
    }
    println!("capped at {}",
            n.load(Ordering::Relaxed));
}

```

What happened:

- Counter starts at **98**; four threads race to increment; at most **2** succeed
-> prints **capped at 100** (never exceeds MAX).

Reading `compare_exchange_weak`:

```
counter.compare_exchange_weak(current, current
    + 1, Ordering::AcqRel, Ordering::Relaxed)
//          ^^^^^^^
//          ^^^^^^^^^^^^^^^^^^^^^^^^^
//          ^^^^^^^^^^^^^^^^^^^^^
//
//          expected  new
//          value     success order     failure order
```

Argument	Value here	Meaning
expected	current (from prior load)	“I think the counter is still this.”
new	current + 1	“If so, write this instead.”
success ordering	AcqRel	If swap succeeds — atomic RMW with acquire+release semantics
failure ordering	Relaxed	If swap fails — only the atomic word is consulted; no extra sync

One CAS attempt, atomically:

```
if counter == expected { counter = new;
    Ok(expected) }
else { Err(actual_value_now) }
```

That check-and-write is **one indivisible step** — unlike load then store separately, where another thread can sneak in between.

Why the loop retries:

Failure reason	What happened
Contention	Another thread incremented between your load and CAS — <code>Err(actual)</code> ; re-read and try again
Cap reached	<code>current >= MAX</code> — exit before CAS
Spurious (<code>_weak</code> only)	Hardware may fail even when values match — retry , same as a lost race

Use **compare_exchange_weak in loops** (cheaper on some CPUs). Use **compare_exchange** (strong) when you need at most one attempt and no spurious failure.

Trace from 98: two threads can reach 99 and 100; the rest see **current** $\geq$ MAX or lose CAS races once value is 100 — cap holds without a mutex.

15.6.4 Level 4 — Medium: Release/Acquire publish pattern

Worker updates **config**, then bumps **version**; reader observes version with **Acquire** before reading config:

```
// Playground
use std::sync::atomic::{AtomicUsize, Ordering};
use std::sync::{Arc, Mutex};
use std::thread;
use std::time::Duration;

fn main() {
    let config = Arc::new(Mutex::new(502u16));
    let version = Arc::new(AtomicUsize::new(0));

    let cfg_w = Arc::clone(&config);
    let ver_w = Arc::clone(&version);
    let writer = thread::spawn(move || {
        {
            let mut p = cfg_w.lock().unwrap();
            *p = 8080;
        } // release lock before publishing
        ver_w.fetch_add(1, Ordering::Release);
    });

    let cfg_r = Arc::clone(&config);
    let ver_r = Arc::clone(&version);
    let reader = thread::spawn(move || {
        loop {
            let seen =
                ver_r.load(Ordering::Acquire);
            if seen > 0 {
```

```
        let p = *cfg_r.lock().unwrap();
        println!("reader saw version {}
                port {}", seen, p);
        break;
    }
    thread::sleep(Duration::from_millis(
        1));
}

writer.join().unwrap();
reader.join().unwrap();
}
```

What happened:

- Writer sets port **8080**, then **fetch_add(1, Release)** publishes.
- Reader **load(Acquire)** on version -> guaranteed to see port **8080**, not stale **502** from before the write.
- Happens-before chain:

```
writer: *config = 8080 --> version Release+1
        --> reader Acquire load --> read config
```

Wrong — Relaxed on version only (visibility bug story):

Using **Relaxed** for both version load and store can let the reader see a new version **without** the config write on some architectures. **Release/Acquire** fixes the handoff. Metrics-only atomics do not need this; **publish patterns do**.

15.6.5 Level 5 — Hard: when Relaxed lies — ordering footgun

Scenario	Ordering	Verdict
Frames-processed counter	Relaxed load/store	OK — standalone stat
Shutdown flag + prior log flush	Relaxed both sides	Risky — reader may not see flushed logs
Shutdown flag + prior log flush	Release store / Acquire load	Correct publish pattern

Scenario	Ordering	Verdict
“I want simple mental model”	SeqCst everywhere	OK at low frequency; slightly slower

Sophisticated edge case — atomic flag + non-atomic payload:

```
// Playground --- conceptual bug pattern (do
    not ship)
use std::sync::atomic::{AtomicBool, Ordering};

struct BadHandoff {
    ready: AtomicBool,
    port: u16,
    // NOT atomic --- paired with ready
}

// Writer: port = 8080; ready.store(true,
    Ordering::Relaxed);
// Reader: if ready.load(Relaxed) { use port }
// may read stale port without Release/Acquire
```

Fix: **Release** after writing `port`, **Acquire** before reading `port` — or protect both under **Mutex** (Chapter 14).

What happened (conceptually): the atomic op is well-defined. **Your program logic** is wrong if ordering does not establish happens-before between the flag and other fields.

15.6.6 Level 6 — Hard: gateway sketch

Metrics (Relaxed) + **shutdown** (Release/Acquire) — same struct works for threads today, async tasks tomorrow:

```
// Playground
use std::sync::atomic::{AtomicBool,
    AtomicUsize, Ordering};
use std::sync::Arc;
use std::thread;
use std::time::Duration;
```

```

struct Gateway {
    polls: AtomicUsize,
    shutdown: AtomicBool,
}

fn main() {
    let gw = Arc::new(Gateway {
        polls: AtomicUsize::new(0),
        shutdown: AtomicBool::new(false),
    });

    let worker_gw = Arc::clone(&gw);
    let worker = thread::spawn(move || {
        while
            !worker_gw.shutdown.load(Ordering::A
            cquire) {
            thread::sleep(Duration::from_millis(
                5));
            worker_gw.polls.fetch_add(1,
                Ordering::Relaxed);
        }
    });

    thread::sleep(Duration::from_millis(25));
    println!("metrics polls={}",
        gw.polls.load(Ordering::Relaxed));
    gw.shutdown.store(true, Ordering::Release);
    worker.join().unwrap();
    println!("supervisor done");
}

```

What happened — trace the run:

Step	Thread	Code	Effect
1	main	Arc::new(Gateway { polls: 0, shutdown: false })	One shared struct on the heap

Step	Thread	Code	Effect
2	main	Arc::clone -> worker_gw, thread::spawn	Worker gets its own handle to the same Gateway
3	worker	load(Acquire) -> false	Keep polling
4	worker	sleep(5ms), polls.fetch_add(1, Relaxed)	Bump metric only — repeats each loop (~5 ms apart)
5	main	sleep(25ms)	Lets worker run several poll cycles
6	main	polls.load(Relaxed) -> prints metrics polls= . . . (often 4-5 , timing-dependent)	Supervisor reads stats — Relaxed OK: counter is standalone
7	main	shutdown.store(true, Release)	“Stop now” — Release publishes the shutdown signal
8	worker	next load(Acquire) -> true	Sees shutdown — Acquire pairs with main’s Release
9	worker	loop exits, thread ends	No more fetch_add
10	main	join() -> supervisor done	Clean shutdown

Why two different orderings in one struct?

Field	Ordering	Reason
polls	Relaxed on fetch_add / load	Only counts polls — no other memory is “published” through this counter
shutdown	Acquire (worker) / Release (main)	Stop signal — same doorbell pattern as Level 1; use when shutdown implies “main is done setting up”

Both fields sit in one Gateway, but each atomic picks the ordering that matches its job — see Ordering cheat sheet.

Async note: swap thread::spawn for tokio::spawn (Chapter 16); keep

Arc<Gateway> and the same load/store/fetch_add calls between .await points.

15.7 Memory ordering — recap

Full table and handoff patterns are in Memory ordering — Ordering cheat sheet at the top. After the examples, the practical rule is:

When Relaxed is wrong: any atomic that **guards other memory** (config version, shutdown + shared buffer) needs **Release/Acquire** (or **Mutex**).

ABA (one line): compare_exchange on **pointers** can succeed spuriously if the same bit pattern reappears after free/realloc. Relevant for lock-free queues. Use Afterparty for depth — not production queues here.

15.8 Atomics vs Mutex

Aspect	Atomics	Mutex (Ch14)
Best for	counters, flags, CAS on one word	multi-field invariants, Vec updates
Composability	single-word ops	arbitrary critical sections
Debugging	subtle ordering bugs	coarser, clearer
Overhead	no lock for hot counter	lock + possible contention

Decision sketch:

one word, simple op (count, flag)? -> atomic

several fields must stay consistent? -> Mutex

stream of messages / commands? ->

channel (Ch14)

custom lock-free queue? -> crate

or Afterparty --- not hand-roll

Port exercise: swap Ch14 Level 4 Arc<Mutex<usize>> for Level 2 AtomicUsize when profiling shows lock contention on a hot path.

15.9 Edge cases and compiler traps

Trap	Symptom	Idiom
Non-atomic shared mut	lost counts, UB in C/unsafe	Atomic* or Mutex
Relaxed for publish flag	stale reads of non-atomic fields	Release / Acquire
CAS without retry loop	spurious compare_exchange_weak failure	loop until is_ok() or give up
Atomics for large struct	wrong tool — one word only	Mutex or channel
False sharing (advanced)	cache line ping-pong between cores	pad/separate hot atomics — Afterparty
ABA on pointer CAS	wrong lock-free pop	epoch / RCU — Afterparty
Confusing panic with race	thread died vs data race	Ch8 join vs atomics

Wrong — compare_exchange once, no retry:

```
// Playground --- may silently fail to
    increment under contention
use std::sync::atomic::{AtomicUsize, Ordering};

fn bad_inc(n: &AtomicUsize) {
    let c = n.load(Ordering::Relaxed);
    let _ = n.compare_exchange(c, c + 1,
        Ordering::Relaxed, Ordering::Relaxed);
    // spurious failure -> lost increment
}
```

Wrong — expect Mutex-less Vec push from many threads:

```
// Playground --- does not compile
use std::sync::Arc;
use std::thread;

fn main() {
    let v = Arc::new(vec![1, 2, 3]);
    let v2 = Arc::clone(&v);
    thread::spawn(move || {
        // v2.push(4);
        // ERROR: cannot borrow as mutable
    });
}
```

Use **Mutex<Vec<_>>**, a **channel**, or a dedicated concurrent crate — not a lone atomic.

15.10 Idiom spotlight

Start with Mutex; switch to atomics when profiling shows lock contention on a hot counter or flag. Use **Relaxed** for metrics-only words; use **Release/Acquire** when another thread must see your **prior non-atomic writes**.

Same **Arc<Atomic*>** behind **OS threads** (Chapter 14) and **async tasks** (Chapter 16). Do not hand-roll lock-free queues without study.

15.11 Go deeper

- [The Rust Book — Fearless Concurrency](#) — mutex-first baseline; atomics go deeper below
- [Atomic types — lock-free primitives](#) — 452
- [Rust nomicon — atomics](#)

15.12 See also

- Chapter 14: Multithreading — Arc, Mutex, threads
- Chapter 16: Async — tasks sharing Arc<AtomicBool>
- Chapter 8: Errors and panic — join() after panic vs race bugs

15.12.1 Afterparty

Use these for lock-free structures and formal memory-model topics not covered above.

15.12.1.1 Concepts and placement

1. **Threads vs async atomics** — “Same Arc<AtomicBool> in thread::spawn vs tokio::spawn — what differs, what stays the same?”
2. **Data race vs logic race** — “Define both; classify lost static mut increment vs fetch_add.”
3. **Ch14 port** — “Rewrite Mutex counter (Ch14 L4) as AtomicUsize; when is each better?”

15.12.1.2 Memory ordering

4. **Ordering quiz** — “Shutdown flag + published config pointer — pick orderings; justify.”
5. **When Relaxed lies** — “Metrics OK, config handoff broken — show Release/Acquire fix.”
6. **Fence intuition** — “Draw happens-before for Release store + Acquire load (Level 4).”
7. **Relaxed polls vs version** — “Why is Relaxed OK for Level 2 polls but wrong for Level 4 version? One sentence each.”

15.12.1.3 CAS and compare_exchange

8. **Counter port** — “Cap counter with CAS loop; explain spurious compare_exchange_weak failure.”
9. **ABA problem** — “Explain ABA in 80 words for pointer compare_exchange — no full queue.”
10. **Retry loop** — “Show single-shot CAS anti-pattern; fix with loop.”
11. **Double-checked locking** — “Show broken lazy init with atomics; fix with OnceLock or explain why not hand-roll.”

15.12.1.4 Races and debugging

12. **Race quiz** — “Mark 6 snippets: safe atomic, UB static mut, ordering bug, needs Mutex.”
13. **Visibility story** — “Writer updates port then Relaxed flag — what can reader see?”

15.12.1.5 Atomics vs Mutex vs channels

14. **When not** — “Three cases atomics are the wrong tool; prefer channels or Mutex.”
15. **Mutex vs RwLock** — “Read-heavy sensor cache — atomic counter vs RwLock vs Mutex.”
16. **Vec push** — “Why many threads cannot push to shared Vec; three fixes.”
17. **Profile-first** — “Gateway uses Mutex for counter; profiler shows lock contention — minimal atomic refactor.”
18. **Channel vs atomic flag** — “When is crossbeam/bounded channel + shutdown better than lone AtomicBool?”

15.12.1.6 Production and Java port

19. **Gateway metrics** — “Design Gateway { polls, shutdown } for async; orderings per field.”
20. **False sharing** — “Two hot atomics on same cache line — problem and mitigation in 60 words.”
21. **Lock-free queue** — “Why this chapter says don’t hand-roll; name crates/patterns instead.”

Chapter 16

Async Rust and Tokio

16.1 Hook

Async/await shows up in many stacks (**Java** `CompletableFuture`, **Python** `asyncio`, JavaScript promises, ...). Rust feels familiar in shape, different in mechanics.

`async fn` returns a **Future** — lazy work until you `.await` or spawn it.

Tokio is the usual runtime for network and concurrent I/O.

16.2 Scope — a brief tour

Practical Tokio intro — not runtime internals or web-framework guides.

This chapter covers	Deferred to See also / Afterparty
<code>async / .await</code> , Future mental model	Pin/Unpin formalism, <code>async</code> trait impls in depth (Ch7)
Tokio: <code>#[tokio::main]</code> , <code>spawn</code> , <code>join!</code> , <code>select!</code> , <code>timeout</code>	<code>async-std</code> , embedded runtimes, runtime tuning
Blocking footgun + <code>spawn_blocking</code>	<code>block_in_place</code> , worker thread counts
Async I/O overview (<code>TcpListener</code> , <code>fs</code>)	Production TCP/Modbus stacks — Chapter 19
Shutdown via <code>Arc<AtomicBool></code> (Ch15 L6)	<code>axum</code> , <code>tonic</code> , stream combinators

16.2.1 What this chapter skips — and where to learn it

The table above is honest scope, not a teaser list. Each deferred row is safe to ignore until you hit the trigger in the right column:

Deferred topic	Learn it when...	Start here
Pin / Unpin	You implement a manual Future, build self-referential async state, or read compiler errors mentioning Pin	Rustonomicon — Pin ; return here after Chapter 7 traits
async fn in traits	A gateway trait needs async methods (connect, poll_device)	Chapter 7 — async fn in traits, Send bounds, dyn limits
Other runtimes / tuning	Tokio defaults starve your workload or you target no_std / embedded	async-std for API comparison; Embassy for embedded; Tokio tuning for worker thread counts
block_in_place	CPU-heavy or blocking work must run <i>on</i> an async worker thread, not the blocking pool	Tokio — block_in_place after you have measured executor latency (Level 5)
Production TCP / Modbus stacks	One echo connection works; you need timeouts, retry, framing, and logging in production	Chapter 19 sync I/O first, then port the poll loop async
axum, tonic, streams	Raw TcpListener + spawn is understood; you want HTTP/gRPC routes or stream pipelines	axum , tonic , tokio_stream combinators
tokio::sync::mpsc + cancellation hygiene	Tasks need async message passing, or select! drops work mid-flight	Tokio sync docs; Level 4 caveat + Cancellation hygiene in Afterparty

You do **not** need every row to ship a Level 6 supervisor. Add depth when a concrete project forces the question.

16.3 What async is

Async is cooperative concurrency: one OS thread hosts many **tasks**, each pausing at `.await` so the executor runs others. Unlike Chapter 14 threads

(OS-preempted), async tasks **yield voluntarily** — which makes blocking calls inside them dangerous.

The [Rust Book — Async, Await, Futures, and Streams](#) covers the std mental model: **futures**, **tasks**, and **executors**. This chapter applies that model with **Tokio** — a third-party runtime the book uses in examples but does not standardize.

Idea	Plain language
async fn	Returns a Future — a state machine describing work; does not run until polled
.await	Pause this task until the step is ready; the executor runs other tasks
Executor (Tokio)	Thread pool + scheduler that polls futures to completion
vs OS thread (Chapter 14)	Threads are preempted by the OS; async cooperates — a blocking call in one task can stall many
Shared state	Same concurrency rules: Arc, atomics (Chapter 15), tokio::sync::Mutex — not std::sync::Mutex held across .await

When a task hits **.await**, control returns to the executor until the waited-on step is ready:

```
caller .await --> executor runs other tasks -->
    future ready --> caller resumes
```

16.3.1 Level 0 — Mental model (sync stand-in, not async)

Before adding Tokio, anchor the **names** (fetch, parse, log) with ordinary sequential Rust. Real async needs a **runtime**; this Playground snippet is **not async** — it runs one step after another on one thread:

```
// Playground --- conceptual; NOT async (no
    executor)
fn main() {
    let tasks = vec!["fetch", "parse", "log"];
    for t in tasks {
        println!("step: {}", t);
```

```
    }
}
```

What happened: prints **step: fetch, parse, log** in order — one thread, no concurrency. An `async fn` would **not** interleave until `.await` yields to Tokio.

All Tokio examples below are **Cargo projects**. Add this dependency once:

```
[dependencies]
tokio = { version = "1", features = ["full"] }
```

16.4 Examples: elementary -> hard

The levels below build from “start a runtime” to a small async supervisor. Work through in order; after each snippet, read **what happened** before moving on.

16.4.1 Level 1 — Elementary: minimal Tokio spawn

Start here: boot Tokio with `#[tokio::main]`, spawn a child task, and **await** its join handle so `main` does not exit early.

```
// Cargo project
use tokio::time::{sleep, Duration};

#[tokio::main]
async fn main() {
    let handle = tokio::spawn(async {
        sleep(Duration::from_millis(50)).await;
        42
    });
    println!("result = {:?}", handle.await);
}
```

What happened:

- `#[tokio::main]` starts the Tokio runtime and runs `async fn main`.
- Child task sleeps 50 ms, returns **42**; `handle.await` waits for **Ok(42)**.
- Without `.await` on the handle, `main` could exit before the task finishes — dropped join handle **aborts** the task.

16.4.2 Level 2 — Elementary: sequential vs concurrent `.await`

Writing two `.awaits` in one function does **not** run them in parallel. Compare sequential awaits with `tokio::join!` to see real async concurrency:

```
// Cargo project
use tokio::time::{sleep, Duration, Instant};

async fn sequential() {
    sleep(Duration::from_millis(100)).await;
    sleep(Duration::from_millis(100)).await;
}

#[tokio::main]
async fn main() {
    let start = Instant::now();
    sequential().await;
    println!("sequential ~{} ms",
        start.elapsed().as_millis());

    let start = Instant::now();
    tokio::join!(
        sleep(Duration::from_millis(100)),
        sleep(Duration::from_millis(100)),
    );
    println!("join! ~{} ms",
        start.elapsed().as_millis());
}
```

What happened:

- **sequential** ~ 200 ms — second sleep starts after the first completes.
- **tokio::join!** ~ 100 ms — both sleeps run **concurrently** on the executor.
- For background work that outlives the caller, use `tokio::spawn` (Level 1).

16.4.3 Level 3 — Medium: **join!** and **timeout**

Services often wait on **multiple** device polls at once and must **give up** when hardware is slow. **join!** waits for all branches; **timeout** caps how long one future may run:

```
// Cargo project
use tokio::time::{sleep, timeout, Duration};

async fn fast_poll() -> u16 {
    sleep(Duration::from_millis(10)).await;
    502
}

async fn slow_poll() -> u16 {
    sleep(Duration::from_millis(500)).await;
    8080
}

#[tokio::main]
async fn main() {
    let (a, b) = tokio::join!(fast_poll(),
                             fast_poll());
    println!("join ports {} {}", a, b);

    match timeout(Duration::from_millis(50),
                  slow_poll()).await {
        Ok(port) => println!("slow ok {}",
                             port),
        Err(_) => eprintln!("slow poll timed
                             out"),
    }
}
```

What happened:

- **join!** runs both **fast_poll** tasks concurrently -> prints **502 502** quickly.
- **timeout(50ms, slow_poll())** returns **Err(_)** — slow poll needs 500 ms. This is the **deadline** pattern for Modbus/serial timeouts.

16.4.4 Level 4 — Medium: **select!** — first branch wins

When you want the **fastest** response and can abandon slower paths, **select!** races branches and **cancels** the losers:

```
// Cargo project
use tokio::time::{sleep, Duration};

async fn fast_cache_hit() -> &'static str {
    sleep(Duration::from_millis(5)).await;
    "cached"
}

async fn slow_device_read() -> &'static str {
    sleep(Duration::from_millis(200)).await;
    "device"
}

#[tokio::main]
async fn main() {
    let source = tokio::select! {
        v = fast_cache_hit() => v,
        v = slow_device_read() => v,
    };
    println!("source = {}", source);
}
```

What happened:

- Prints **source = cached** — fast branch completes first; **slow branch is cancelled** (dropped future).
- **Caveat:** cancelled work may stop mid-flight — ensure partial I/O is safe; see Afterparty for cancellation hygiene.

16.4.5 Level 5 — Hard: **blocking footgun + spawn_blocking**

Tokio runs many async tasks on a **small** set of OS worker threads. When a task hits `.await`, it **steps aside** so the worker can run other tasks. **Blocking** code never steps aside — it holds the worker until it finishes.

What you call	What actually happens
<code>thread::sleep(100ms)</code> inside <code>async fn</code>	Worker thread is stuck for 100 ms. Every other task on that worker waits .
<code>tokio::time::sleep(100ms).await</code>	Task releases the worker. Other tasks run during the wait.
<code>std::fs::read</code> / heavy CPU on hot path	Same as <code>thread::sleep</code> — blocks the worker.
<code>tokio::task::spawn_blocking(...)</code>	Moves blocking work to a separate thread pool. Async workers stay free.

Restaurant analogy: one waiter (worker) serves many tables (tasks). `.await` = “I’ll check back when the kitchen is ready.” `thread::sleep` = waiter stands frozen at one table while others go ignored.

Under load, a few `thread::sleep` or `std::fs::read` calls in `async` handlers can slow **every** connection — not just the one that blocked.

The snippet below runs two experiments with `tokio::join!`: first with `std::thread::sleep` (bad), then with `tokio::time::sleep` (good), then `spawn_blocking` for work you cannot make `async`.

```
// Cargo project
use std::thread;
use std::time::Duration as StdDuration;
use tokio::time::{sleep, Duration, Instant};

async fn bad_blocking() {
    thread::sleep(StdDuration::from_millis(100))
        ; // stalls executor worker
}

async fn good_async_sleep() {
    sleep(Duration::from_millis(100)).await;
    // yields --- other tasks run
}

#[tokio::main]
async fn main() {
    let start = Instant::now();
    tokio::join!(bad_blocking(),
```

```
        sleep(Duration::from_millis(10));
println!("bad path elapsed ~{} ms (10 ms
        task delayed)",
        start.elapsed().as_millis());

let start = Instant::now();
tokio::join!(good_async_sleep(),
        sleep(Duration::from_millis(10)));
println!("good path elapsed ~{} ms",
        start.elapsed().as_millis());

let n = tokio::task::spawn_blocking(|| {
    thread::sleep(StdDuration::from_millis(5
        0));
    99
});
.await
.unwrap();
println!("spawn_blocking -> {}", n);
}
```

Quick reference for what belongs inside an async task:

Call	In async task?	Effect
std::thread::sleep	Bad	blocks executor worker — other tasks wait
tokio::time::sleep().await	Good	yields until deadline
std::fs::read on hot path	Risky	blocks — prefer tokio::fs or spawn_blocking
tokio::task::spawn_blocking	Good	runs closure on blocking thread pool

What happened:

- **bad_blocking + 10 ms task** — prints elapsed ~ **100+ ms**. The 10 ms sleep **started late** because `thread::sleep` hogged the worker. Two “concurrent” tasks ran **one after another**.
- **good_async_sleep + 10 ms task** — prints elapsed ~ **100 ms**. Both tasks **shared** the worker during waits; wall time is ~max(100, 10), not 100 + 10.

- **spawn_blocking** — prints **spawn_blocking -> 99**. The 50 ms `thread::sleep` ran on a **blocking** thread; `async` workers were not stuck.

Rule of thumb: in Chapter 14, `thread::sleep` on its own OS thread only blocks **that** thread. Inside Tokio async code, it blocks a **shared** worker — so it hurts **all** tasks on that worker.

16.4.6 Level 6 — Hard: async gateway supervisor

This mirrors Chapter 15 Level 6, but the worker is a **Tokio task** instead of an OS thread:

Piece	Role
Gateway	Shared state: poll counter + shutdown flag
Arc<Gateway>	Lets main and the worker both hold the same struct safely
run_worker	Loop: sleep 5 ms -> increment polls -> repeat until shutdown
tokio::spawn	Starts the worker in the background without blocking main
shutdown flag	main sets true; worker sees it and exits the loop cleanly

main is the supervisor; **run_worker** is the poll loop on a device gateway. Supervisor runs for 25 ms, prints poll count, then stops the worker and waits for a clean exit.

```

main                                worker (spawned
    task)
|                                  |
+- create Arc<Gateway>            |
+- spawn run_worker ----->| loop: if
    !shutdown
|                                  | sleep
    5ms (.await --- yields!)
|                                  | polls
    += 1
+- sleep 25ms (.await)            | (keeps
    looping. . . )
+- print polls count              |

```

```

+- shutdown = true ----->| sees
    shutdown, exits loop
+- await join handle <-----+ returns
    Ok(())
+- print "supervisor done"      |

```

Why atomics again? Same as Chapter 15: `polls` is a **metric** (Relaxed is fine). `shutdown` is a **signal** — Release when main writes, Acquire when the worker reads.

Why `sleep(...).await` in the worker? Each 5 ms pause **yields** to Tokio. `thread::sleep` here would block a worker on every iteration (Level 5 footgun).

```

// Cargo project
use std::sync::atomic::{AtomicBool,
    AtomicUsize, Ordering};
use std::sync::Arc;
use tokio::time::{sleep, Duration};

struct Gateway {
    polls: AtomicUsize,
    shutdown: AtomicBool,
}

async fn run_worker(gw: Arc<Gateway>) ->
    Result<(), &'static str> {
    while !gw.shutdown.load(Ordering::Acquire) {
        sleep(Duration::from_millis(5)).await;
        gw.polls.fetch_add(1,
            Ordering::Relaxed);
    }
    Ok(())
}

#[tokio::main]
async fn main() -> Result<(), &'static str> {
    let gw = Arc::new(Gateway {
        polls: AtomicUsize::new(0),
        shutdown: AtomicBool::new(false),

```

```

});

let worker_gw = Arc::clone(&gw);
let handle = tokio::spawn(async move {
    run_worker(worker_gw).await });

sleep(Duration::from_millis(25)).await;
println!("metrics polls={}",
    gw.polls.load(Ordering::Relaxed));

gw.shutdown.store(true, Ordering::Release);
handle.await.map_err(|_| "worker join
    failed")??;
println!("supervisor done");
Ok(())
}

```

What happened (summary):

1. Worker polls concurrently with main while supervisor sleeps 25 ms.
2. **polls.load(Relaxed)** prints something like **metrics polls=4** (~25 ms / 5 ms per loop).
3. **shutdown.store(true, Release)** stops the worker after the current iteration.
4. **handle.await** waits for the task to finish. The **??** applies ? to both the join result and **run_worker**'s Result.
5. **supervisor done** — no dangling background task.

Compared to Chapter 15: replace `thread::spawn + thread::sleep` with `tokio::spawn + tokio::time::sleep().await`. Atomics and shutdown pattern stay the same — only the runtime changes.

main() -> Result (Chapter 8): errors bubble to one place instead of scattering `unwrap()` in the supervisor.

16.4.7 Level 7 — Medium: bounded concurrency with Semaphore

Unbounded `tokio::spawn` on thousands of devices can overwhelm databases or serial buses. A **Semaphore** caps how many in-flight operations run at once:


```
// Cargo project
use std::sync::Arc;
use tokio::sync::Semaphore;
use tokio::time::{sleep, Duration};

async fn probe_device(id: u32) -> u32 {
    sleep(Duration::from_millis(20)).await;
    id
}

#[tokio::main]
async fn main() {
    let sem = Arc::new(Semaphore::new(2));
    // at most 2 probes at once
    let mut handles = Vec::new();

    for id in 0..5u32 {
        let permit = Arc::clone(&sem);
        handles.push(tokio::spawn(async move {
            let _permit =
                permit.acquire().await.unwrap();
            // held until task finishes
            probe_device(id).await
        })));
    }

    for h in handles {
        println!("id {}", h.await.unwrap());
    }
}
```

What happened:

- Five tasks are spawned, but **acquire().await** blocks when two permits are already taken.
- The **_permit** guard releases the slot when the task ends — no manual release.
- Same pattern protects MongoDB cursors, HTTP fan-out, and Modbus batch reads.

16.4.8 Level 8 — Medium: batch spawn and `await??`

Fan out independent async work with `tokio::spawn`, then collect results. Each `handle.await` yields `Result<T, JoinError>` — use `??` when the inner task also returns `Result` (Chapter 8):

```
// Cargo project
use tokio::time::{sleep, Duration};

async fn fetch_reading(seed: u32) ->
    Result<f64, &'static str> {
    sleep(Duration::from_millis(10)).await;
    if seed == 3 {
        Err("device offline")
    } else {
        Ok(seed as f64 * 1.5)
    }
}

#[tokio::main]
async fn main() -> Result<(), Box<dyn
    std::error::Error>> {
    let seeds = [1u32, 2, 3];
    let mut handles = Vec::new();

    for seed in seeds {
        handles.push(tokio::spawn(fetch_reading(
            seed)));
    }

    for handle in handles {
        match handle.await? {
            Ok(v) => println!("reading {}", v),
            Err(e) => eprintln!("skip: {}", e),
        }
    }
    Ok(())
}
```

Layer	Type	? / ??
Outer	JoinError from panicked/cancelled task	first ? on handle.await
Inner	Result<T, E> from your async fn	second ? when you want to propagate

Use **match** on the inner **Result** when one failure should not abort the whole batch (as above for seed 3).

16.4.9 Pipeline sketch — reader task, channel, workers

Long-running services decouple **read throughput** from **write latency** with bounded channels. One task pulls documents/frames; workers consume batches:

```
// Cargo project --- outline
use tokio::sync::mpsc;

async fn reader_task(tx: mpsc::Sender<Vec<u8>>())
    -> Result<(), String> {
    for i in 0u8..3 {
        tx.send(vec![i]).await.map_err(|e|
            e.to_string())?;
    }
    Ok(())
}

async fn worker_task(mut rx:
    mpsc::Receiver<Vec<u8>>()) -> Result<(),
    String> {
    while let Some(batch) = rx.recv().await {
        println!("process {} bytes",
            batch.len());
    }
    Ok(())
}

#[tokio::main]
async fn main() -> Result<(), String> {
```

```

let (tx, rx) = mpsc::channel(4);
// backpressure when full
let worker = tokio::spawn(worker_task(rx));
reader_task(tx).await?;
// dropping tx closes the channel
worker.await.unwrap()?;
Ok(())
}

```

Bounded capacity (4 here) applies backpressure: a fast reader blocks on `send().await` when workers fall behind — preferable to unbounded memory growth.

16.4.10 Polling loop — long-running HTTP or job status

APIs that return **202 + poll URL** fit a **loop**, **sleep().await**, and exit on success or timeout:

```

// Cargo project
use tokio::time::{sleep, timeout, Duration,
Instant};

async fn job_ready(tick: u32) ->
Option<&'static str> {
    if tick >= 3 {
        Some("done")
    } else {
        None
    }
}

async fn poll_until_ready(max_wait: Duration)
-> Result<&'static str, &'static str> {
    let start = Instant::now();
    let mut tick = 0;
    loop {
        if let Some(status) =
            job_ready(tick).await {

```

```
        return Ok(status);
    }
    if start.elapsed() >= max_wait {
        return Err("timed out waiting for
                    job");
    }
    tick += 1;
    sleep(Duration::from_millis(50)).await;
}
}

#[tokio::main]
async fn main() {
    match timeout(Duration::from_secs(1),
        poll_until_ready(Duration::from_secs(5))
    ).await {
        Ok(Ok(s)) => println!("status = {}", s),
        Ok(Err(e)) => eprintln!("{}", e),
        Err(_) => eprintln!("outer timeout"),
    }
}
```

Wrap the whole poll in `timeout(...)` (Level 3) so a stuck remote cannot loop forever.

16.5 Techniques at a glance

Use this table as a cheat sheet while reading Tokio code elsewhere. Each row maps to a level above.

Technique	One-line use	Level
<code>async fn / .await</code>	lazy futures; yield points	0-2
<code>#[tokio::main]</code>	start Tokio runtime	1
<code>tokio::spawn</code>	concurrent background task	1, 6, 8
<code>tokio::join!</code>	wait for all branches	2-3
<code>tokio::select!</code>	first ready branch	4
<code>tokio::time::timeout</code>	deadline / device timeout	3, poll loop
<code>spawn_blocking</code>	sync I/O or <code>thread::sleep</code> off executor	5

Technique	One-line use	Level
Semaphore	cap in-flight async work	7
mpsc channel	decouple reader and workers	pipeline sketch
async_trait	async methods on dyn Trait	testing boundary
Arc<AtomicBool>	shutdown flag across tasks	6, Ch15

Mutex choice: use `tokio::sync::Mutex` if the lock may be held across `.await`. Holding `std::sync::MutexGuard` across `.await` makes the future **!Send** — the compiler rejects it (see edge cases below).

16.6 Async I/O (brief)

Networking and file I/O are where async pays off: many connections wait on kernel I/O while a small thread pool keeps working. Tokio mirrors std I/O with **async** APIs — `.await` instead of blocking the thread:

Sync (Ch19)	Async (Tokio)
<code>std::fs::read</code>	<code>tokio::fs::read</code>
<code>TcpListener::accept</code>	<code>tokio::net::TcpListener::accept().await</code>

The sketch below accepts one connection, reads bytes, and echoes them back — a minimal pattern before a multi-connection server:

```
// Cargo project --- outline
use tokio::io::{AsyncReadExt, AsyncWriteExt};
use tokio::net::TcpListener;

#[tokio::main]
async fn main() -> Result<(), Box<dyn
    std::error::Error>> {
    let listener =
        TcpListener::bind("127.0.0.1:0").await?;
    let (mut socket, _) =
        listener.accept().await?;
    let mut buf = [0u8; 64];
    let n = socket.read(&mut buf).await?;
    socket.write_all(&buf[..n]).await?;
```

```
Ok(())
}
```

Many connections -> **one process, many tasks** — each accept can `tokio::spawn` a handler. Full protocols and error handling: Chapter 19 + Afterparty.

16.7 Async vs OS threads

You already have OS threads from Chapter 14. Async is not a replacement — it is a different tool for **many concurrent waits** on a small thread pool:

Aspect	OS threads (Ch14)	Async tasks (Tokio)
Best for	CPU-bound parallel work	Many I/O waits (TCP, timers)
Stack / memory	higher per thread	many tasks on thread pool
Blocking call	blocks one thread only	blocks executor worker — hurts all tasks on that worker
Shared shutdown	<code>Arc<AtomicBool></code> (Ch15)	same atomics behind <code>tokio::spawn</code>

When choosing for a device gateway or network service:

- many idle TCP/serial waits? -> async
- CPU-bound parallel compute? -> threads (or rayon --- Ch14 Afterparty)
- one simple PLC poll loop? -> single thread may suffice
- 1000 connections, one process? -> async

16.8 Edge cases and compiler traps

These mistakes show up constantly in real Tokio code. The table lists the symptom and the fix; the snippets below are **wrong on purpose** so you recognize them in the wild.

Trap	Symptom	Idiom
Forgot <code>.await</code>	logic never runs; unused future warning	always <code>.await</code> or <code>spawn</code>
No runtime	<code>async fn main</code> won't run futures	<code>#[tokio::main]</code>
<code>std::thread::sleep</code> in <code>async</code>	latency spikes for all tasks	<code>tokio::time::sleep / spawn_blocking</code>
<code>MutexGuard</code> across <code>.await</code>	Future not Send compile error	<code>tokio::sync::Mutex</code> ; drop guard before <code>await</code>
Blocking <code>std::fs</code> on hot path	executor starvation	<code>tokio::fs / spawn_blocking</code>
<code>select!</code> cancels losing branch	partial work / leaked state	scope cancellation; abort handles — Afterparty

Wrong — `std::sync::MutexGuard` across `.await`:

```
// Cargo project --- does not compile
use std::sync::Mutex;
use tokio::time::{sleep, Duration};

async fn bad(m: Mutex<i32>) {
    let _guard = m.lock().unwrap();
    sleep(Duration::from_millis(10)).await;
    // ERROR: future cannot be sent between
    //       threads safely (MutexGuard not Send)
}

// fn main() {
//     tokio::runtime::Runtime::new().unwrap().block_on(
//         bad(Mutex::new(0));
//     );
// }
```

Wrong — drop future without `.await`:

```
// Cargo project
async fn important() {
    println!("never runs if not awaited");
}

#[tokio::main]
async fn main() {
    important();
}
```



```
// warning: unused Future --- does NOT run
// important().await; // correct
}
```

16.9 Idiom spotlight

One paragraph to carry into production code:

Async for many I/O waits; threads for CPU-heavy or a single simple poll loop. Never block the executor — `tokio::time::sleep`, not `thread::sleep`. Share shutdown with `Arc<AtomicBool>` (Chapter 15). Bubble errors with `?` and `main() -> Result` at the boundary (Chapter 8).

Cap fan-out with Semaphore. Decouple stages with `bounded mpsc. async_trait` for swappable I/O in tests.

16.10 Go deeper

When this chapter's ladder is not enough, these links cover fundamentals, channels, I/O, and the official Tokio walkthrough:

- [The Rust Book — Async, Await, Futures, and Streams](#)
- [async fn and .await fundamentals](#) — 321
- [Async channels mpsc](#) — 328
- [Async I/O](#) — 342, 921
- [Tokio tutorial](#)

16.11 See also

Related chapters — read these when you need threads, atomics, errors, sync I/O, or async traits in depth:

- Chapter 14: Multithreading — when threads beat async
- Chapter 15: Atomics — `Arc<AtomicBool>` shutdown in async
- Chapter 8: Errors — `Result` boundary in `main`
- Chapter 19: I/O — sync I/O and processes
- Chapter 7: Traits — async traits (advanced)

16.11.1 Afterparty

Use these for runtime internals and web-framework topics not covered above.

16.11.1.1 Concepts and mental model

Prompts to solidify how futures, polling, and scope fit together:

1. **Future diagram** — “Draw state machine for `async fn` with two `.await` points.”
2. **Poll vs await** — “Who polls the Future — caller, executor, or Tokio runtime?”
3. **Executor vs thread** — “One paragraph: cooperative async vs pre-emptive OS threads.”
4. **Forgotten await** — “Show unused Future bug; fix with `.await` or `spawn`.”

16.11.1.2 Tokio basics

Runtime setup, spawn lifecycle, and error boundaries:

5. **Tokio scaffold** — “Minimal `#[tokio::main] + spawn + join handle` — explain each line.”
6. **Spawn vs await** — “What if main drops join handle without `.await`?”
7. **Join handle ??** — “Explain the ?? on `handle.await` in Level 6 — outer join Result vs inner `run_worker` Result.”

16.11.1.3 Level 5-6 drills

Reinforce blocking vs yielding and the async gateway supervisor:

8. **Bad join timing** — “Walk through Level 5 bad path: why does a 10 ms task wait ~100 ms when paired with `thread::sleep`?”
9. **Ch15 port** — “Compare Level 6 to Ch15 L6 — what changes with `tokio::spawn`, what stays the same?”
10. **Supervisor timeline** — “Trace Level 6 step by step: when does the worker stop, and why must it use `tokio::time::sleep` not `thread::sleep`?”

16.11.1.4 Concurrency primitives

When to wait for all vs first, deadlines, and cancellation:

11. **join vs select** — “Two slow tasks: when join! vs select!? One automation example each.”
12. **select! scenario** — “Cancel slow request when fast path returns — full select! sketch.”
13. **Timeout drill** — “Wrap Modbus read `async fn` with 100 ms timeout; handle `Elapsed`.”
14. **Cancellation hygiene** — “What runs when `select!` drops the losing branch? Modbus read vs cache hit example.”

16.11.1.5 Blocking and I/O

Keeping the executor healthy while doing real device and file work:

15. **Blocking fix** — “Audit `async` snippet with `thread::sleep`, `std::fs::read`; fix each.”
16. **spawn_blocking** — “When `tokio::fs` vs `spawn_blocking` for config file read?”
17. **Tcp echo** — “Expand Async I/O outline to multi-connection echo with `spawn` per `accept`.”

16.11.1.6 Async vs threads

Architecture choices for gateways and connection counts:

18. **async vs thread** — “1000 Modbus polls — argue `async` vs `thread` pool for latency.”
19. **200 TCP connections** — “One process — `async` vs `thread`-per-connection memory story.”
20. **When thread enough** — “Single serial port poll — justify `thread` loop over `Tokio`.”

16.11.1.7 Atomics and shared state (Ch15)

Bridging lock-free flags from Chapter 15 into `async` tasks:

21. **Async sleep in worker** — “Why must Level 6’s poll loop use `tokio::time::sleep().await` on every iteration, not `thread::sleep`?”
22. **tokio Mutex** — “Rewrite bad `std::sync::MutexGuard` across `await` with `tokio::sync::Mutex`.”

Part III

Metaprogramming and Unsafe

Chapter 17

Metaprogramming

17.1 Hook

Metaprogramming varies by ecosystem — **Java** annotations and reflection, **Python** metaclasses, Rust macros at compile time. Rust metaprogramming is mostly **compile-time**. Macros expand source **before** type checking — zero runtime cost for `macro_rules!` and `derive`.

Approach	When it runs	Typical cost
Java reflection	Runtime	Slow; errors at runtime
Lombok / annotation processors	Compile time	IDE-dependent
Python metaclass	Import / class creation	Runtime setup
Rust <code>macro_rules!</code> / <code>derive</code>	Compile time	No runtime overhead
Rust <code>const fn</code> / <code>const</code>	Compile time	Baked into binary

17.2 Scope — a brief tour

Intro to `macro_rules!`, `derive`, and `const` — not proc-macro authoring with `syn/quote`.

This chapter covers	Deferred to See also / Afterparty
<code>macro_rules!</code> , fragments, repetition, hygiene	Writing proc macros with <code>syn/quote</code>

This chapter covers	Deferred to See also / Afterparty
#[derive] syntax, annotations vs decorators, std + ecosystem + custom derives, Clone/Copy , serde JSON round-trip	Proc-macro authoring with syn/quote, build.rs codegen
const / const fn / env! / include_str!	generic_const_exprs, advanced const traits
Forbidden matcher clauses, edge cases , why macros are hard to trace	Miri, macro fuzzing, proc-macro testing

17.3 Compile-time pipeline

Mental model: the compiler sees your source as **tokens**, expands every macro invocation, then type-checks the **expanded** program.

source tokens -> macro expansion -> type check
 + borrow check -> LLVM

Macros match **syntax trees** (tokens), not types. A macro can compile while the **expanded** code fails. Errors often say in expansion of macro

17.4 Under the hood — what the compiler actually does

Brief peek inside — enough to make restrictions and debugging feel logical:

[Diagram omitted in print edition — see the web repository.]

Declarative (macro_rules!):

- The **matcher** compares token trees against patterns with fragment specifiers (expr, ty, ...).
- The **transcriber** pastes captured fragments into a template — substitution, not evaluation of Rust code.
- No access to types, trait bounds, or name resolution at match time — only token shape.

Procedural ([derive], sql!, ...):

- The compiler calls a proc-macro function with a **TokenStream** (tokens + span metadata).

- The macro runs as **normal Rust at compile time** (often using `syn + quote`) and returns another `TokenStream`.
- A **proc-macro = true crate** may export **only** proc macros. Helpers live in a sibling crate — a hard platform rule.

Hygiene (why names “disappear”):

- Macro-generated identifiers get a fresh **syntax context**. They do not collide with caller locals, but also **won’t resolve** to caller bindings unless passed as `$x:expr`.
- `$crate` exists so exported macros refer to **their** crate’s paths, not the caller’s.

17.5 macro_rules!

```
// Playground
macro_rules! say_hello {
    () => { println!("Hello!"); };
    ($name:expr) => { println!("Hello, {}!",
                             $name); };
}

fn main() {
    say_hello!();
    say_hello!("Automation");
}
```

What happened: the macro matches **token patterns** — empty `()` or one expression — and expands to `println!` calls before type checking.

17.5.1 Fragment specifiers

Specifier	Matches	Use when
<code>expr</code>	expression	values, calls, literals
<code>ident</code>	identifier	names, field keys
<code>ty</code>	type	generic params, type slots
<code>pat</code>	pattern	match arms in macro-generated code
<code>stmt</code>	statement (no trailing <code>;</code>)	one-line statements
<code>block</code>	<code>{ ... }</code> block	multi-statement bodies

Specifier	Matches	Use when
item	fn, struct, mod, ...	generating whole items
literal	42, "hi", true	numeric / string constants
tt	token tree	escape hatch — match almost anything

17.5.2 Repetition

```
// Playground
macro_rules! vec_str {
    ($($x:expr),*) => {{
        let mut v = Vec::new();
        $( v.push(String::from($x)); )*
        v
    }};
}

fn main() {
    let v = vec_str!["a", "b"];
    println!("{:?}", v);
}
```

What happened:

- `$(...)*` repeats zero or more times — `vec_str![]` would work with `*`, not with `+`.
- `$( ... )*` in the output expands once per matched input.
- Trailing comma: `$(x),*` allows `"a", "b"`, — common Rust style.

Modifier	Meaning
*	zero or more
+	one or more
?	zero or one

17.5.3 Level 2 — automation: register map

Map symbolic register names to addresses — typical Modbus-style tables:


```
// Playground
macro_rules! register_map {
    ($($name:ident = $addr:literal),* $(,)? ) =>
    {{
        fn addr_of(name: &str) -> Option<u16> {
            match name {
                $( stringify!($name) =>
                    Some($addr), )*
                _ => None,
            }
        }
        addr_of
    }};
}

fn main() {
    let lookup = register_map!(temp = 0x01,
        pressure = 0x02);
    println!("{:?}", lookup("temp"));
}
```

What happened: `ident` captures names; `literal` captures `u16` values; repetition builds a match over `stringify!` keys — all at compile time, zero runtime parsing of a table file.

Reading the matcher — `($($name:ident = $addr:literal),* $(,)? ) => {{:`

`register_map!(temp = 0x01, pressure = 0x02)`
 `+- one repetition +- +-`
 `second +-`

Piece	Meaning
<code>\$(...),*</code>	Repeat zero or more times, separated by commas
<code>\$name:ident</code>	Capture an identifier token (<code>temp</code> , <code>pressure</code>)
<code>=</code>	Literal <code>=</code> in the input — not a capture
<code>\$addr:literal</code>	Capture a literal (<code>0x01</code> , <code>0x02</code> , <code>502</code> , ...)
<code>\$(,)?</code>	Optional trailing comma after the last entry (<code>temp = 0x01</code> , is OK)

Piece	Meaning
<code>=> {{ ... }}</code>	Expand to a block that defines <code>addr_of</code> and returns that function

For `register_map!(temp = 0x01, pressure = 0x02)`, the macro expands roughly to:

```
{
  fn addr_of(name: &str) -> Option<u16> {
    match name {
      "temp" => Some(0x01),
      // stringify!(temp) + $addr
      "pressure" => Some(0x02),
      _ => None,
    }
  }
  addr_of
  // block's value --- caller gets the lookup
  // fn
}
```

Each `$( stringify!($name) => Some($addr), )*` in the macro body runs **once per** `name = addr` pair in the invocation — that is how one macro call generates many match arms.

17.5.4 Hygiene and `$crate`

Exported macros should use `$crate::` for paths inside **your** crate so re-exports and dependency trees resolve correctly. Macro-generated `let` names are **hygienic** — they won't accidentally shadow or capture caller locals.

Debug expanded code:

```
cargo install cargo-expand
cargo expand --bin my_app # in a Cargo project
```

Conceptual `cargo expand` on `#[derive(Debug)]` for `Point` emits an `impl Debug for Point { fn fmt(...) { ... } }` — the `derive` proc macro wrote that `impl`; you never see it in source unless you expand.

17.6 Syntax forbidden or restricted in macros — and why

The macro matcher is a **separate, weaker parser** that must stay **unambiguous today and in future Rust editions** (follow-set rules).

Important nuance: follow-set rules limit what can follow `$e:expr` in a **matcher** — not what a macro can **emit**. You can still generate `for` loops in expanded code.

What is restricted: (1) keywords **after** `$e:expr` / `$s:stmt` in a matcher, and (2) expanding **multiple statements** where Rust expects a **single expression**.

17.6.1 Follow-set: keywords after `expr` / `stmt`

For `$e:expr` and `$s:stmt`, the only tokens allowed **immediately after** the fragment in a matcher are `=>`, `,`, and `;`. Nothing else — not `=`, not `for`, not `let`, not `if`, not `>>`.

Fragment	May only be followed by	Forbidden example	Why
<code>expr, stmt</code>	<code>=>, ,, ;</code>	<code>\$a:expr for \$i:ident in \$r:expr</code>	<code>for</code> could start a trailing expression — parser won't commit
<code>expr, stmt</code>	<code>=>, ,, ;</code>	<code>\$a:expr let \$x:ident = \$v:expr</code>	same follow-set rule
<code>expr, stmt</code>	<code>=>, ,, ;</code>	<code>\$a:expr >> \$b:expr</code>	<code>>></code> could be part of a future expression
<code>pat, pat_param</code>	<code>=>, ,, =, \, , if, in, ...</code>	<code>\$p:pat + \$q:pat</code>	<code>+</code> not in follow-set — but <code>in</code> is allowed
<code>ty, path</code>	<code>=>, ,, =, \, , ;;, :, >, >>, '{, {, as, where, ... \$t:ty + \$u:ty '+' might start a different grammar production</code>		

Wrong — `$e:expr` followed by `for` (matcher fails at macro definition):

```
// Playground --- does not compile
macro_rules! bad_foreach_matcher {
    ($e:expr for $i:ident in $r:expr) => { $e };
}
// error: ` $e:expr ` is followed by `for`, which
//       is not allowed for `expr` fragments
// note: allowed there are: `=>`, `,` or `;`
```

Wrong — `$e:expr` followed by `=`:

```
// Playground --- does not compile
macro_rules! set_reg {
    ($addr:expr = $val:expr) => { /* ... */ };
}
// error: ` $addr:expr ` is followed by `=`,
//       which is not allowed for `expr` fragments
// FIX: set_reg!(0x01, 42) or set_reg! { 0x01
//       => 42 }
```

Right — `$p:pat` followed by `in` (foreach-style matcher):

The keyword `in` is in the follow-set for `pat` / `pat_param`, so this pattern is **legal**. Literal `for` comes *before* the pattern, not after an `expr`:

```
// Playground
macro_rules! foreach {
    ($p:pat in $r:expr => $body:expr) => {
        for $p in $r { $body }
    };
}

fn main() {
    foreach!(i in 0..3 => println!("tick {}"),
            i);
}
```

What happened: the matcher uses `$p:pat in $r:expr` — allowed because `in` follows `pat`. The transcriber emits a normal `for` loop. This is how you write “mini `foreach`” macros without breaking follow-set rules.

17.6.2 Expression position vs statement expansion (**let**, **for**, multiple lines)

A macro invoked where Rust expects an **expression** (`let x = mac!();`, `mac!() + 1`) must expand to **one** expression. A bare `{ let a = 1; for i in 0..n { ... } }` block is a **block expression** and can work.

A transcribe template with **multiple statements** separated at the top level often breaks.

Wrong — `let` + `for` in expression position (single-brace body):

```
// Playground --- does not compile when used as
// expression
macro_rules! poll_twice {
    () => {
        let n = 2;
        for i in 0..n {
            println!("poll {}", i);
        }
    };
}

fn main() {
    let _x = poll_twice!();
    // error: expected expression, found `let`
    // statement
    // note: macro expansion ignores keyword
    // `for` and any tokens following
}
```

Fix — double braces `{{ ... ; last_expr }}`:

```
// Playground
macro_rules! poll_twice {
    () => {{
        let n = 2;
        for i in 0..n {
            println!("poll {}", i);
        }
    }}
    ()
    // block's value --- unit, or return a
```

```
        // count/Result
    });
}

fn main() {
    let _x = poll_twice!();
}
```

What happened: outer { ... } is the macro’s transcribe delimiter. Inner { ... ; () } is a **block expression** — one value Rust can assign to `_x`. Without the inner block, `let` and `for` are loose statements, invalid in expression position.

Situation	Works?	Pattern
for loop only, used as expression	often yes (loop is an expression)	=> { for ... { } }
let then for, used as expression	no (single braces)	=> {{ let ...; for ...; value }}
for after <code>\$e:expr</code> in matcher	never	use <code>\$p:pat</code> in <code>\$r:expr</code> or change DSL punctuation
for in macro output at statement position	yes	<code>poll_twice!();</code> as its own statement

Same logic applies to **let**, **while**, and **match** when they appear **after** an `expr` fragment in a matcher. It also applies to **let-heavy expansions** in expression context.

Other important limits:

Rule	What breaks	Why
Forwarded fragments are opaque	Inner macro can’t match literals inside outer <code>\$e:expr</code>	Fragment is an AST blob
No type introspection in <code>macro_rules!</code>	“Derive Clone only for Clone fields”	Macro sees tokens, not types — needs <code>proc macro</code>
Recursion depth cap (~128)	recursion limit exceeded	Protects compile time
ident excludes <code>\$crate</code>	Reserved for path injection	Hygiene

Escape hatch: `tt` (token tree) matches delimited chunks — **you** validate structure (often in a `proc macro` with `syn`).

17.7 Standard library macros you already use

Macro	Role
<code>vec!</code> , <code>format!</code> , <code>println!</code>	collection / formatting
<code>matches!</code>	pattern guard sugar ([Chapter 6 — Advanced patterns])
<code>env!</code> , <code>option_env!</code>	build-time strings (Preface)
<code>include_str!</code> , <code>concat!</code> , <code>stringify!</code>	embed / stringify at compile time
<code>cfg!</code> , <code>assert!</code> , <code>debug_assert!</code>	conditional compile / tests
<code>todo!</code> , <code>unimplemented!</code>	stub markers

```
// Playground
fn main() {
    let name = env!("CARGO_PKG_NAME");
    // compile-time; needs Cargo project
    let msg = format!("running {}", name);
    assert!(matches!(msg.as_str(), s if
        s.starts_with("running")));
    println!("{}", msg);
}
```

17.8 Derive attributes

`#[derive(...)]` is **core Rust syntax** — not optional decoration. You place it **above** a `struct` or `enum` to auto-implement one or more **traits** before type checking runs. You will see it from the first error enum (Chapter 3) through config types (Chapter 13), `thiserror` (Chapter 8), and `serde` below.

```
// Playground
#[derive(Debug, Clone, PartialEq)]
struct Point {
    x: i32,
    y: i32,
}

fn main() {
    let a = Point { x: 1, y: 2 };
}
```

```
let b = a.clone();
println!("{:?} == {:?} ? {}", a, b, a == b);
}
```

Comma-separated trait names in one attribute: `#[derive(Debug, Clone)]`. Field-level helpers (e.g. `#[serde(rename = "...")]`) sit on individual fields and are interpreted by the same ecosystem crate — see Serde JSON below.

17.8.1 Not annotations or decorators

If you come from **Java**, **Python**, **C#**, or **TypeScript**, do **not** read `#[derive]` as an annotation or decorator. The mental model is different.

Habit from other languages	What you might assume	What Rust <code>#[derive]</code> actually does
Java <code>@Override</code> , <code>@SuppressWarnings</code>	Hints to compiler / IDE	Generates Rust source (impl blocks) at compile time
Java Spring <code>@Autowired</code> , JAX-RS annotations	Runtime wiring via reflection	No runtime reflection — expanded code is normal Rust
Python <code>@property</code> , <code>@staticmethod</code>	Wraps or replaces the function at call time	Does not wrap your type — emits trait impls once at compile time
Python <code>@dataclass</code>	Runtime class builder	Closest cousin — but Rust expand is before type check, zero runtime cost
C# / TypeScript decorators	Metadata or method interception at runtime	Rust derive is not interception — it is codegen
Lombok <code>@Data</code> (compile-time Java)	Closest analogy	Still different: Rust errors are in your crate's type check, not a separate processor pass

Rules of thumb for migrants:

- `#[derive(Name)]` -> “run the proc macro registered for Name” — usually emits impl for a trait also called Name (compile time).
- `#[some_attr]` on a **function** (e.g. `#[test]`, `#[tokio::main]`) -> attribute proc macro — also compile time, but not derive; see [Proc macros — three kinds](#).

- **No attribute runs when you call a function** unless you explicitly invoked a macro that generated that call.

17.8.2 Mechanics

1. `#[derive(Name)]` looks up a **registered derive proc macro** for that identifier — built into the compiler (`Debug`, `Clone`, ...) or shipped in a dependency (`Serialize`, `Error`, ...).
2. The macro **expands** and typically emits `impl SomeTrait for YourType { ... }` (and sometimes extra items). **Name is usually the same as `SomeTrait`**, but the compiler invokes the **macro registration**, not “any trait you defined.”
3. The compiler type-checks the expanded code like hand-written Rust.

Not automatic: writing `trait Foo { ... }` in your application module does **not** enable `#[derive(Foo)]`. A proc-macro = true crate must register `#[proc_macro_derive(Foo)]` ([custom derives below](#)). You can always `impl Foo for Bar { ... }` by hand without a derive.

No derive for every trait: `Display` is a real std trait with **no** `#[derive(Display)]`. Unknown names fail at compile time: *cannot find derive macro Foo in this scope*.

Inspect expansion when confused: `cargo expand` ([Go deeper](#)).

Field rule: derive succeeds only when **every field** (or every variant’s fields) already implements the trait. Otherwise: “field X doesn’t implement Y”.

Enum vs struct: enums get per-variant match arms in generated `Debug`, `Clone`, `PartialEq`, ...; structs get field-wise impls.

Chained requirements:

Derive	Also requires
Copy	Clone (all fields Copy)
Eq	PartialEq
Ord	PartialEq + Eq
Hash	usually paired with Eq for map keys

What happened in the `Point` example: `Clone` generated `fn clone(&self) -> Self { ... }` field-wise. `PartialEq` generated `==`

field-wise.

Manual vs derive: derive when default behaviour fits. Hand-write for redacted Debug, custom Clone (share Arc instead of deep copy), or enum Default with #[default] on one variant (Rust 1.62+). Chapter 13 lists which std traits can be derived vs must be written by hand (Display has no derive).

Wrong — Eq on floats:

```
// Playground --- does not compile
// #[derive(Eq, PartialEq)]
// struct Reading { value: f64 }
// FIX: integer fixed-point, `PartialEq` only,
//      or ordered-float crate --- see Ch7
```

17.8.3 Std derives, ecosystem derives, and custom derives

Three sources — same **syntax**, different implementers:

Source	Examples	Provided by
Standard library	Debug, Clone, Copy, PartialEq, Eq, Hash, Default, Ord	Compiler-built-in proc macros
Ecosystem crates	Serialize, Deserialize, Error, Parser, instrument, ...	serde, thiserror, clap, tracing, ... (Derive ecosystem)
Custom (your org / crate)	#[derive(RegisterMap)], ...	Your proc-macro = true crate (syn + quote)

instrument is **ecosystem proc-macro tooling** from tracing, but it is an **attribute** on fn ([tracing::instrument]), not a name inside [derive(...)]. Same crate family, different proc-macro kind — **Proc macros — three kinds**.

You **cannot register** a new derive macro name in a normal application module. Custom derives live in a **dependency** crate with proc-macro = true. **Using** a registered name looks identical to std — the lookup at step 1 above is the same:

```
// Cargo.toml: my-derive = { path =
    "../my-derive" }
// my-derive crate registers
    #[proc_macro_derive(RegisterMap)] and
    defines trait RegisterMap

// #[derive(RegisterMap)]
// struct DeviceId(u16);
// expands to: impl RegisterMap for DeviceId {
    ... }
```

Defining `trait RegisterMap` only in your app without the proc-macro crate gives you manual `impl` — not `#[derive(RegisterMap)]`.

17.9 Clone and Copy — derive in practice

Cloning ties directly to ownership (Chapter 1) and shows up in config reload, channel messages, and `async` task state.

Aspect	Copy	Clone
Mechanism	implicit bitwise copy on move/assign	explicit <code>.clone()</code>
Trait relation	Copy: Clone	standalone
Typical types	i32, bool, small arrays	String, Vec, most structs/enums
Derive	<code>#[derive(Copy, Clone)]</code> if all fields Copy	<code>#[derive(Clone)]</code> if all fields Clone
Cost	free (stack)	may heap-allocate (deep copy)

When to derive **Clone** in automation:

- Config structs from TOML passed into worker threads (Chapter 14 `move` closures).
- Command enums — clone before send if both sides need a copy.
- Snapshot sensor readings for logging without holding locks.

When not to derive blindly:

- Large buffers — prefer `Arc<[u8]>` or `&T`.

- Types with `Mutex/RefCell` inside — usually not `Clone` (Chapter 10).
- Cheap shared clone — manual `Clone` calling `Arc::clone`, not derived deep copy.

```
// Playground
#[derive(Clone)]
struct Label(String);

#[derive(Copy, Clone)]
struct RegisterAddr(u16);

fn main() {
    let a = Label("sensor-A".into());
    let b = a.clone(); // heap string duplicated
    let r1 = RegisterAddr(0x01);
    let r2 = r1; // Copy --- no .clone() needed
    println!("{}", b.0, r2.0);
}
```

Wrong — Copy on heap field:

```
// Playground --- does not compile
// #[derive(Copy, Clone)]
// struct Bad { name: String }
// String is not Copy
```

Trap	Symptom	Fix
Clone in hot poll loop	CPU + allocator churn	borrow, <code>Arc</code> , or move
Deep clone of nested graph	accidental megabyte copies	<code>Arc</code> , <code>intern</code> , or move
Java-style dyn <code>Clone</code>	not in <code>std</code>	enum of variants or <code>T: Clone</code> (Chapter 7)
Copy on non- <code>Copy</code> field	compile error	drop <code>Copy</code> ; keep <code>Clone</code>

17.10 Derive ecosystem — existing solutions

Std derives (compact reference):

Derive	Use for
Debug	logs, tests, REPL-style printing

Derive	Use for
Clone / Copy	duplicates — see above
PartialEq / Eq	equality (not f64 total order — Ch7)
Hash	HashMap keys
Default	.. Default::default() fills; enum needs #[default] variant
Ord / PartialOrd	sorting when total order exists

Automation-relevant proc-macro crates (consumer view):

Crate / macro	Typical use
serde Serialize/Deserialize	config TOML/JSON, log payloads, IPC
thiserror Error	library error enums (Chapter 8)
clap Parser, Subcommand, Args	CLI gateways
tracing instrument	spans around poll loops
tokio main, select!, join!	async runtime (Chapter 16)
strum / enum-as-inner	enum <-> string for protocols
async-trait	object-safe async traits (escape hatch)

```
// Cargo project --- composite sketch
// serde = { version = "1", features =
//     ["derive"] }
// thiserror = "2"
// clap = { version = "4", features =
//     ["derive"] }

use clap::Parser;
use serde::Deserialize;
use thiserror::Error;

#[derive(Debug, Clone, Deserialize)]
struct GatewayConfig {
    port: u16,
}

#[derive(Error, Debug)]
enum GatewayError {
```

```

    #[error("config read failed")]
    ConfigRead,
}

#[derive(Parser)]
struct Cli {
    #[arg(long, default_value = "502")]
    port: u16,
}

```

Version coupling: `serde derive` feature must match `serde` version; same for `clap` majors.

Serde + Clone pairing: `#[derive(Debug, Clone, Deserialize)]` — `deserialize` once, then `clone()` hands copies to threads without re-reading the file.

17.10.0.1 Serde JSON — serialize and deserialize

`serde` defines the traits; **`serde_json`** is the format crate for JSON text and bytes. Derive **`Serialize`** and **`Deserialize`**, then call **`to_string`** / **`from_str`** at boundaries — config files, log lines, HTTP bodies (Chapter 19).

```

// Cargo project
// [dependencies]
// serde = { version = "1", features =
//     ["derive"] }
// serde_json = "1"

use serde::{Deserialize, Serialize};

#[derive(Debug, Clone, Serialize, Deserialize)]
struct SensorReading {
    device_id: String,
    value: f64,
}

fn main() -> Result<(), serde_json::Error> {
    let reading = SensorReading {
        device_id: "plc1".into(),
    }
}

```

```

        value: 23.5,
    };

    let json = serde_json::to_string(&reading)?;
    println!("{}", json);
    // {"device_id":"plc1","value":23.5}

    let back: SensorReading =
        serde_json::from_str(&json)?;
    println!("{}", back);
    Ok(())
}

```

Serialize turns Rust values into JSON text. **Deserialize** parses JSON back into typed structs. Failures return `serde_json::Error` — propagate with `?` or map into your app error enum (Chapter 8).

API	Input	Typical use
<code>serde_json::to_string</code>	<code>&T</code> where <code>T: Serialize</code>	log line, HTTP response body
<code>serde_json::from_str</code>	<code>&str</code>	config file read into <code>String</code> , API payload
<code>serde_json::from_slice</code>	<code>&[u8]</code>	socket/frame buffer before UTF-8 validation

For files: `std::fs::read_to_string(path)?` then `from_str` (Chapter 19).
 For outgoing logs: `to_string` and write one line.

Field names in JSON often differ from Rust idioms — use `serde` attributes instead of manual string parsing:

```

// Cargo project
// serde = { version = "1", features =
//     ["derive"] }
// serde_json = "1"

use serde::{Deserialize, Serialize};

#[derive(Debug, Serialize, Deserialize)]
struct PollConfig {

```

```
#[serde(rename = "pollIntervalMs")]
poll_ms: u64,
}

fn main() -> Result<(), serde_json::Error> {
    let json = r#"{"pollIntervalMs":250}"#;
    let cfg: PollConfig =
        serde_json::from_str(json)?;
    println!("poll every {} ms", cfg.poll_ms);

    let out = serde_json::to_string(&cfg)?;
    println!("{}", out);
    // {"pollIntervalMs":250}
    Ok(())
}
```

Rename attrs keep Rust field names idiomatic while matching external JSON. After a refactor, add an integration test with a fixture file — silent drift is a common production trap (see Afterparty **Serde rename**).

17.11 Proc macros — three kinds

[Derive attributes](#) above is the **derive** kind in depth. Two other proc-macro shapes share the same compile-time expansion pipeline:

Kind	Syntax	Example	Differs from derive how?
derive	<code>#[derive(Trait)]</code> on type	Debug, Deserialize, Error	Emits impl Trait for Type
attribute	<code>#[attr(...)]</code> on item	<code>#[tokio::main]</code> , <code>#[test]</code> , <code>#[tracing::instrument]</code>	Rewrites the annotated item (function, module, ...)
function-like	<code>name!(...)</code>	<code>include_bytes!</code> , <code>sqlx::query!</code>	Expands at the call site like <code>macro_rules!</code>

Do not mix these up with **annotations/decorators** from other languages — all three kinds expand to Rust source **before** runtime ([Not annotations or decorators](#)).

Proc macros live in separate `proc-macro = true` crates — `rust-analyzer` sometimes stops at the macro boundary. **Authoring proc macros is out of scope here.** Know how to **consume** and **debug** them with `cargo expand`.

17.12 const and compile-time evaluation

Complementary to macros — typed computation without generating new syntax.

Form	Role
<code>const X: T = ...</code>	compile-time constant embedded in binary
<code>static X: T = ...</code>	fixed memory location; 'static lifetime
<code>const fn</code>	function callable in const context

const fn limits (conservative): no I/O, no threads, no `Mutex`; heap allocation limited — prefer simple numeric / bitwise logic.

```
// Playground
const MAX_REGISTERS: usize = 128;

const fn crc_nibble(x: u8) -> u8 {
    x ^ 0x0F
}

fn main() {
    const CRC: u8 = crc_nibble(0xA5);
    println!("max {} crc {:02x}",
        MAX_REGISTERS, CRC);
}
```

env! vs runtime: `env!("CARGO_PKG_NAME")` is substituted at **compile time** (Preface). `std::env::var("HOME")` reads the process environment at **runtime**. Different tools.

When to use which:

- need typed compile-time value? -> `const` / `const fn`
- repeated syntax across types? -> `derive` or

```
macro_rules!  
single generic algorithm?      -> fn +  
  generics + traits (Ch7)  
external schema/codegen file? -> build.rs  
  or include! --- Afterparty
```

17.13 When custom metaprogramming pays off

Benefit	Example
DRY impl explosion	<code>#[derive(Error)]</code> vs 40 lines manual
Domain DSL at zero runtime cost	<code>register_map!</code> -> <code>const</code> + match arms
Compile-time validation	<code>env!</code> missing -> build fails
Type-safe codegen	field rename breaks compile, not silent JSON drift
API ergonomics	<code>command!</code> subcommand tables
Cross-cutting hooks	<code>tracing::instrument</code>

17.14 Restrictions — what macros cannot do

Restriction	Consequence
Token matching, not types	macro ok; expanded code fails confusingly
Recursion limit (~128)	recursion limit exceeded
Hygiene	can't see caller locals unless <code>\$x:expr</code>
Proc-macro crate isolation	helpers in sibling crate
Orphan rule on generated <code>impl</code>	<code>impl ForeignTrait</code> for <code>LocalType</code> still illegal
<code>const</code> eval bounds	no file I/O, no threads
Follow-set violation	<code>`\$x:expr`</code> is followed by <code>`X`</code> , which is not allowed
Derive bound failure	field doesn't implement trait
IDE / rust-analyzer	jump-to-def stops at macro

Wrong — type check after expansion:

```
// Playground --- macro parses; expanded code  
fails type check
```

```
macro_rules! bad_impl {
    ($t:ty) => { impl Clone for $t { fn
        clone(&self) -> Self { unimplemented!()
        } } };
}
// bad_impl!(i32);
// ERROR: conflicting Clone impl or incomplete
// --- i32 already has Clone
```

17.15 Why macro code is hard to trace

Yes — genuinely harder than hand-written Rust, for structural reasons:

Reason	What you experience	Mitigation
Expansion indirection	in expansion of macro_rules! foo	cargo expand; read call site first
No stable source file	generated code not in repo	expand to stdout; test public API
Stacked expansions	derive inside macro inside attribute	expand one layer at a time
Hygiene	grep misses generated names	search macro definition; expand
IDE limits	no “go to impl” through derive	cargo expand side buffer
Proc macro opacity	serde logic in another crate	read crate docs/tests
Debuggers	breakpoints at call site or odd spans	test expanded or public API

Java contrast: annotation processors also generate code, but IDE culture around generated sources is mature. Rust macro output is **invisible unless you expand**. Same compile-time idea, rougher tooling surface.

Team guidance: macros at **stable boundaries** (serde, thiserror, documented internal DSLs) are fine. Business logic buried in macro layers nobody expands is a liability.

17.16 Edge cases and compiler traps

17.16.1 Follow-set: **for**, **let**, and expression position

Same rules as Syntax forbidden — wrong vs right examples below.

17.16.1.1 Edge 1 — **\$e:expr then for** in the matcher (illegal)

The macro **definition** fails — you never get to call it.

```
// Playground --- does not compile
macro_rules! repeat_body {
    ($body:expr for $i:ident in $range:expr) =>
    {
        for $i in $range { $body }
    };
}
// error: ` $body:expr ` is followed by `for`,
// which is not allowed for `expr` fragments
```

17.16.1.2 Edge 2 — **\$e:expr then let** in the matcher (illegal)

Same follow-set as **for** — only **=>**, **,**, **;** may follow **expr**.

```
// Playground --- does not compile
macro_rules! init_then {
    ($setup:expr let $x:ident = $v:expr) => {
        $setup; let $x = $v; };
}
// error: ` $setup:expr ` is followed by `let`,
// which is not allowed for `expr` fragments
```

17.16.1.3 Edge 3 — **\$e:expr then while** in the matcher (illegal)

```
// Playground --- does not compile
macro_rules! while_wrap {
    ($cond:expr while $c:expr) => { while $c {
        $cond } };
}
// error: ` $cond:expr ` is followed by `while`,
// which is not allowed for `expr` fragments
```

17.16.1.4 Edge 4 — legal foreach matcher: \$p:pat in \$r:expr

Literal for sits **before** the pattern; **in** after **pat** is allowed.

```
// Playground
macro_rules! foreach {
    ($p:pat in $r:expr => $body:expr) => {
        for $p in $r { $body }
    };
}

fn main() {
    foreach!(reg in 0u16..3 => println!("read
        {}", reg));
}
```

What happened: compiles and prints three lines — the matcher never put **for** after an **expr** fragment.

17.16.1.5 Edge 5 — let + for expansion in expression position (illegal, single braces)

The macro **definition** is fine; the **call site** fails when you need a value back.

```
// Playground --- macro ok; main fails
macro_rules! poll_twice {
    () => {
        let n = 2;
        for i in 0..n {
            println!("poll {}", i);
        }
    };
}

fn main() {
    let _done = poll_twice!();
    // error: expected expression, found `let`
    //         statement
    // note: macro expansion ignores keyword
    //       `for` and any tokens following
```

```
}

```

17.16.1.6 Edge 6 — same macro as a statement (legal)

Same `poll_twice!` body — but no assignment — works.

```
// Playground
macro_rules! poll_twice {
    () => {
        let n = 2;
        for i in 0..n {
            println!("poll {}", i);
        }
    };
}

fn main() {
    poll_twice!(); // ok --- statement position
}
```

17.16.1.7 Edge 7 — `let` + `for` in expression position: double-brace fix

```
// Playground
macro_rules! poll_twice {
    () => {{
        let n = 2;
        for i in 0..n {
            println!("poll {}", i);
        }
        n
        // block tail expression --- value
        // returned to caller
    }};
}

fn main() {
    let count = poll_twice!();
    println!("polled {} times", count);
}
```

```
}
```

What happened: prints poll 0, poll 1, then polled 2 times — inner { ... ; n } is one block expression.

17.16.1.8 Edge 8 — lone for as expression (often legal)

A **single** for loop (no leading let) can expand where an expression is expected — the loop itself is the expression.

```
// Playground
macro_rules! count_to {
    ($n:expr) => {
        for i in 0..$n {
            println!("{}", i);
        }
    };
}

fn main() {
    let _ = count_to!(3);
    // ok --- for-loop expression (value is ())
}
```

17.16.1.9 Edge 9 — let inside \$e:expr argument (legal)

The restriction is on matcher **grammar**, not on what tokens the caller passes. This compiles:

```
// Playground
macro_rules! id {
    ($e:expr) => { $e };
}

fn main() {
    id!({
        let x = 10;
        x + 1
    });
    // ok --- one block expression passed as
    $e:expr
}
```

}

What happened: caller wrapped `let + value` in `{ ... }` — one `expr` fragment. Contrast Edge 5, where the **macro body** emitted bare `let` without a wrapping block expression.

Edge	Layer	Fails?	Takeaway
1-3	matcher	macro def	no keyword after <code>\$e:expr</code> except <code>=></code> , <code>..</code> , ;
4	matcher	no	use <code>\$p:pat</code> in <code>\$r:expr</code> , literal <code>for</code> in transcriber
5	expansion	call site	<code>let + for</code> need <code>{{ ... ; tail }}</code> for <code>let x = mac!()</code>
6	expansion	no	statement-position call needs no return value
7	expansion	no	double braces + tail <code>expr</code>
8	expansion	no	single <code>for</code> is often enough
9	caller	no	<code>block { let ...; expr }</code> is one expression

17.16.2 macro_rules! (other traps)

Trap	Symptom	Fix
<code>\$e:expr</code> <code>for ...</code> in matcher	<code>`\$e:expr`</code> is followed by <code>`for`</code> at macro definition	use <code>\$p:pat</code> in <code>\$r:expr</code> — see Syntax forbidden
<code>let + for</code> in <code>expr</code> position	expected expression, found <code>let</code> ; ignores keyword <code>for</code>	<code>{{ let ...; for ...; value }}</code>
<code>\$(*)</code> on empty input	no rules expected this token on <code>my_macro!()</code>	use <code>\$(*)*</code> or add <code>() => ... arm</code>
Expression-position macro	expected expression, found <code>statement</code>	wrap body in <code>{{ ... }}</code>
Arm overlap	wrong arm fires	most-specific arm first
<code>#[macro_export]</code>	macro at crate root path	document path; use <code>\$crate</code>

Trap	Symptom	Fix
stringify! vs eval	stringify!(a + b) -> "a + b"	diagnostics only

```
// Playground --- WRONG with `` on empty call
macro_rules! my_vec {
    ($($x:expr),+) => { vec![$($x),*] };
}
// my_vec!() fails --- FIX: use `` or add ()
=> { vec![] }
```

17.16.3 Derive / attributes

Trap	Symptom	Fix
Default on enum	derive error	#[default] on one variant
Manual + derived same trait	E0119 conflicting impls	remove one
Generic T in struct	T: Clone not satisfied	bound on struct definition
Recursive enum	infinite size	Box, Arc, manual Clone
#[serde(default)] without Default	odd runtime defaults	add Default derive

```
// Playground --- does not compile
// #[derive(Clone)]
// struct Id(u64);
// impl Clone for Id { fn clone(&self) -> Self
//     { Self(self.0) } }
// ERROR: conflicting implementations of trait
// `Clone`
```

17.16.4 Clone / build-time / automation

Trap	Symptom	Fix
Arc<String> derived Clone	cheap pointer bump, not deep string	know semantics
Clone before channel send	double alloc when move suffices	send(value) not send(value.clone())

Trap	Symptom	Fix
env!("VAR") missing in CI	hard compile error	option_env! or document env
#[cfg] + macro	code “missing” in release	cfg-gated code not typechecked when off
Serde rename + refactor	silent config mismatch	integration test with fixture TOML
Debug on secrets	keys in logs	redacted manual Debug

```
// Playground --- conceptual; needs
    compile-time env or fails in playground
// let version = env!("RELEASE_VERSION");
// prefer:
    option_env!("RELEASE_VERSION").unwrap_or("dev")
    in a const context pattern
```

17.17 Cons, warnings, and team hygiene

Warning	Symptom	Mitigation
Macro when fn suffices	opaque errors	fn first
Proc-macro sprawl	slow builds, version hell	minimal derives; pin versions
DSL too clever	onboarding cost	document expansion
Attribute magic	logic hidden from grep	explicit calls in hot paths
Trust proc macros	build-time arbitrary code	audit deps
Over-derive	compile bloat	derive only what you use
Clone everywhere	hidden perf cost	audit hot paths
Skipping cargo expand	guesswork debugging	expand on confusion
Deep macro stacks	untraceable errors	max 1-2 layers in app code

17.18 Idiom spotlight

Macros for syntax repetition; generics/traits for logic reuse. If a macro could be a function, prefer the function.

Derive Clone on small config/command types; reach for .clone() sparingly in poll loops — prefer move or borrow.

When follow-set rules fight your DSL, change punctuation or use a proc macro. **Audit expanded code** before shipping custom macros.

17.19 Go deeper

- [The Rust Book — Macros](#)
- [macro_rules! basics](#) — 411
- [Derive macro concepts](#) — 422
- [The Little Book of Rust Macros](#)
- [Rust Reference — Follow-set ambiguity](#)
- [Rust Reference — Macros](#)
- [cargo-expand](#)
- [serde derive, serde_json, thiserror, clap derive](#)

17.20 See also

- [Preface — env! vs runtime env](#)
- [Chapter 3: Functions — first #\[derive\(Debug\)\] on error enums](#)
- [Chapter 6: Enums and pattern matching — matches!, advanced patterns](#)
- [Chapter 7: Traits — derive usage, float/Eq](#)
- [Chapter 13: Standard traits — which traits to derive vs implement \(Display, Debug, cheat sheet\)](#)
- [Chapter 8: Errors — #\[derive\(Error\)\] with thiserror](#)
- [Chapter 14: Multithreading — Clone + channels](#)
- [Chapter 16: Async — Tokio attribute macros](#)
- [Chapter 18: Unsafe](#)
- [Chapter 19: I/O — config files](#)

17.20.1 Afterparty

Use these for proc-macro authoring and advanced DSL topics not covered above.

17.20.1.1 Concepts and mental model

1. **Expansion order** — “List compiler phases from tokens to LLVM; where do macros run?”

2. **Tokens vs types** — “Why can a macro compile but expanded code fail? One example.”
3. **Follow-set why** — “Explain in 80 words why `$a:expr = $b:expr` is forbidden in matchers.”
4. **Scope honesty** — “List 5 metaprogramming topics Ch17 skips and where to learn each.”
5. **Trace checklist** — “Give 6 reasons macro code is hard to trace and one mitigation each.”

17.20.1.2 `macro_rules!` drills

6. **Macro vs fn** — “Rewrite macro as generic fn if possible; when impossible, say why.”
7. **Fragment picker** — “I describe a DSL shape; you pick `expr/ident/tt` for each slot.”
8. **Expr equals fix** — “Fix `set_reg!($addr = $val)` matcher; show two valid surface syntaxes.”
9. **For after expr** — “Why is `$e:expr for $i:ident in $r:expr` illegal? Show legal `$p:pat in $r:expr` `foreach` macro.”
10. **Double-brace fix** — “Fix `poll_twice!` macro that mixes `let` and `for` for use in `let x = poll_twice!()`.”
11. **Trailing comma** — “Explain `$(x),*` vs `$(x),+` on empty input; show failing and fixed macro.”
12. **Hygiene** — “Explain macro hygiene in 60 words with `$crate` mention.”
13. **Register DSL** — “Extend `register_map!` with a third register; explain `stringify! arm`.”

17.20.1.3 How `derive` works

14. **Debug expand** — “Walk me through `cargo expand` on `derive Debug` output (conceptual).”
15. **Clone expand** — “Show conceptual expanded `impl Clone` for struct with two `i32` fields.”
16. **Enum vs struct** — “How does derived `PartialEq` differ for `enum` vs `struct`? Sketch match arms.”
17. **Field bound failure** — “Struct with `Mutex<i32> field + #[derive(Clone)]` — quote error and fix.”
18. **Redacted Debug** — “When hand-write `Debug` instead of `derive` on command `enum` with secrets.”

17.20.1.4 Clone and Copy

19. **Copy vs Clone quiz** — “Classify 8 types: Copy, Clone only, or neither; justify.”
20. **Hot-loop clone audit** — “Audit poll loop with `.clone()` each tick; suggest `move/Arc/borrow`.”
21. **Arc vs derive Clone** — “Explain cheap Arc clone vs deep String clone with one snippet.”

17.20.1.5 Derive ecosystem

22. **derive need** — “List derives I want for config struct loaded from TOML — justify each.”
23. **Serde rename** — “Field `poll_ms` in JSON as `pollIntervalMs` — show attr; trap on refactor.”
24. **thiserror vs manual** — “Same error enum: count lines derive vs hand-written (Ch8 style).”
25. **Float Eq trap** — “Show `#[derive(Eq)]` on `f64` field failure; two fixes from Ch7.”

17.20.1.6 Debugging and tooling

26. **cargo expand walkthrough** — “Step-by-step: install, run, read output for one derive.”
27. **In expansion of** — “Decode a 3-note compiler error chain from nested macro + derive.”
28. **tt vs expr escape** — “When switch matcher from `expr` to `tt`; tradeoffs in 80 words.”
29. **Three-layer trace** — “Derive inside attribute inside `macro_rules` — how to debug layer by layer.”

17.20.1.7 const vs macro

30. **Port to const fn** — “Replace tiny numeric macro with `const fn`; when macro still needed?”
31. **env vs var** — “Compare `env!`, `option_env!`, `std::env::var` — table with one use case each.”
32. **include_str config** — “Embed default TOML with `include_str!`; deserialize at startup sketch.”

17.20.1.8 Pros / cons decisions

- 33. **Macro vs fn audit** — “Mark 6 snippets: should be macro, derive, or plain fn — justify.”
- 34. **When not proc macro** — “Three scenarios where proc macro is overkill; alternative each.”
- 35. **Minimal derive set** — “Gateway config + error + CLI: smallest derive list that still ships.”

17.20.1.9 Edge case audit

- 36. **Trap quiz** — “Mark 8 snippets: empty +, double brace, Default enum, Arc Clone, env!, duplicate impl, cfg macro, serde rename.”
- 37. **Duplicate register DSL** — “Design compile-time error for duplicate keys in register_map!”
- 38. **Serde refactor test** — “Integration test plan after renaming TOML field with serde attrs.”

17.20.1.10 Derive and attribute proc macros

- 39. **Derive vs decorator** — “I come from Python/Java. Explain why `#[derive(Debug)]` is not a decorator or annotation that runs at call time — what actually happens at compile time?”
- 40. **Three kinds quiz** — “Classify 8 snippets: derive proc macro, attribute proc macro, function-like macro, compiler attribute (`#[inline]`, `#[cfg]`), field meta parsed inside a derive (`#[serde(rename)]`).”
- 41. **tokio::main expand** — “Sketch the conceptual expansion of `#[tokio::main] async fn main() { ... }`. Why is this an attribute proc macro, not `#[derive]`?”
- 42. **Custom attribute inputs** — “For `#[my_attr(some = \"config\"])` `fn poll() { ... }`, what two token streams does the proc macro receive? Give two things the macro might emit.”
- 43. **Field attr vs item attr** — “Contrast `#[serde(rename = \"pollIntervalMs\"])` on a struct field vs `#[tracing::instrument]` on `fn poll` — same proc-macro kind or not?”
- 44. **Custom attribute when** — “Three scenarios (poll-loop tracing, type-level JSON mapping, wrapping `main` with a runtime). Pick: custom attribute proc macro, derive, or plain helper fn — justify each.”
- 45. **Runtime myth** — “Does `#[tracing::instrument]` or `#[test]` run every time I call the function? Explain what runs at compile time vs

run time.”

17.20.1.11 Automation capstone

46. **DSL sketch** — “Design tiny command! macro for CLI subcommands — tokens only.”
47. **Modbus table macro** — “Spec register table macro generating lookup + const max address.”
48. **Derive soup review** — “I paste 40-line struct with 12 derives; trim to minimal set with reasons.”

Chapter 18

Unsafe and When to Stop

18.1 Hook

Every ecosystem has escape hatches — **Java** `sun.misc.Unsafe`, **Python** C extensions, Rust **unsafe blocks**. Inside `unsafe`, the borrow checker is off. You must uphold invariants the compiler cannot verify.

18.2 Scope — a brief tour

Practical intro to reading `unsafe`, wrapping FFI, and knowing when to stop — not the full `Nomicon`.

This chapter covers	Deferred to See also / Afterparty
Four powers of <code>unsafe</code> + why it exists	Full Rustonomicon
Soundness vs safety	Custom allocators, <code>Pin</code> / <code>Unpin</code> formalism
Safe wrappers over raw pointers	Lock-free data structures (Chapter 15)
<code>unsafe fn</code> , <code>unsafe trait</code> , <code>Send/Sync</code> proof sketch	Writing proc macros with <code>unsafe</code> traits (Chapter 17)
FFI orientation + checklist	Full <code>bindgen</code> / <code>cxx</code> walkthrough
When not to use <code>unsafe</code> + <code>Miri</code> intro	<code>Miri</code> deep dives, fuzzing proc macros

18.3 Why unsafe exists — the aim

Rust’s default contract: **safe Rust that compiles cannot cause undefined behavior (UB)**. That promise is **soundness**. `unsafe` implements that promise under the hood. It is also how **you** opt in when the compiler cannot see the full story.

Role	Plain language
Implement safe std	<code>Vec::push</code> , <code>str</code> , <code>Box</code> — internally use <code>unsafe</code> so your call sites stay safe
FFI / C ABI	Call vendor PLC drivers, <code>libc</code> , OS APIs — memory layout and ownership cross a boundary
Certify invariants	<code>unsafe impl Send</code> for a thread-safe file descriptor wrapper the compiler cannot prove
Rare performance	After profiling , sometimes a hot path needs intrinsics or hand-tuned memory — last resort

Language	Escape hatch	Who certifies safety?
Java	<code>sun.misc.Unsafe</code> , JNI	JVM spec + you at JNI boundary
Python	C extension modules	Extension author; CPython users trust maintainers
Rust	<code>unsafe blocks</code> / <code>fns</code> / <code>traits</code>	You document invariants; safe API must still be sound

Key idea: `unsafe` does **not** turn off all rules. Safe functions that call `unsafe` must uphold Rust’s global soundness. Callers of your `safe open_port()` must never get UB from correct use.

18.4 What unsafe allows

Four operations only possible inside `unsafe` (or in an `unsafe fn` body):

Power	Rust	Typical Java / Python parallel
Dereference raw pointers	<code>*const T, *mut T</code>	JNI pointers; ctypes address
Call unsafe fn	Vec growth, <code>from_raw_parts</code>	Native method calls
unsafe impl trait	Send / Sync for custom handles	@GuardedBy — manual proof
Mutable static	global state (avoid in app code)	static fields; module-level globals

Safe Rust **around** unsafe must still be **sound**. A safe public API must not let callers trigger UB through normal use.

18.4.1 Soundness vs safety

Term	Meaning
Safety	No unsafe in <i>this</i> function body — compiler enforces borrow rules
Soundness	<i>Entire program</i> cannot UB if only safe APIs are used — includes your unsafe invariants

A **safe** wrapper can still be **unsound** if its unsafe block is wrong:

```
// Playground --- unsound safe API (do not ship)
fn always_five() -> i32 {
    let x = 5;
    let p = &x as *const i32;
    unsafe { *p } // safe fn, but...
    // x dropped here --- p would dangle if we
    // returned *p
}

fn main() {
    println!("{}", always_five());
    // OK today only because we copy before drop
}
```

What happened: compiles because we **copy** `*p` before `x` is dropped. Returning `&i32` or storing `p` past `x`’s lifetime would be UB — the safe

signature would lie. Sound wrappers **narrow** unsafe to a proof you document.

18.5 Examples: elementary -> hard

Work through in order. After each snippet: **run it**, then read **what happened**.

18.5.1 Level 1 — Elementary: deref a stack pointer

```
// Playground
fn main() {
    let n = 5;
    let p = &n as *const i32;
    unsafe {
        println!("{}", *p);
    }
}
```

What happened:

- `&n as *const i32` **coerces** a valid borrow to a raw pointer — no allocation change.
- **unsafe block** is required to **dereference** `p`; outside the block, `*p` is forbidden.
- `n` is still alive on the stack; `*p` reads 5. The borrow checker is **off inside** the block — you must ensure `n` outlives the deref.

18.5.2 Level 2 — Elementary: viewing bytes + dangling trap

Safe pattern: slice from **valid** `&[u8]` — no unsafe needed for everyday buffer work:

```
// Playground
fn as_hex_preview(buf: &[u8], max: usize) ->
    String {
    let take = buf.len().min(max);
    buf[..take]
        .iter()
```

```

        .map(|b| format!("{b:02x}"))
        .collect::<Vec<_>>()
        .join(" ")
    }

fn main() {
    let frame = [0x01, 0x03, 0x00, 0x10, 0xFF];
    println!("{}", as_hex_preview(&frame, 3));
}

```

What happened: prints **01 03 00** — Modbus-style frames are handled with **safe slices** most of the time.

When **unsafe** appears: `slice::from_raw_parts(ptr, len)` — **your** invariants must guarantee `ptr` is valid for `len` bytes:

```

// Playground --- conceptual; invariants listed
    in comments
use std::slice;

/// INVARIANTS caller must ensure:
/// 1. ptr is non-null and aligned for u8
/// 2. ptr..ptr+len is initialized and readable
/// 3. no mutable alias to same memory while
    &slice lives
/// 4. len does not exceed allocation
/// 5. memory outlives returned slice (or copy
    out)
unsafe fn view_bytes(ptr: *const u8, len:
    usize) -> &'static [u8] {
    slice::from_raw_parts(ptr, len)
}

fn main() {
    let data = [10u8, 20, 30];
    let p = data.as_ptr();
    let s = unsafe { view_bytes(p, 3) };
    println!("{}", s);
}

```

What happened: works here because `data` is stack-allocated and lives for `'static` in this toy. -> `&'static` is **misleading** in real code. Production wrappers return `&[u8]` tied to an **owner** lifetime, or **copy** into `Vec`.

Wrong (UB — do not run): use `p` after `data` is dropped -> dangling deref. Afterparty drills invariant lists.

18.5.3 Level 3 — Intermediate: `unsafe fn` in a small module

Encapsulate unsafe in the **smallest** module; expose only safe constructors:

```
// Playground
mod ring {
    pub struct RingBuf {
        data: Vec<u8>,
        len: usize,
    }

    impl RingBuf {
        pub fn new(cap: usize) -> Self {
            Self { data: vec![0; cap], len: 0 }
        }

        /// INVARIANT: len <= data.len()
        pub unsafe fn set_len_unchecked(&mut
            self, len: usize) {
            self.len = len;
        }

        pub fn push_byte(&mut self, b: u8) ->
            bool {
            if self.len >= self.data.len() {
                return false;
            }
            self.data[self.len] = b;
            // Safe because we just wrote index
            `len` and len < cap
            unsafe {
                self.set_len_unchecked(self.len
```

```

        + 1) };

    true
}

pub fn as_slice(&self) -> &[u8] {
    &self.data[..self.len]
}
}

fn main() {
    let mut rb = ring::RingBuf::new(4);
    rb.push_byte(0xAA);
    rb.push_byte(0xBB);
    println!("{:02x?}", rb.as_slice());
}

```

What happened: prints **[170, 187]** (0xAA, 0xBB). `set_len_unchecked` is unsafe because a wrong `len` makes `as_slice()` read **uninitialized** or **out-of-bounds** memory. Same idea as `Vec`'s internal length updates.

18.5.4 Level 4 — Hard: unsafe impl Send for an opaque handle

Chapter 14: types moved into `thread::spawn` must be **Send**. Raw pointers are not **Send**; OS handles sometimes need a **manual proof**:

```

// Playground
use std::sync::Arc;
use std::thread;

/// Wraps a fictional OS file descriptor (i32).
/// PROOF for Send: fd is owned, close-on-drop
/// only on owning thread,
/// and we never share &mut access --- only Arc
/// move into one thread at a time.
struct SerialHandle {
    fd: i32,
}

```

```

unsafe impl Send for SerialHandle {}

impl SerialHandle {
    fn open_fake() -> Self {
        Self { fd: 3 }
    }
    fn read_line(&self) -> String {
        format!("fd={} ok", self.fd)
    }
}

fn main() {
    let h = Arc::new(SerialHandle::open_fake());
    let h2 = Arc::clone(&h);
    let t = thread::spawn(move ||
        h2.read_line());
    println!("main: {}", h.read_line());
    println!("thread: {}", t.join().unwrap());
}

```

What happened: both threads print **fd=3 ok**. Without `unsafe impl Send`, `thread::spawn(move || h2...)` would **not compile**. Production code often uses crates like `serialport` (safe API, unsafe inside). You rarely write this yourself — but you may **read** it in driver wrappers.

`Sync` is a separate proof (sharing `&T` across threads) — often `Arc<Mutex<...>>` instead of hand-rolled `unsafe impl Sync`.

18.5.5 Level 5 — Hard: FFI sketch (Cargo only)

Calling C from Rust — needs `libc`, link flags, and ABI discipline:

```

// Cargo only --- conceptual; needs libc in
// Cargo.toml
// use std::ffi::{CStr, CString};
// use std::os::raw::c_char;
//
// extern "C" {
//     fn strlen(s: *const c_char) -> usize;
// }

```

```
//
// fn main() -> Result<(), Box<dyn
//     std::error::Error>> {
//     let msg = CString::new("PING")?;
//     let len = unsafe { strlen(msg.as_ptr())
// };
//     println!("C strlen = {len}");
//     Ok(())
// }
```

What happened (conceptually): CString owns a nul-terminated buffer; as_ptr() borrows it for the call. **unsafe** because C may read arbitrary memory if invariants fail. Never call after drop(msg); mind **who frees** (Rust owns CString; C must not free it unless documented).

Tool	When
bindgen	Generate extern "C" from .h
cxx	Safe-ish C++ interop bridge
libc	Common C types and functions

18.6 Practical cases in services and embedded

Situation	Typical approach
Serial / GPIO / async I/O	Use serialport , tokio , rppal — safe API, unsafe inside crate
Parse JSON/TOML config	serde — no hand-written pointer tricks
Vendor C PLC SDK	Thin safe Rust module; extern "C" + ownership docs; or ask for Rust bindings
Shared-memory ring buffer	unsafe + atomics (Chapter 15); Miri + tests
“Speed up JSON”	Profile -> better algorithm -> simd-json / serde features -> unsafe last

Decision flow:

- Need C library or OS API?
- +– no -> stay in safe Rust
 - +– yes -> safe wrapper crate exists?


```

+- yes -> use crate (preferred)
+- no   -> bindgen/cxx + checklist below

```

Hot path slow?

```

+- profile first
+- algorithm / batching / fewer allocations
+- then SIMD crate or expert-reviewed unsafe

```

CRC / Modbus example: prefer the Rust `crc` crate or pure-Rust protocol code over pasting C `unsafe`. Use C only when a hardware vendor **requires** it and the library is audited.

18.7 `unsafe fn` and `unsafe trait`

- **`unsafe fn`:** caller must uphold preconditions (e.g. `set_len_unchecked`). Mark the **contract** in doc comments.
- **`unsafe trait`:** `Send`, `Sync` — implementing asserts thread-safety the compiler cannot derive.
- **Std library:** `Vec::push`, `String::from_utf8_unchecked` (in std internals) — pattern: **unsafe inside, safe outside**.

Ecosystem Chapter 17 derives may emit `unsafe impl` in generated code — application authors rarely write those by hand.

18.8 FFI — checklist and pitfalls

Step	Check
ABI	extern "C" unless docs say otherwise
Strings	CString / CStr; nul-terminated; encoding (UTF-8 vs locale)
Ownership	Who allocates? Who frees? Same allocator?
Pointers	Valid for entire call; not dangling after Rust drop
Panics	Unwinding across C is UB — <code>panic=abort</code> or <code>catch_unwind</code> at boundary
Threads	Is the C API thread-safe? Match with Send/Sync proofs
Errors	<code>errno</code> vs return codes — map to <code>Result</code> in safe wrapper

Java JNI parallel: local/global refs, exception checking, and “who owns the buffer” mirror Rust’s pointer + lifetime discipline. Rust catches more at compile time in safe code. **FFI is still manual proof.**

18.9 Miri — when to run it

Miri is an interpreter that detects undefined behavior in unsafe code (use-after-free, aliasing violations, etc.).

```
rustup +nightly component add miri
cargo +nightly miri test
```

When	Why
After adding/changing unsafe	Catch UB tests miss
Before merging FFI wrapper PR	Cheap extra audit
Teaching / learning	See <i>why</i> a pattern is wrong

Miri does not replace code review or fuzzing — it complements them. See Afterparty for drill prompts.

18.10 When not to use unsafe

Bad reason	Better move
“Borrow checker annoys me”	Restructure ownership (Chapter 5), Arc/Mutex (Chapter 14)
“Faster without measuring”	cargo bench, flamegraph, fewer allocations
“Avoid learning lifetimes”	Fix API shape; unsound shortcuts break production
“Replace Mutex with raw pointers”	Data races -> UB; use atomics or channels (Chapter 15)

Most application code **never** needs `unsafe` in *your* crate — depend on maintained libraries instead.

18.11 Edge cases and compiler traps

Trap	Symptom	Idiom
Dangling raw pointer	UB, Miri failure	Tie pointer to owner lifetime; copy if needed
<code>from_raw_parts</code> wrong len	read past buffer	Assert <code>len <= cap</code> ; test boundary
Two <code>*mut</code> aliases + write	UB (stacked borrows)	Mutex, single owner, or proven uniqueness
<code>static mut</code> + threads	data race UB	Atomic* or Mutex (Chapter 15)
Unwinding into C	UB	<code>panic=abort</code> or no panic across FFI
“Safe” API returns dangling ref	unsound library	code review + Miri

18.12 Idiom spotlight

Encapsulate `unsafe` in the smallest module; document invariants in comments; test aggressively. Prefer safe crates maintained by experts for FFI.

Profile before `unsafe` for speed. A safe algorithm beats a wrong `unsafe` patch.

Safe wrapper, documented proof: every `unsafe` block should name what safe callers rely on.

18.13 Go deeper

- [The Rust Book — Unsafe Rust](#)
- [Rustonomicon](#) — ownership, FFI, unwinding
- [Miri](#) — UB detection
- [Procedural macro intro](#) — 423 (boundary with unsafe traits)

18.14 See also

- Chapter 1: Paradigm shift — raw pointers vs references
- Chapter 8: Errors and testing — no panic across unattended service boundaries
- Chapter 14: Multithreading — Send / Sync

- Chapter 15: `Atomics` — `static mut` vs `atomics`
- Chapter 17: `Metaprogramming` — `derive-generated unsafe impl`
- Chapter 19: `I/O` — `Read/Write`, processes, binary frames

18.14.1 Afterparty

18.14.1.1 Why unsafe and soundness

1. **Invariant list** — “For raw pointer to buffer + length, list 5 invariants a safe wrapper must enforce.”
2. **Soundness** — “Explain ‘safe Rust can’t cause UB’ vs `unsafe` — one paragraph; include unsound safe wrapper example.”
3. **Scope honesty** — “List 6 topics Ch18 skips and where to learn each (nomicon, Miri, Pin, ...).”
4. **Aim table** — “Fill: why `Vec` needs `unsafe` internally while `push` stays safe for callers.”
5. **Promise diagram** — “Draw safe API -> `unsafe block` -> invariants -> caller cannot UB; label soundness.”

18.14.1.2 Raw pointers and safe wrappers

6. ***const vs &T quiz** — “Give 5 snippets: legal ref, needs `unsafe block`, compile error; I classify each.”
7. **from_raw_parts design** — “Design `fn view_frame(ptr, len) -> Result<&[u8], Error>` without `&'static`; list invariants.”
8. **Dangling audit** — “Show stack pointer used after drop; I explain UB; you show Miri-style symptom.”
9. **Modbus buffer** — “Register table as `&[u8]` vs `from_raw_parts` — when is each idiomatic in a gateway?”
10. **Hex preview port** — “Port Level 2 `as_hex_preview` to return `Result` on empty buffer; no `unwrap`.”

18.14.1.3 unsafe fn, Send, and Sync

11. **set_len contract** — “Document pre/post conditions for `set_len_unchecked`; what breaks `as_slice` if violated?”
12. **Send proof** — “I claim `Rc<*mut u8>` is `Send`; you disprove with compiler error quote.”
13. **SerialHandle Sync** — “When would `SerialHandle` need `unsafe impl Sync` vs `Arc<Mutex<...>>`? Two sentences each.”

14. **Ch14 port** — “Rewrite Level 4 spawn example using only safe types — when is it impossible?”
15. **Proc-macro boundary** — “Why do `serde/tokio` crates use `unsafe impl` you don’t write? Link Ch17.”

18.14.1.4 FFI and automation

16. **FFI checklist** — “Checklist for calling a C Modbus library from a Rust binary; include panic and ownership rows.”
17. **CString trap** — “Show `into_raw` forgotten `from_raw` leak; fix with RAI pattern sketch.”
18. **Vendor SDK** — “Diagram ownership: Rust owns config, C owns connection, callback pointer — boxes and arrows only.”
19. **serialport hide** — “Where does `unsafe` live in a typical serial crate vs my application code?”
20. **CRC decision** — “C `crc16` vs Rust `crc` crate vs hand-rolled — decision tree for production gateway.”
21. **Java JNI** — “Compare JNI pitfalls (refs, exceptions, pinning) to Rust FFI ownership rules.”

18.14.1.5 Miri, testing, and review

22. **Miri** — “What is Miri and when should I run it relative to `unsafe` changes? Include one command.”
23. **Trap quiz** — “Mark 6 snippets: safe, UB, ordering bug, needs Miri, needs Mutex, unsound safe API.”
24. **Review rubric** — “10-point code-review checklist for an `unsafe` PR in an automation repo.”
25. **Test plan** — “Unit + Miri + integration tests for new extern ‘C’ wrapper — bullet list only.”

18.14.1.6 When not to use unsafe

26. **Avoid** — “Review use case: speed up JSON — `unsafe` vs `simd-json` vs algorithm; pick with justification.”
27. **Borrow checker fight** — “I paste fight-the-borrow-checker code; you refactor to safe Rust without `unsafe`.”
28. **static mut** — “Compare `static mut` counter vs `AtomicUsize` from Ch15 — UB vs defined behavior.”

18.14.1.7 Capstone

29. **PlcDriver API** — “Design safe `PlcDriver` Rust API over fictional `extern 'C'` — types, `Result`, no raw pointers in public API.”
30. **Level ladder recap** — “Explain Levels 1-5 in one paragraph each for a Java teammate who knows JNI.”

Part IV

Standard Library I/O

Chapter 19

I/O, Processes, and Bits

19.1 Hook

I/O APIs differ by language (**Python** `open()`/subprocess, **Java** `Files/ProcessBuilder`, ...). Rust unifies files, sockets, serial ports, and pipes behind **Read** and **Write**. It uses **Command** for subprocesses with explicit error types. CLIs, services, and protocol code all live here.

19.2 Scope — a brief tour

Practical capstone for files, processes, and binary frames — not kernel or embedded-HAL depth.

This chapter covers	Deferred to See also / Afterparty
Read / Write, buffering, <code>io::Result</code>	Zero-copy I/O, <code>io_uring</code> , <code>mmap</code>
File and line processing (sync)	Full logging/observability stacks
Command, pipes, shell vs direct exec	Job orchestration (Kubernetes, <code>systemd</code>)
Bit packing, endianness, CRC sketch	Full Modbus/TCP stacks, <code>nom</code> parsers
Serial orientation (<code>serialport</code>)	GPIO/MCU bare-metal
Sync vs async I/O choice	Tokio tuning (Chapter 16)

19.3 Why Read and Write — the aim

Application code follows the same patterns: **pull bytes in, push bytes out, handle errors, don't panic on bad input.**

Medium	Rust surface	Same trait?
File	File	Read / Write
stdin / stdout	std::io::stdin() / stdout()	yes
In-memory test double	Cursor<[u8]>, Vec<u8>	yes
TCP socket (sync)	std::net::TcpStream	yes
Serial port	serialport open handle	yes
Async TCP	tokio::net::TcpStream	.await variants (Chapter 16)

Aim: write `one fn process<R: Read, W: Write>(...)` and reuse it for files, pipes, and test buffers. Java `InputStream/OutputStream` and Python file objects play the same role. Rust encodes errors in `io::Result` instead of exceptions.

19.4 Read and Write essentials

Method	Contract
<code>read(&mut buf)</code>	Up to <code>buf.len()</code> bytes; Ok(0) = EOF
<code>read_exact(&mut buf)</code>	Fills <code>buf</code> or returns <code>UnexpectedEof</code>
<code>write(&mut buf)</code>	May write partial slice
<code>write_all(&mut buf)</code>	Loops until all bytes sent or error

Prefer **write_all** and **read_exact** when the protocol has fixed-size fields. Use **read** in loops when length is unknown until EOF.

19.5 Examples: elementary -> hard

Work through in order. After each snippet: **run it**, then read **what happened**.

19.5.1 Level 1 — Elementary: generic echo

```
// Playground --- in-memory stand-in (no
    filesystem)
use std::io::{self, Cursor, Read, Write};

fn echo(mut r: impl Read, mut w: impl Write) ->
    io::Result<()> {
    let mut buf = [0u8; 64];
    loop {
        let n = r.read(&mut buf)?;
        if n == 0 { break; }
        w.write_all(&buf[..n])?;
    }
    Ok(())
}

fn main() -> io::Result<()> {
    let input = b"sensor:42\n";
    let mut reader = Cursor::new(&input[..]);
    let mut out = Vec::new();
    echo(&mut reader, &mut out)?;
    println!("{}",
        String::from_utf8_lossy(&out));
    Ok(())
}
```

What happened: prints **sensor:42** — `Cursor` implements `Read` over a byte slice; `Vec` implements `Write`. Same `echo` works for files or sockets without changing the loop.

19.5.2 Level 2 — Elementary: line-oriented config

```
// Playground
use std::io::{BufRead, BufReader, Cursor};

fn parse_kv_lines(data: &str) -> Vec<(String,
    String)> {
    let mut out = Vec::new();
    let reader =
```

```

        BufReader::new(Cursor::new(data.as_bytes
        ())),
    for line in reader.lines() {
        let Ok(line) = line else { continue };
        let Some((k, v)) = line.split_once('=')
            else { continue };
        out.push((k.trim().into(),
            v.trim().into()));
    }
    out
}

fn main() {
    let cfg = "poll_ms=100\nport=502\n#
comment\n";
    for (k, v) in parse_kv_lines(cfg) {
        println!("{k} -> {v}");
    }
}

```

What happened: prints **poll_ms -> 100** and **port -> 502** — BufReader reduces syscalls for real files; # comment line skipped (no =). In production, return `io::Result` and map errors (Chapter 8) instead of expect on lines.

19.5.3 Level 3 — Intermediate: frame struct + CRC

Wrap bytes in a type once the layout stabilizes:

```

// Playground
#[derive(Debug, PartialEq)]
struct Frame {
    id: u8,
    value: u16,
    valid: bool,
    alarm: bool,
}

impl Frame {
    fn encode(&self) -> [u8; 5] {
        let vb = self.value.to_be_bytes();
    }
}

```

```

    let mut status = 0u8;
    if self.valid { status |= 1; }
    if self.alarm { status |= 2; }
    let body = [self.id, vb[0], vb[1],
                status];
    let crc = body.iter().fold(0u8, |a, &b|
                               a ^ b);
    let mut out = [0u8; 5];
    out[..4].copy_from_slice(&body);
    out[4] = crc;
    out
}

fn decode(buf: &[u8; 5]) -> Option<Self> {
    let body = &buf[..4];
    if body.iter().fold(0u8, |a, &b| a ^ b)
       != buf[4] {
        return None;
    }
    let value =
        u16::from_be_bytes([body[1],
                             body[2]]);
    Some(Self {
        id: body[0],
        value,
        valid: body[3] & 1 != 0,
        alarm: body[3] & 2 != 0,
    })
}

}

fn main() {
    let f = Frame { id: 0x01, value: 1025,
                    valid: true, alarm: false };
    let bytes = f.encode();
    println!("{:02x?}", bytes);
    println!("{:?}", Frame::decode(&bytes));
}

```

What happened: prints hex frame and `Some(Frame { ... })` — XOR CRC is toy-grade; real Modbus uses CRC-16. Pattern: **encode/decode on struct**, not loose indices.

19.5.4 Level 4 — Intermediate: subprocess output

Python	Java	Rust
<code>subprocess.run</code>	<code>ProcessBuilder</code>	<code>Command::new(...).output()</code>
<code>shell=True</code>	—	<code>sh -c "..."</code> (use sparingly)

Cargo only:

```
use std::process::Command;

fn main() -> std::io::Result<> {
    let out = Command::new("echo").arg("hello
        automation").output()?;
    println!("status: {}", out.status);
    println!("stdout: {}",
        String::from_utf8_lossy(&out.stdout));
    Ok(())
}
```

What happened: `output()` waits for the child, captures stdout/stderr into `Vec<u8>`, returns exit status. `from_utf8_lossy` handles non-UTF8 tool output without panicking.

Safer than shell: pass arguments with `.arg()` — no injection surface. Use `sh -c` only when you need `|`, globs, or shell variables.

19.5.5 Level 5 — Hard: pipeline sketch (Cargo only)

Wire **program A** -> **program B** with pipes (conceptual):

```
// Cargo only --- Unix-oriented; needs
    std::process::{Command, Stdio}
// use std::process::{Command, Stdio};
//
// fn main() -> std::io::Result<> {
//     let mut first = Command::new("cat")
```

```
//      .arg("input.txt")
//      .stdout(Stdio::piped())
//      .spawn()?;
//      let first_out =
//          first.stdout.take().expect("piped");
//      let status = Command::new("grep")
//          .arg("ERROR")
//          .stdin(first_out)
//          .status()?;
//      first.wait()?;
//      println!("pipeline status: {status}");
//      Ok(())
// }
```

What happened (conceptually): first child's **stdout** becomes second child's **stdin**. Always wait() children to avoid zombies. For complex graphs, consider duct crate — std is verbose but explicit.

19.5.6 Level 6 — Hard: sync file transform (Cargo only)

```
// Cargo only
use std::fs::File;
use std::io::{BufRead, BufReader, Write};

fn process_file(input: &str, output: &str) ->
    std::io::Result<()> {
    let reader =
        BufReader::new(File::open(input)?);
    let mut out = File::create(output)?;
    for line in reader.lines() {
        let line = line?;
        writeln!(out, "{}",
            line.trim().to_uppercase());
    }
    out.flush()?;
    Ok(())
}
```

What happened: `?` propagates `io::Error` from `open`, `read`, or `write` — `main` can use `fn main() -> io::Result<>`. **flush** before `exit` ensures buffered bytes hit disk (important for logs and PLC command files).

19.6 File I/O patterns

Pattern	When
<code>File::open + BufReader</code>	Line or chunk input
<code>read_to_string</code>	Small text configs only
<code>BufWriter + write_all</code>	Many small writes
<code>main() -> io::Result<></code>	CLI tools — map to exit code at boundary

Boundary discipline (Chapter 8): use internal `fn load_config() -> Result<Config, Error>`. In `main`, print `eprintln!("{e}")` and `exit 1`. Never unwrap on user-supplied paths in unattended services.

19.7 Bitwise and binary protocols

Register and status words on the wire are **bit fields**, not separate `bool` variables:

Technique	Use
<code>value & = 1 << n</code>	Set bit <i>n</i>
<code>value & (1 << n) != 0</code>	Test bit <i>n</i>
<code>to_be_bytes / from_be_bytes</code>	Modbus-style big-endian registers
<code>to_le_bytes / from_le_bytes</code>	Intel layouts, many CAN/USB payloads
<code>struct + encode/decode</code>	Stable frame layout

Use **unsigned** types (`u8`, `u16`, `u32`) for shifts — signed shifts are a footgun.

19.8 Sync vs async I/O

Situation	Prefer
CLI tool, batch file transform	sync <code>std::fs</code> , <code>BufReader</code>

Situation	Prefer
One serial poll loop, blocking read with timeout	sync + dedicated thread (Chapter 14)
Many TCP connections, one process	async Tokio (Chapter 16)
Blocking read inside async fn	Bad — stalls executor; tokio::fs or spawn_blocking

Same **traits** mentally. Async adds `.await` and a runtime. Chapter 16 covers production protocol code.

19.9 Practical I/O scenarios

Task	Approach
Read CSV/TOML config	BufReader + parse, or serde + file (Chapter 17)
JSON config / log line / REST body	read_to_string or buffer -> serde_json::from_str / to_string (Chapter 17)
Run systemctl / vendor CLI	Command direct args; capture stdout; check status
Modbus-style register	Frame struct, BE u16, CRC per spec
Serial sensor / PLC	serialport crate — same Read/Write
Vendor C driver only	Safe Rust wrapper (Chapter 18)

Serial orientation (Cargo only):

```
# Cargo.toml
# [dependencies]
# serialport = "4"

// Cargo only --- list ports; optional open
//   when arg provided
// use std::io::{Read, Write};
// use std::time::Duration;
//
// fn main() -> Result<(), Box<dyn
//   std::error::Error>> {
//   for p in serialport::available_ports()? {
```



```
//      println!("{}", p.port_name);
//    }
//
//    serialport::new("/dev/ttyUSB0",
//    9600).timeout(...).open()?
//
//    port.write_all(b"PING\n"); port.read(&mut
//    buf)?;
//    Ok(())
// }
```

Poll loops: set **read timeout**, log the port on error, and **retry with backoff** (Afterparty P361). Do not spin tight on failure.

19.10 CLI utility — paths, env, and time

One tool thread: resolve config path from env -> check file size -> read key=value lines.

19.10.1 Path and env

```
// Cargo only --- conceptual main fragment
// use std::path::{Path, PathBuf};
// use std::env;
//
// fn config_path() -> PathBuf {
//     env::var("APP_CONFIG")
//         .map(PathBuf::from)
//         .unwrap_or_else(|_| {
//
//             Path::new(&env::var("HOME").unwrap_or_else(|_| ".").into())
//                 .join(".config")
//                 .join("gateway.toml")
//         })
// }
```

Path / PathBuf join segments without manual slash handling. Accept impl AsRef<Path> in helpers (Chapter 13).

19.10.2 File metadata before read

```
// Cargo only
// use std::fs;
//
// fn read_if_small(path: &Path) ->
//     std::io::Result<String> {
//     let meta = fs::metadata(path)?;
//     if meta.len() > 1_000_000 {
//         return Err(std::io::Error::new(
//             std::io::ErrorKind::InvalidData,
//             "config too large",
//         ));
//     }
//     fs::read_to_string(path)
// }
```

Check `metadata().len()` before allocating a huge buffer — cheap guard for untrusted paths.

19.10.3 Stdin prompt alongside args

Not a Playground example — `read_line` blocks until Enter; the online Playground has no interactive stdin and will hang. Run locally with Cargo instead.

Playground — **args path only** (pass a fake port via how you invoke — in Playground, hard-code or skip stdin):

```
// Playground
fn main() {
    let port =
        std::env::args().nth(1).unwrap_or_else(|
            | "502".into());
    println!("using port {}", port);
}
```

Cargo only — **full pattern** (run `cargo run -- 8080` for args, or `cargo run` then type at the prompt):

```
// Cargo only
use std::io::{self, Write};
```

```
fn main() -> io::Result<()> {
    let port = if let Some(p) =
        std::env::args().nth(1) {
        p
    } else {
        print!("port: ");
        io::stdout().flush()?;
        let mut buf = String::new();
        io::stdin().read_line(&mut buf)?;
        buf.trim().to_string()
    };
    println!("using port {}", port);
    Ok(())
}
```

Args for scripts and automation; stdin when a human runs the binary in a terminal — same program, two entry paths.

19.10.4 Timeout with Duration

```
// Playground
use std::time::Duration;

fn retry_delay(attempt: u32) -> Duration {
    let secs: u64 = std::env::var("RETRY_SECS")
        .ok()
        .and_then(|s| s.parse().ok())
        .unwrap_or(2);
    Duration::from_secs(secs.saturating_mul(attempt as u64))
}

fn main() {
    println!("attempt 3 sleep {:?}",
        retry_delay(3));
}
```

Duration pairs with thread sleep and serial timeouts — env var gives ops

a knob without recompile.

19.10.5 CLI I/O edge cases

Wrong — assume path is UTF-8 string on all OSes: use Path end to end; convert to str only when displaying.

Line endings: read_line keeps \n; call .trim() before parsing numbers.

19.11 PTY (note)

Interactive CLI tools sometimes need a **pseudo-terminal** (line discipline, echo). Not in std; crates like pty-process or expectrl fill the gap. Start with Command; add PTY when the tool checks isatty.

19.12 Edge cases and compiler traps

Trap	Symptom	Idiom
read once assuming full buffer	short read, corrupt parse	loop read or read_exact
write without write_all	partial send on socket	always write_all for protocols
Wrong endianness	register value off by 256x	document BE vs LE per spec
unwrap on lines()	panic on bad UTF-8	? + error type
shell -c with user input	command injection	.arg() list
Forgot flush	lost tail bytes on crash	flush before exit
Blocking serial in async task	stalled Tokio worker	dedicated thread or async crate

19.13 Idiom spotlight

Parse bytes with types, not magic indices forever. Once a frame layout stabilizes, wrap it in struct + encode/decode methods.

Generic over Read/Write for testability — Cursor in unit tests, real File in integration tests.

Treat I/O as Result at boundaries — long-running binaries should survive bad paths and timeouts without panic (Chapter 8).

19.14 Go deeper

- [Csv parser / writer](#) — 958-959
- [Rust book](#) — I/O

19.15 See also

- Chapter 8: Errors — `io::Error`, `?`, no panic in production loops
- Chapter 14: Multithreading — blocking I/O on worker threads
- Chapter 16: Async I/O — `tokio::fs`, `TcpListener`
- Chapter 17: Metaprogramming — `include_str!`, `serde` JSON serialize/deserialize
- Chapter 18: Unsafe — FFI and safe wrappers over I/O handles
- Chapter 20: Production standards — review checklist before merge

19.15.1 Afterparty

19.15.1.1 Read / Write and buffering

1. **Trait refactor** — “Refactor file copy loop to generic `copy<R: Read, W: Write>`; discuss error propagation.”
2. **read vs read_exact** — “Give 3 protocol shapes; I pick `read` loop vs `read_exact` each time; you verify.”
3. **Cursor test** — “Write unit test for `parse_kv_lines` using `Cursor` — no filesystem.”
4. **BufReader why** — “Explain in 60 words why `BufReader` matters for 10k-line log files.”

19.15.1.2 Files and CLI

5. **CSV tool** — “Spec for CLI: read two-column CSV, emit `name=value`; I implement with `BufRead`.”
6. **Boundary errors** — “Sketch main mapping `io::Error` to exit code 1 with context path — no `unwrap`.”
7. **read_to_string trap** — “When is `read_to_string` wrong for automation configs? Give size threshold rule.”

19.15.1.3 Binary frames and endianness

8. **Packet layout** — “Add CRC byte to 4-byte packet; update encode/decode with XOR — show tests.”
9. **Endian trap** — “Quiz: 3 scenarios pick LE vs BE for Modbus-style register.”
10. **Bit field port** — “Java status int with flags — port to Rust encode with |= and & masks.”
11. **CRC upgrade** — “Replace XOR toy CRC with CRC-16-Modbus — outline steps, no full crate required.”

19.15.1.4 Processes and pipelines

12. **Command safety** — “Review shell=True style command; rewrite without shell when possible.”
13. **Pipeline** — “Design program A | program B using only Rust std (two processes, pipe).”
14. **Exit status** — “Child exited 2 — how should gateway log and retry? Table: fatal vs transient.”
15. **Env and cwd** — “Show Command with .env(“PORT”, “502”) and .current_dir — when needed?”

19.15.1.5 Sync vs async and serial

16. **Sync vs async pick** — “Three gateway designs: I pick sync thread vs Tokio per scenario; you justify.”
17. **Serial debug** — “I get timeout on read; give systematic checklist (baud, cable, permissions).”
18. **serialport traits** — “Explain how serialport maps to Read/Write — diagram only.”
19. **Blocking in async** — “Show wrong std::fs::read inside async fn; fix with tokio::fs or spawn_blocking.”

19.15.1.6 Capstone and operations

20. **Capstone scaffold** — “Generate module tree and function signatures for sensor_gateway; no bodies.”
21. **Retry policy** — “Design exponential backoff for Modbus-style poll errors; Rust pseudocode.”
22. **Log schema** — “Propose JSON log lines for sensor events with timestamp and error codes.”

- 23. **GPIO next step** — “After serial works on Pi, outline migration to gpio-cdev for one LED.”
- 24. **Code review** — “I paste capstone main loop; review for panic risks and missing flush.”
- 25. **Gateway capstone** — “End-to-end: config file -> serial poll -> JSON log line — module list and error types only.”
- 26. **Ch16 bridge** — “Same echo server: sketch sync thread version vs Tokio version — tradeoffs table.”

19.15.1.7 Paths, env, and CLI

- 27. **Path join** — “Build config path from HOME + .config/app.toml — show Path::join vs string concat trap.”
- 28. **Env default** — “TIMEOUT env var with default 30 — var().unwrap_or_else pattern.”
- 29. **Metadata guard** — “Reject config files over 1MB before read_to_string — sketch metadata().len() check.”
- 30. **Stdin fallback** — “Port from argv or interactive prompt — one main with both paths.”
- 31. **Line parser** — “BufRead lines, skip #, parse key=value — handle trailing newline.”
- 32. **Capstone CLI** — “End-to-end: env path -> metadata check -> line parse -> print port — function list only.”

Part V

Production Standards

Chapter 20

Production Rust Standards

20.1 Hook

You can write Rust that compiles and still miss what experienced reviewers look for: typed boundaries, errors as data, minimal cloning, and panic-free production paths. This chapter is a **review checklist** — the habits teams enforce in code review and the patterns worth asking an AI to verify after every Rust change.

20.2 Scope — a brief tour

This chapter covers	Assumes you have read
Review checklist with examples + motivation	Ch 6-8 types and errors
Ownership, allocation, and pointer idioms	Ch 1, Ch 10
Workspace and test conventions	Ch 9, Ch 8

Run these checks **only when Rust code changed**. No Rust diff -> skip this chapter.

20.3 How to use the checklist

Each section follows the same shape:

1. **Anti-pattern** (or weak pattern) — what reviewers reject.
2. **Idiomatic fix** — runnable example.
3. **Why** — short motivation: what breaks in production if you ignore it.

At the end, a **one-page table** collects every rule for quick scans.

20.4 Types — newtypes and enums

Production code often smuggles meaning through **raw numbers**: two `u8` parameters that look interchangeable, and status codes checked with `== 0`. Newtypes and enums make that meaning visible to the compiler.

20.4.1 Weak — primitives hide two different mistakes

```
// Playground --- weak
fn write_register(unit_id: u8, register: u8,
    value: u16) {
    let _ = (unit_id, register, value);
}

fn is_healthy(status_code: u8) -> bool {
    status_code == 0
    // 0 = ok, 1 = fault --- magic number at
    // the call site
}

fn main() {
    write_register(3, 1, 100);
    // unit 3, register 1
    write_register(1, 3, 100);
    // compiles --- same types, wrong order

    let status_code = 1;
    if !is_healthy(status_code) {
```

```
println!("device fault");
}
}
```

Two separate problems:

Problem	What goes wrong
unit_id: u8 and register: u8	Swapping arguments compiles — both are u8.
status_code: u8 with == 0	Callers pass undocumented magic numbers; easy to typo a valid u8.

Different primitive types (u16 vs String) already prevent some swaps — the real footgun is **two parameters of the same type**, plus **open-coded status values**.

20.4.2 Misleading — type aliases rename, they do not create types

A **type alias** (type UnitId = u8) is only a synonym. The compiler still treats UnitId, Register, and u8 as the **same** type — nicer names in signatures, but none of the swap protection below.

```
// Playground --- misleading
type UnitId = u8;
type Register = u8;

fn write_register(unit: UnitId, reg: Register,
    value: u16) {
    let _ = (unit, reg, value);
}

fn main() {
    write_register(3, 1, 100);
    write_register(1, 3, 100);
    // still compiles --- both args are u8
}
```

Use aliases when you want **shorter spelling of one identity** — e.g. pub type Result<T> = std::result::Result<T, GatewayError> (Chapter 8).

Use **struct newtypes** when two domain concepts share a representation but must stay distinct.

20.4.3 Strong — newtypes for identity, enums for state

```
// Playground
struct UnitId {
    id: u8,
}

struct Register {
    address: u8,
}

enum DeviceStatus {
    Healthy,
    Fault { code: u8 },
}

fn write_register(unit: UnitId, reg: Register,
    value: u16) {
    println!(
        "unit {} reg {} := {}",
        unit.id, reg.address, value
    );
}

fn describe(status: DeviceStatus) -> &'static
    str {
    match status {
        DeviceStatus::Healthy => "healthy",
        DeviceStatus::Fault { .. } => "fault",
    }
}

fn main() {
    write_register(
        UnitId { id: 3 },
        Register { address: 1 },
    )
}
```

```
        100,
    );
    // write_register(
    //     Register { address: 1 },
    //     UnitId { id: 3 },
    //     100,
    // );
    // ERROR: expected UnitId, found Register

    let status = DeviceStatus::Fault { code:
        0x07 };
    println!("{}", describe(status), match
        status {
            DeviceStatus::Fault { code } =>
                format!("code 0x{:02X}", code),
            DeviceStatus::Healthy => "no
                code".into(),
        });
}
```

Fix	Effect
UnitId / Register	Swapping arguments is a type error , not a silent wrong write.
DeviceStatus enum	States are named variants — add Degraded and the compiler lists every match to update.

Struct newtypes vs type aliases for the same underlying u8:

Capability	type X = u8	struct X { ... }
Prevent same-primitive argument swaps	No — still u8	Yes — distinct nominal types
Separate impl blocks and traits	No — one impl for u8	Yes — impl UnitId, impl Display for Register, ...
Implement foreign traits (orphan rule)	No	Yes — wrap and impl on your struct (Chapter 7)
Validation at construction	Any u8 passes through	UnitId::new(n)?, TryFrom<u8>, private fields

Why: newtypes attach **identity** to values that share a representation (`u8`, `u16`, `String`). Enums attach **behaviour** to state instead of scattered integer checks. Together they turn whole classes of field and wiring bugs into compile-time failures (Chapter 6, Chapter 7).

20.5 Lifetimes — correct and minimal

Weak: `'static` everywhere because it silences the borrow checker.

```
// Playground --- smell: unnecessary 'static
fn label(_s: &'static str) -> &'static str {
    "fixed"
}
```

Strong: the **shortest** lifetime that describes the data — often elided on simple helpers.

```
// Playground
fn first_token(s: &str) -> Option<&str> {
    s.split_whitespace().next()
}

fn main() {
    let line = String::from("temp 22.5");
    println!("{:?}", first_token(&line));
}
```

Why: over-long lifetimes hide bugs (returning references to temporaries) and force callers to leak or allocate. Minimal signatures stay reusable and honest (Chapter 5). Add explicit lifetimes only when the compiler asks or when two inputs must outlive the same scope.

20.6 Option and Result — not null-like sentinels

Weak: magic values (`-1`, empty string) or nullable references for “missing”.

```
// Playground --- weak
fn find_port(config: &str) -> i32 {
    if config.is_empty() {
        -1
    } else {
        502
    }
}
```

Strong: Option for absence, Result for failure with a reason.

```
// Playground
fn find_port(config: &str) -> Option<u16> {
    if config.is_empty() {
        None
    } else {
        Some(502)
    }
}

fn parse_port(config: &str) -> Result<u16,
    &'static str> {
    let p = find_port(config).ok_or("missing
        port"?);
    if p < 1024 {
        Err("privileged port")
    } else {
        Ok(p)
    }
}

fn main() {
    println!("{:?} {:?}", find_port("modbus"),
        parse_port("8080"));
}
```

Why: sentinels collide with valid data and skip the type system. Option/Result force callers to handle absence and failure — the core of reliable automation (Chapter 6, Chapter 8).

20.7 Avoid `unwrap` and `expect` in production

Weak: I/O and parsing that panic on bad input.

```
// Playground --- production anti-pattern
fn load_port(s: &str) -> u16 {
    s.parse().unwrap()
}
```

Strong: propagate or handle at a boundary.

```
// Playground
fn load_port(s: &str) -> Result<u16, String> {
    s.parse().map_err(|e| e.to_string())
}

fn main() {
    match load_port("502") {
        Ok(p) => println!("port {}", p),
        Err(e) => eprintln!("skip tick: {}", e),
    }
}
```

Why: `unwrap/expect` **panic** — they stop the thread (often the whole gateway) on expected failures like bad config or a dropped device. Reserve them for **tests**, **prototypes**, or truly impossible states after invariants you control (Chapter 8).

20.8 Panic audit — indexing, `panic!`, and production paths

Weak: unchecked indexing and explicit panics on user data.

```
// Playground --- can panic in production
fn first_byte(frame: &[u8]) -> u8 {
    frame[0] // panics if empty
}
```



```
fn validate_tag(tag: &str) {
    if tag.is_empty() {
        panic!("empty tag");
    }
}
```

Strong: safe access and Result for invalid input.

```
// Playground
fn first_byte(frame: &[u8]) -> Option<u8> {
    frame.first().copied()
}

fn validate_tag(tag: &str) -> Result<(),
    &'static str> {
    if tag.is_empty() {
        Err("empty tag")
    } else {
        Ok(())
    }
}

fn main() {
    println!("{:?}", first_byte(&[]),
        validate_tag("temp"));
}
```

Why: every `vec[i]`, `unwrap`, `expect`, and `panic!` on a production path is an **unhandled incident** waiting for bad wire data. Tests may panic freely; field code should return `Err` and let the supervisor retry or alarm.

20.9 Propagate with ? consistently

Weak: nested match noise in fallible pipelines.

```
// Playground --- verbose
fn load(s: &str) -> Result<u16, String> {
    let t = match s.parse::<u16>() {
```

```

    Ok(v) => v,
    Err(e) => return Err(e.to_string()),
};
Ok(t)
}

```

Strong: ? on the railway (Chapter 8).

```

// Playground
fn load(s: &str) -> Result<u16, String> {
    let t = s.parse::<u16>().map_err(|e|
        e.to_string());
    Ok(t)
}

fn main() {
    println!("{:?}", load("502"));
}

```

Why: consistent ? keeps helpers thin and makes the **happy path** readable. Reviewers spot missing error handling when ? is the default and match is reserved for recovery.

20.10 Custom errors — **thiserror** in libraries, **anyhow** in small binaries

Weak: library API returning `Result<T, String>` or using `anyhow` in public interfaces.

```

// Playground --- weak public API
pub fn connect(_host: &str) -> Result<,
    String> {
    Err("timeout".into())
}

```

Strong: focused enum + **thiserror** in library crates; **anyhow** only in binary/main helpers.

```
// Cargo project --- library pattern (thiserror
    = "2")
use thiserror::Error;

#[derive(Debug, Error)]
pub enum ConnectError {
    #[error("timeout after {ms} ms")]
    Timeout { ms: u64 },
    #[error("invalid host")]
    InvalidHost,
}

pub fn connect(_host: &str) -> Result<(),
    ConnectError> {
    Err(ConnectError::Timeout { ms: 500 })
}

// Cargo project --- binary only (anyhow = "1")
use anyhow::Context;

fn run() -> anyhow::Result<> {
    let _port =
        std::env::var("PORT").context("PORT
        must be set");
    Ok(())
}
```

Why: typed errors are **matchable** for recovery; strings are not. **thiserror** removes boilerplate while keeping a stable library contract. **anyhow** is fine for **one-off scripts and main**, not for code other crates depend on (Chapter 8).

20.11 Errors store data, not messages

Weak: formatting errors into strings — context lost, no structured logging.

```
// Playground --- weak
enum AppError {
```

```

    Io(String),
    // "failed to read /etc/gateway.toml:
    // permission denied"
}

```

Strong: variants carry **fields**; display text comes from `thiserror` or `Display`.

```

// Playground
#[derive(Debug)]
enum AppError {
    ConfigRead {
        path: String,
        source: std::io::Error,
    },
}

fn main() {
    let e = AppError::ConfigRead {
        path: "gateway.toml".into(),
        source:
            std::io::Error::new(std::io::ErrorKind::
                NotFound, "missing"),
    };
    println!("{:?}", e);
    // logs match on kind + path, not string
    // grep
}

```

Why: structured variants support **metrics**, **retries**, and **tests** (match on `ErrorKind::NotFound`). Stringly errors force fragile substring checks and duplicate messages across call sites.

20.12 Focused error enums per layer

Weak: one mega `AppError` for validation, Mongo, config, and serial — returned from a string trim helper.

```
// Playground --- weak: validation returns
    unrelated variants
enum AppError {
    InvalidPort,
    MongoTimeout,
    ConfigParse,
}

fn trim_port(s: &str) -> Result<u16, AppError> {
    s.trim().parse().map_err(|_|
        AppError::InvalidPort)
    // why would MongoTimeout appear here?
}
```

Strong: each **module or concern** owns a small enum; map at boundaries.

```
// Playground
#[derive(Debug)]
enum ValidateError {
    InvalidPort,
    OutOfRange(u16),
}

#[derive(Debug)]
enum StorageError {
    Timeout { ms: u64 },
}

fn trim_port(s: &str) -> Result<u16,
    ValidateError> {
    let p: u16 = s.trim().parse().map_err(|_|
        ValidateError::InvalidPort)?;
    if p < 1024 {
        Err(ValidateError::OutOfRange(p))
    } else {
        Ok(p)
    }
}
```

```
fn save(_port: u16) -> Result<(), StorageError>
{
    Err(StorageError::Timeout { ms: 500 })
}

fn main() {
    println!("{:?} {:?}", trim_port("8080"),
        save(8080));
}
```

Why: callers of `trim_port` should not handle database timeouts. Focused enums document **who can recover from what** and keep match arms honest. Compose or wrap at the service boundary (Chapter 8 — error aggregation).

20.13 Cloning — appropriate, not a band-aid

Weak: `.clone()` to silence borrow checker errors without understanding ownership.

```
// Playground --- weak: clone hides that you
// could borrow
fn print_twice(s: &String) {
    println!("{}", s.clone(), s.clone());
}
```

Strong: clone when you **need** an owned copy; borrow otherwise.

```
// Playground
fn print_twice(s: &str) {
    println!("{}", s, s);
}

fn store_label(s: &str) -> String {
    s.to_string()
    // one intentional allocation for storage
}
```

```
fn main() {  
    let label = String::from("sensor-1");  
    print_twice(&label);  
    let owned = store_label(&label);  
    println!("{}", owned);  
}
```

Why: unnecessary clones cost CPU and memory in hot paths (parsers, poll loops). Cloning to “make it compile” often means the API should take `&str`, return owned data once, or use Cow (Chapter 13).

20.14 Avoid unnecessary `.clone()` and `.to_owned()`

Weak: cloning collections on every iteration.

```
// Playground --- weak  
fn sum_tags(tags: &Vec<String>) -> usize {  
    tags.clone().iter().map(|t| t.len()).sum()  
}
```

Strong: iterate by reference.

```
// Playground  
fn sum_tags(tags: &[String]) -> usize {  
    tags.iter().map(|t| t.len()).sum()  
}  
  
fn main() {  
    let tags = vec!["a".into(), "bb".into()];  
    println!("{}", sum_tags(&tags));  
}
```

Why: each stray clone duplicates heap data. Accept `&[T]` / `&str` in helpers; clone only when crossing threads, storing long-term, or transforming into an owned value.

20.15 Rc::clone / Arc::clone — not .clone() on the smart pointer

Weak: `arc.clone()` — ambiguous (clone the pointer or the inner value?).

```
// Playground --- works but unclear in review
use std::sync::Arc;

fn share(data: Arc<String>) -> Arc<String> {
    data.clone()
}
```

Strong: explicit `Arc::clone(&data)` (same for `Rc::clone`).

```
// Playground
use std::sync::Arc;

fn share(data: Arc<String>) -> Arc<String> {
    Arc::clone(&data)
}

fn main() {
    let a = Arc::new("config".to_string());
    let b = share(Arc::clone(&a));
    println!("{}", a, b);
}
```

Why: `Arc::clone` increments the **reference count** only — cheap and obvious in code review. `.clone()` on `Arc` does the same thing but reads like a deep copy of the inner `String`. Team style guides often require the explicit form (Chapter 10).

20.16 Borrow instead of move when you only read

Weak: taking `String` by value when the caller still needs it.

```
// Playground --- weak
fn log_name(name: String) {
```



```

    println!("{}", name);
}

fn main() {
    let name = String::from("plc-1");
    log_name(name);
    // println!("{}", name); // ERROR: moved
}

```

Strong: `&str` / `&T` for read-only use (Chapter 3).

```

// Playground
fn log_name(name: &str) {
    println!("{}", name);
}

fn main() {
    let name = String::from("plc-1");
    log_name(&name);
    println!("still have {}", name);
}

```

Why: moving values forces clones at call sites or loses data after one call. Borrowing keeps ownership with the caller and documents read-only intent.

20.17 Heap allocations — only when needed

Weak: `Box`, `Vec`, and `String` for tiny stack-sized data.

```

// Playground --- weak
fn port() -> Box<u16> {
    Box::new(502)
}

```

Strong: stack types and iterators; heap when size is dynamic or ownership must move.

```

// Playground
fn port() -> u16 {

```

```

502
}

fn readings(count: usize) -> Vec<f64> {
    (0..count).map(|i| i as f64).collect()
}

fn main() {
    println!("{}", port(), readings(3));
}

```

Why: heap allocations add latency and fragmentation. Prefer stack values, `&str`, and `impl Iterator` return types until you need dynamic size or trait objects (Chapter 3, Chapter 7).

20.18 `Vec::with_capacity` when size is bounded

Weak: repeated push on an empty `Vec` — many reallocations.

```

// Playground --- weak for known upper bound
fn build_tags(n: usize) -> Vec<String> {
    let mut v = Vec::new();
    for i in 0..n {
        v.push(format!("tag-{}", i));
    }
    v
}

```

Strong: `with_capacity` when max length is known.

```

// Playground
fn build_tags(n: usize) -> Vec<String> {
    let mut v = Vec::with_capacity(n);
    for i in 0..n {
        v.push(format!("tag-{}", i));
    }
    v
}

```

```

}

fn main() {
    println!("len={}", build_tags(100).len());
}

```

Why: pre-allocation avoids $O(n)$ realloc copies in parsers and batch builders. If capacity is unknown, start empty; if you know “at most N registers”, reserve N (Chapter 11).

20.19 Workspace dependencies — `{ workspace = true }`

Weak: duplicate version pins in every crate of a workspace.

```

# member Cargo.toml --- weak
[dependencies]
tokio = { version = "1.40", features = ["full"]
    }
serde = "1.0.210"

```

Strong: centralize in the **root** `Cargo.toml`, inherit in members.

```

# root Cargo.toml
[workspace.dependencies]
tokio = { version = "1", features = ["full"] }
serde = "1"
thiserror = "2"

```

```

# member Cargo.toml
[dependencies]
tokio = { workspace = true }
serde = { workspace = true }
thiserror = { workspace = true }

```

Why: one bump updates every crate; CI and local builds stay on the same versions. Exception: a member-only dependency that is large and unrelated to the rest — pin it locally with a comment (Chapter 9).

20.20 Tests — equality, not field-by-field drift

Weak: asserting each field manually — tests break on every harmless derive change.

```
// Playground --- brittle
#[derive(Debug, PartialEq)]
struct Reading { tag: String, value: f64 }

#[cfg(test)]
mod tests {
    use super::*;

    #[test]
    fn reading() {
        let r = Reading { tag: "t".into(),
                          value: 1.0 };
        assert_eq!(r.tag, "t");
        assert_eq!(r.value, 1.0);
    }
}

fn main() {}
```

Strong: `assert_eq!` on the whole value when `PartialEq` is implemented; use `pretty_assertions` in Cargo projects for diffs.

```
// Playground
#[derive(Debug, PartialEq)]
struct Reading { tag: String, value: f64 }

#[cfg(test)]
mod tests {
    use super::*;

    #[test]
    fn reading() {
        let got = Reading { tag: "t".into(),
                           value: 1.0 };
        let want = Reading { tag: "t".into(),
```

```

        value: 1.0 };
        assert_eq!(got, want);
    }
}

fn main() {}

# Cargo --- dev-dependencies
[dev-dependencies]
pretty_assertions = "1"

// Cargo test module
use pretty_assertions::assert_eq;

```

Why: struct equality catches **any** field drift in one failure. `pretty_assertions` prints side-by-side diffs for nested data — essential for parser golden tests (Chapter 8).

20.21 Tests — avoid time and global state pitfalls

Weak: tests that sleep real time or mutate static globals — flaky in CI.

```

// Playground --- flaky pattern (conceptual)
// static mut COUNTER: u32 = 0;
// #[test]
// fn depends_on_wall_clock() {
//
//     std::thread::sleep(Duration::from_secs(1));
//     assert!(true);
// }

```

Strong: inject clocks and stores; use temp dirs in integration tests.

```

// Playground
trait Clock {
    fn now_ms(&self) -> u64;
}

```

```

struct FixedClock(u64);

impl Clock for FixedClock {
    fn now_ms(&self) -> u64 {
        self.0
    }
}

fn is_stale(c: &impl Clock, last: u64, max_age:
u64) -> bool {
    c.now_ms().saturating_sub(last) > max_age
}

#[cfg(test)]
mod tests {
    use super::*;

    #[test]
    fn stale_after_threshold() {
        let c = FixedClock(1000);
        assert!(is_stale(&c, 0, 500));
        assert!(!is_stale(&c, 900, 500));
    }
}

fn main() {}

```

Why: wall-clock sleeps slow CI and fail under load. Global mutable state makes tests **order-dependent**. Fake clocks and per-test fixtures keep suites deterministic (Chapter 8).

20.22 Quick reference — full checklist

#	Rule	One-line test
1	Newtypes / enums for domain meaning	Same-type args swappable? Status as raw u8/bool?

#	Rule	One-line test
2	Minimal lifetimes	Any 'static that could be elided?
3	Option / Result, not sentinels	Any magic -1 / empty meaning “error”?
4	No unwrap / expect in production paths	Only tests and impossible invariants?
5	Panic audit	Any [], panic!, or unwrap on external input?
6	Consistent ?	Fallible helpers return Result?
7	thiserror in libs; anyhow in bins/scripts	Public API free of anyhow?
8	Errors carry data	Variants have fields, not preformatted strings?
9	Focused error enums per layer	Does validation return DB errors?
10	Clone deliberately	Clone fixes borrow errors -> fix API first?
11	No stray .clone() / .to_owned()	Could this be &str / &[T]?
12	Rc::clone / Arc::clone	Smart-pointer clones explicit?
13	Borrow for read-only	Parameters own String without storing?
14	Heap when needed	Box<u16>-style noise removed?
15	with_capacity when bounded	Known max len -> reserve?
16	{ workspace = true }	Member deps aligned with root?
17	Tests use assert_eq! on values	Field-by-field asserts replaced?
18	pretty_assertions for rich diffs	Dev-dependency in workspace?
19	No flaky time / global state	Clocks and stores injected?

20.23 Idiom spotlight

Production Rust is typed, panic-averse, and layered. Model meaning with newtypes and enums; propagate with `?` and **focused** `thiserror` enums; borrow by default; allocate with intent; test with **equality** and deterministic fixtures. Review Rust diffs against this table before merge.

20.24 Go deeper

- [Rust API Guidelines](#)
- [The Rust Book — Error Handling](#)
- [The Rust Book — Writing Automated Tests](#)
- [Clippy lints](#) — many rules overlap this checklist

20.25 See also

- Chapter 8: Errors and testing — `Result`, `thiserror`, mocks
- Chapter 7: Structs and traits — newtypes, trait boundaries
- Chapter 9: Modules and crates — workspaces
- Chapter 10: Smart pointers — `Arc` / `Rc`
- Chapter 11: Collections — capacity and maps
- Chapter 13: Standard traits — `From`, `AsRef`, `PartialEq`

20.25.1 Afterparty

20.25.1.1 Checklist drills

1. **Diff review** — “Paste a 30-line Rust PR; I mark each checklist row pass/fail with one sentence.”
2. **Mega-error refactor** — “Split one `AppError` into `ValidateError` + `StorageError`; show boundary mapping.”
3. **Panic hunt** — “Find five panic sources in a snippet; replace with `Option/Result`.”
4. **Clone audit** — “Remove three unnecessary clones by fixing signatures to `&str` / `&[T]`.”
5. **Arc style** — “Rewrite `arc.clone()` call sites to `Arc::clone(&arc)`; explain review benefit.”

20.25.1.2 Workspace and tests

6. **Workspace.toml** — “Sketch root + two members; all shared deps use `{ workspace = true }`.”
7. **Golden test** — “Parser output: one `assert_eq!(got, want)` + `pretty_assertions`; no per-field asserts.”
8. **Flaky test fix** — “Replace `thread::sleep` in test with injected `Clock` trait.”
9. **Pre-merge gate** — “Checklist-only review of gateway `main.rs` + `lib.rs` — findings only.”
10. **AI review prompt** — “One paragraph prompt for an assistant to verify the Ch20 rules on a Rust diff.”

Appendices

Appendix A

Playground Guide

A.1 When to use the playground

Use playground	Use Cargo locally
Learning syntax, ownership, types	Files, Command, serial ports
Quick experiments (< 50 lines)	External crates (tokio, serialport)
Sharing a link in chat or docs	Integration tests, multi-file crates

Playground: <https://play.rust-lang.org/>

A.2 Rules (matches STYLE_GUIDE.md)

1. **One file** with `fn main()` — no `mod` tree in book snippets marked **Playground**.
2. **std only** — playground cannot add `Cargo.toml` dependencies.
3. **No filesystem** — use `Cursor<u8>`, `Vec<u8>`, or `&str` instead of `File::open`.
4. **No subprocess** — simulate with strings or skip to Cargo lab.
5. **Output via println!** — playground has no CLI args; hardcode sample input.

A.3 Sharing code

1. Paste the snippet into the editor.
2. Click **Share** to get a permalink.
3. In your notes, store: Playground: <url> next to the chapter example.

A.4 Edition and channel

Book examples target **Edition 2021**, **stable** channel — same as playground default.

A.5 Playground stand-ins

Real API	Playground substitute
<code>File::open</code>	<code>Cursor::new(bytes)</code> or <code>include_str!</code> in local Cargo only
<code>Command::output</code>	Print mock stdout string
<code>tokio::spawn</code>	Describe in comment; run Cargo lab
Serial port	Encode/decode bytes in memory

A.6 Local run for Cargo labs

```
cd your_project
cargo run
cargo test
cargo run --example name # if using examples/
```

A.7 Troubleshooting

Problem	Fix
cannot find crate	You need a Cargo lab — add dependency to <code>Cargo.toml</code>
Playground timeout	Infinite loop or huge print; reduce data
Different output on Mac/Linux	Line endings or <code>echo</code> vs <code>cmd</code> — expected for process labs

A.8 See also

- CONTENTS.md
- AI Prompt Index

Appendix B

Java / Python / Rust — Mental Model Map

Optional one-page orientation — especially handy if you know **Java** or **Python**. Not required to read the notes; details live in linked chapters.

B.1 Execution and memory

Topic	Java	Python	Rust
Run model	JVM bytecode	Interpreter / VM	Native binary
Memory	GC, heap objects	GC + refcounts	Ownership, stack + heap
Null	null	None	Option<T>
Exceptions	throw / catch	raise	Result + panic!
Immutability default	fields optional	variables rebind	let immutable

-> Ch 1 Paradigm

B.2 Types and polymorphism

Topic	Java	Python	Rust
Classes	class + inheritance	class, duck typing	struct + impl, no inheritance
Interfaces	interface	informal protocol	trait
Generics	type erasure + bounds	hints optional	monomorphized, zero-cost
Runtime polymorphism	virtual methods	duck typing	dyn Trait or generics

-> Ch 7 Traits

B.3 Collections and loops

Topic	Java	Python	Rust
Dynamic array	ArrayList	list	Vec<T>
Map	HashMap	dict	HashMap<K,V>
Loop style	indexed for	for x in	for x in iter / iterators
Comprehensions	streams (Java 8+)	list comp	iterator adapters

-> Ch 4 Iterators · Ch 11 Collections (Vec, HashMap)

B.4 Concurrency

Topic	Java	Python	Rust
Threads	Thread, executors	threading (GIL)	std::thread
Shared state	synchronized, locks	locks, GIL limits	Mutex, Arc, atomics
Async	virtual threads, CompletableFuture	asyncio	async/await + Tokio
Data races	possible at runtime	possible (C ext)	prevented in safe code

-> Ch 14-16 (threads, atomics, async)

B.5 I/O and systems

Topic	Java	Python	Rust
Files	Files, streams	open()	File + Read/Write traits
Subprocess	ProcessBuilder	subprocess	Command
Binary data	ByteBuffer	bytes, struct	u8, slices, to_be_bytes

-> Ch 19

B.6 Package management

Java	Python	Rust
Maven / Gradle	pip / poetry	Cargo + crates.io

-> Ch 3 Functions · Ch 9 Modules · Ch 2 Types

B.7 Habits to unlearn

1. **Shared mutable aliasing** — default to one owner; borrow briefly.
2. **Null checks everywhere** — use `Option` and `match`.
3. **Catch-all exceptions** — use `Result` and typed errors.
4. **Clone by habit** — borrow first; clone when semantics require a copy.
5. **Ignore return values** — `Result` must be used or explicitly dropped with `intent`.

B.8 AI Afterparty

Use `AI_PROMPT_INDEX.md` with this map: if you know Java or Python, ask the model to translate a snippet from either into idiomatic Rust using the right column.

Appendix C

AI Prompt Index

All **Afterparty** prompts from the book, numbered for reuse. Paste into any AI assistant after reading the linked chapter.

C.1 How to use Afterparty

Full workflow (starter context, one chat per chapter, verification): Preface — Afterparty.

Quick loop (read and run examples first — see How to work through a chapter):

1. Open a **new chat** for this chapter; paste the starter context from the preface.
2. Paste **one** Afterparty prompt; read the answer and follow up until you understand it — verify code in the playground or with **cargo check** before the next prompt.
3. Push back if the answer is vague or wrong.

Find a prompt by ID (P046 -> table below), open the linked chapter for context, paste the prompt into the **same chapter chat**, and keep correcting until the model matches the compiler.

C.2 Preface

ID	Prompt
P001	Learning plan — I already program in at least one language. Based on this preface, build me a 2-week study plan using only Rust Core’s chapter list; 45 minutes per day.
P002	Gap check — Ask me five quick questions to see if I should skip straight to Chapter 5 (lifetimes) or read Chapters 1-4 first.
P003	Prompt practice — Give me one sample Afterparty-style question about ownership; I will answer; you grade like a Rust teacher.
P004	Motivation map — List three real projects (automation, CLI, service) where Rust’s ownership model wins over GC; no hype.
P005	Glossary seed — Define in one sentence each: ownership, borrow, trait, async, atomics — I will refine after reading Part I.

C.3 Chapter 1 — Paradigm shift

ID	Prompt
P006	Move drill — Give 6 tiny Rust snippets mixing String, Vec, and i32. I predict ok or compile error; you reveal answers, quote the error message, and give the smallest fix.
P007	Use-after-move decoder — Show one println! after a move that fails to compile. I explain the error in plain English; you refine my explanation and show the three fixes: borrow, .clone(), or restructure scope.
P008	Return transfers ownership — Write a function fn take(s: String) -> String and a main that calls it. I trace who owns the heap buffer at each line, including after the return.

ID	Prompt
P009	Move into a call — Snippet: <code>process(build_label())</code> where <code>build_label() -> String</code> . Explain when the heap allocation happens, when ownership moves into process, and when drop runs if process takes String by value.
P010	Scope and drop — Give a nested-block snippet with two Strings and one Vec. I mark the exact line where each heap buffer is freed vs where stack slots disappear; you correct and explain drop order.
P011	Single-owner principle — State Rust's rule 'every value has exactly one owner' in one sentence, then give one String example where breaking that rule would double-free. Use the key-handoff metaphor.
P012	Stack vs heap quiz — For 10 declarations (<code>i32</code> , <code>bool</code> , <code>[u8; 4]</code> , <code>String</code> , <code>Vec<i32></code> , <code>&str</code> , <code>&String</code> , (<code>i32</code> , <code>f64</code>), <code>Box<i32></code> , <code>()</code>), I say stack-only, heap involved, or both; you draw the pointer picture for any I miss.
P013	String layout sketch — For <code>let s = String::from("hi")</code> , I describe stack fields (<code>ptr</code> , <code>len</code> , <code>cap</code>) and heap bytes; you correct and extend to <code>let v = vec![1, 2, 3]</code> .
P014	Stack frame drill — I give a 3-function call chain with <code>i32</code> locals and one String passed by value. Trace stack frame push/pop, when each slot dies, and when the String heap buffer is dropped.
P015	Nested block live ranges — Snippet with <code>{ let inner = ... }</code> inside <code>main</code> . I list which bindings are alive on each line; you explain why shorter live ranges matter for borrowing later.
P016	Borrow without heap copy — Show <code>let s = String::from("x"); let r = &s;</code> — I explain what is on stack vs heap and why <code>r</code> does not duplicate the buffer; you add one line that would fail because of move/borrow conflict.

ID	Prompt
P017	& vs &mut pick — Five tasks (log a label, increment a tick counter, parse into a temp struct, share read-only config, swap two buffers in place). I choose pass-by-value, &T, or &mut T; you explain owner count and alias rules.
P018	Create the borrow — Given let mut n = 10; and let s = String::from("plc");, I write the types and expressions for one & and one &mut borrow; you verify the owner is still usable afterward.
P019	When is * required? — Four expressions mixing &i32, &mut i32, and println!. I mark where explicit *r is needed vs where auto-deref handles it; you correct with compiled examples.
P020	Mutate through * — Fill in fn reset(n: &mut u32) { ... } and a main that calls it. I must use * to zero the caller's value; you show a broken version that tries to reassign n instead.
P021	Reference copy vs move — After let r1 = &s; let r2 = r1; vs let s2 = s1;; I compare heap owner count, which bindings stay valid, and whether heap data was copied.
P022	Type of *r — Bindings r: &i32, m: &mut i32, b: Box<i32>. I name the type of *r, *m, and whether *m = 5 mutates the owner; you extend with one &T where *r = 5 fails.
P023	Borrow blocks mutation — Snippet: immutable let r = &s; then s.push('!'). I explain the compile error; you fix by shrinking r's scope with a nested block.
P024	Automation counter — Sketch a 1 kHz loop with ticks: u64 and a helper fn maybe_rollover(t: &mut u64). I write the call site and one *t mutation inside the helper; you review without full thread code.

ID	Prompt
P025	Move or &mut quiz — Six tasks (append to a reusable log line, send a message on a channel, fill a frame <code>Vec<u8></code> in a loop, store a device name in a struct, transform-and-return a label, increment a counter). I pick <code>fn take(T) move vs fn tweak(&mut T)</code> ; you explain who owns the heap data after the call.
P026	After the call — Two functions: <code>consume(String)</code> and <code>append_bang(&mut String)</code> . I trace which bindings in <code>main</code> are valid after each call and whether the heap buffer was dropped, reused, or returned.
P027	Fix the signature — Show code that moves a <code>String</code> into a helper but then tries to <code>println!</code> it in <code>main</code> . I explain the error and choose the smallest fix: change to <code>&mut String</code> , restructure with a return value, or <code>.clone()</code> — you rank the idiomatic options.
P028	Loop reuse — A serial parser reuses <code>let mut buf = Vec::with_capacity(256)</code> every tick. I explain why <code>parse_frame(buf)</code> (<code>move</code>) is wrong and write <code>parse_frame(&mut buf)</code> instead; you add one line showing the caller reading updated length after <code>parse</code> .
P029	Transform-and-return — Task: uppercase a <code>String</code> and give it back. I write <code>fn upper(s: String) -> String</code> and call site; you contrast with an anti-pattern that takes <code>&mut String</code> when ownership should transfer.
P030	i32 vs String move — Same pattern with <code>fn add_one(n: i32)</code> and <code>fn add_bang(s: String)</code> : I explain why the caller can still use <code>n</code> after the call but not <code>s</code> ; you map each to <code>Copy</code> vs heap move.
P031	Call it twice — Snippet that needs to pass the same <code>String</code> to two helpers in one scope. I show why two by-value calls fail, then fix with <code>&str/&mut</code> borrows or one move plus <code>.clone()</code> ; you flag the smell.

ID	Prompt
P032	Copy eligibility quiz — Quiz me on 12 types (<code>i32</code> , <code>String</code> , <code>&str</code> , <code>Vec<i32></code> , <code>[u8; 8]</code> , <code>(i32, String)</code> , <code>Box<i32></code> , <code>fn()</code> , <code>bool</code> , <code>char</code> , <code>Rc<i32></code> , struct with only <code>i32</code> fields). Copy, move, or ‘Copy only if derived’? Cite the rule each time.
P033	Copy vs Drop paradox — Why can’t a type be both Copy and Drop? Use a <code>File</code> or <code>socket handle</code> : show what goes wrong if assignment bitwise-copies the handle.
P034	Semantic copy test — For <code>i32</code> , <code>&T</code> , and <code>String</code> : if I duplicate the stack bits, is the result always semantically identical? When does it fail, and what does Rust do instead?
P035	Double-free thought experiment — Hypothetical: <code>String</code> were Copy. Walk line-by-line through <code>let a = ...; let b = a; }</code> end of block — show both drops freeing the same address. Contrast with real move semantics.
P036	When to derive Copy — I put <code>#[derive(Copy, Clone)]</code> on a struct with a <code>String</code> field — explain the error. Then give three struct shapes: safe to derive Copy, must stay move-only, and where <code>.clone()</code> belongs in the public API.
P037	Reference is Copy — Four snippets: <code>let r2 = r1</code> for <code>&String</code> vs <code>let s2 = s1</code> for <code>String</code> , plus one with <code>&mut</code> . I predict ok/error; you count owners of the heap buffer after each line.
P038	.clone() judgment — Five scenarios (store config <code>String</code> , pass into <code>fn</code> twice, cache key in a map, loop append, return from <code>fn</code>). For each, say move, borrow, or <code>.clone()</code> — and flag when <code>.clone()</code> is a design smell.

C.4 Chapter 2 — Types and expressions

ID	Prompt
P039	Integer pick — Quiz me: for 8 scenarios (Modbus port, byte buffer, collection index, money cents, timestamp millis, hash output, loop counter, enum discriminant), I pick u8/u16/u32/u64/i32/usize; you correct and explain overflow risk.
P040	char vs byte — Explain why 'A' is not the same type as 65u8. Show one valid char literal and one invalid escape; mention UTF-8 only when discussing str/String.
P041	Tuple vs array — When do I use (u16, u16) vs [u16; 2] for a coordinate pair? Give one idiomatic example of each.
P042	Array bounds — Snippet with [u8; 4] and an index from user input. I explain compile-time vs runtime safety; you show bounds checking with .get() vs [i].
P043	&str vs String API — Five function signatures for logging labels. I choose &str or String for each; you explain ownership and allocation cost.
P044	Parse drill — Give let s = \"502\"; — I write two ways to get u16 (unwrap version and proper Result version); you grade idiomatic error handling.
P045	Slice from array — Given [u8; 8] with a 2-byte header and 6-byte payload, I write range expressions for header, payload, and full as &[u8]; you check half-open bounds.
P046	.get vs [i] — Three indexing scenarios (fixed compile-time index, user-supplied index, loop variable). I pick [i] or .get(i) for each; you explain panic risk.
P047	UTF-8 length trap — For \"naïve\" and \"hi (crab)\", I predict .len() vs .chars().count(); you show one bad string slice that panics and the safe fix (.chars() or careful byte ranges).
P048	Coercion quiz — Five call sites passing &str, String, &String, and a literal into fn log(msg: &str). I say ok or what coercion happens; you quote the effective type.

ID	Prompt
P049	Empty slice — Explain whether <code>&frame[0..0]</code> is valid, what <code>.len()</code> returns, and how if <code>slice.is_empty()</code> reads in a parser guard.
P050	Lines and config — Multiline config string with comments and blank lines. I sketch a loop using <code>.lines()</code> and <code>.trim().starts_with('#')</code> ; you extend with one <code>strip_prefix</code> for <code>KEY=</code> rows.
P051	Range quiz — For half-open vs inclusive ranges, I predict output of four for loops using <code>..</code> and <code>..=</code> ; you correct with printed values.
P052	if expression types — Show an if where branches return different types and fail to compile. I explain the error; you fix with a unified type (same type in both arms or wrap in enum preview).
P053	match warm-up — Extend a match on HTTP status codes with 3xx, 4xx, 5xx groupings using range patterns; I write arms; you check exhaustiveness.
P054	Loop choice — Four iteration tasks (infinite retry with break, consume iterator, index 0..len, early exit on condition). I pick loop/while/for; you confirm idiomatic choice.

C.5 Chapter 3 — Functions and methods

ID	Prompt
P055	Parameter audit — Five function signatures for logging: <code>&str</code> , <code>String</code> , <code>&String</code> , <code>Cow<str></code> , <code>impl AsRef<str></code> . I pick one per use case; you explain ownership cost.
P056	Move vs borrow — Snippet calls <code>process(s)</code> then uses <code>s</code> again. I explain the error and show two fixes (<code>&s</code> , <code>clone</code>).
P057	impl block — Struct <code>Timer</code> with <code>start</code> , <code>elapsed</code> , <code>reset</code> . I write <code>impl</code> ; you check <code>&self</code> vs <code>&mut self</code> .

ID	Prompt
P058	Associated fn — When is <code>Type::new()</code> idiomatic vs <code>Default::default()</code> ? One example each.
P059	Consume self — Method <code>fn into_inner(self) -> Vec<u8></code> — why must it take <code>self</code> by value?
P060	Semicolon trap — Three tiny functions: one returns <code>i32</code> correctly, two fail due to <code>;</code> . I fix them.
P061	Early return — Rewrite nested <code>if</code> in a parser as early <code>return None / ?</code> style.
P062	Unit vs value — Which functions should return <code>()</code> vs <code>bool</code> vs <code>Option<T></code> ? Three CLI helper names, I choose.
P063	Generic bounds — Fix <code>fn max(a: T, b: T)</code> without bounds; add <code>T: Ord</code> or <code>PartialOrd</code> .
P064	Result signature — Design <code>fn read_config(path: &str) -> Result<Config, ...></code> ; list error variants, no body.
P065	impl Iterator return — Write <code>top_n(vals: &[f64], n: usize) -> impl Iterator<Item = f64></code> — explain why two different iterator types in <code>if arms</code> fail.
P066	mem take drain — Buffer struct drains <code>Vec<u8></code> to caller via <code>mem::take</code> — show before/after inner field.
P067	where clause — Rewrite cluttered <code>fn f<T: A + B + C>(x: T)</code> with <code>where</code> block — same behaviour.
P068	Self return — Method <code>fn into_inner(self) -> Vec<u8></code> on wrapper — why <code>Self</code> not concrete type name?
P069	const fn — One <code>const fn</code> port validator <code>fn is_valid(p: u16) -> bool</code> — what can and cannot run at compile time?
P070	Error constructors — Add <code>missing_field(name, record)</code> on a parse error enum with <code>impl Into<String></code> ; compare to spelling the variant at each site.

ID	Prompt
P071	Config holder — HTTP poll client: struct holds timeout/retries/endpoint; one build_request(device_id) — no builder crate.
P072	Extension trait — Add TrimOrEmpty for &str; use in three parser call sites vs free functions.

C.6 Chapter 4 — Iterators

ID	Prompt
P073	for desugar quiz — Four for loops (0..n, &vec, vec, &mut vec). I say which calls iter, into_iter, or range; you correct and show owner state after the loop.
P074	iter vs into_iter — Give 4 snippets using Vec; I predict whether v is usable after the loop; you explain move vs borrow.
P075	iter_mut drill — Task: double every element in Vec<f64> in place without allocating a new Vec. I write the loop; you review *n and borrow rules.
P076	Range vs collect — When is for i in 0..n better than (0..n).collect:: <vec<_>>()? (retry="" automation="" count="" examples="" give="" indices).<="" materializing="" td="" two="" vs=""></vec<_>>()?>
P077	Adapter chain — Task: parse lines, trim, keep non-empty, parse as u16. I sketch .lines().map(...).filter(...).collect(); you refine.
P078	zip pairs — Two Vec<u16>: register IDs and values. Build Vec<(u16, u16)> with .zip(); I write it; you handle length mismatch policy.
P079	take and skip — Paginate a log: skip first 100 lines, take next 20. I use .skip().take() on .lines(); you show one pitfall if the source is not recomputed.

ID	Prompt
P080	chain iterators — Concatenate header rows and body rows (two <code>&[u8]</code> slices) without copying bytes into one array first — sketch with <code>.iter().chain()</code> .
P081	enumerate vs index — Same sum-over-evens task twice: index for vs <code>.enumerate()</code> . Compare readability and bounds-check risk.
P082	Lazy vs eager — Explain when <code>filter().map()</code> allocates vs when <code>.collect()</code> forces work. One example with <code>println!</code> in <code>map</code> showing evaluation order.
P083	collect turbofish — Three <code>collect()</code> calls that fail without hints — I add type annotation or turbofish; you verify.
P084	find and Option — Wire scan: <code>Vec<&str></code> of lines, find first containing <code>ERROR</code> . I return <code>Option<&str></code> with <code>.find()</code> ; you contrast with <code>.filter().next()</code> .
P085	fold vs sum — Compute max and count in one pass with <code>.fold()</code> vs calling <code>.max()</code> and <code>.len()</code> separately — when is fold worth it?
P086	any and all — Validate a batch: all ports in <code>1..=65535</code> , any line starts with <code>#</code> . I write <code>.all()</code> / <code>.any()</code> predicates; you fix one double-reference mistake.
P087	Moved Vec mistake — Show code that does for <code>x</code> in <code>v</code> then uses <code>v</code> again. I explain the error and fix with <code>.iter()</code> or <code>clone</code> ; you rank fixes.
P088	Borrow in chain — Snippet: build <code>Vec<&str></code> from <code>String</code> lines then drop the <code>String</code> . I explain why it fails; you fix with owned <code>String</code> or different lifetime design.
P089	Modbus-style scan — List of raw register values <code>Vec<u16></code> : filter evens, map to f64 scale 0.1, sum. I write the iterator chain; you check overflow and types.
P090	Zero-cost check — Does <code>nums.iter().filter(...).map(...).sum()</code> allocate intermediate <code>Vecs</code> ? Answer for <code>--release</code> and what to measure conceptually.

ID	Prompt
P091	Trap sheet drill — Give 6 snippets mixing for x in v, for x in &v, into_iter, and collect type errors. I predict compile ok or fail and why; you show the fixed line.
P092	&&i32 decoder — One .iter().filter().map() chain: I label the closure parameter type at each step; you correct and show x , \&x , and \&\&x versions that compile.
P093	Empty iterator policy — Three tasks (sum, find, all) on possibly empty Vec. I state the result and whether it is a domain bug; you correct (e.g. empty all is true).
P094	zip truncation — IDs len 5, values len 100 — I write zip collect; you explain silent loss and sketch zip + length check or enumerate on the longer vec.
P095	RangeCounter impl — Implement Iterator for integers 1..=5; collect to Vec and sum with .sum() — show type Item and fn next only.
P096	Skip blanks — Config string with empty lines — write NonEmptyLines iterator that trims and skips ' '; collect keys before =.
P097	Infinite take — Counter from 0 without end — why must you .take(n) before .collect()? Show hang vs bounded collect.
P098	IntoIterator pair — Same struct: implement Iterator and IntoIterator so both scan.next() and for p in scan work — minimal impl blocks.
P099	Stateful parser — Byte buffer iterator yielding complete 4-byte frames; partial frame stays in struct — sketch next() state machine.
P100	Capstone iterator — CSV line iterator: split fields, parse col 2 as u16, filter > 0 — custom struct + one consumer chain.

C.7 Chapter 5 — Lifetimes

ID	Prompt
P101	Error archaeology — I paste a ‘lifetime may not live long enough’ error; walk me through owner vs reference diagram.
P102	Return type choice — For API fn <code>title(book: &Book) -> ???</code> compare <code>&str</code> vs <code>String</code> trade-offs for a library.
P103	Struct lifetime — Design <code>ConfigParser</code> holding <code>&str</code> slices into input buffer — when is it sound vs use owned <code>String</code> ?
P104	Elision quiz — Add explicit lifetimes to 4 function signatures where elision fails.
P105	Fix mine — I return <code>&String</code> built inside function; show three idiomatic fixes ranked by simplicity.
P106	static trap — Function returns <code>&str</code> from <code>format!</code> — show error and owned fix.
P107	two lifetimes — Write fn <code>first<'a, 'b>(x: &'a str, y: &'b str) -> &'a str</code> — drop <code>y</code> while result lives.
P108	Config struct — Parse <code>host:port</code> into <code>Config<'a></code> — when must caller keep line alive?
P109	Owned refactor — Same parser returning owned <code>Config { host: String, port: u16 }</code> — tradeoffs in 3 bullets.
P110	T: 'a bound — Explain struct <code>Holder<'a, T: 'a> { value: &'a T }</code> — what fails if <code>T</code> is shorter-lived?
P111	Elision fail — Four signatures where elision works vs fails — I label each.
P112	Iterator borrow — Collect <code>Vec<&str></code> from <code>String</code> lines — why drop order matters (link Ch 4).

C.8 Chapter 6 — Enums and pattern matching

ID	Prompt
P113	Null replacement — Translate 5 Java methods returning null into Option Rust; explain callsite changes.
P114	unwrap audit — Paste 20 lines with 4 unwrap() calls; I mark panic risk; you rewrite with match or ?.
P115	Combinator chain — Parse port Option<u16>, double if Some, default 502 — I write map/unwrap_or; you add and_then version.
P116	let-else port — Rewrite nested match on parse() into let Ok(x) = ... else { ... }; preserve behavior.
P117	Result railway — Chain parse -> validate -> compute with ?; I fill blanks, you verify.
P118	Err arm missing — Show match on Result without Err arm; I quote error and fix; add boundary eprintln! pattern.
P119	unwrap vs ? — Same parser twice: unwrap vs fn -> Result with ?; compare panic risk and signature honesty.
P120	Exhaustive match — Enum with 4 variants; I write match; you add variant Safe and show compile error until I fix.
P121	Wildcard footgun — Explain why _ on your own enum hides new variants; show explicit arms vs _ refactor story.
P122	Partial move — enum with String field: match moves field; I fix with ref or match &e; you show error text.
P123	State machine — TCP/serial states as enum; connect/send/close return Result<(), IllegalTransition>.
P124	Python Union — Python int str parameter -> Rust enum + match; no dyn.
P125	if let vs match — Three tasks: one variant, two variants, five variants — I pick if let or match each time.
P126	Match on ref — Given owned Status, I choose match s vs match &s; predict move errors; you diagram ownership.

ID	Prompt
P127	Guard drill — Classify i32 with guards (<0, ==0, >100); I write arms; you check exhaustiveness.
P128	Opcode table — Design enum for 3 frame types + match returning u8 opcode; add fourth type as compile-break exercise.
P129	ReadOutcome extend — Add Disconnected to automation ReadOutcome; show all match sites compiler lists.
P130	Config sentinel — Rewrite fn port() -> i32 returning -1 on failure to Option<u16> + match in main.
P131	Checklist drill — Match 6 Chapter 6 compiler errors to snippets; I name fix (match arm, ref, ?, return type).
P132	Java enum map — Java enum State { IDLE, RUN } with method — port to Rust enum + match + impl; contrast nullability.
P133	Slice split — Parse POST /api/v1/run HTTP/1.1 with [method, path @ .., _ver] — I write; you handle single-token input.
P134	matches refactor — Replace 6-arm match that returns bool with matches! — show before/after on Mode enum.
P135	if let chain — Parse host:port:extra — chain should reject extra segments; fix my broken parser.
P136	@ binding quiz — Three arms with @ ranges for port classes — I label which values hit which arm.
P137	Exhaustive slice — [a, b] on Vec of len 1 or 3 — predict _ arm vs bug; suggest .. rest pattern.
P138	matches vs match — When must you keep full match instead of matches!? Give one example returning String.
P139	**Guard + @** — Match port n @ 1024..=65535 with guard n % 2 == 0 — sketch arm.
P140	Capstone parse — Frame header [sync, len @ 1..=255, payload @ ..] from byte slice — sketch match arms only.

C.9 Chapter 7 — Structs, traits, generics

ID	Prompt
P141	new vs default — When is <code>Sensor::new</code> idiomatic vs <code>Default</code> + field update? One automation example each.
P142	Method receiver — Same logic three ways: <code>fn f(self)</code> , <code>fn f(&self)</code> , <code>fn f(&mut self)</code> on a struct; I predict what calls compile.
P143	Enum trait impl — Add variant <code>Fault</code> to <code>Status</code> ; show every compile error until <code>trait impl</code> and inherent methods match.
P144	Struct vs enum layout — Modbus/OpcUa device registry: justify <code>struct Device { kind: Enum }</code> vs <code>enum Device { Modbus(...), OpcUa(...) }</code> ; sketch types.
P145	SensorReading port — Python <code>Union[TempReading, PressReading, Skipped]</code> -> Rust <code>enum</code> + struct payloads + <code>Measurable</code> trait; show match in impl.
P146	Two impl blocks — Same type: add inherent <code>is_valid()</code> and trait <code>Summary</code> ; explain which call sites see which methods.
P147	Partial move fix — Given <code>enum Packet { Raw(String), Empty }</code> , reproduce ‘partially moved value’ and rewrite with <code>&self</code> or one match.
P148	matches! drill — Rewrite <code>!matches!(self, Skipped { .. })</code> as longhand match; then back to <code>matches!</code> ; link to <code>macro_rules</code> conceptually.
P149	Default override — Trait <code>HasCode</code> with default <code>label()</code> ; override for one <code>enum</code> variant only; use <code>HasCode::label(self)</code> for the rest.
P150	Recursion trap — Show infinite recursion when <code>override</code> calls <code>self.label()</code> instead of <code>HasCode::label(self)</code> ; fix it.
P151	One trait per impl — Show <code>impl HasCode, Display for T</code> compile error; split into two blocks; add <code>fn show<T: HasCode + Display>(x: T)</code> .

ID	Prompt
P152	Command + log row — enum Command + struct SetSpeedLog + to_log(); list which derives each type needs and why.
P153	Eq on floats — Add Analog(f64) variant; show #[derive(Eq)] failure; fix with integer fixed-point or PartialEq only.
P154	Generic bounds — Fix compiler error: T needs Display + Clone; minimal bound set on fn duplicate_and_print<T>(x: T).
P155	largest pitfalls — Why does largest need non-empty slice? Add Option return or document panic; compare to Java generics erasure story.
P156	where clause — Rewrite fn f<T: A + B + C>(x: T) with a where block; same signature, longer trait list.
P157	dyn vs impl quiz — Four scenarios (plug-in Vec, single helper fn, closed protocol, factory return) — pick impl, dyn, or enum each time.
P158	AlarmSink registry — Implement &[dyn AlarmSink] for log + metrics; then refactor to enum Sink if set is closed — compare trade-offs.
P159	Factory return — greeter_for(lang) -> Box<dyn Greeter> vs -> impl Greeter — show why impl fails when arms return En and Fr.
P160	Driver three ways — Same poll() behaviour with &impl Driver, &[&dyn Driver], and enum Device; benchmark story without running code.
P161	Box vs borrow — When is &dyn Trait enough vs Box<dyn Trait> required? Vec of mixed types + dangling return examples.
P162	Object safety audit — Mark each trait dyn-safe or not: no-self method, generic method, -> Self, trait Foo: Sized.
P163	Reader fix — Trait with fn read() -> f64 fails as dyn; redesign for &dyn Reader or use enum.
P164	Clone not dyn — Why no Box<dyn Clone> pattern for heterogenous clone list; suggest enum or generic T: Clone instead.

ID	Prompt
P165	Send + Sync — Alarm handler shared across threads: write type as <code>Arc<dyn AlarmSink + Send + Sync></code> ; what breaks if handler holds <code>Rc</code> ?
P166	Unsize trap — Show three snippets that fail: <code>let g: dyn Greeter = En, Vec<dyn Greeter></code> , returning <code>&dyn</code> from locals; fix each.
P167	Orphan rule — Why <code>impl Display for Vec<u8></code> fails; fix with newtype struct <code>Frame(pub Vec<u8>) + impl Display for Frame</code> .
P168	External trait — Wrap third-party struct; implement your trait on the wrapper; call from automation main.
P169	Checklist drill — Match 8 Chapter 7 compiler errors to snippets (non-exhaustive match, partial move, orphan, not dyn compatible, unsize, moved self, <code>Eq+f64</code> , multi-trait impl).
P170	PLC message model — Design full model: enum frames, struct payloads, two traits, one dyn registry for sinks, derive list — no code over 80 lines.
P171	Refactor story — Start with <code>Vec<Box<dyn Driver>></code> ; protocol closes to two devices; refactor to enum; list what the compiler now catches.
P172	Item type quiz — Change <code>PortScan</code> 's type <code>Item</code> from <code>u16</code> to <code>(u16, bool)</code> — list every call site in a <code>.map().collect()</code> chain that breaks and why.
P173	Summarizable design — Add <code>Summarizable</code> with type <code>Output = String</code> for three sensor structs; one <code>fn report(r: &impl Summarizable)</code> — no duplicate return types in the trait.
P174	Associated vs generic — Same cache API twice: <code>trait Get<T></code> vs <code>trait Get { type Value; }</code> — when is each painful at call sites?
P175	Supertrait bounds — Trait <code>Exportable: Display + Debug</code> with default <code>fn export</code> — show impl for <code>Port(u16)</code> ; what fails if <code>Display</code> is missing?

ID	Prompt
P176	UFCS fix — Type implements A and B, both define name() — show ambiguous call error and UFCS fix with A: :name(&x).

C.10 Chapter 8 — Errors and testing

ID	Prompt
P177	? chain — Refactor nested match on Results to ? railway; explain each desugared return Err.
P178	map / and_then — Same parser with .map_err + .map vs .and_then(validate); when to use each.
P179	Wrong return type — Show ? compile error in fn -> u32; fix signature and boundary match.
P180	Poll loop — Rewrite unwrap Modbus config parse to log-and-continue; process must survive bad line.
P181	Internal vs boundary — Mark 8 functions in a gateway crate: bubble with ? or handle in main?
P182	Unwind vs Result — Diagram stack for open()? vs unwrap() on missing file; who runs Drop?
P183	panic audit — Mark 10 unwrap/expect sites: keep (test/invariant) vs Result (I/O/config).
P184	Drop double panic — Explain why panicking in Drop aborts; sketch safe cleanup pattern.
P185	catch_unwind scope — When is catch_unwind appropriate vs abuse? One automation anti-example.
P186	abort strategy — Embedded firmware: panic = abort — what cleanup is skipped vs unwind?
P187	Std limits — List 4 pain points of io::Error + String errors in a Modbus gateway; no FUD.

ID	Prompt
P188	Manual vs thiserror — Same AppError twice: hand-written Display/Error/From vs #[derive(Error)]; count lines saved.
P189	thiserror design — Design GatewayError with ConfigRead, ParsePort, Timeout, Serial sub-enum; full derive attrs.
P190	anyhow vs thiserror — Pick for automation binary vs library crate; show main -> anyhow::Result + .context().
P191	Why not String — Explain why pub fn connect() -> Result<(), String> is weak; propose enum API.
P192	Error enum design — Design AppError for config + serial + timeout; variant shapes + recovery match.
P193	Recovery match — Poll loop: Timeout -> retry, ParsePort -> alert, DeviceOffline -> safe state — sketch match.
P194	Nested SerialError — Add AppError::Serial(SerialError); show propagation with #[from].
P195	Exhaustive trap — Add DeviceOffline variant; quote non-exhaustive match errors until fixed.
P196	Boundary pattern — run() -> Result + main maps to exit code; no unwrap in between.
P197	map_err drill — Convert ParseIntError -> AppError::Parse with and without From / #[from].
P198	Table-driven ports — Tests for parse_port: valid, zero, non-numeric, too large for u16.
P199	Integration test — Sketch tests/config_load.rs that expects Err on missing file without panicking.

C.11 Chapter 9 — Modules, paths, and crates

ID	Prompt
P200	File tree — Design module tree for a CLI that reads config and runs commands. Directories + mod lines only, no bodies.
P201	Nested mod.rs — Draw ui/theme/themes/ and ui/theme/palettes/ as a directory tree. When is foo/mod.rs better than foo.rs?
P202	Path quiz — From <code>crate::service::worker::run</code> , how do I reach <code>crate::config::load</code> ? Show use and fully qualified call.
P203	use crate:: — In <code>app/events.rs</code> , import <code>AppState</code> from <code>app/state.rs</code> and <code>PanelLocation</code> from <code>core/location.rs</code> — write both use lines.
P204	super:: siblings — <code>dsp/flowgraph.rs</code> needs <code>silence::silenced</code> from a sibling file under <code>dsp/</code> . Show the use line (not a path from crate root).
P205	lib vs bin — What belongs in <code>main.rs</code> vs <code>lib.rs</code> for a tool with 500 lines of logic?
P206	Binary-only — No <code>lib.rs</code> : list top-level mod lines in <code>main.rs</code> for a TUI with <code>core</code> , <code>ui</code> , <code>app</code> , and <code>util</code> modules. Where does <code>pub(crate)</code> fit?
P207	pub audit — List items that should be <code>pub</code> vs <code>private</code> in a library crate exposing <code>Client::connect</code> .
P208	pub(super) — <code>dsp/mod.rs</code> must call <code>flowgraph::run</code> , but <code>sdr.rs</code> must not reach it. Show <code>mod flowgraph</code> ; and <code>pub(super) fn run</code> — who can call <code>run</code> ?
P209	Facade module — <code>ui/mod.rs</code> has <code>mod renderer</code> ; and <code>pub use renderer::Renderer</code> ; . Why not <code>pub mod renderer</code> ?
P210	Barrel re-export — <code>config/mod.rs</code> exposes <code>Station</code> and <code>load_stations</code> without callers writing <code>config::stations::</code> . Sketch <code>pub mod</code> + <code>pub use</code> lines.
P211	pub(crate) — Name three helpers that should be <code>pub(crate)</code> in a binary crate (shared across <code>app/</code> , <code>ui/</code> , <code>main</code>) but must not be <code>pub</code> .

ID	Prompt
P212	pub(crate) mod — theme/mod.rs keeps palettes visible inside the crate only. Show pub(crate) mod palettes; vs pub mod palettes — what breaks for external callers?
P213	pub mod vs mod — In browser/mod.rs, viewer_image is private but viewer is pub mod. What can code outside browser/ import?
P214	Re-export dep — Sketch pub use so users see my_crate::Error but you wrap thiserror internally.
P215	Workspace split — Two crates: core library + cli binary. Write Cargo.toml dependency path only.
P216	Integration test — Where does tests/smoke.rs live and how does it use the library?
P217	Orphan fix — I want Display on Vec<u8> — show newtype wrapper module layout.
P218	Lib name split — Tauri package sdr_fm uses [lib] name = \"sdr_fm_lib\" and main calls sdr_fm_lib::run(). Why not name the lib sdr_fm?
P219	Domain layers — Split a file manager TUI into core/ (no ratatui), ui/ (drawing), app/ (state + events). What must never live in core/?
P220	Split monolith — Given one main.rs with config + parser + runner, name three modules and what each owns.
P221	cfg test — Explain why mod tests uses #[cfg(test)] and use super::*.
P222	Leaf tests — dsp/mod.rs and core/settings.rs each have #[cfg(test)] mod tests. Why co-locate tests in submodule files instead of only at lib.rs?
P223	Capstone — Generate src/ tree for sensor_core library + sensor_cli binary in one workspace; I implement.
P224	Feature flag — Add serial feature gating mod serial_io — write Cargo.toml [features] and one #[cfg] line.

ID	Prompt
P225	cfg vs cfg! — Same debug log twice: <code>#[cfg(debug_assertions)]</code> block vs <code>if cfg!(debug_assertions)</code> — what stays in release binary?
P226	Optional dep — Wire optional tokio behind feature <code>async</code> — show <code>[features]</code> and <code>dep:tokio</code> line.
P227	Platform module — Sketch <code>#[cfg(target_os = \"linux\")] pub mod linux_home_trash; in core/mod.rs</code> — what happens on macOS builds?
P228	Platform stub fn — Same <code>fn process_is_root() -> bool</code> on Unix (real check) and Windows (always false). Show the <code>#[cfg(unix)] / #[cfg(not(unix))]</code> pair.
P229	Empty stub fn — <code>disable_spellcheck()</code> runs real code on macOS and <code>pub fn disable_spellcheck() {}</code> elsewhere. Why compile the module on all targets instead of gating the whole file?
P230	Target deps — Add <code>libc</code> only on Unix in <code>Cargo.toml</code> : show <code>[target.'cfg(unix)'.dependencies]</code> block.
P231	Integration layout — Draw tree for <code>tests/load_config.rs</code> calling <code>public load()</code> — what is invisible to the test?
P232	pub use prelude — Users should call <code>my_crate::connect</code> not <code>my_crate::internal::connect</code> — show re-export.
P233	Module docs — Top of <code>core/mod.rs</code> describes what the module owns. Show a <code>//!</code> inner doc comment (two lines) vs <code>///</code> on a single <code>pub fn</code> .
P234	doc hidden — When to mark helper <code>#[doc(hidden)]</code> on a public re-export surface?
P235	Capstone crate — Design gateway crate: serial feature, integration test, <code>///</code> on public <code>parse fn</code> — tree + TOML only.

C.12 Chapter 10 — Smart pointers and interior mutability

ID	Prompt
P236	Box why — When is <code>Box<T></code> better than <code>Vec<T></code> on the stack? Two cases.
P237	Recursive list — Draw memory for <code>Cons(1, Box::new(Cons(2, Nil)))</code> — stack vs heap boxes.
P238	Move out of Box — What happens after <code>let s = *box_string</code> ? When is that idiomatic vs keeping the <code>Box</code> ?
P239	Trait object box — Three plugin types implement <code>Plugin</code> — sketch <code>Vec<Box<dyn Plugin>></code> factory; why not <code>Vec<Plugin></code> ?
P240	Rc cycle — Explain why <code>Rc</code> cycles leak; contrast with Python reference cycles and <code>Weak</code> fix.
P241	Handle vs deep clone — Audit snippet with <code>Rc<String></code> and both <code>Rc::clone</code> and <code>(*rc).clone()</code> — label cost of each.
P242	Arc vs Rc — Thread spawn with <code>Rc</code> — show error; fix with <code>Arc</code> ; explain atomic count overhead in one sentence.
P243	Weak upgrade — Parent/child with <code>Rc</code> parent and <code>Weak</code> child back-ref — I sketch types; you explain cycle break.
P244	strong_count debug — <code>strong_count</code> stays at 2 after I thought I dropped all refs — list 5 places handles hide.
P245	RefCell trap — Show double borrow_mut panic; fix with scoped borrows.
P246	Immutable then mut — Hold <code>let r = cell.borrow()</code> and call <code>borrow_mut</code> — explain panic; fix with nested block.
P247	Cell vs RefCell — Counter <code>u32</code> vs <code>Vec</code> cache — I pick <code>Cell</code> or <code>RefCell</code> each; you correct.
P248	Rc RefCell graph — Two nodes share <code>Rc<RefCell<Node>></code> — one updates field, one reads — sketch borrow rules on one thread.

ID	Prompt
P249	Compile vs runtime — Same overlapping-mut pattern: show compile error with <code>&mut</code> and runtime panic with <code>RefCell</code> side by side.
P250	Deref coercion — Why does <code>fn takes_str(s: &str)</code> accept <code>&String</code> , <code>&Box<String></code> , and <code>&Rc<String>?</code> Trace steps.
P251	Drop order — Three <code>Drop</code> structs in one function — I predict print order; you confirm reverse declaration rule.
P252	Rc last handle — Two <code>Rc</code> clones dropped at different times — when does inner <code>Drop</code> run? Step through with <code>println</code> in <code>Drop</code> .
P253	Drop panic — Explain double-panic abort if <code>Drop</code> panics during unwind — link to Ch8.
P254	Arc Mutex sketch — Diagram thread-safe cache with <code>Arc<Mutex<HashMap>></code> — no full code.
P255	Pick pointer — Five scenarios (AST node, thread cache, GUI callback graph, config string, plugin list) — I pick <code>Box/Rc/Arc/RefCell/Weak</code> ; you grade.
P256	Java heap map — Map Java ‘everything is reference’ to Rust ownership — when <code>Arc</code> , when plain <code>&</code> , when neither.
P257	Refactor to smart ptr — I paste struct with <code>Box</code> , <code>Rc</code> , or raw <code>Vec</code> tree — you suggest minimal smart pointer fix and justify.
P258	Leak hunt — Sketch <code>Rc</code> cycle in observer pattern; refactor one edge to <code>Weak</code> and explain count after each drop.

C.13 Chapter 11 — Collections

ID	Prompt
P259	Pick collection — Five tasks (dedup, sorted range scan, FIFO queue, index by id, min-key lookup) — I pick Vec/HashMap/BTree/VecDeque/HashSet each.
P260	Hash vs BTree — Same 10k insert + range scan workload — when HashMap wins vs BTreeMap; one sentence each.
P261	Queue anti-pattern — Review while !v.is_empty() { v.remove(0) } — cost and fix with VecDeque.
P262	Loop port — Rewrite C-style indexed loop as iterator chain; preserve behavior.
P263	get vs index — Four access patterns — I pick [i] vs .get(i) vs .get_mut vs if let Some.
P264	sort dedup — Dedup [3,1,4,1,5] wrong vs right — show sort + dedup pipeline.
P265	retain vs filter — Remove evens in-place vs new Vec — compare retain and filter().collect().
P266	Borrow push trap — Explain let r = &v[0]; v.push(1) error; fix with scope.
P267	entry drill — Word frequency from Vec<&str> using only .entry — no double lookup.
P268	or_insert_with — Lazy cache: expensive Vec built once per key — sketch with or_insert_with.
P269	HashMap merge — Two maps of scores — merge by max per key; iterator + entry style.
P270	insert overwrite — Track old value on port remap 502 -> 503 using insert return.
P271	Set ops — Tags on two records — union, intersection, difference with HashSet.
P272	Stable dedup — Unique String lines preserving first-seen order — no HashSet-only collect.
P273	BTree range — List keys in BTreeMap<u32, _> between 100 and 200 inclusive.

ID	Prompt
P274	Windows — Detect rising edges in <code>Vec<f64></code> with <code>.windows(2)</code> ; extend to <code>.windows(3)</code> for slope.
P275	chunks vs windows — Parse byte stream into 4-byte frames — <code>chunks(4)</code> vs <code>windows(4)</code> when?
P276	collect types — Three <code>collect()</code> calls that need type hints — fix with <code>turbofish</code> .
P277	Duplicate keys — <code>collect</code> to <code>HashMap</code> from duplicate-key pairs — predict final map; explain overwrite rule.
P278	Performance myth — Do Rust iterators optimize to loops? When might they not?
P279	Capacity hint — Read 1M lines into <code>Vec</code> — when <code>with_capacity</code> matters; rough sizing rule.
P280	Capstone — Design in-memory store: register id <code>u16</code> -> last reading <code>f64</code> , need range scan by id — pick map type, list three API methods, no impl.

C.14 Chapter 12 — Closures and the `Fn` traits

ID	Prompt
P281	Fn quiz — Four closures: I label each <code>Fn</code> / <code>FnMut</code> / <code>FnOnce</code> ; you correct and explain capture.
P282	move drill — Thread spawn snippet missing <code>move</code> — show compile error and fix.
P283	Iterator chain — <code>.filter</code> closure that uses <code>&config</code> — why <code>Fn</code> not <code>FnMut</code> ?
P284	RefCell bump — Closure mutates <code>RefCell<u32></code> counter — which <code>Fn</code> trait and why?
P285	fn vs closure — When can you pass <code>fn()</code> vs <code>impl Fn()</code> to the same helper?
P286	Box dyn Fn — Store heterogeneous callbacks in a <code>Vec</code> — sketch trait object version.

ID	Prompt
P287	Return closure — Write <code>make_multiplier(f: f64) -> impl Fn(f64) -> f64</code> and explain move.
P288	Callback registry — Three log filters in <code>Vec<Box<dyn Fn(&str) -> bool>></code> — all must match signature; I add a wrong one; you fix.
P289	Loop move trap — Building <code>Vec<Box<dyn Fn()>></code> in a for loop over <code>String</code> — show move error and clone fix.
P290	Double reference — Fix <code>.iter().filter(x ...)</code> type error on <code>Vec<String></code> — show <code> s </code> vs <code> \&s </code> patterns.
P291	sort_by — Sort <code>Vec<(String, u32)></code> by count descending with <code>sort_by</code> closure.
P292	sort_by_key — Same sort with <code>sort_by_key</code> — when is key extraction cleaner?
P293	retain valid — Drop invalid <code>SensorReading</code> rows in-place with <code>.retain</code> — <code>FnMut</code> bound.
P294	for_each vs for — Same side-effect loop twice: <code>for</code> vs <code>.for_each(...)</code> — style tradeoffs.
P295	Fn bound too strict — Helper takes <code>impl Fn()</code> but caller passes closure that mutates — fix signature.
P296	Thread move — spawn closure borrows <code>String</code> — show error without move and fix.
P297	Send on Box Fn — When does <code>Box<dyn Fn() + Send></code> matter for thread pool callbacks?
P298	Capstone — Pipeline: read lines, filter non-empty, parse <code>u16</code> , sum — all with closures; I write; you review <code>Fn</code> bounds.

C.15 Chapter 13 — Standard traits and conversions

ID	Prompt
P299	Debug vs Display — When derive Debug only vs implement Display for a CLI status line?
P300	Redacted Debug — Struct with <code>api_key: String</code> — sketch manual Debug with [REDACTED].
P301	Pretty debug — Same struct with <code>{:#?}</code> vs <code>{:?}</code> — when does pretty-print help in tests?
P302	Display impl — Implement Display for <code>Port(u16)</code> showing <code>Port(8080)</code> — I write <code>fmt</code> , you review.
P303	Default enum — Three-variant mode enum — derive Default with <code>#[default]</code> on Auto; show update syntax.
P304	Derive set quiz — Map key, log line, sortable row, error enum — I list derives each needs.
P305	Eq on floats — Struct with <code>f64</code> field — show Eq derive failure; three fixes.
P306	HashMap key — Why does <code>UserId(String)</code> need Eq + Hash for HashMap keys?
P307	Ord sort — Sort <code>Vec<MyRecord></code> by timestamp field — bounds needed on <code>MyRecord</code> ?
P308	From chain — <code>String -> MyLabel</code> via <code>From</code> ; add <code>From<&str></code> without duplicating logic.
P309	TryFrom port — Port validation with custom enum error <code>OutOfRange</code> instead of <code>&str</code> .
P310	parse vs TryFrom — When <code>s.parse::<u16>()</code> vs <code>u16::try_from(x)</code> vs custom <code>FromStr</code> ?
P311	FromStr type — Parse <code>host:port</code> into struct — sketch <code>FromStr</code> with split and two parse steps.
P312	Silent cast trap — Show <code>70000i32</code> as <code>u16</code> vs <code>TryFrom</code> — predict values; argue for validation.
P313	From in errors — Wire <code>ParseIntError</code> into <code>AppError</code> with <code>From</code> so <code>? works</code> — list impl only.

ID	Prompt
P314	AsRef drill — Rewrite three functions taking <code>&String</code> to <code>impl AsRef<str></code> .
P315	AsRef bytes — Function logging wire payload — signature with <code>impl AsRef<[u8]></code> ; accept <code>Vec, &[u8]</code> , array.
P316	Borrow lookup — Explain <code>HashMap<String, V>.get(&str)</code> — role of <code>Borrow<str></code> .
P317	Cow API — Normalize slug: accept <code>Cow<str></code> , return borrowed if already valid else owned.
P318	into_owned — When caller needs <code>String</code> after your <code>Cow</code> helper — where call <code>into_owned()</code> ?
P319	to_owned vs clone — Three snippets: <code>&str->store</code> , <code>&String->store</code> , <code>already-String</code> — pick <code>to_owned</code> , <code>borrow</code> , or <code>move</code> ; explain each.
P320	Newtype Display — Wrap <code>Vec<u8></code> as <code>HexBytes</code> — implement <code>Display</code> without orphan violation.
P321	Derive audit — Config with secrets + TOML — list safe vs unsafe derives.
P322	Mini crate API — Public <code>HostPort { host, port }</code> with <code>Display</code> , <code>TryFrom<&str></code> for <code>host:port</code> — list <code>impl</code> blocks only.
P323	Capstone — Design public API for <code>RateLimit { max: u32, window_secs: u64 }</code> : parsing, display, equality — traits only, no bodies.

C.16 Chapter 14 — Multithreading

ID	Prompt
P324	Race quiz — Which snippets are data races in C++ but rejected by Rust compiler?
P325	Move fix — Show <code>spawn</code> without <code>move</code> that fails; fix with <code>move</code> or <code>clone</code> .

ID	Prompt
P326	Join panic — Worker panics; rewrite main to match join() and keep supervisor alive.
P327	Detached threads — Why is mem::forget(handle) after spawn dangerous? Better pattern?
P328	Channel design — Worker pool with mpsc: I describe throughput; sketch thread count + channel shape.
P329	Drop tx footprint — Main exits before worker sends — diagram who holds tx/rx.
P330	Multiple producers — Clone tx to two workers; main receives merged stream — sketch code.
P331	Bounded vs unbounded — When does unbounded mpsc blow memory in automation? Bounded alternative?
P332	Mutex vs RwLock — Read-heavy sensor cache — pick primitive and why.
P333	Hold lock briefly — Refactor bad code that calls network I/O while holding Mutex lock.
P334	Deadlock sketch — Two mutexes, lock order A then B vs B then A — show hang scenario.
P335	Poison recovery — Thread panics holding lock; show PoisonError and into_inner() recovery.
P336	Send fix — I try to move Rc into thread; show fix with Arc.
P337	RefCell trap — Why Arc<RefCell<T>> is not Sync; fix pattern for shared mutation.
P338	PLC gateway layout — Sketch poll thread + command channel + main supervisor; no code over 40 lines.
P339	Lock latency — Modbus cycle 20 ms — max time holding mutex for register cache update?
P340	Level ladder recap — Explain Levels 1-6 in one paragraph each for a Java teammate.
P341	RwLock cache — Read-heavy sensor map — sketch Arc<RwLock<HashMap>>; when does write starve readers?

ID	Prompt
P342	Mutex vs RwLock — Same cache with 50% writes — pick Mutex or RwLock and justify in two sentences.
P343	OnceLock init — <code>get_or_init</code> fails second init with different value — show <code>OnceLock</code> behaviour.
P344	Scope borrow — Parallel sum over <code>Vec<[u8;512]></code> with <code>thread::scope</code> — why plain <code>spawn</code> fails on <code>&chunk</code> .
P345	Poison recovery — Writer thread panics holding <code>RwLock</code> — show poisoned <code>read()</code> error and recovery options.
P346	Capstone sync — Design: lazy config (<code>OnceLock</code>), shared cache (<code>RwLock</code>), scoped batch workers — list types only.

C.17 Chapter 15 — Atomics

ID	Prompt
P347	Threads vs async atomics — Same <code>Arc<AtomicBool></code> in <code>thread::spawn</code> vs <code>tokio::spawn</code> — what differs, what stays the same?
P348	Data race vs logic race — Define both; classify lost static mut increment vs <code>fetch_add</code> .
P349	Ch14 port — Rewrite Mutex counter (Ch14 L4) as <code>AtomicUsize</code> ; when is each better?
P350	Ordering quiz — Shutdown flag + published config pointer — pick orderings; justify.
P351	When Relaxed lies — Metrics OK, config handoff broken — show <code>Release/Acquire</code> fix.
P352	Fence intuition — Draw happens-before for <code>Release store</code> + <code>Acquire load</code> (Level 4).
P353	Relaxed polls vs version — Why is Relaxed OK for Level 2 polls but wrong for Level 4 version? One sentence each.

ID	Prompt
P354	Counter port — Cap counter with CAS loop; explain spurious compare_exchange_weak failure.
P355	ABA problem — Explain ABA in 80 words for pointer compare_exchange — no full queue.
P356	Retry loop — Show single-shot CAS anti-pattern; fix with loop.
P357	Double-checked locking — Show broken lazy init with atomics; fix with OnceLock or explain why not hand-roll.
P358	Race quiz — Mark 6 snippets: safe atomic, UB static mut, ordering bug, needs Mutex.
P359	Visibility story — Writer updates port then Relaxed flag — what can reader see?
P360	When not — Three cases atomics are the wrong tool; prefer channels or Mutex.
P361	Mutex vs RwLock — Read-heavy sensor cache — atomic counter vs RwLock vs Mutex.
P362	Vec push — Why many threads cannot push to shared Vec; three fixes.
P363	Profile-first — Gateway uses Mutex for counter; profiler shows lock contention — minimal atomic refactor.
P364	Channel vs atomic flag — When is crossbeam/bounded channel + shutdown better than lone AtomicBool?
P365	Gateway metrics — Design Gateway { polls, shutdown } for async; orderings per field.
P366	False sharing — Two hot atomics on same cache line — problem and mitigation in 60 words.
P367	Lock-free queue — Why this chapter says don't hand-roll; name crates/patterns instead.

C.18 Chapter 16 — Async and Tokio

ID	Prompt
P368	Future diagram — Draw state machine for async fn with two <code>.await</code> points.
P369	Poll vs await — Who polls the Future — caller, executor, or Tokio runtime?
P370	Executor vs thread — One paragraph: cooperative async vs preemptive OS threads.
P371	Forgotten await — Show unused Future bug; fix with <code>.await</code> or <code>spawn</code> .
P372	Tokio scaffold — Minimal <code>#[tokio::main] + spawn + join</code> handle — explain each line.
P373	Spawn vs await — What if main drops join handle without <code>.await</code> ?
P374	Join handle ?? — Explain the ?? on <code>handle.await</code> in Level 6 — outer join Result vs inner <code>run_worker</code> Result.
P375	Bad join timing — Walk through Level 5 bad path: why does a 10 ms task wait ~100 ms when paired with <code>thread::sleep</code> ?
P376	Ch15 port — Compare Level 6 to Ch15 L6 — what changes with <code>tokio::spawn</code> , what stays the same?
P377	Supervisor timeline — Trace Level 6 step by step: when does the worker stop, and why must it use <code>tokio::time::sleep</code> not <code>thread::sleep</code> ?
P378	join vs select — Two slow tasks: when <code>join!</code> vs <code>select!?</code> One automation example each.
P379	select! scenario — Cancel slow request when fast path returns — full <code>select!</code> sketch.
P380	Timeout drill — Wrap Modbus read async fn with 100 ms timeout; handle <code>Elapsed</code> .
P381	Cancellation hygiene — What runs when <code>select!</code> drops the losing branch? Modbus read vs cache hit example.
P382	Blocking fix — Audit async snippet with <code>thread::sleep</code> , <code>std::fs::read</code> ; fix each.
P383	spawn_blocking — When <code>tokio::fs</code> vs <code>spawn_blocking</code> for config file read?

ID	Prompt
P384	Tcp echo — Expand Async I/O outline to multi-connection echo with spawn per accept.
P385	async vs thread — 1000 Modbus polls — argue async vs thread pool for latency.
P386	200 TCP connections — One process — async vs thread-per-connection memory story.
P387	When thread enough — Single serial port poll — justify thread loop over Tokio.
P388	Async sleep in worker — Why must Level 6's poll loop use <code>tokio::time::sleep().await</code> on every iteration, not <code>thread::sleep</code> ?
P389	tokio Mutex — Rewrite bad <code>std::sync::MutexGuard</code> across <code>await</code> with <code>tokio::sync::Mutex</code> .

C.19 Chapter 17 — Metaprogramming

ID	Prompt
P390	Expansion order — List compiler phases from tokens to LLVM; where do macros run?
P391	Tokens vs types — Why can a macro compile but expanded code fail? One example.
P392	Follow-set why — Explain in 80 words why <code>\$a:expr = \$b:expr</code> is forbidden in matchers.
P393	Scope honesty — List 5 metaprogramming topics Ch17 skips and where to learn each.
P394	Trace checklist — Give 6 reasons macro code is hard to trace and one mitigation each.
P395	Macro vs fn — Rewrite macro as generic fn if possible; when impossible, say why.
P396	Fragment picker — I describe a DSL shape; you pick <code>expr/ident/tt</code> for each slot.

ID	Prompt
P397	Expr equals fix — Fix <code>set_reg!(\$addr = \$val)</code> matcher; show two valid surface syntaxes.
P398	For after expr — Why is <code>\$e:expr</code> for <code>\$i:ident</code> in <code>\$r:expr</code> illegal? Show legal <code>\$p:pat</code> in <code>\$r:expr</code> <code>foreach</code> macro.
P399	Double-brace fix — Fix <code>poll_twice!</code> macro that mixes <code>let</code> and <code>for</code> for use in <code>let x = poll_twice!()</code> .
P400	Trailing comma — Explain <code>\$(x),*</code> vs <code>\$(x),+</code> on empty input; show failing and fixed macro.
P401	Hygiene — Explain macro hygiene in 60 words with <code>\$crate</code> mention.
P402	Register DSL — Extend <code>register_map!</code> with a third register; explain <code>stringify!</code> arm.
P403	Debug expand — Walk me through <code>cargo expand</code> on derive <code>Debug</code> output (conceptual).
P404	Clone expand — Show conceptual expanded <code>impl Clone</code> for <code>struct</code> with two <code>i32</code> fields.
P405	Enum vs struct — How does derived <code>PartialEq</code> differ for <code>enum</code> vs <code>struct</code> ? Sketch match arms.
P406	Field bound failure — <code>Struct</code> with <code>Mutex<i32> field + #[derive(Clone)]</code> — quote error and fix.
P407	Redacted Debug — When hand-write <code>Debug</code> instead of <code>derive</code> on command <code>enum</code> with secrets.
P408	Copy vs Clone quiz — Classify 8 types: <code>Copy</code> , <code>Clone</code> only, or neither; justify.
P409	Hot-loop clone audit — Audit <code>poll</code> loop with <code>.clone()</code> each tick; suggest <code>move/Arc/borrow</code> .
P410	Arc vs derive Clone — Explain cheap <code>Arc clone</code> vs deep <code>String clone</code> with one snippet.
P411	derive need — List derives I want for <code>config struct</code> loaded from <code>TOML</code> — justify each.

ID	Prompt
P412	Serde rename — Field <code>poll_ms</code> in JSON as <code>pollIntervalMs</code> — show attr; trap on refactor.
P413	thiserror vs manual — Same error enum: count lines derive vs hand-written (Ch8 style).
P414	Float Eq trap — Show <code>#[derive(Eq)]</code> on <code>f64</code> field failure; two fixes from Ch7.
P415	cargo expand walkthrough — Step-by-step: install, run, read output for one derive.
P416	In expansion of — Decode a 3-note compiler error chain from nested macro + derive.
P417	tt vs expr escape — When switch matcher from <code>expr</code> to <code>tt</code> ; tradeoffs in 80 words.
P418	Three-layer trace — Derive inside attribute inside <code>macro_rules</code> — how to debug layer by layer.
P419	Port to const fn — Replace tiny numeric macro with <code>const fn</code> ; when macro still needed?
P420	env vs var — Compare <code>env!</code> , <code>option_env!</code> , <code>std::env::var</code> — table with one use case each.
P421	include_str config — Embed default TOML with <code>include_str!</code> ; deserialize at startup sketch.
P422	Macro vs fn audit — Mark 6 snippets: should be macro, derive, or plain <code>fn</code> — justify.
P423	When not proc macro — Three scenarios where <code>proc macro</code> is overkill; alternative each.
P424	Minimal derive set — Gateway config + error + CLI: smallest derive list that still ships.
P425	Trap quiz — Mark 8 snippets: empty <code>+</code> , double brace, Default enum, Arc Clone, <code>env!</code> , duplicate <code>impl</code> , <code>cfg</code> macro, <code>serde</code> rename.

ID	Prompt
P426	Duplicate register DSL — Design compile-time error for duplicate keys in <code>register_map!</code>
P427	Serde refactor test — Integration test plan after renaming TOML field with <code>serde</code> attrs.
P428	Derive vs decorator — I come from Python/Java. Explain why <code>#[derive(Debug)]</code> is not a decorator or annotation that runs at call time — what actually happens at compile time?
P429	Three kinds quiz — Classify 8 snippets: <code>derive proc macro</code> , <code>attribute proc macro</code> , <code>function-like macro</code> , <code>compiler attribute</code> (<code>#[inline]</code> , <code>#[cfg]</code>), <code>field meta</code> parsed inside a <code>derive</code> (<code>#[serde(rename)]</code>).
P430	tokio::main expand — Sketch the conceptual expansion of <code>#[tokio::main]</code> <code>async fn main() { ... }</code> . Why is this an <code>attribute proc macro</code> , not <code>#[derive]</code> ?
P431	Custom attribute inputs — For <code>#[my_attr(some = \"config\")] fn poll() { ... }</code> , what two token streams does the <code>proc macro</code> receive? Give two things the macro might emit.
P432	Field attr vs item attr — Contrast <code>#[serde(rename = \"pollIntervalMs\"]</code> on a <code>struct</code> field vs <code>#[tracing::instrument]</code> on <code>fn poll</code> — same <code>proc-macro</code> kind or not?
P433	Custom attribute when — Three scenarios (<code>poll-loop tracing</code> , <code>type-level JSON mapping</code> , wrapping <code>main</code> with a <code>runtime</code>). Pick: <code>custom attribute proc macro</code> , <code>derive</code> , or <code>plain helper fn</code> — justify each.
P434	Runtime myth — Does <code>#[tracing::instrument]</code> or <code>#[test]</code> run every time I call the function? Explain what runs at compile time vs run time.
P435	DSL sketch — Design tiny <code>command!</code> macro for CLI subcommands — tokens only.

ID	Prompt
P436	Modbus table macro — Spec register table macro generating lookup + const max address.
P437	Derive soup review — I paste 40-line struct with 12 derives; trim to minimal set with reasons.

C.20 Chapter 18 — Unsafe

ID	Prompt
P438	Invariant list — For raw pointer to buffer + length, list 5 invariants a safe wrapper must enforce.
P439	Soundness — Explain ‘safe Rust can’t cause UB’ vs <code>unsafe</code> — one paragraph; include unsound safe wrapper example.
P440	Scope honesty — List 6 topics Ch18 skips and where to learn each (nomicon, Miri, Pin, ...).
P441	Aim table — Fill: why <code>Vec</code> needs <code>unsafe</code> internally while <code>push</code> stays safe for callers.
P442	Promise diagram — Draw safe API -> <code>unsafe block</code> -> invariants -> caller cannot UB; label soundness.
P443	*const vs &T quiz — Give 5 snippets: legal ref, needs <code>unsafe block</code> , compile error; I classify each.
P444	from_raw_parts design — Design fn <code>view_frame(ptr, len) -> Result<&[u8], Error></code> without <code>&'static</code> ; list invariants.
P445	Dangling audit — Show stack pointer used after drop; I explain UB; you show Miri-style symptom.
P446	Modbus buffer — Register table as <code>&[u8]</code> vs <code>from_raw_parts</code> — when is each idiomatic in a gateway?
P447	Hex preview port — Port Level 2 <code>as_hex_preview</code> to return <code>Result</code> on empty buffer; no <code>unwrap</code> .

ID	Prompt
P448	set_len contract — Document pre/post conditions for <code>set_len_unchecked</code> ; what breaks <code>as_slice</code> if violated?
P449	Send proof — I claim <code>Rc<*mut u8></code> is Send; you disprove with compiler error quote.
P450	SerialHandle Sync — When would <code>SerialHandle</code> need <code>unsafe impl Sync</code> vs <code>Arc<Mutex<...>></code> ? Two sentences each.
P451	Ch14 port — Rewrite Level 4 spawn example using only safe types — when is it impossible?
P452	Proc-macro boundary — Why do <code>serde/tokio crates</code> use <code>unsafe impl</code> you don't write? Link Ch17.
P453	FFI checklist — Checklist for calling a C Modbus library from a Rust binary; include panic and ownership rows.
P454	CString trap — Show <code>into_raw</code> forgotten <code>from_raw</code> leak; fix with RAII pattern sketch.
P455	Vendor SDK — Diagram ownership: Rust owns config, C owns connection, callback pointer — boxes and arrows only.
P456	serialport hide — Where does <code>unsafe</code> live in a typical serial crate vs my application code?
P457	CRC decision — C <code>crc16</code> vs Rust <code>crc</code> crate vs hand-rolled — decision tree for production gateway.
P458	Java JNI — Compare JNI pitfalls (refs, exceptions, pinning) to Rust FFI ownership rules.
P459	Miri — What is Miri and when should I run it relative to <code>unsafe</code> changes? Include one command.
P460	Trap quiz — Mark 6 snippets: safe, UB, ordering bug, needs Miri, needs Mutex, unsound safe API.
P461	Review rubric — 10-point code-review checklist for an <code>unsafe</code> PR in an automation repo.

ID	Prompt
P462	Test plan — Unit + Miri + integration tests for new extern 'C' wrapper — bullet list only.
P463	Avoid — Review use case: speed up JSON — unsafe vs simd-json vs algorithm; pick with justification.
P464	Borrow checker fight — I paste fight-the-borrow-checker code; you refactor to safe Rust without unsafe.
P465	static mut — Compare static mut counter vs AtomicUsize from Ch15 — UB vs defined behavior.
P466	PlcDriver API — Design safe PlcDriver Rust API over fictional extern 'C' — types, Result, no raw pointers in public API.
P467	Level ladder recap — Explain Levels 1-5 in one paragraph each for a Java teammate who knows JNI.

C.21 Chapter 19 — I/O and processes

ID	Prompt
P468	Trait refactor — Refactor file copy loop to generic copy<R: Read, W: Write>; discuss error propagation.
P469	read vs read_exact — Give 3 protocol shapes; I pick read loop vs read_exact each time; you verify.
P470	Cursor test — Write unit test for parse_kv_lines using Cursor — no filesystem.
P471	BufReader why — Explain in 60 words why BufReader matters for 10k-line log files.
P472	CSV tool — Spec for CLI: read two-column CSV, emit name=value; I implement with BufRead.
P473	Boundary errors — Sketch main mapping io::Error to exit code 1 with context path — no unwrap.

ID	Prompt
P474	read_to_string trap — When is read_to_string wrong for automation configs? Give size threshold rule.
P475	Packet layout — Add CRC byte to 4-byte packet; update encode/decode with XOR — show tests.
P476	Endian trap — Quiz: 3 scenarios pick LE vs BE for Modbus-style register.
P477	Bit field port — Java status int with flags — port to Rust encode with = and & masks.
P478	CRC upgrade — Replace XOR toy CRC with CRC-16-Modbus — outline steps, no full crate required.
P479	Command safety — Review shell=True style command; rewrite without shell when possible.
P480	Pipeline — Design program A program B using only Rust std (two processes, pipe).
P481	Exit status — Child exited 2 — how should gateway log and retry? Table: fatal vs transient.
P482	Env and cwd — Show Command with .env("PORT", "502") and .current_dir — when needed?
P483	Sync vs async pick — Three gateway designs: I pick sync thread vs Tokio per scenario; you justify.
P484	Serial debug — I get timeout on read; give systematic checklist (baud, cable, permissions).
P485	serialport traits — Explain how serialport maps to Read/Write — diagram only.
P486	Blocking in async — Show wrong std::fs::read inside async fn; fix with tokio::fs or spawn_blocking.
P487	Capstone scaffold — Generate module tree and function signatures for sensor_gateway; no bodies.
P488	Retry policy — Design exponential backoff for Modbus-style poll errors; Rust pseudocode.

ID	Prompt
P489	Log schema — Propose JSON log lines for sensor events with timestamp and error codes.
P490	GPIO next step — After serial works on Pi, outline migration to gpio-cdev for one LED.
P491	Code review — I paste capstone main loop; review for panic risks and missing flush.
P492	Gateway capstone — End-to-end: config file -> serial poll -> JSON log line — module list and error types only.
P493	Ch16 bridge — Same echo server: sketch sync thread version vs Tokio version — tradeoffs table.
P494	Path join — Build config path from HOME + .config/app.toml — show Path::join vs string concat trap.
P495	Env default — TIMEOUT env var with default 30 — var().unwrap_or_else pattern.
P496	Metadata guard — Reject config files over 1MB before read_to_string — sketch metadata().len() check.
P497	Stdin fallback — Port from argv or interactive prompt — one main with both paths.
P498	Line parser — BufRead lines, skip #, parse key=value — handle trailing newline.
P499	Capstone CLI — End-to-end: env path -> metadata check -> line parse -> print port — function list only.

C.22 Chapter 20 — Production standards

ID	Prompt
P500	Diff review — Paste a 30-line Rust PR; I mark each checklist row pass/fail with one sentence.

ID	Prompt
P501	Mega-error refactor — Split one <code>AppError</code> into <code>ValidateError</code> + <code>StorageError</code> ; show boundary mapping.
P502	Panic hunt — Find five panic sources in a snippet; replace with <code>Option/Result</code> .
P503	Clone audit — Remove three unnecessary clones by fixing signatures to <code>&str / &[T]</code> .
P504	Arc style — Rewrite <code>arc.clone()</code> call sites to <code>Arc::clone(&arc)</code> ; explain review benefit.
P505	Workspace.toml — Sketch root + two members; all shared deps use <code>{ workspace = true }</code> .
P506	Golden test — Parser output: one <code>assert_eq!(got, want)</code> + <code>pretty_assertions</code> ; no per-field asserts.
P507	Flaky test fix — Replace <code>thread::sleep</code> in test with injected <code>Clock</code> trait.
P508	Pre-merge gate — Checklist-only review of gateway <code>main.rs</code> + <code>lib.rs</code> — findings only.
P509	AI review prompt — One paragraph prompt for an assistant to verify the Ch20 rules on a Rust diff.

Total: 509 prompts (P001-P509).

C.23 By theme

Theme	IDs
Preface / study plan	P001-P005
Ownership / borrow (Ch 1)	P006-P038
Types / expressions (Ch 2)	P039-P054
Functions / methods (Ch 3)	P055-P072
Iterators (Ch 4)	P073-P100
Lifetimes (Ch 5)	P101-P112
Enums / match (Ch 6)	P113-P140
Structs / traits (Ch 7)	P141-P176

Theme	IDs
Errors / testing (Ch 8)	P177-P199
Modules / crates (Ch 9)	P200-P235
Smart pointers (Ch 10)	P236-P258
Collections (Ch 11)	P259-P280
Closures (Ch 12)	P281-P298
Standard traits (Ch 13)	P299-P323
Multithreading (Ch 14)	P324-P346
Atomics (Ch 15)	P347-P367
Async / Tokio (Ch 16)	P368-P389
Metaprogramming (Ch 17)	P390-P437
Unsafe (Ch 18)	P438-P467
I/O and processes (Ch 19)	P468-P499
Production standards (Ch 20)	P500-P509

C.24 See also

- `PLAYGROUND_GUIDE.md`
- `JAVA_PYTHON_RUST_MAP.md`
- `CONTENTS.md`